Mastering Cisco Catalyst 9000 Switches
Layer 2 and Layer 3

Packet Walk and Deep Dive Troubleshooting – Like a TAC Engineer

Lakshmi Ganesh Kondaveeti
Technical Leader - Enterprise

Sandesh Sawant
Technical Leader - Enterprise

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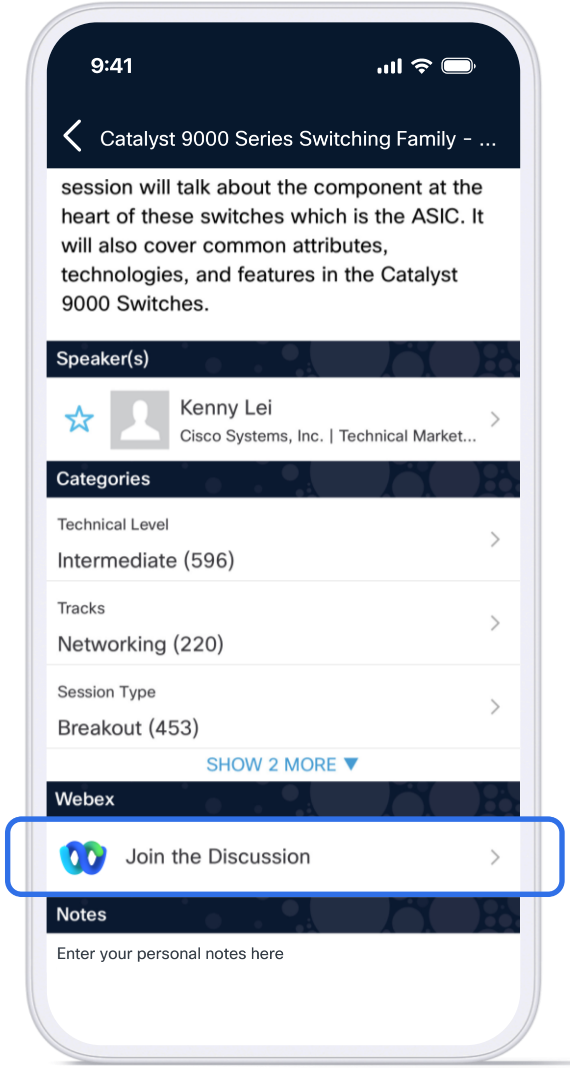


Agenda

Silicon 1 is The FUTURE

Enterprise
Campus

This session will make you FUTURE aware.

Who are we?

Sandesh [TL, CCIE - Creator/Teacher]

Professional Experience

- Over 15 years of TAC experience.
- Deep expertise in Enterprise Switching and Routing.
- Successfully handled 10,000+ cases across diverse customer environments.

Professional Role

- Senior Technical Leader – Enterprise LAN Switching.
- Driving engineering engagement and solving complex network challenges.

Key Project Highlight

- TAC Technical Skills Enablement Programs.
- Leading Silicon One collaboration between TAC and Business Units.

A seasoned TAC professional with 15+ years of experience driving stability, efficiency, and resilience in enterprise switching environments .*Outside the workplace, I prioritize travel and family time to maintain balance and perspective.*

Who are we?

Lakshmi Ganesh Kondaveeti (**LG**)

Professional Experience

- Overall 20 years with 15 years of TAC experience.
- Deep expertise in Enterprise Switching ,Routing and Cisco Catalyst Centre, SDA Fabric including wired and wireless
- Dual CCIE certified: Enterprise Infrastructure & Service Provider

Professional Role

- Senior Technical Leader - Catalyst Centre, SDA Fabric
- Passionate about building scalable, resilient, and future-ready networks

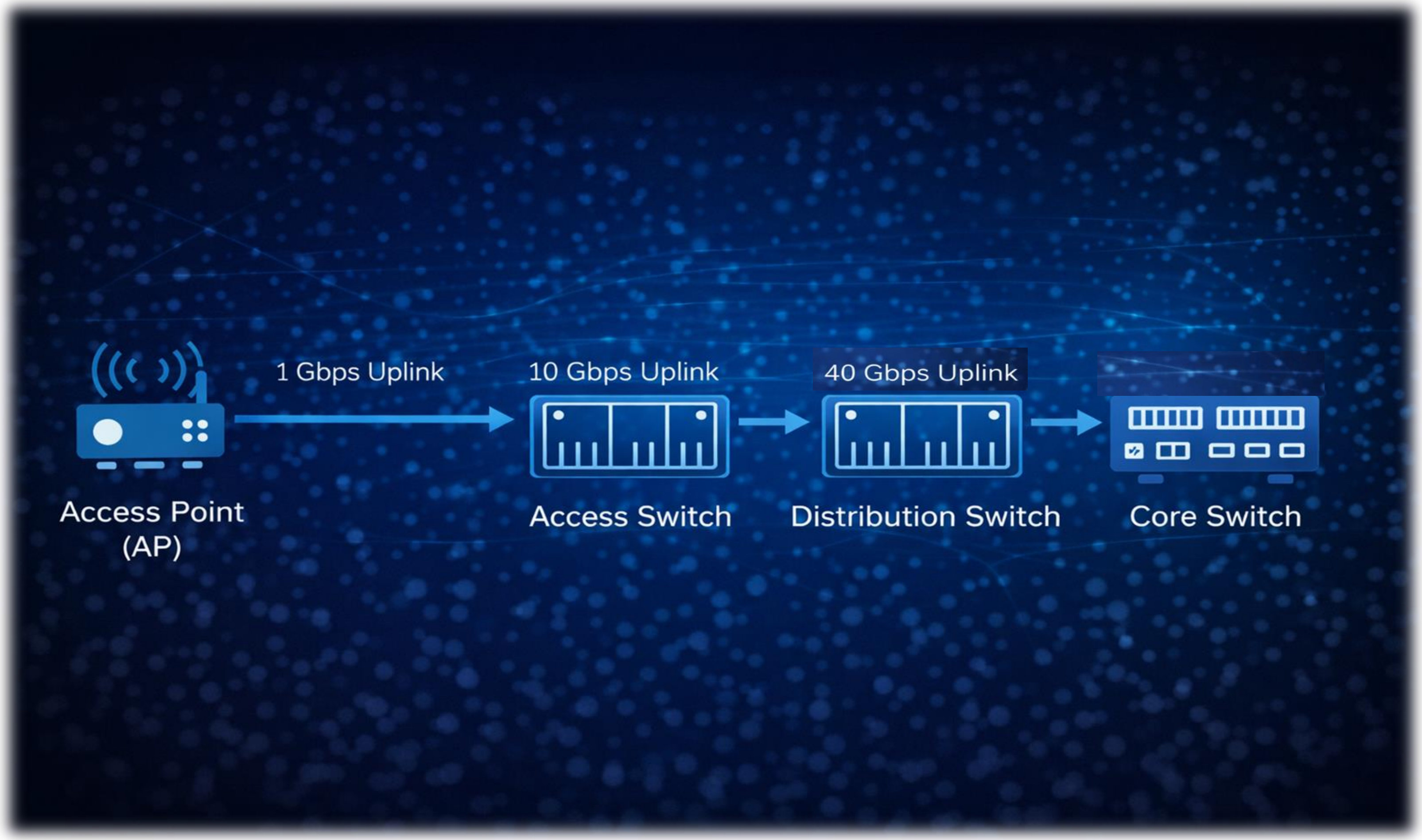
Key Highlight

- Cisco Inventor, 4 issued patents, with a strong focus on innovation and engineering excellence.

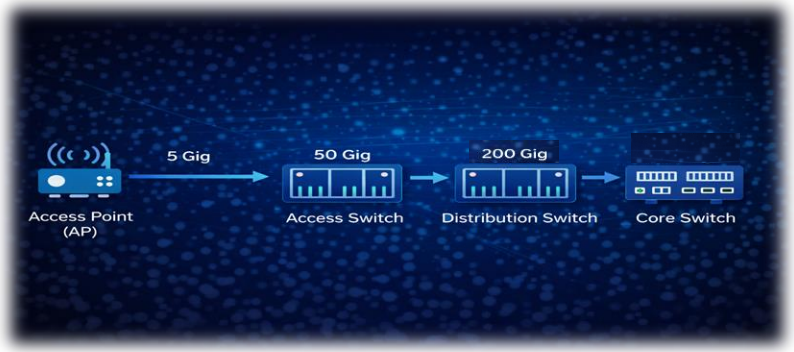
Modern Switching Trends Driving Need for New ASIC

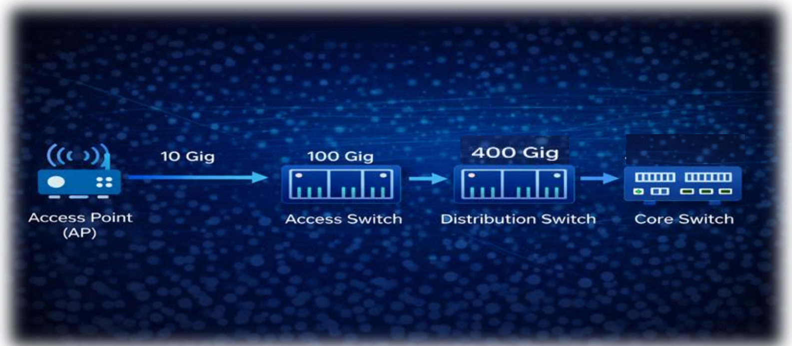
Enterprise Switching

Enterprise Switching



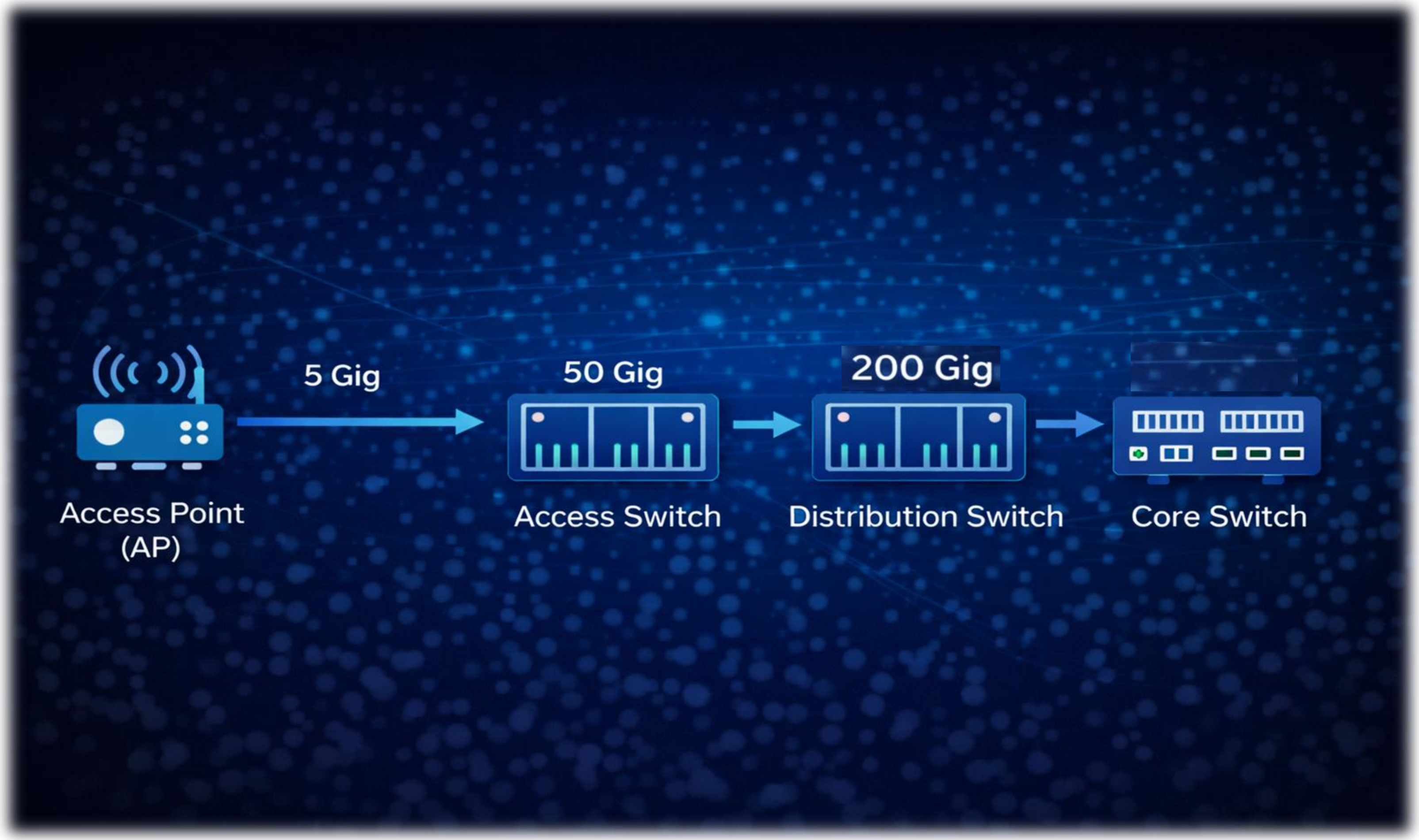
Yesterday

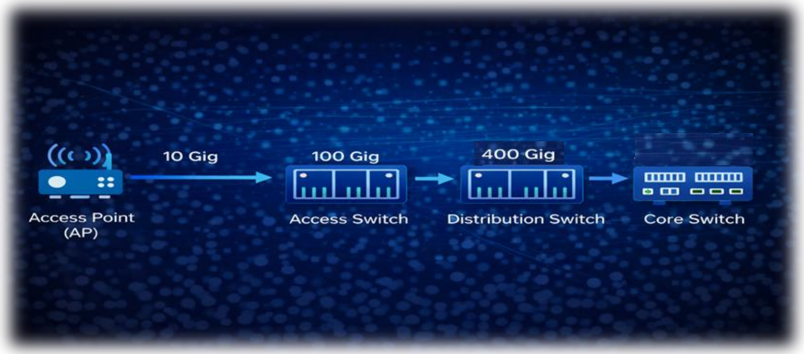




Enterprise Switching

Today

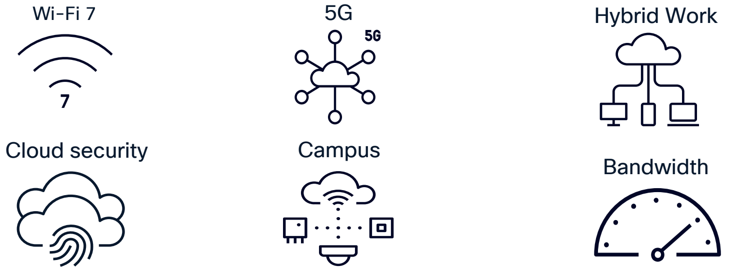


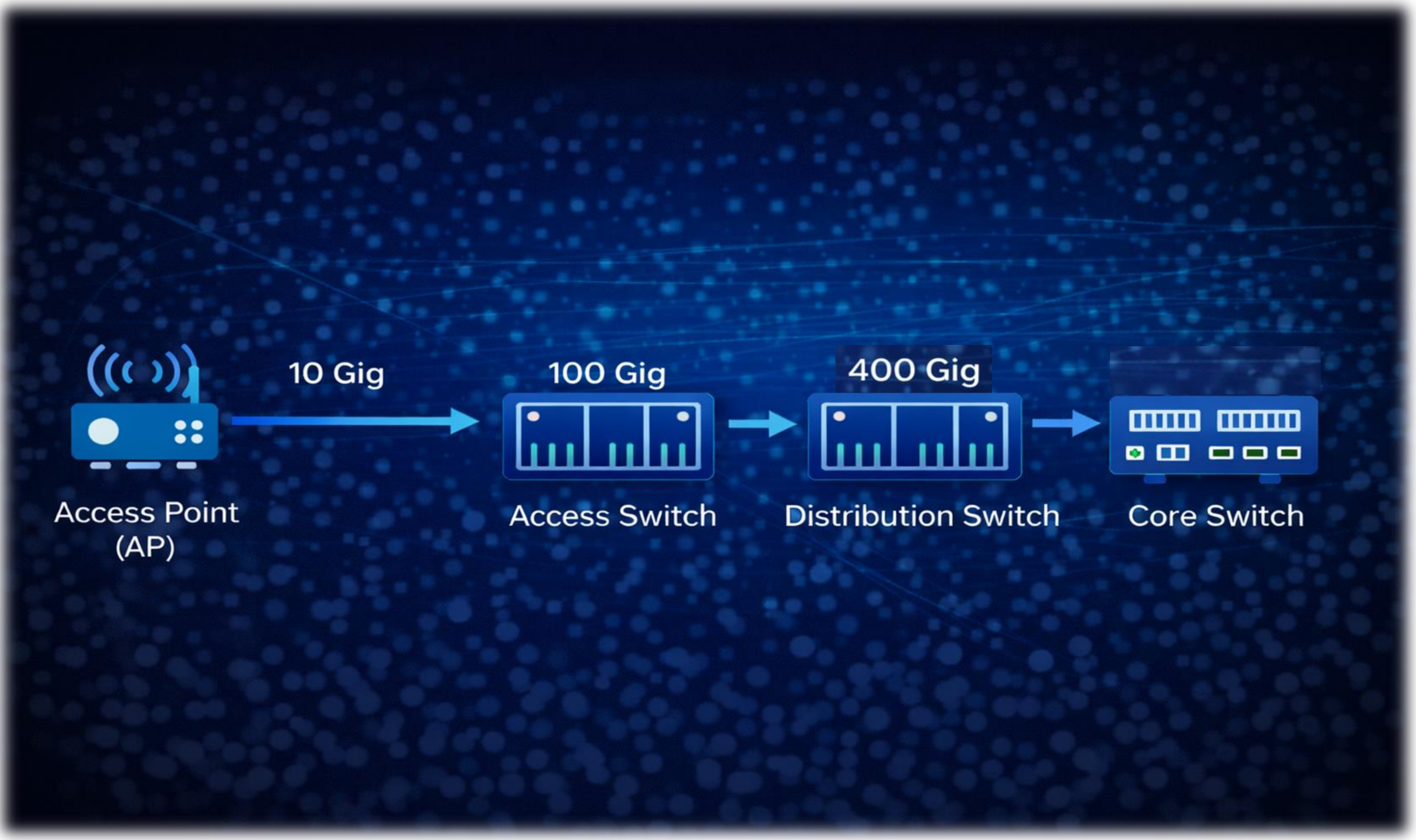


Enterprise Switching

Tomorrow

- Wi-Fi 6/6E/7, Private 5G, & Video needs on Access.
- Smart Buildings & **Hybrid Work** use cases driving **IoT and 90W PoE**
- End-User, Transport, & **Cloud Security** in campus
- **mGig** Speed in Access driving **25G/100G/400G** speed in **Distribution & Core**





Addressing the Campus Core + Edge Market

More than just "Speeds & Feeds"

System Capacity

- **Higher Port Speeds**
  enabling newer Ethernet speeds and types, such as 50G SFP and 400G QSFP
- **More Access Bandwidth**
  higher density of 802.11ac moving to 11ax, HD Video (1080p) moving to UHD (4K)
- **Non-Blocking**
  Core needs non-blocking (1:1) performance

System Scale

- **High-Scale Routing**
  support for full IPv4 and IPv6 Internet + LAN + VRF routing
- **Flexible Transport Options**
  support for MPLS-VPN, SD-Access and BGP-EVPN with 4K VRFs
- **Centralized Wireless**
  support for high MAC and ARP/NDP scale

Port Density

- **More Uplinks**
  support for multiple QSFP uplinks, with LR/ZR optics for redundant edge connections
- **More Downlinks**
  support for hundreds of SFP ports/breakouts
- **Port Options**
  mix of SFP & QSFP ports, at various interface speeds

QoS & Buffers

- **Low-Latency Buffers**
  support for local optimized low-latency shared memory
- **High-Bandwidth Memory**
  support large buffers for micro-burst & congestion
- **Virtual Output Queuing**
  eliminate head-of-line blocking, with dedicated output queues for every egress port

Flexibility

- **Programmable ASIC**
  Enable new features.
- **Model-driven Microcode/APIs**
  hardware functions use
- **App Hosting Infrastructure**
- **Architecture-level HA**
  L2/L3 redundancy with Stackwise Virtual and GIR

Core + Distro

Core + Edge

Silicon 1 is The FUTURE

Enterprise Campus

This session will make you FUTURE aware.

Doppler And Polaris Architecture

What is IOS XE? What is Polaris?

- IOS-XE = Linux based modular operating system. IOS & other processes operate as daemons (IOSd = Linux background process)
- 2 flavours of IOS-XE 15.x in the past:-

  BinOS for Routers: ASR, ISR, CSR

  Nova for Catalyst Switches: Cat4k, Cat3850/3650

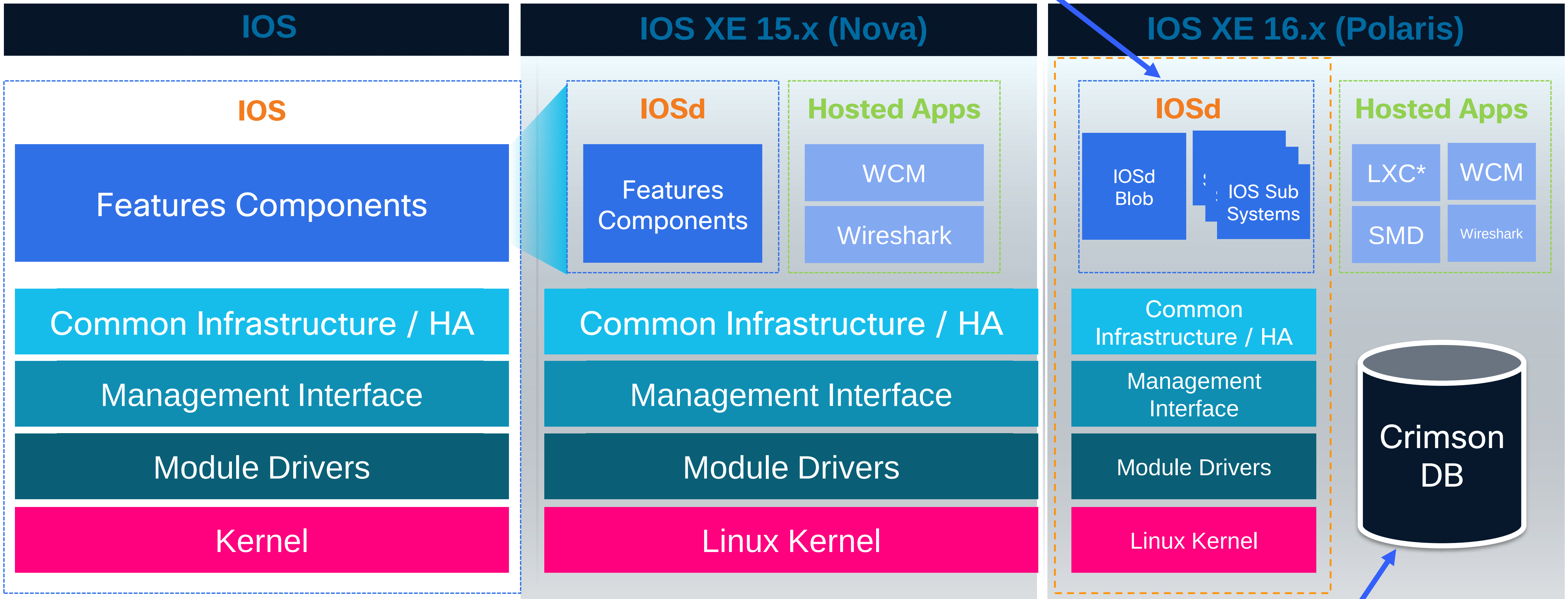
- Polaris IOS XE 16.x. It is a consolidation of the 2 flavours and is based on the BinOS variant. Polaris runs on Catalyst switches, routers, & next generation wireless controllers & IoT platforms
- Polaris software is designed to be platform agnostic. It is only the bottom layers that will need to be adapted to new hardware

IOS XE Evolution

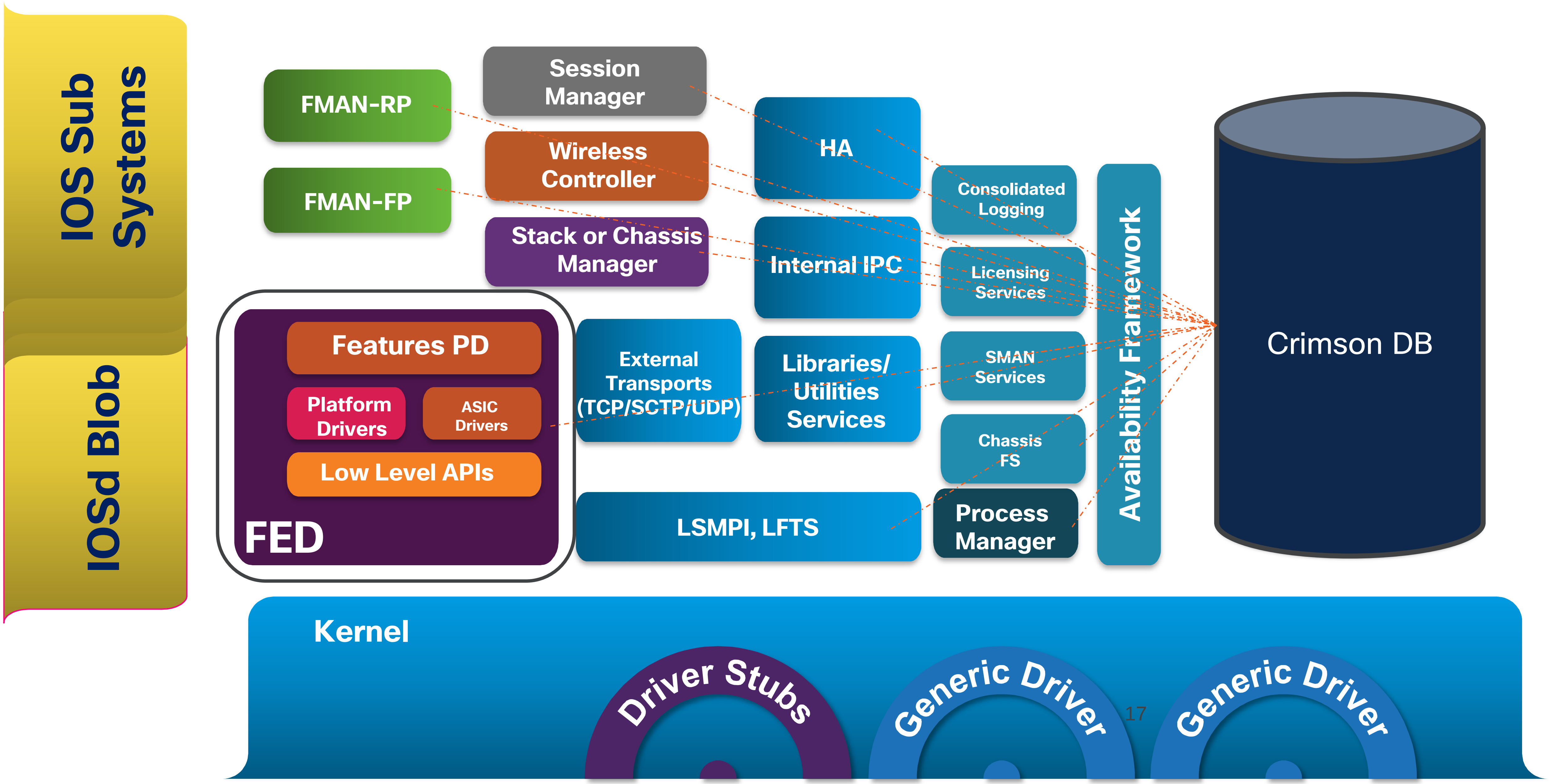
- Same Look & Feel, More Powerful Architecture

More modular IOS is broken into more sub systems within IOSd for future bug patching

LXC = Linux Containers
WCM = Wireless Client Manager
SDM = Session Manager Daemon

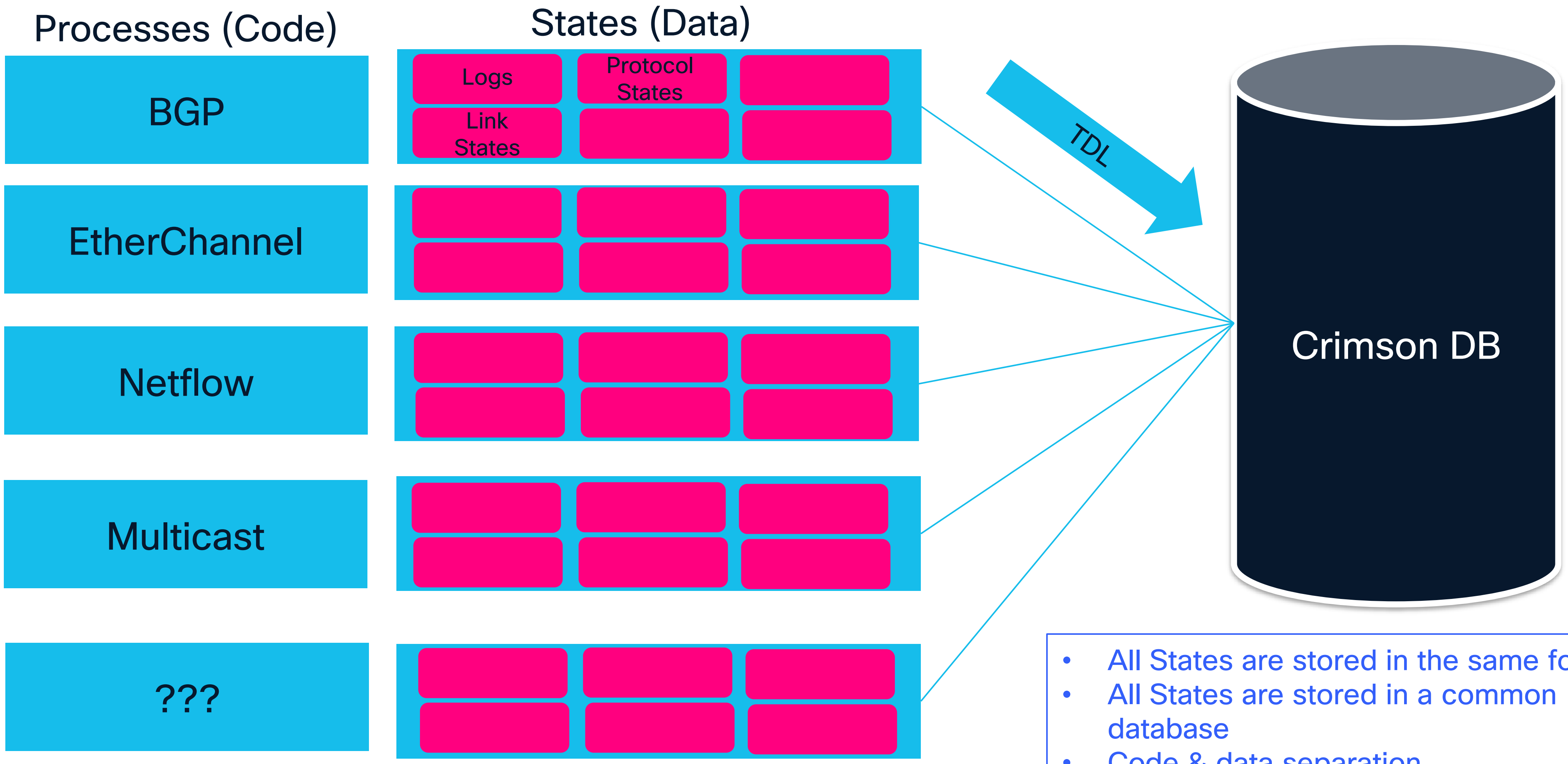


Crimson Database stores operational state of all operating system features in a consistent format



IOS Sub Systems
IOSd Blob
FMAN-RP
FMAN-FP
Session Manager
Wireless Controller
Stack or Chassis Manager
HA
Internal IPC
Consolidated Logging
Licensing Services
SMAN Services
Chassis FS
Process Manager
Availability Framework
Crimson DB
Features PD
Platform Drivers
ASIC Drivers
Low Level APIs
FED
External Transports (TCP/SCTP/UDP)
Libraries/ Utilities Services
LSMPI, LFTS
Kernel
Driver Stubs
Generic Driver
Generic Driver

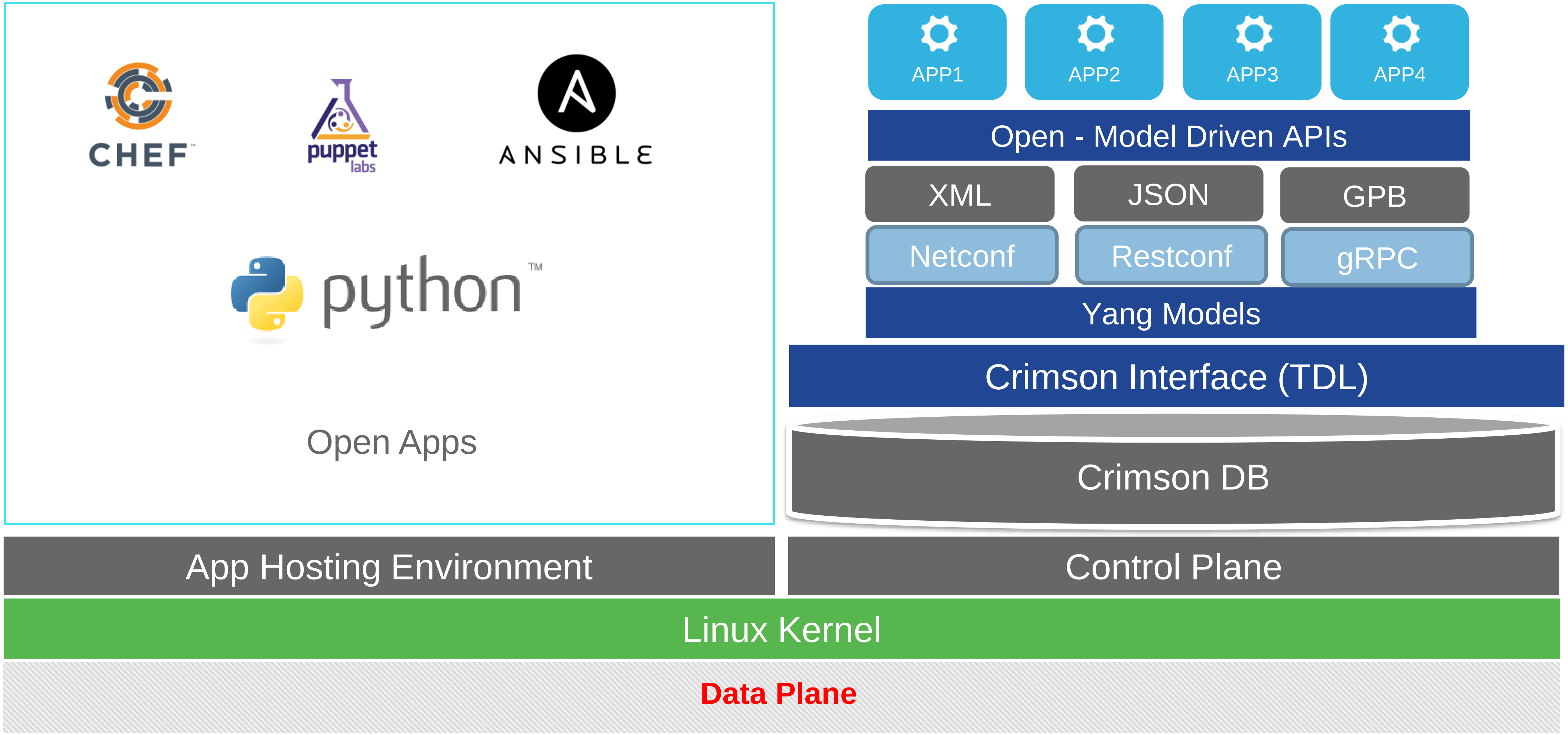
Distributed Database – Code & Data Separation



- All States are stored in the same format
- All States are stored in a common database
- Code & data separation

- TDL (Type Definition language) = data model format & messaging

Crimson Database – Enabling Programmability



Show Process CPU

```
show processes cpu sorted
CPU utilization for five seconds: 1%/0%; one minute: 1%; five minutes: 1%
 PID Runtime(ms)   Invoked    uSecs   5Sec   1Min   5Min TTY Process
 111      7102      60635      117  0.15%  0.02%  0.00%   0 IOSXE-RP Punt Se
--More--
```

IOSd sub-systems only

```
show processes cpu platform sorted
CPU utilization for five seconds:  3%, one minute:  3%, five minutes:  5%
Core 0: CPU utilization for five seconds:  2%, one minute:  2%, five
minutes:  5%
Core 1: CPU utilization for five seconds:  2%, one minute:  2%, five
minutes:  6%
   Pid    PPid    5Sec    1Min    5Min  Status        Size  Name
--------------------------------------------------------------------------------
 18443   17122     11%     11%     11%  S      2741907456  fed main event
 30360   29744      2%      1%      2%  S      1834418176  linux_iosd-imag
 --More–
```

Entire system

```
show platform software process slot switch active r0 monitor
top - 17:22:39 up 5 days,  7:08,  0 users,  load average: 0.37, 0.38, 0.33
Tasks: 274 total,   1 running, 273 sleeping,   0 stopped,   0 zombie
Cpu(s):  3.1%us,  0.8%sy,  0.0%ni, 96.0%id,  0.0%wa,  0.0%hi,  0.0%si,  0.0%st
Mem:   3958028k total,  3307744k used,   650284k free,   219428k buffers
Swap:        0k total,        0k used,        0k free,  1265628k cached
```

Linux TOP output

(TOP = Table of Processes)

Show Process Memory

show processes memory sorted

```
Processor Pool Total:  886455648 Used:  318788992 Free:  567666656
 lsmpi_io Pool Total:    6295128 Used:    6294296 Free:        832

 PID TTY  Allocated      Freed    Holding    Getbufs    Retbufs Process
   0   0  309748936   45140504  242348368          0          0 *Init*
  72   0   34039760    1848960   31709808          0          0 IOSD ipc task
--More--
```

IOSd sub-systems only

show processes memory platform sorted

```
System memory: 3958028K total, 2339848K used, 1618180K free,
Lowest: 1618180K
   Pid    Text      Data   Stack   Dynamic       RSS     Total           Name
--------------------------------------------------------------------------------
18443      286    405652     132    128612    405652   2677644   fed main event
 30360  147803    624916     136        80    624916   1791424  linux_iosd-imag
 --More–
```

Entire system

show platform software process slot switch active r0 monitor

```
top - 17:22:39 up 5 days,  7:08,  0 users,  load average: 0.37, 0.38, 0.33
Tasks: 274 total,   1 running, 273 sleeping,   0 stopped,   0 zombie
Cpu(s):  3.1%us,  0.8%sy,  0.0%ni, 96.0%id,  0.0%wa,  0.0%hi,  0.0%si,  0.0%st
Mem:   3958028k total,  3307744k used,   650284k free,   219428k buffers
Swap:        0k total,        0k used,        0k free,  1265628k cached
  --More–
```

Linux TOP output

(TOP = Table of Processes)

True free memory is a combination of free, buffers, & cached

Memory

- Committed memory > 95% for warnings

***Mar 31 19:55:33.384 PDT: %PLATFORM-4-LEMENT_WARNING:Switch 1 R0/0: smand: 4/RP/0: Committed Memory value 96% exceeds warning level 95%**

- Committed memory > 99% for critical errors

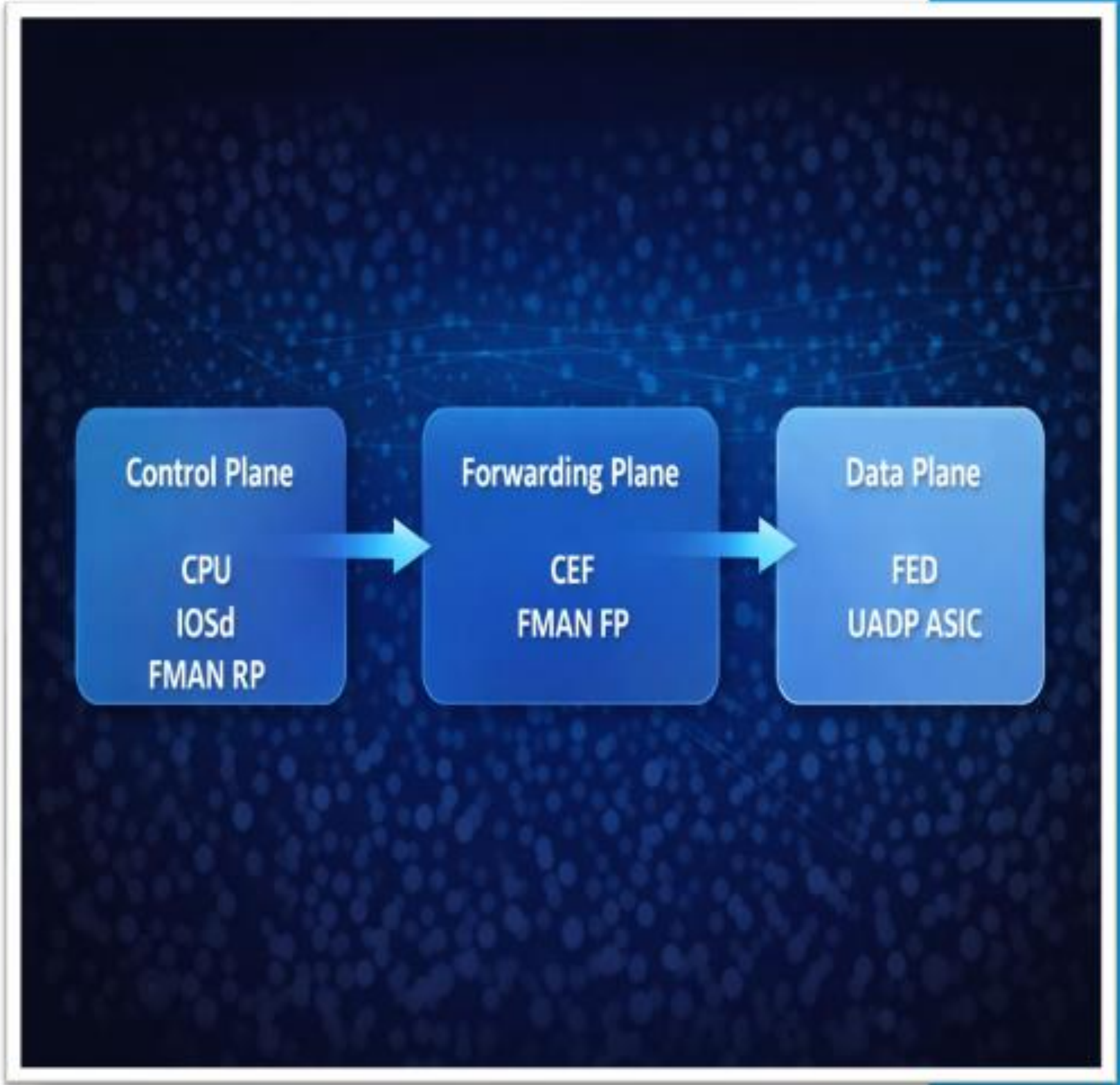
show platform software status control-processor brief

```
Load Average
 Slot  Status  1-Min  5-Min 15-Min
1-RP0 Healthy   0.40   0.40   0.35

Memory (kB)
 Slot  Status    Total     Used (Pct)     Free (Pct) Committed (Pct)
1-RP0 Healthy  3958028  2544852 (64%)  1413176 (36%)   3288040 (83%)

CPU Utilization
 Slot  CPU   User System   Nice   Idle    IRQ   SIRQ IOwait
1-RP0    0   2.40   0.50   0.00  97.10   0.00   0.00   0.00
         1   1.80   0.30   0.00  97.90   0.00   0.00   0.00
         2   1.10   0.40   0.00  98.50   0.00   0.00   0.00
```

Polaris

Architecture Overview

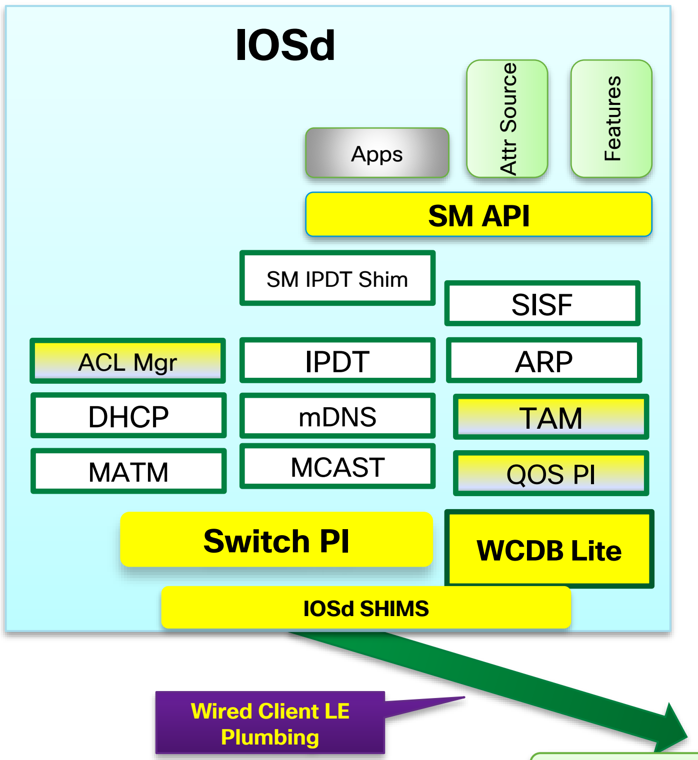
Key Architecture Components



IOSd

New/Modified Modules **New/Modified flows**

(Compared to Nova 15.x)



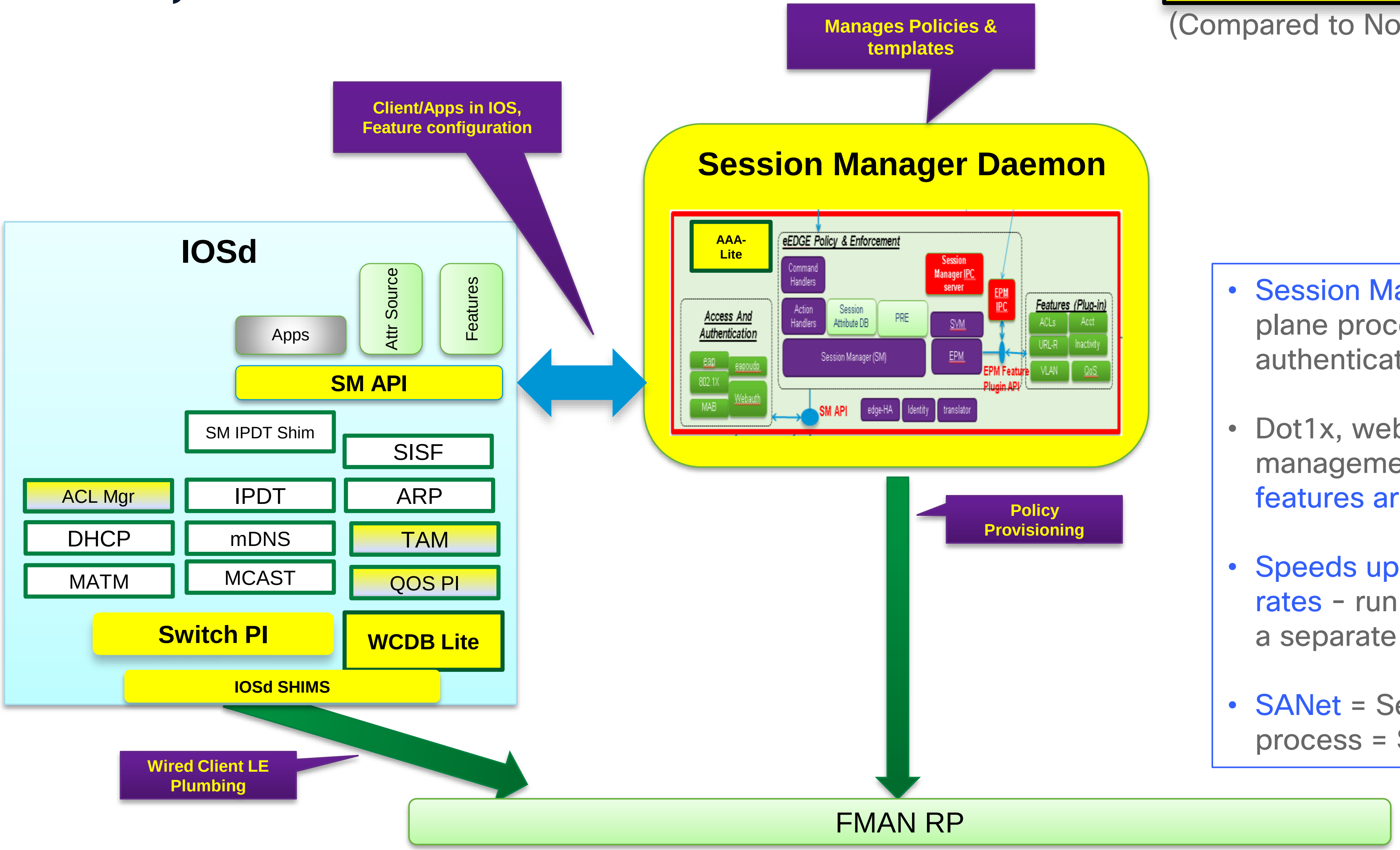
- The IOSd process has L2, L3, QoS, etc sitting inside it
- The IOSd process can only run on 1 CPU core. There is no pinning of a process to a specific core
- IOSd Shim messaging interfaces the legacy IOSd environment to the new Polaris FMAN RP environment.
- FMAN RP = Feature Manager on Route Processor (active stack switch)
- FMAN programs hardware (Doppler ASIC) via the Forwarding Engine Driver (FED)
- FMAN replaces Forwarding and Feature Manager (FFM) that was used in Nova 15.x
- PI = Platform Independent; MATM = MAC Address Table Manager; IPDT = IP Device Tracking; SISF = Switch Integrated Security Feature; WCDB = Wireless Controller Data Base; SM = Session Manager

IOSd, SMD

New/Modified Modules

New/Modified flows

(Compared to Nova 15.x)



- Session Manager Daemon - new control plane process to handle edge authentication type services
- Dot1x, web auth, mab, aaa, policy management and other security session features are pulled out of IOSd
- Speeds up wired and wireless client join rates - run as a separate Linux process on a separate CPU core compared to IOSd
- SANet = Session Aware Networking process = SMD

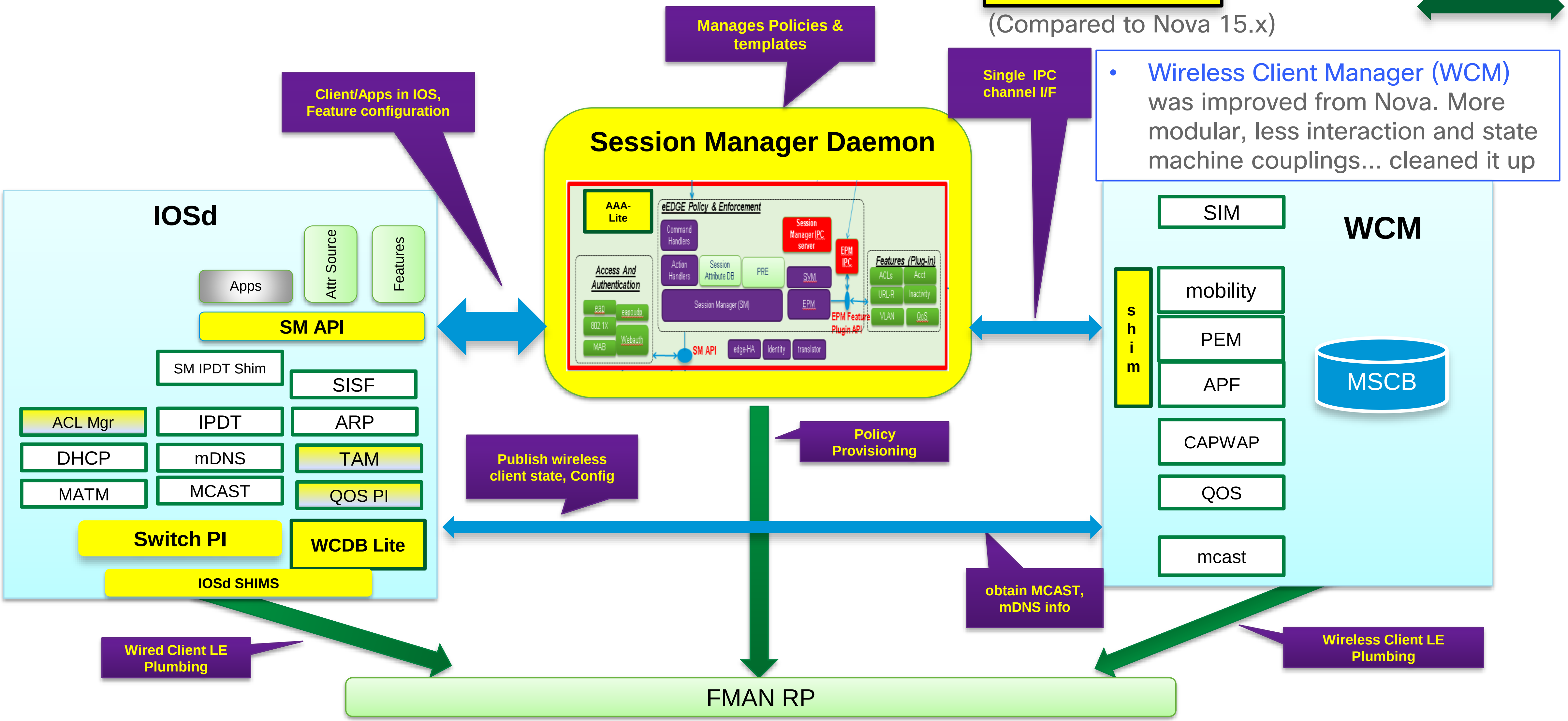
IOSd, SMD, WCM

New/Modified Modules

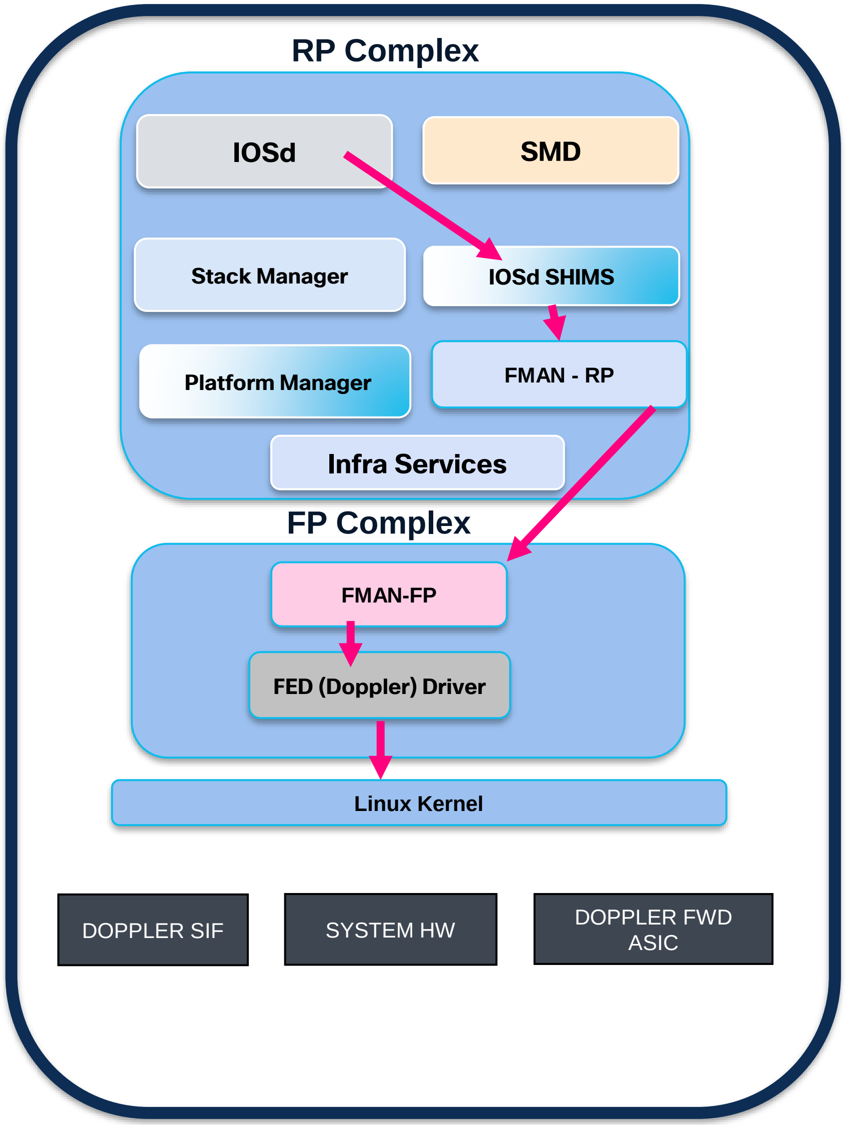
(Compared to Nova 15.x)

New/Modified flows

- Wireless Client Manager (WCM) was improved from Nova. More modular, less interaction and state machine couplings... cleaned it up

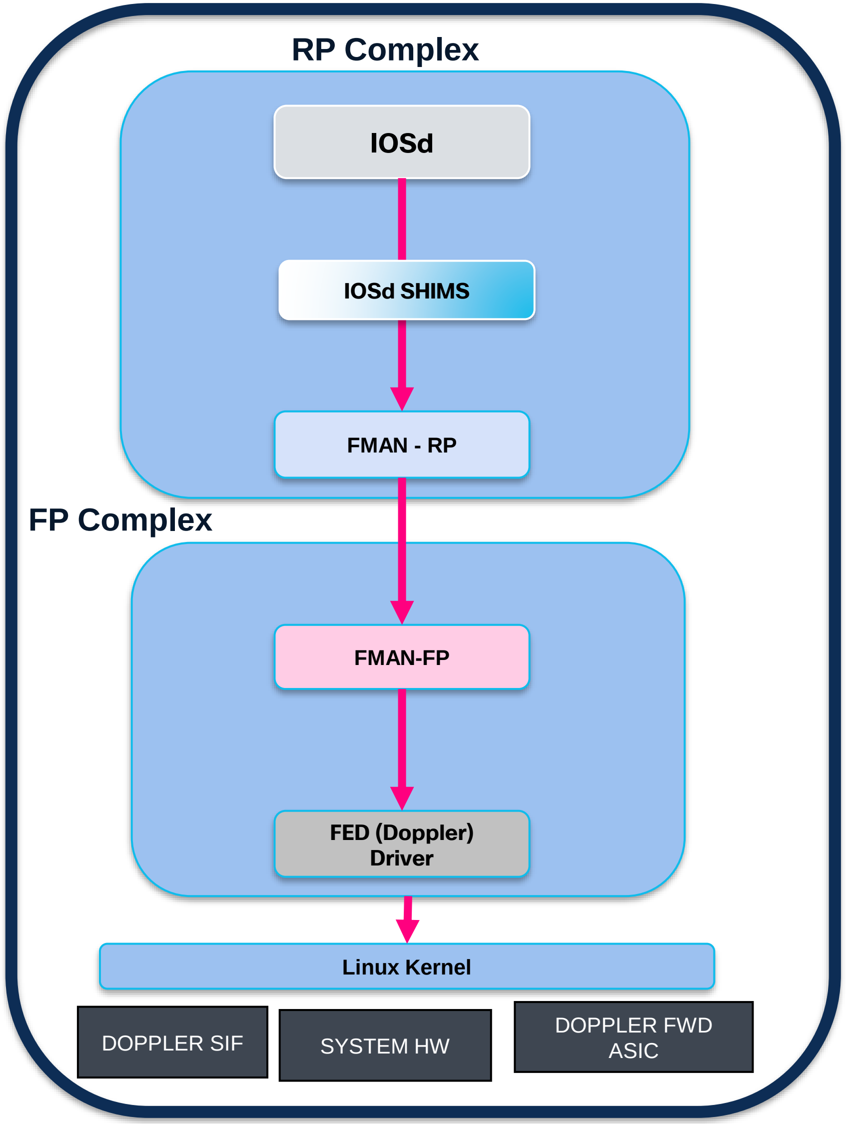


High Level Polaris Architecture



- RP complex – Route Processor. Control Plane software processes that need to run on the active and standby switch in a switch stack & other infrastructure services
- FP complex – Forwarding Processor. Data Plane forwarding & data path software processes used to program the hardware (Doppler ASIC)
- Feature manager (FMAN) is split in 2. The RP portion is a control plane process. IOSd, SMD, WCM share config state with FMAN RP
- FMAN RP talks to FMAN FP
- FMAN FP validates feature dependencies and then downloads state to FED. (for example, a mac address is dependent on an interface and a vlan. FMAN FP cannot push a mac address to FED if the interface and vlan do not exist there 1st)
- Forwarding Engine Driver (FED) programs the Doppler forwarding ASIC

High Level Polaris Architecture - Object Model



IOSd – CLI and Protocol builds configuration and state

IOSd SHIMS - Configuration is processed in active switch IOSd and sent to FMAN RP through IOSd shim

FMAN – RP - OM (Object Manager) - Active FMAN RP maintains persistent state (across process restarts) for features

FMAN-FP - AOM (Asynchronous Object Manager) - Asynchronously checks dependencies on what FMAN FP received from FMAN RP before pushing it to FED

FED - Processes received state and ACKs request back to FMAN FP to mark object create/update/delete. These ACKs are sent back till IOSd

ACK'less

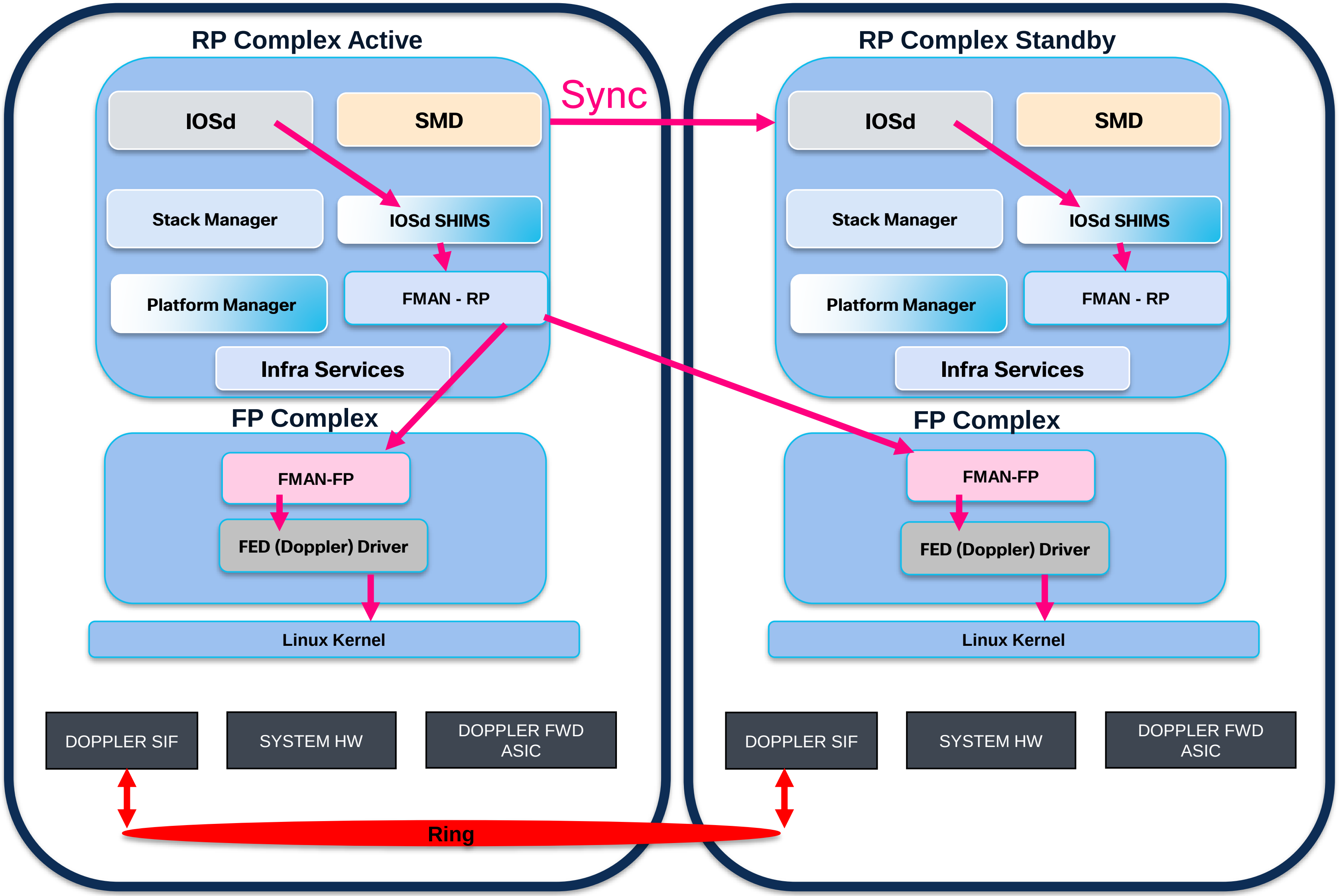
ACK

Configuration Flow – Object Model

- IOSd has CLI (Command Line Interface) and protocols that build up state and configuration
- Configuration is processed in active switch IOSd and sent to FMAN RP through IOSd shim
- OM (Object Manager) - active FMAN RP maintains persistent state (across process restarts) for features via an object library contained in a TDL (Type Definition Language) shared memory database (Crimson)
- OM sends state from active FMAN RP to all FMAN FPs. Mostly an ACK'less model at this point for performance reasons (some exceptions for synchronization)
- AOM (Asynchronous Object Manager) - FMAN FP has 2 Asynchronous interfaces (northbound to FMAN RP & southbound to FED). AOM object library also manages RP switchover reconciliation & stale object processing
- AOM is asynchronous in that it checks dependencies on what FMAN FP received from FMAN RP before pushing it to FED (Example: for a new mac object, make sure interface and vlan objects are already there)
- FED processes received state and ACKs request back to FMAN FP to mark object create/update/delete as complete. FMAN FP forwards the ACK to FMAN RP (that tracks all FP's) which in turn can forward the ACK to IOSd

- Polaris 16.x - All forwarding config & state is sent to the data plane from IOSd and SMD via FMAN RP & FP
- Nova 15.x - L2 features were sent from IOSd direct to FED without FFM. Only L3, ACL, QoS, FNF went through FFM
- Polaris & Nova are both object model (OM) based to maintain feature state

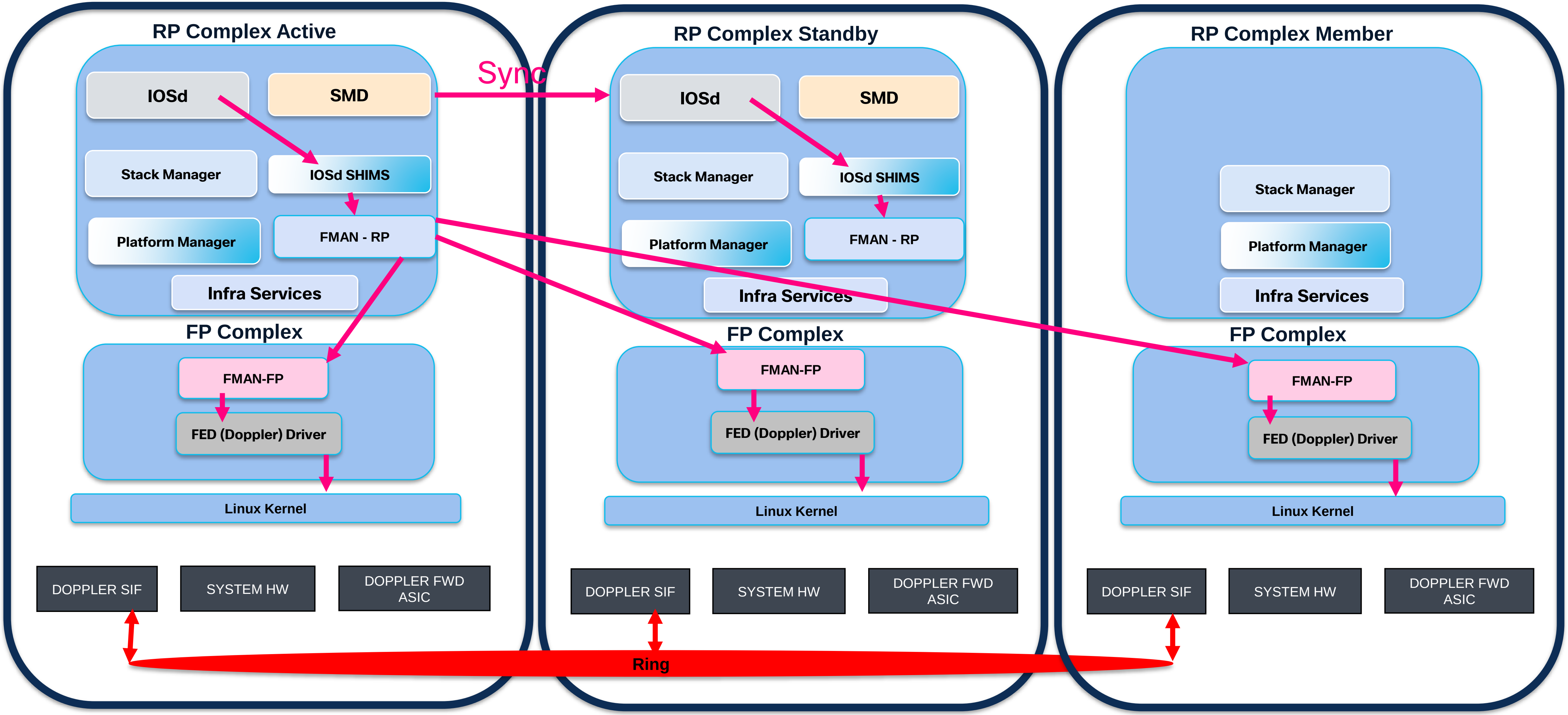
High Level Polaris Architecture Stacking



- Control plane processes talk to FMAN RP on the active stack switch
- FMAN RP on the active switch pushes info to all the FM FP's on all switches in the switch stack
- FMAN RP on standby is only standing by in a hot state and not talking to any of the FMAN FP's. FMAN RP on standby is fed from the standby IOSd & SMD who's state is synched from active IOSd & SMD.
- FMAN FP in each stack switch programs FED which then programs the Doppler ASIC
- Platform Manger – a separate process handling PoE, fan, power supply, system and board level management (functionality is the same as in Nova 15.x)

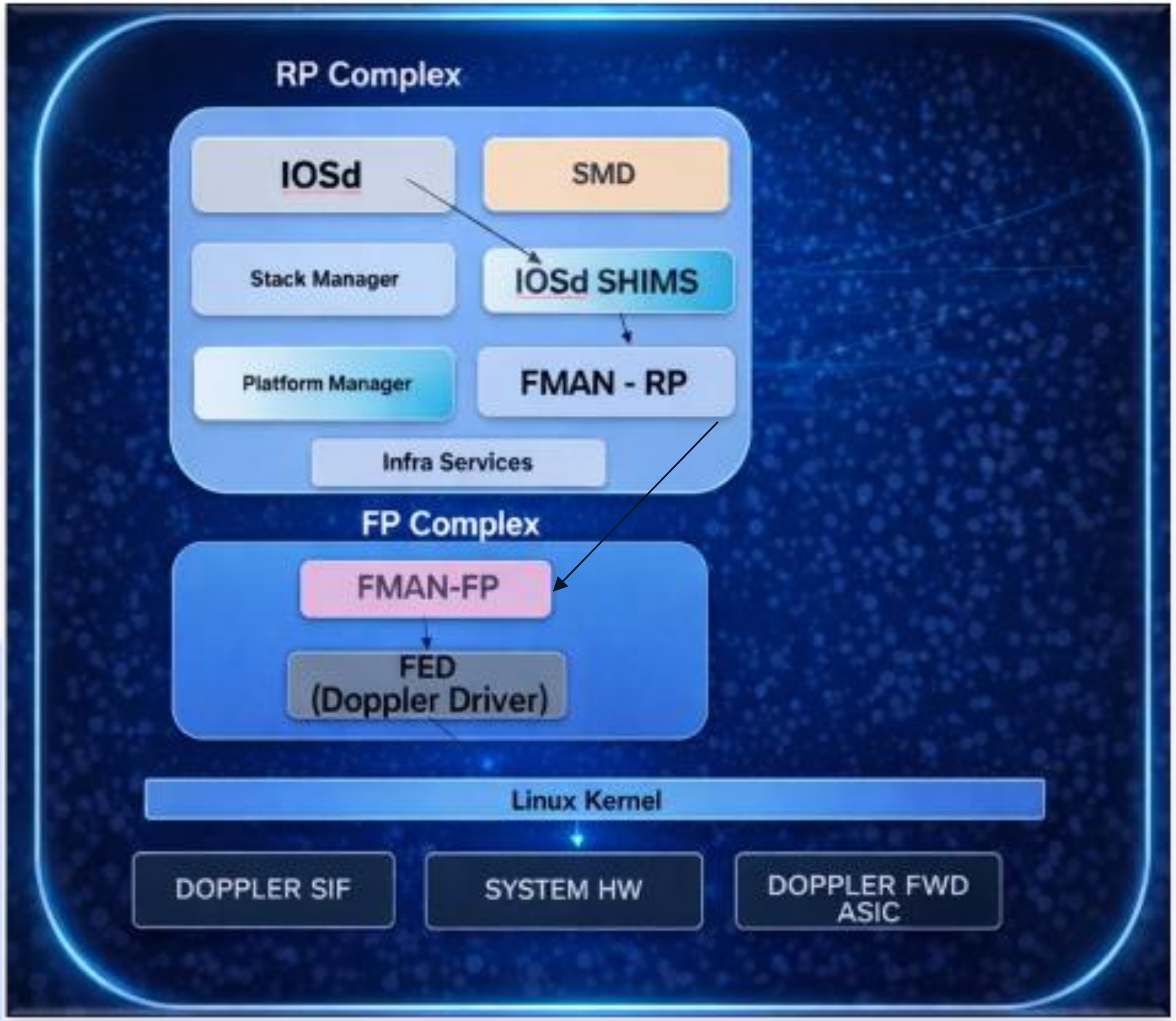
- Arrows represent logical communication flow. Actual communication is via the stack ring

High Level Polaris Architecture Stacking



High Level **Polaris** Architecture

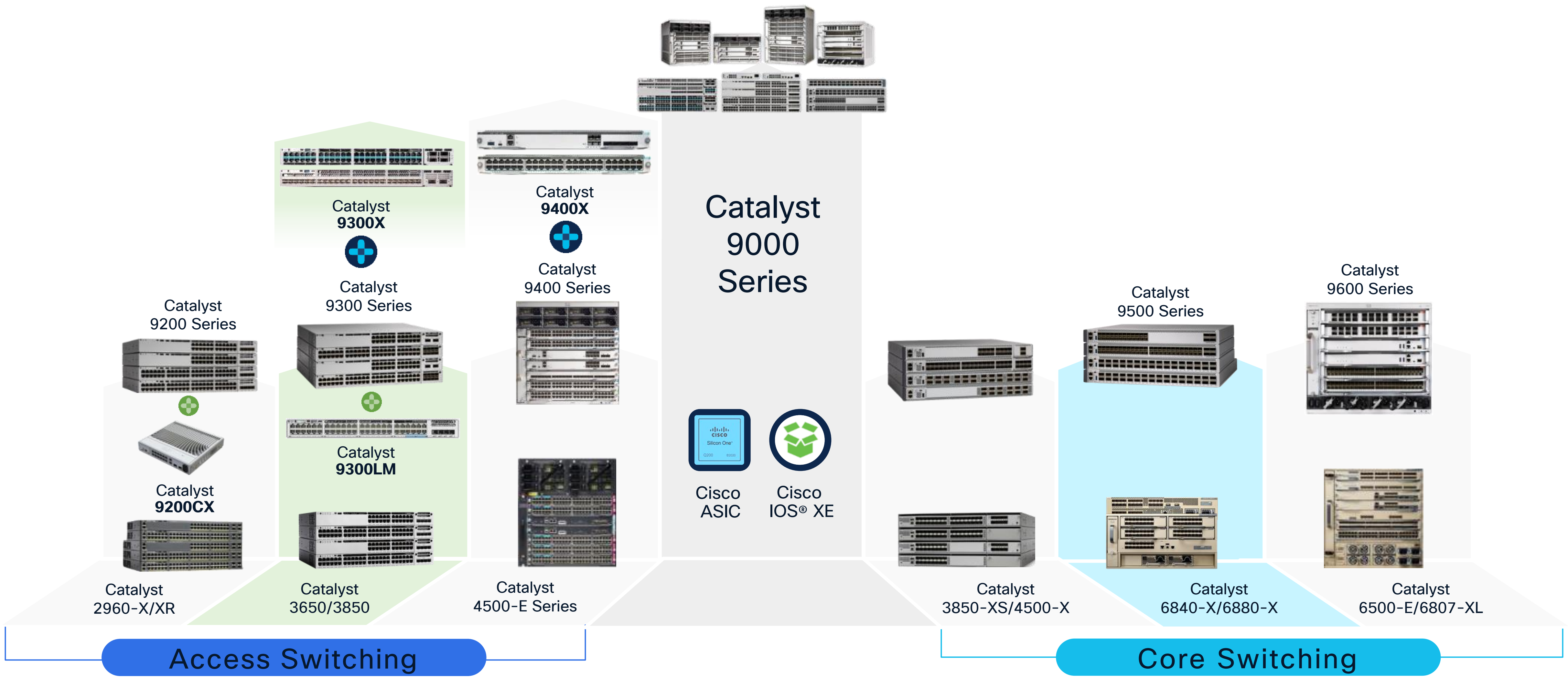
- **RP complex** – Route Processor. Control Plane
- **FP complex** – Forwarding Processor. Data Plane.
- **Feature manager (FMAN)** is split in 2. The RP portion is a control plane process.
- **FMAN RP talks to FMAN FP** – FMAN FP validates feature dependencies and then downloads state to FED.
- **Forwarding Engine Driver (FED)** programs the Doppler forwarding ASIC



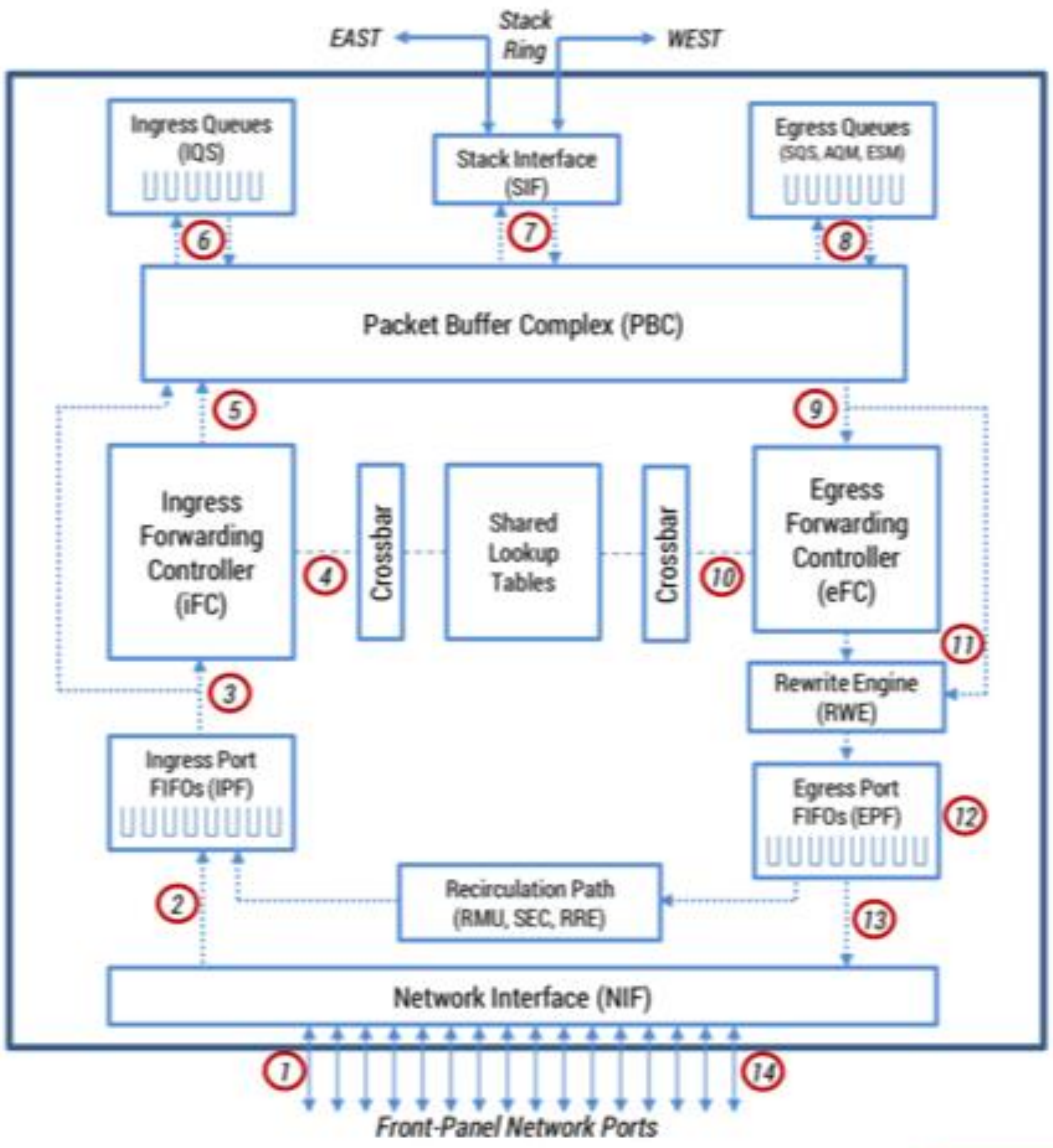
UADP/Doppler Architecture

Cisco Catalyst 9000 Switching Portfolio : Doppler

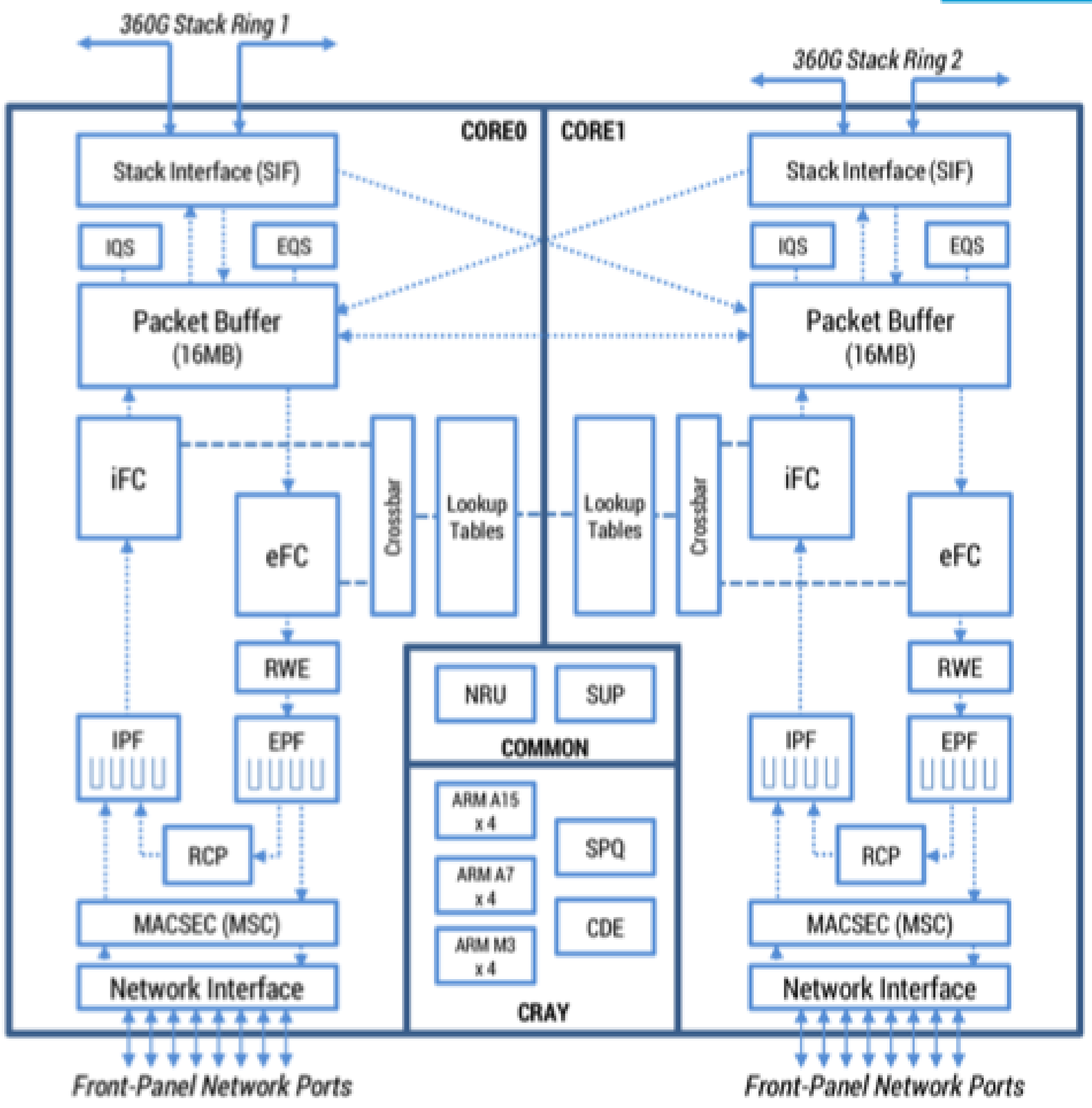
One Family from Access to Core – Common Hardware & Software



High Level **Doppler** Architecture



Doppler D



360G Stack Ring 1
360G Stack Ring 2
CORE0
CORE1
Stack Interface (SIF)
IQS
EQS
Packet Buffer (16MB)
iFC
eFC
Crossbar
Lookup Tables
RWE
IPF
EPF
RCP
MACSEC (MSC)
Network Interface
NRU
SUP
COMMON
ARM A15 x 4
ARM A7 x 4
ARM M3 x 4
SPQ
CDE
CRAY
Front-Panel Network Ports

Doppler E

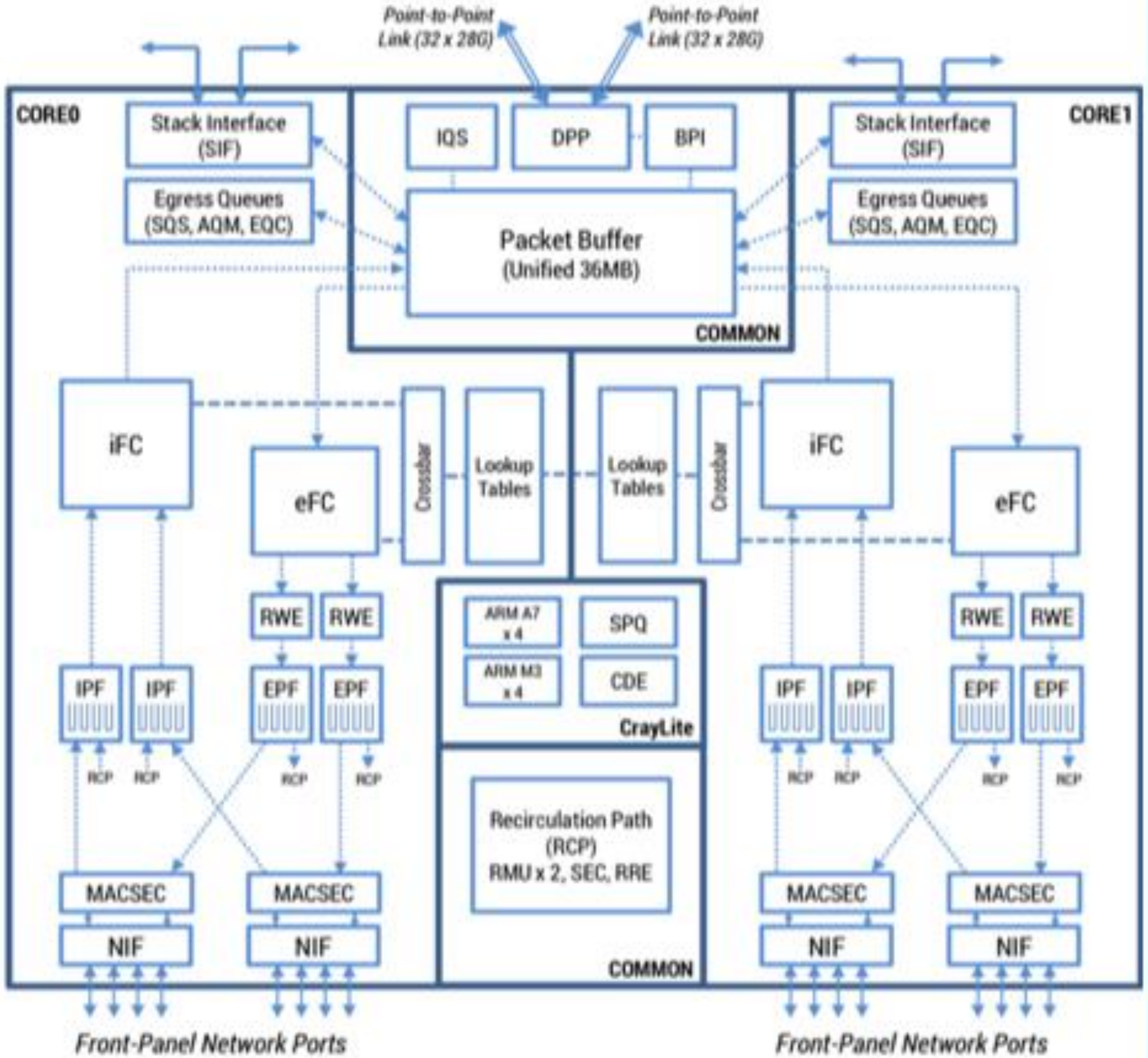


FIGURE 18. Block diagram of DopplerE

Cisco Unified Access Data-Plane (UADP®)

Common ASIC Architecture for Switching Access, Distribution & Core

- Multiple generations and formats, same architecture
- Rich flexible forwarding & services memories
- First fully programmable microcode network

- Multiple functions: system-on-chip or line-card
- Multiple form factors: fixed or modular
- Multiple places: Access, Distribution and Core

Cisco Unified Access Data-Plane (UADP®)

Common ASIC Architecture for Switching Access, Distribution & Core

Catalyst
9200,9200L,9200CX

[UADP 2.0m]
120 Gbps
16nm FinFET
1.3B Transistors
1 Core + ARM
CPU

Catalyst
9300,9300L,9400

[UADP 2.0/XL]
240 Gbps
28nm FinFET
7.6B Transistors
2 Core

Catalyst
9300X/9300LM

[UADP 2.0sec]
480 Gbps
16nm FinFET
7.6B Transistors
1 Core 2 + SEC

Catalyst
9500H/9600

[UADP 3.0]
1.6 Tbps
16nm FinFET
192.B Transistors
2 Core

Catalyst 9400X

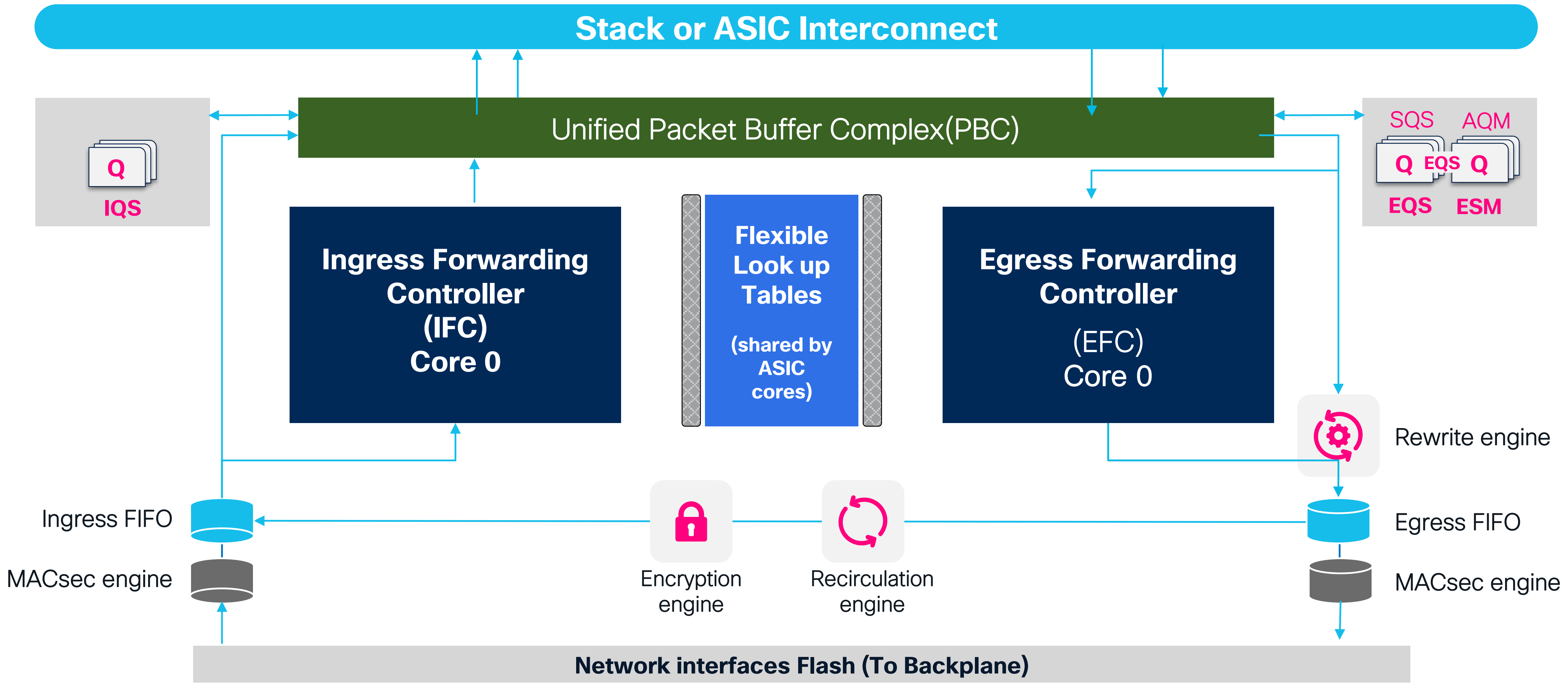
[UADP 3.0sec]
1.6 Tbps
16nm FinFET
192.B Transistors
2 Core + SEC

- Multiple generations and formats, same architecture
- Rich flexible forwarding & services memories
- First fully programmable microcode network

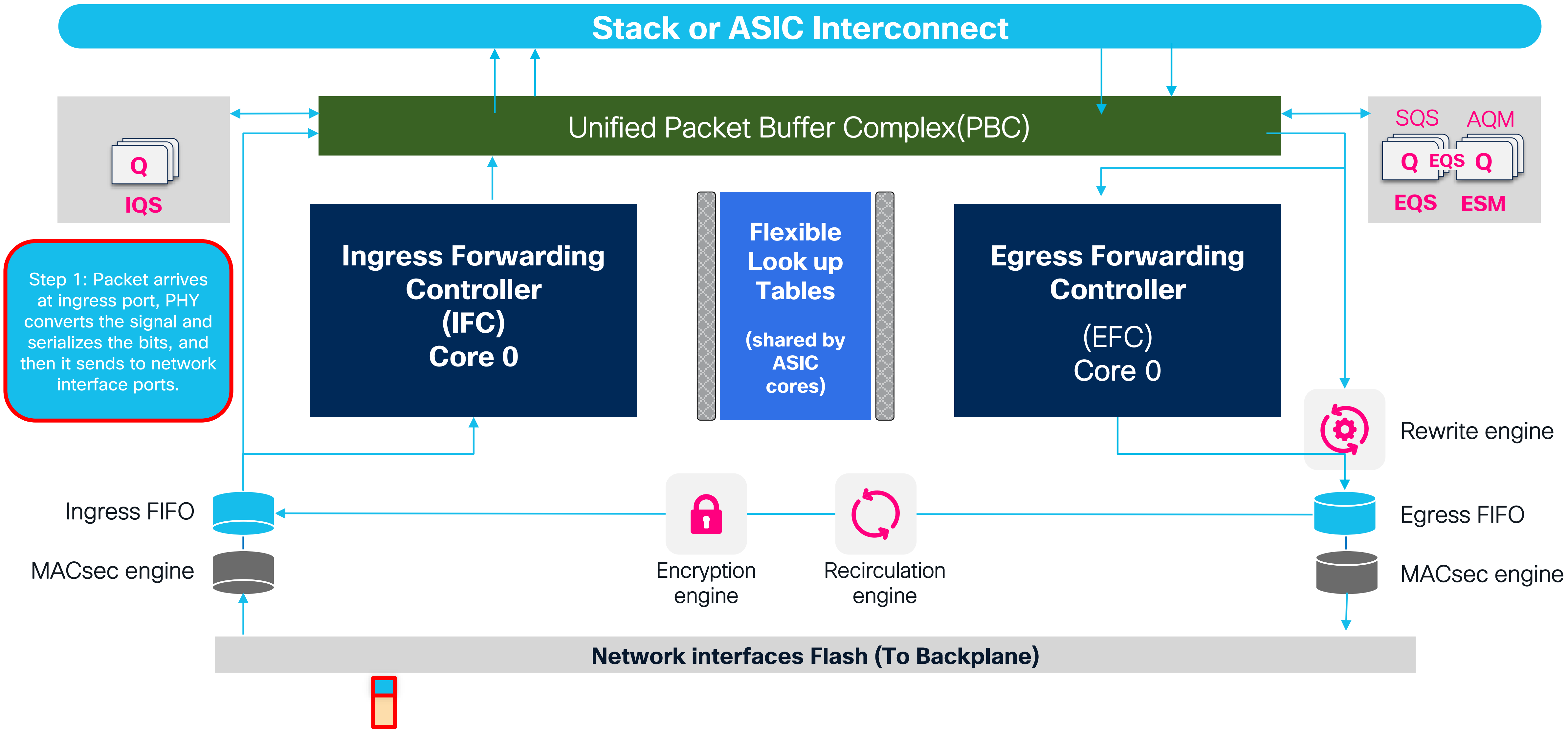
- Multiple functions: system-on-chip or line-card
- Multiple form factors: fixed or modular
- Multiple places: Access, Distribution and Core

Doppler Packet Walk

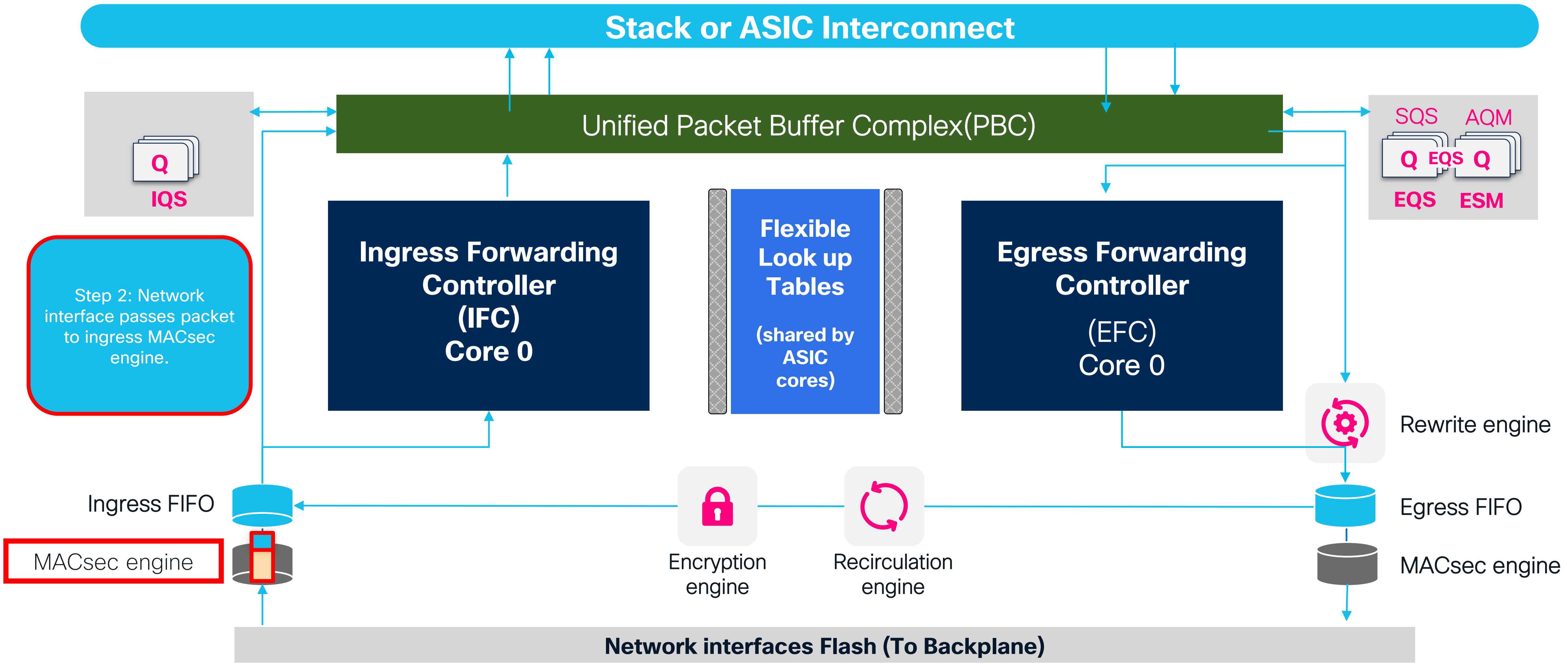
Generic Unicast Packet Walk – UADP ASIC



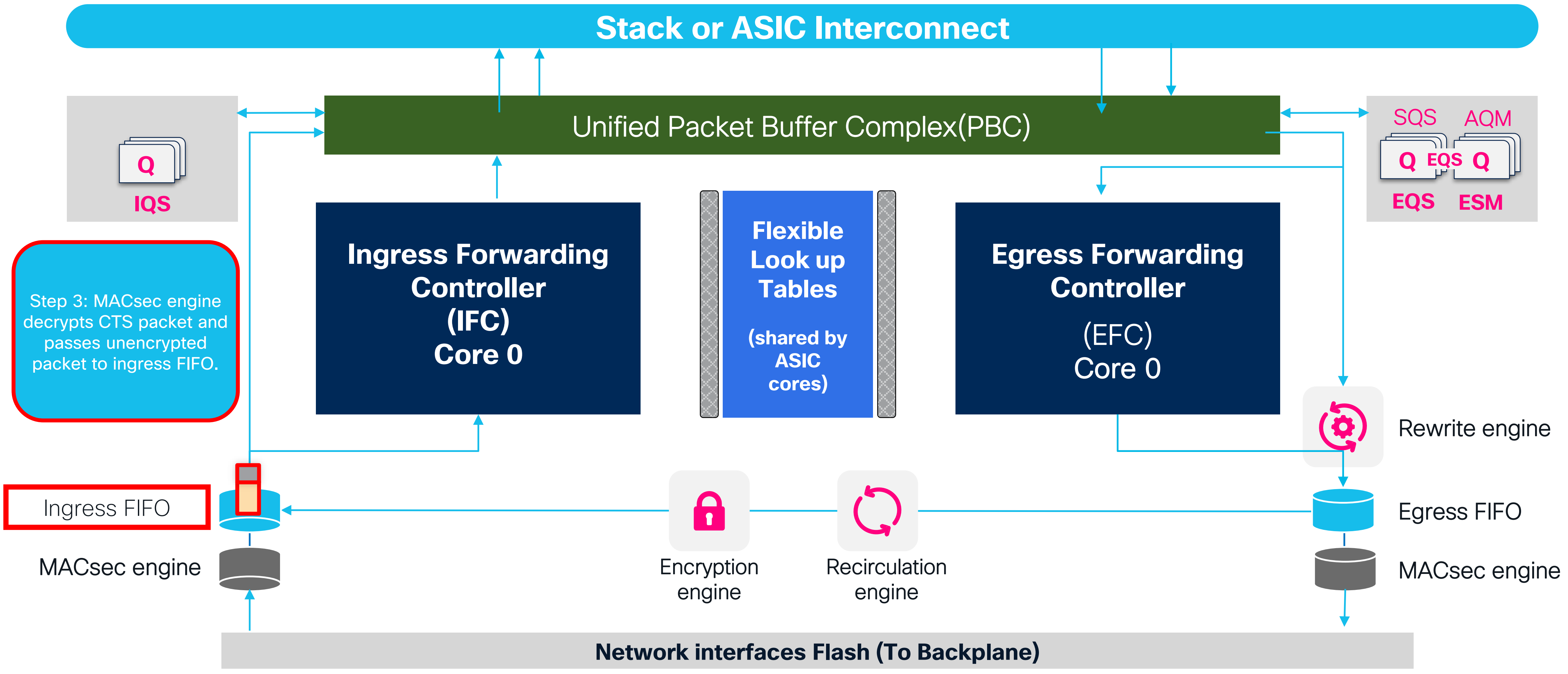
Generic Unicast Packet Walk – UADP ASIC



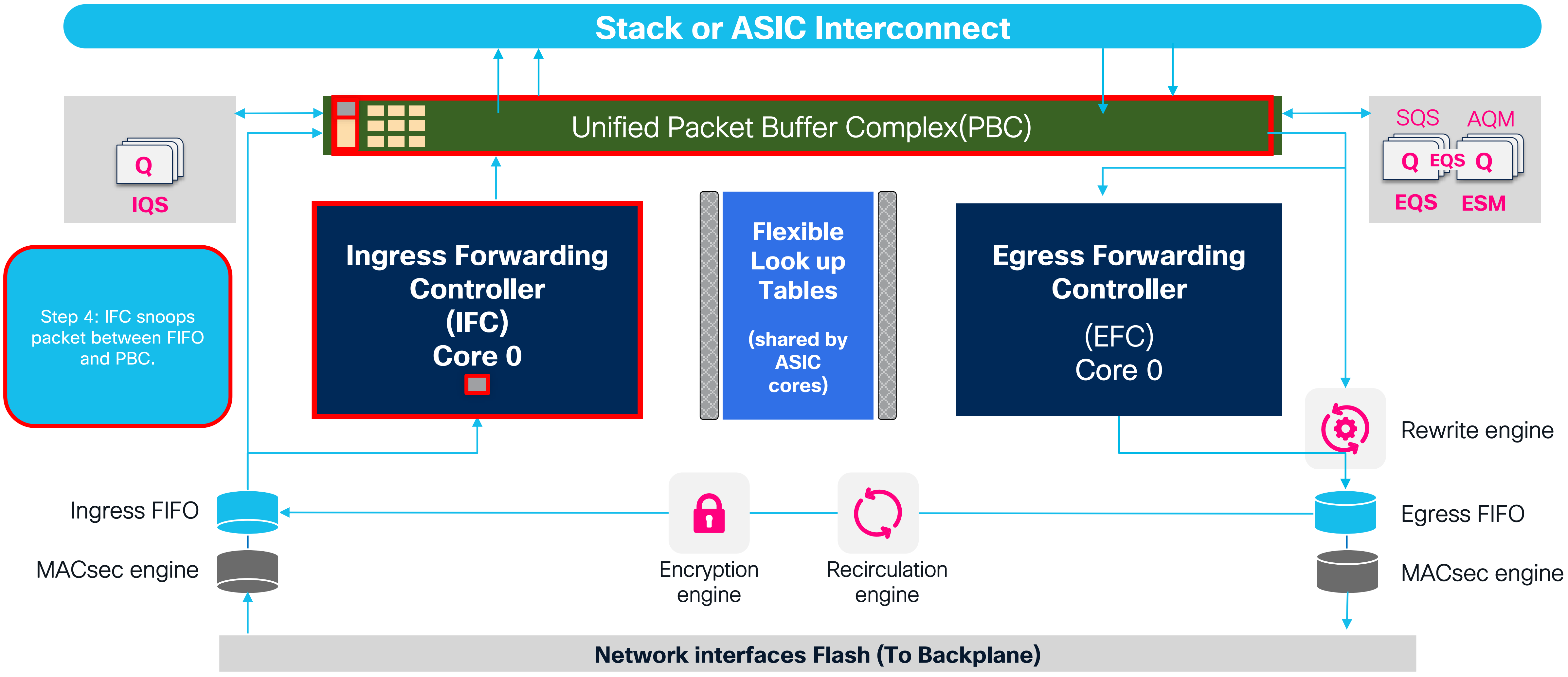
• Generic Unicast Packet Walk – UADP ASIC



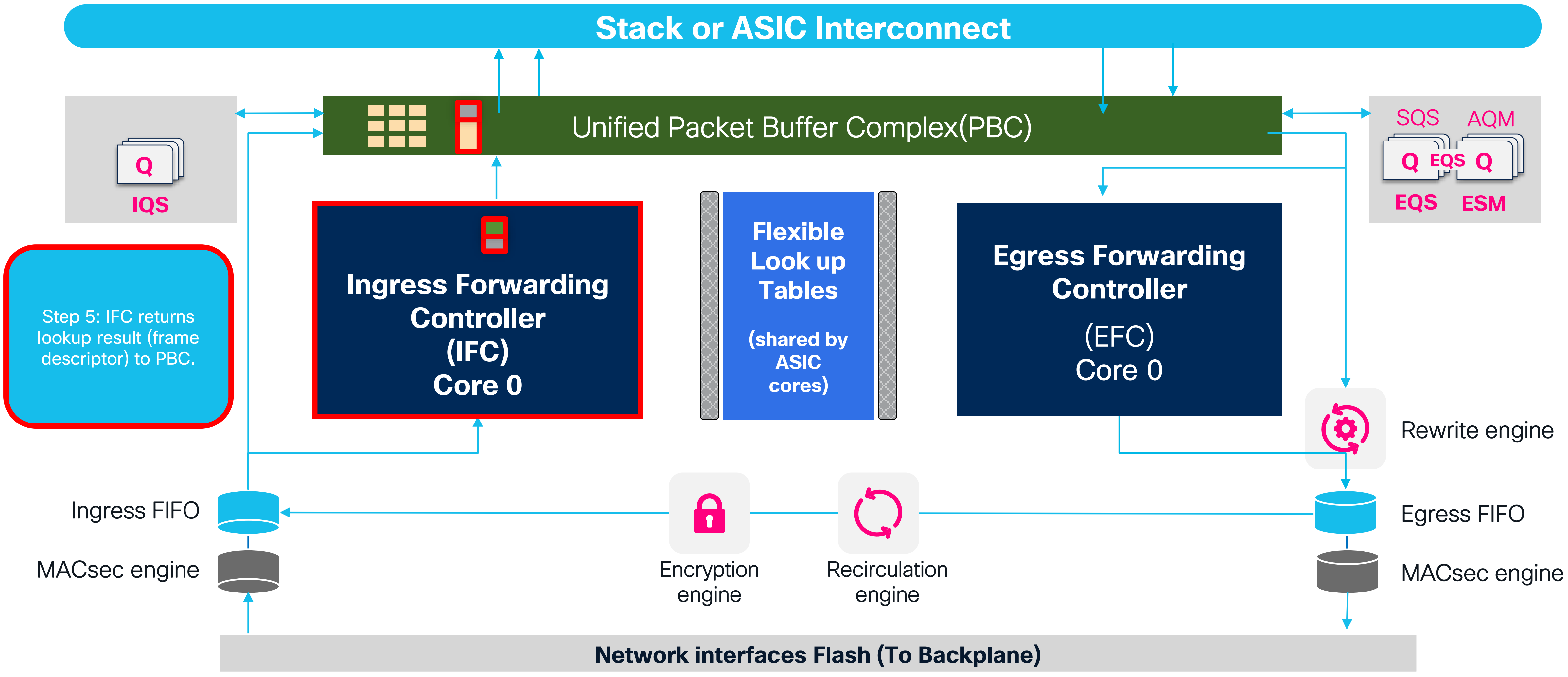
- Generic Unicast Packet Walk – UADP ASIC



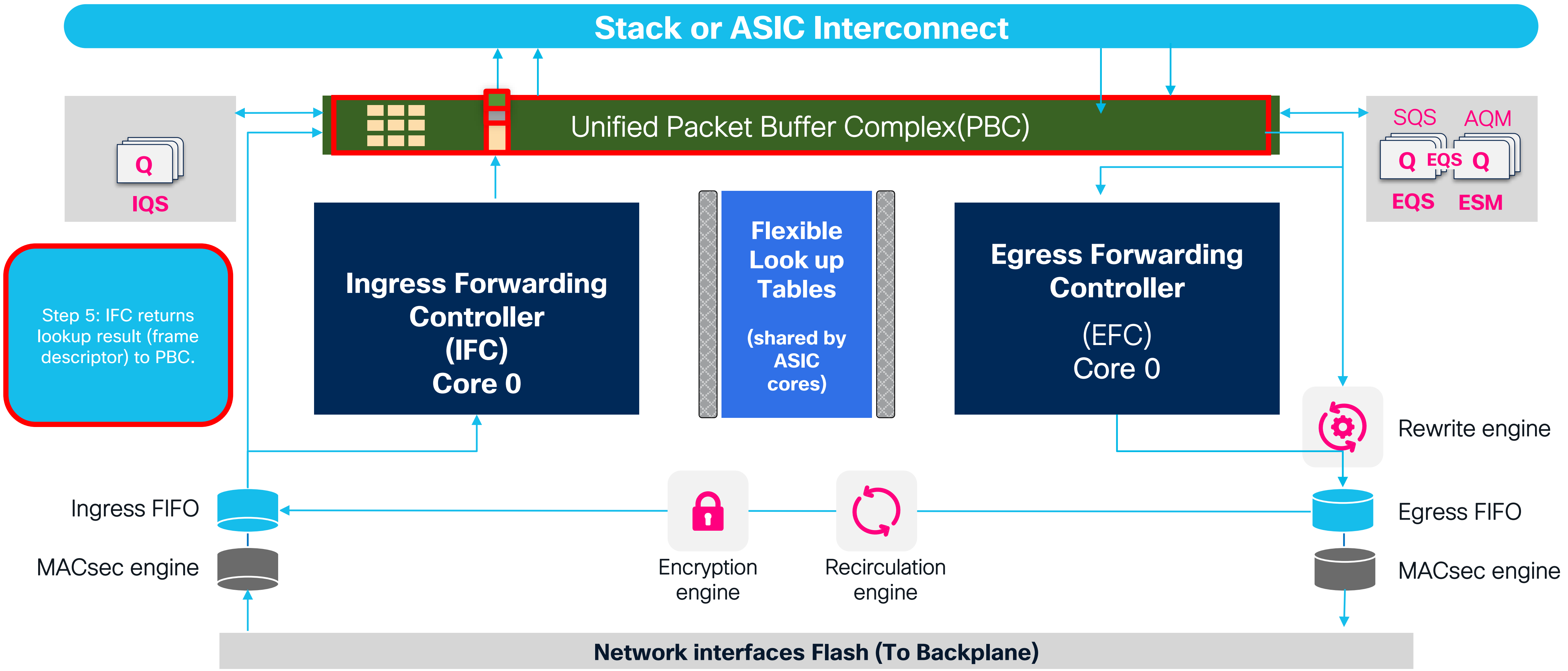
• Generic Unicast Packet Walk – UADP ASIC



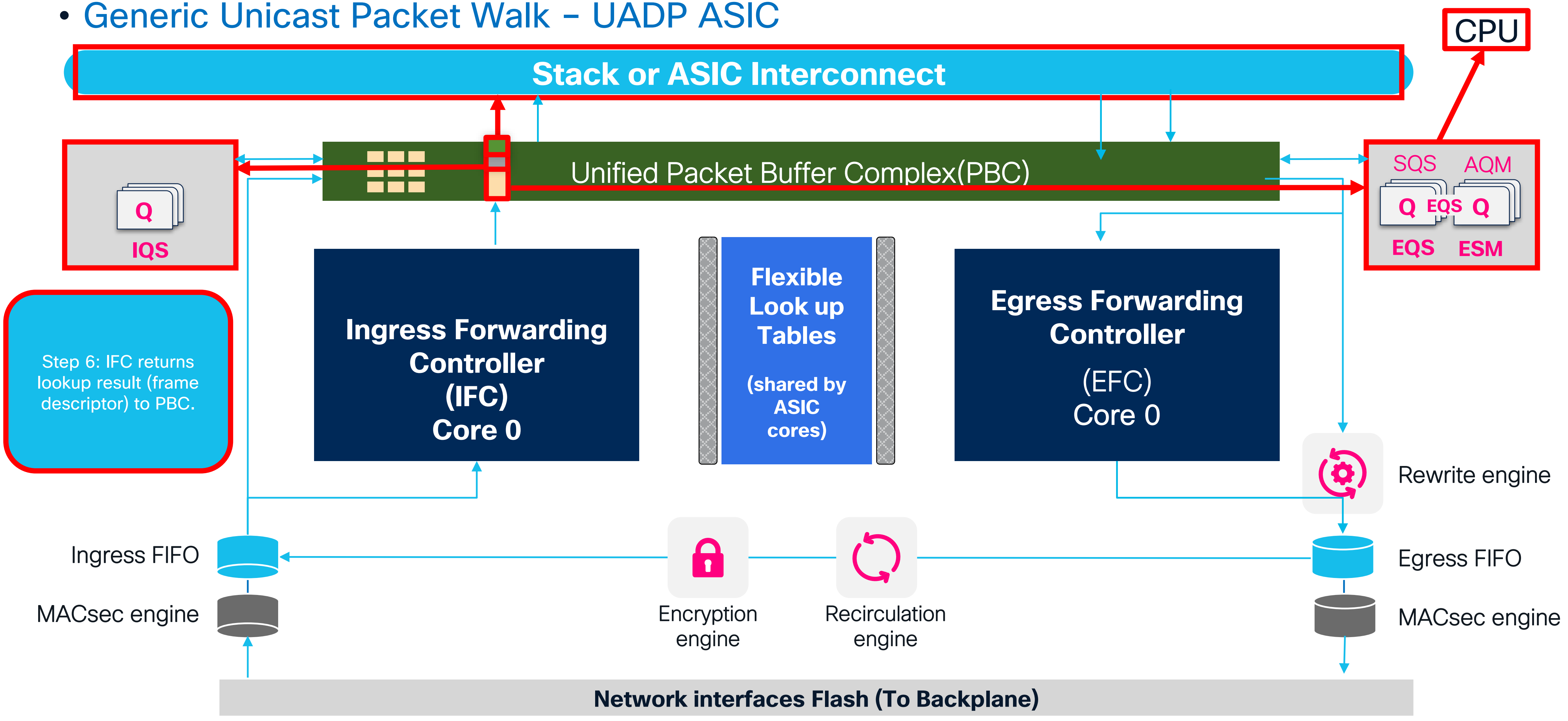
• Generic Unicast Packet Walk – UADP ASIC



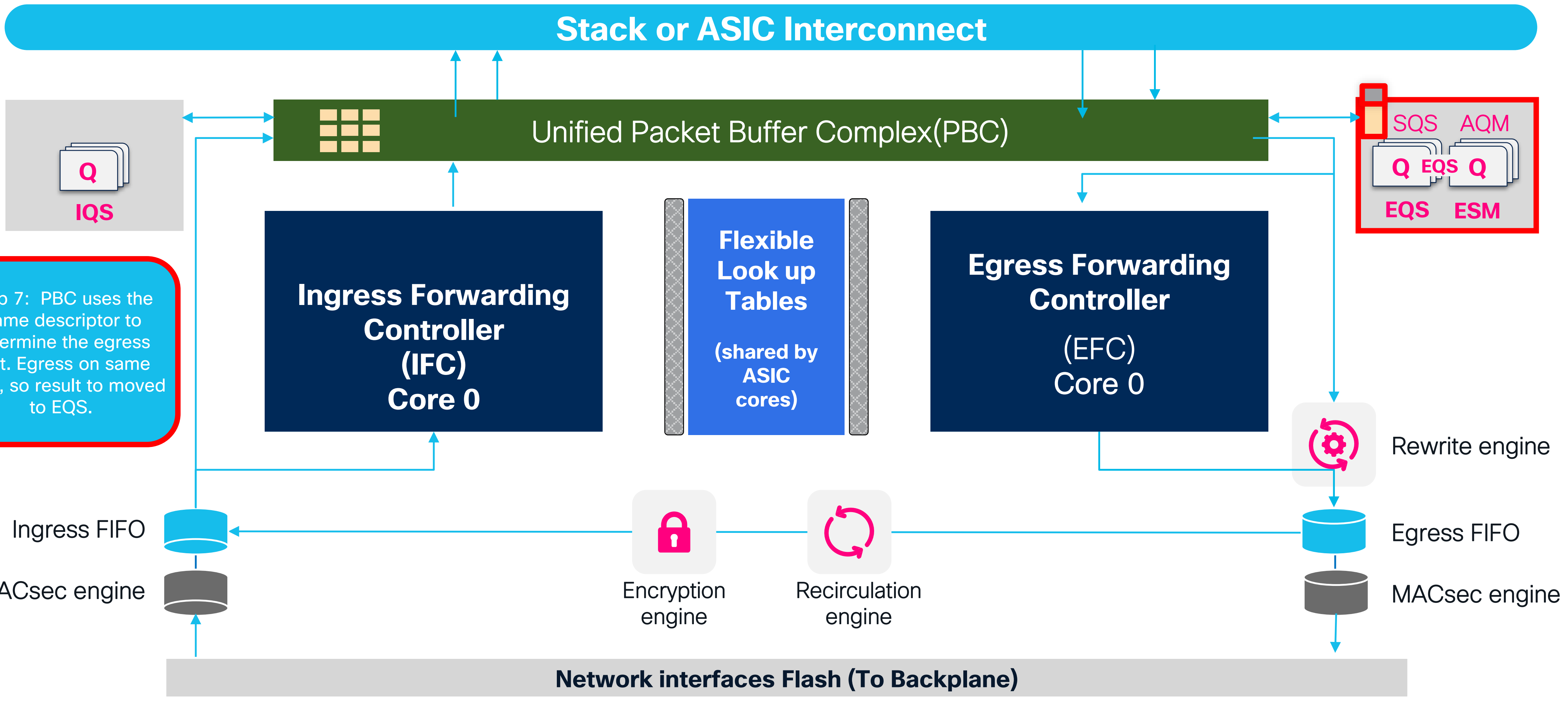
- Generic Unicast Packet Walk – UADP ASIC



- Generic Unicast Packet Walk – UADP ASIC

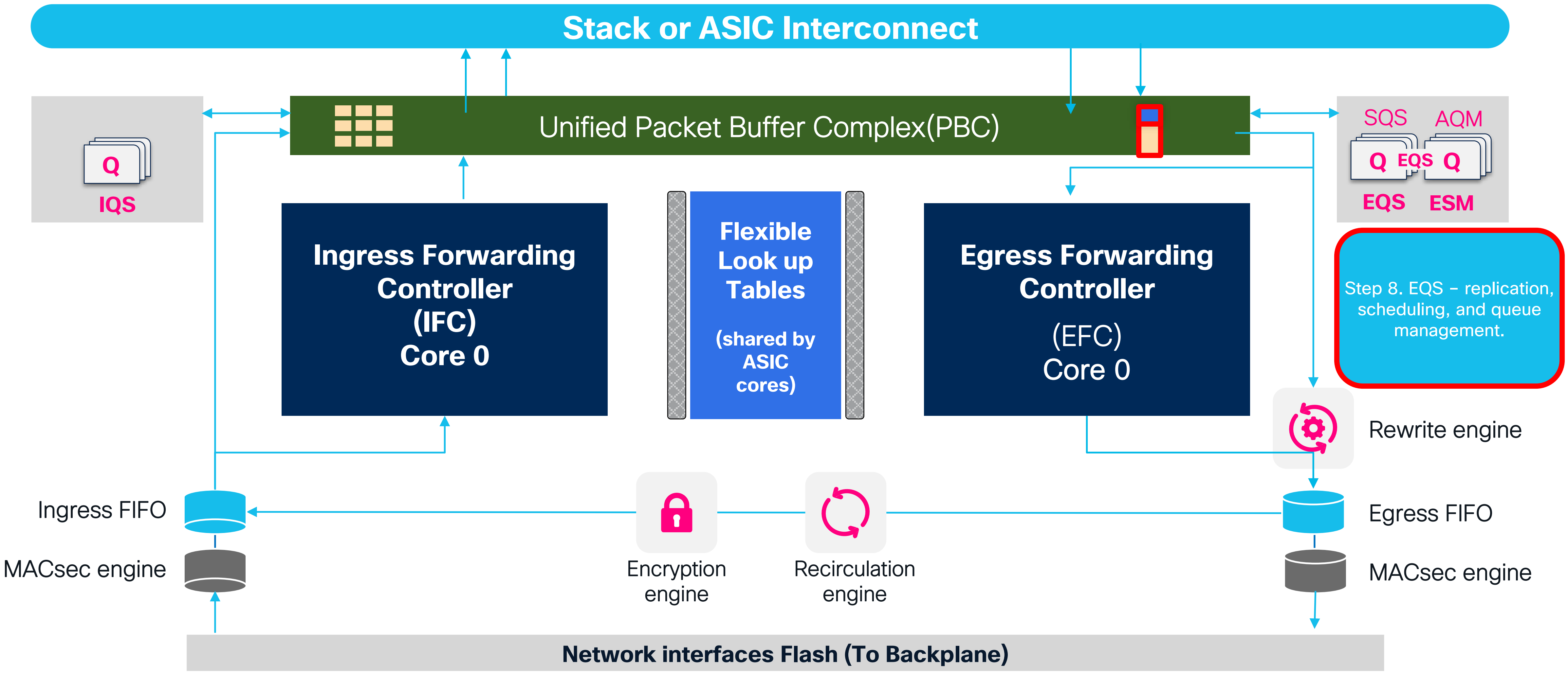


- Generic Unicast Packet Walk – UADP ASIC

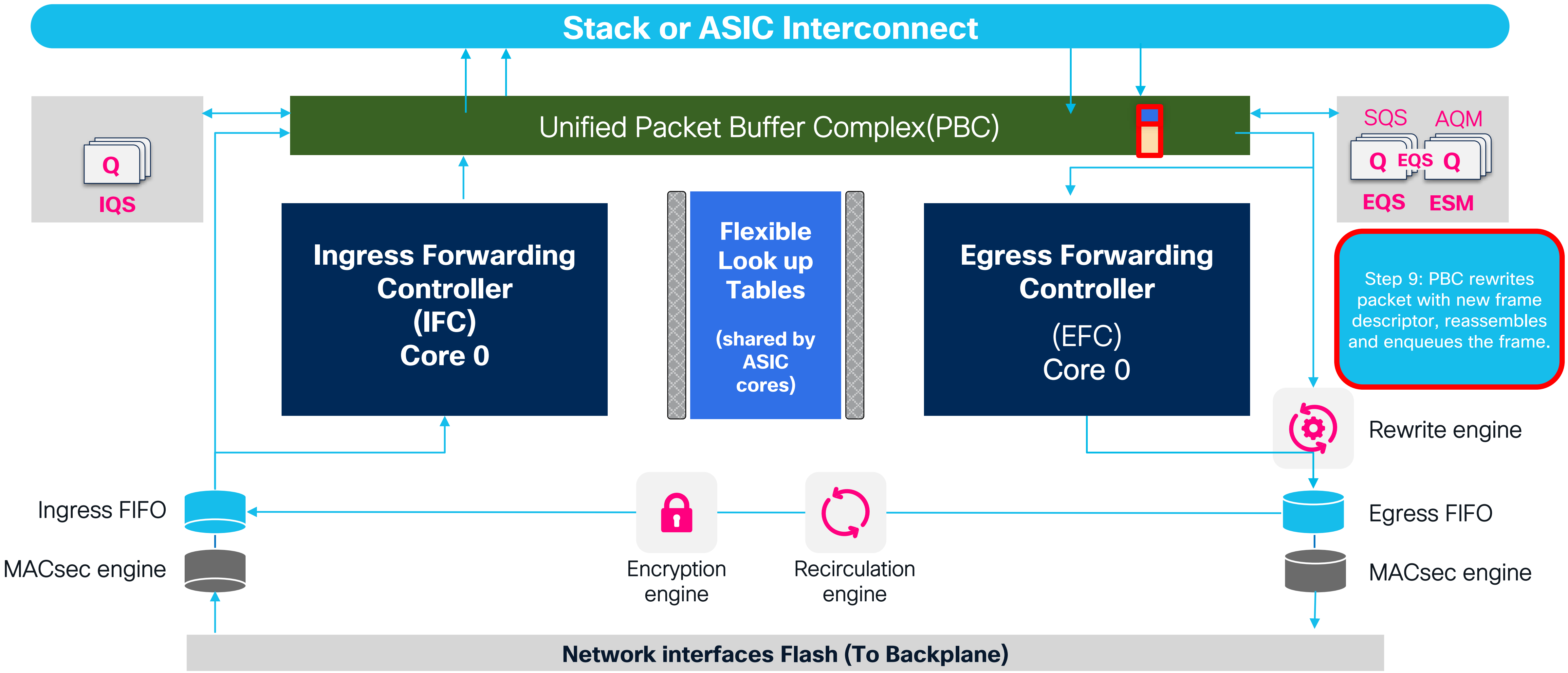


Step 7: PBC uses the frame descriptor to determine the egress port. Egress on same ASIC, so result to moved to EQS.

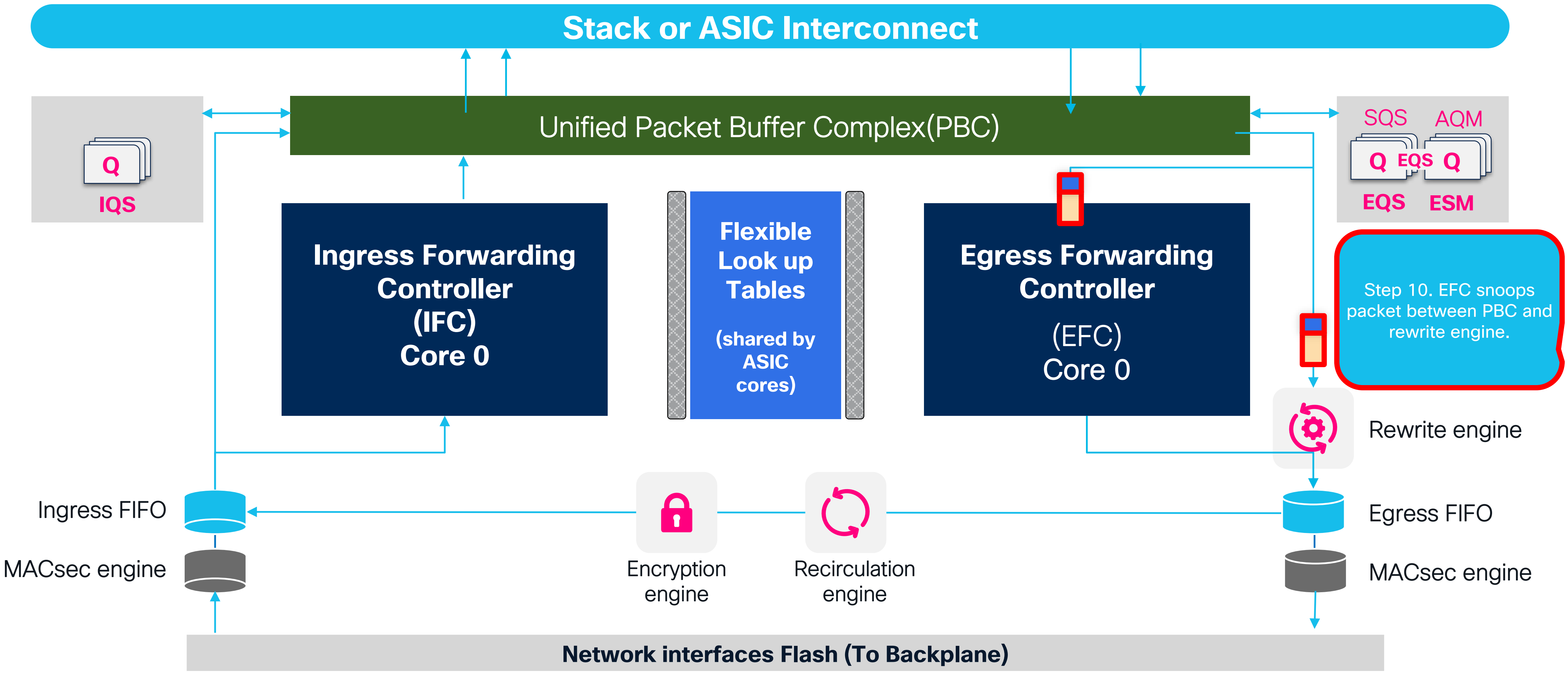
• Generic Unicast Packet Walk – UADP ASIC



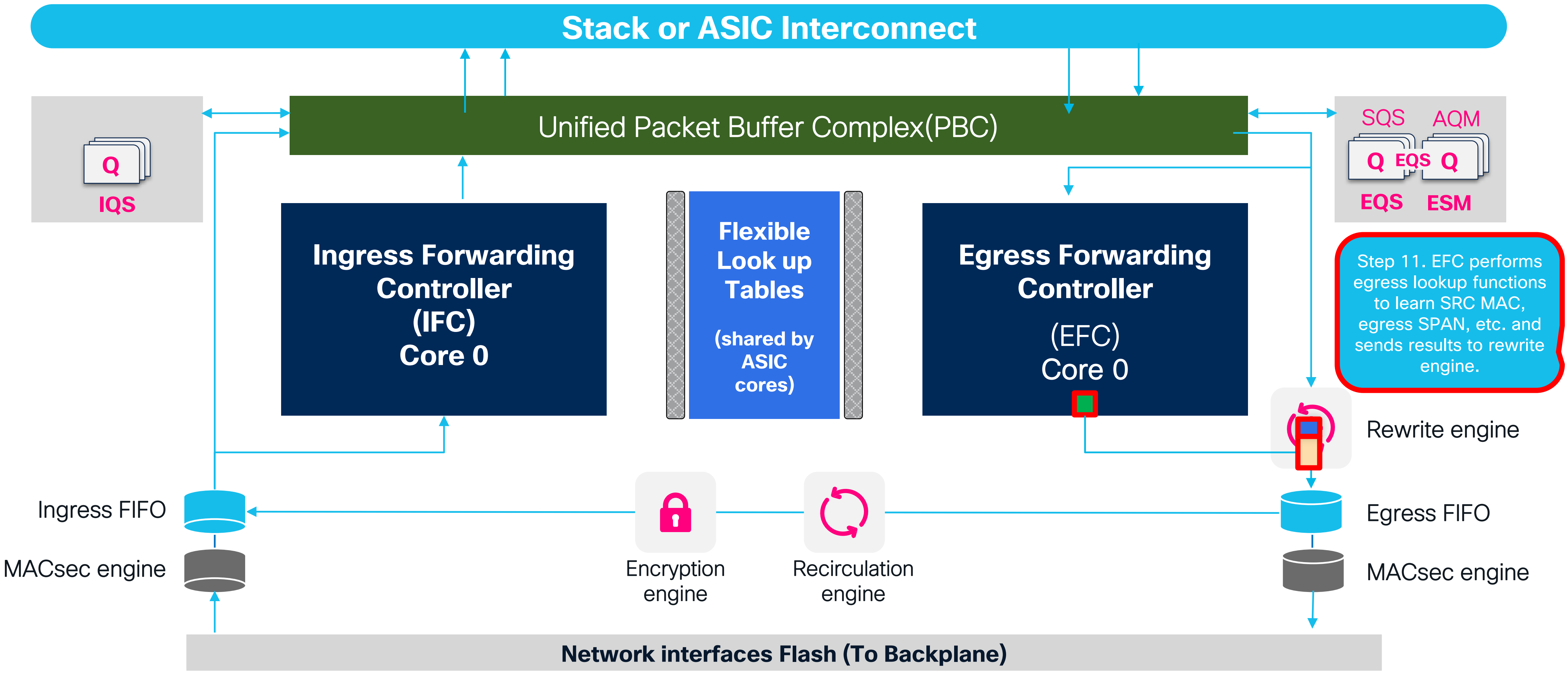
• Generic Unicast Packet Walk – UADP ASIC



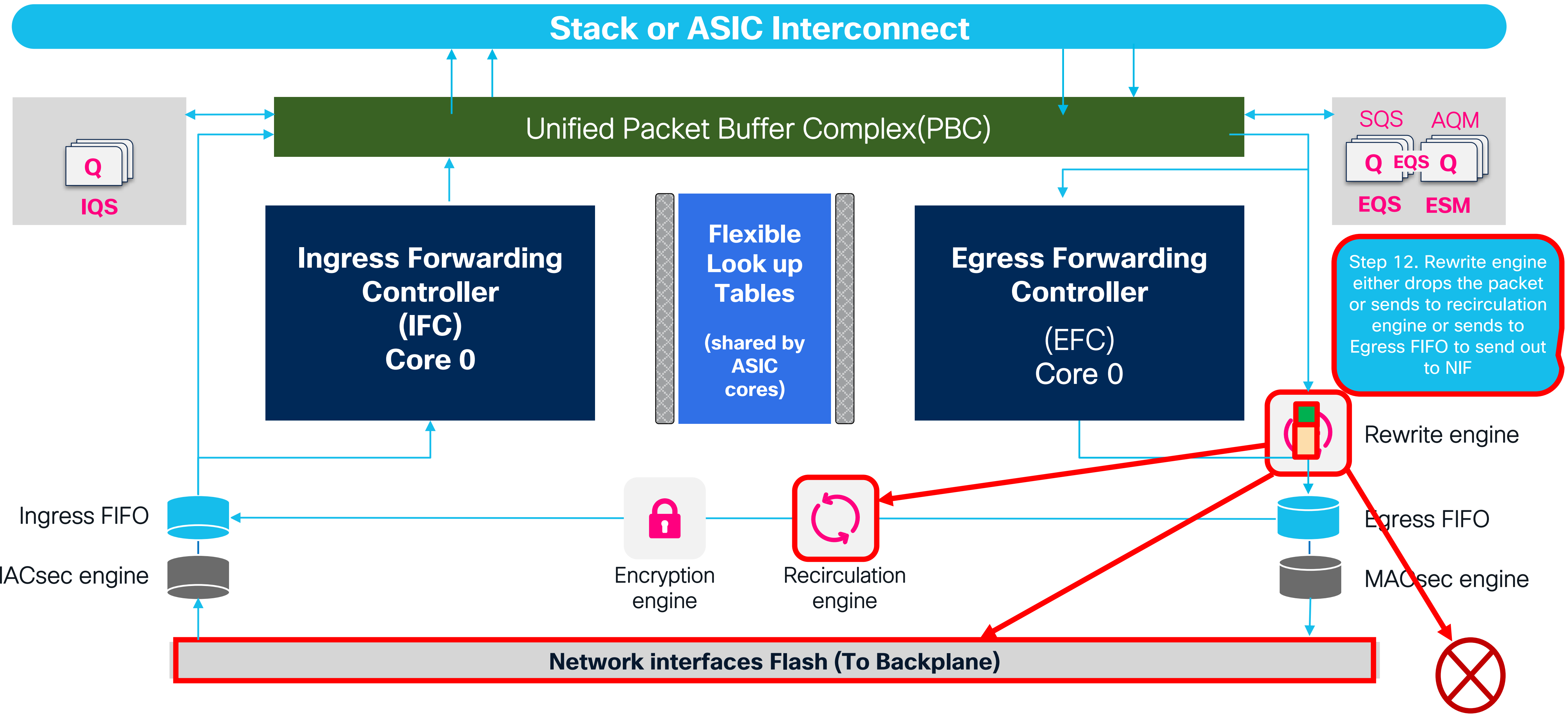
• Generic Unicast Packet Walk – UADP ASIC



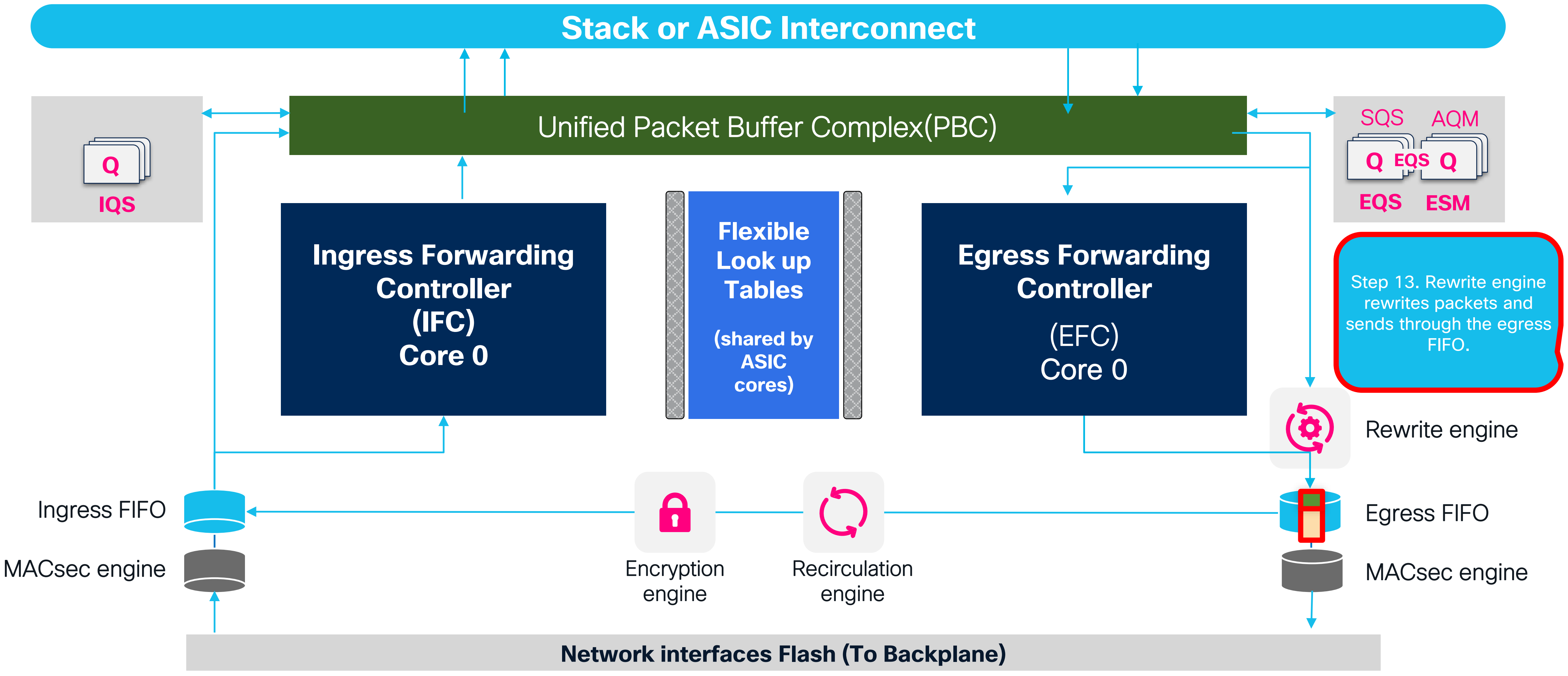
- Generic Unicast Packet Walk – UADP ASIC



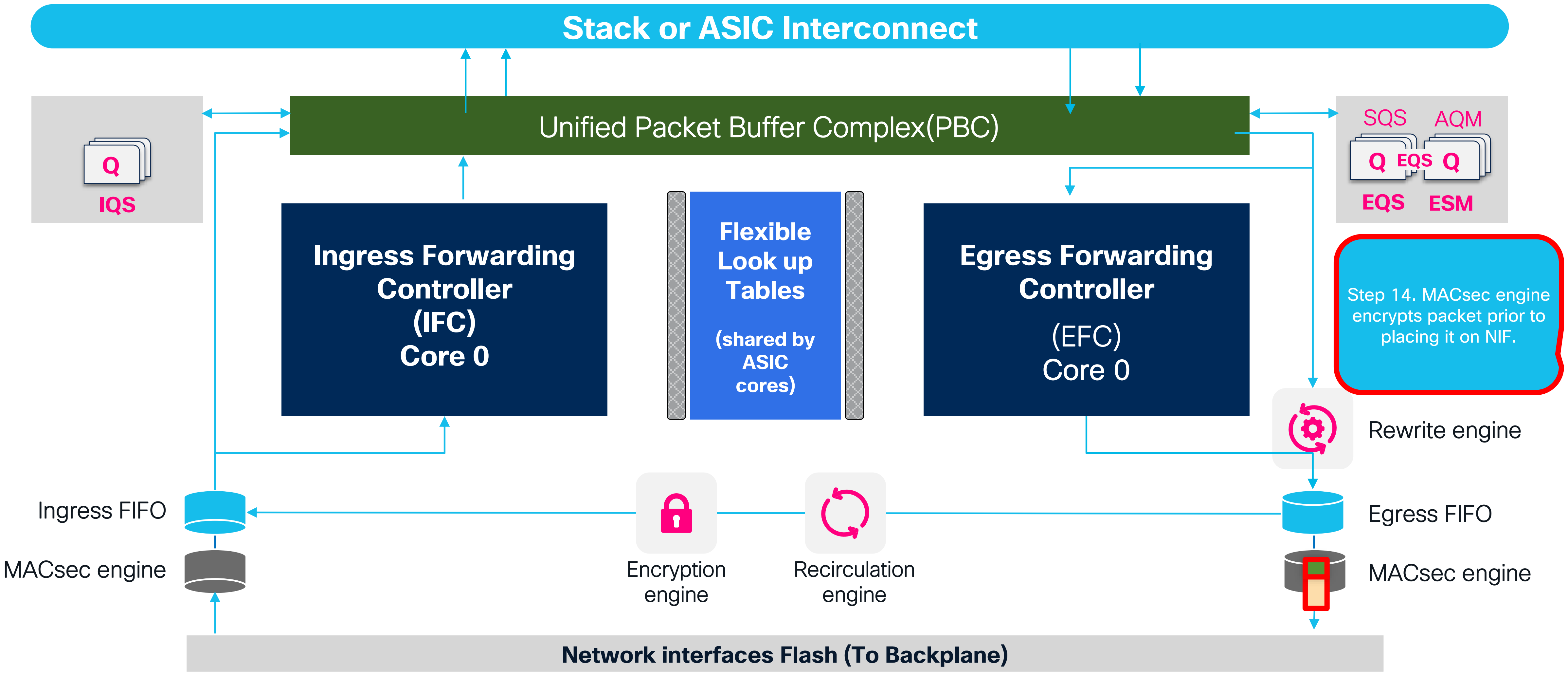
- Generic Unicast Packet Walk – UADP ASIC



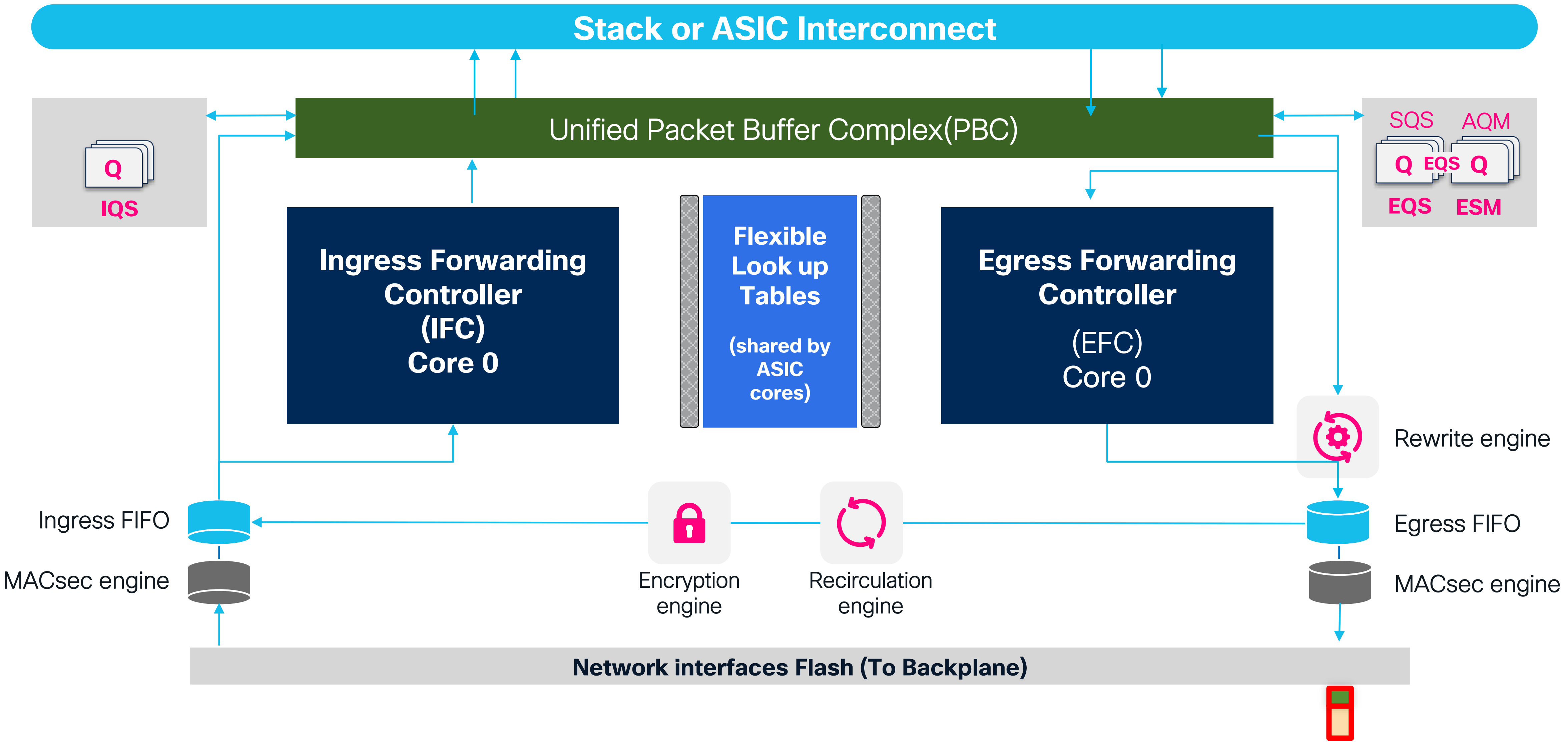
• Generic Unicast Packet Walk – UADP ASIC



• Generic Unicast Packet Walk – UADP ASIC



• Generic Unicast Packet Walk – UADP ASIC



Catalyst 9000 Family PID decoder

Product ID: C9300LM-48UX-4Y

Product Family
C9200
C9300
C9400 (C9404, C9407, C9410)
C9500
C9600 (C9606)

Product sub-family
L – Fixed uplink
C – Compact
X - Expansion
M – Mini (Shallow Depth)
R – Modular Chassis with Redundant Supervisor

# of ports and port type		
Copper (RJ45)		Fiber
T – Data only		S – 1Gbps (SFP)
P – PoE (30W)		X – 10Gbps (SFP+)
U – UPoE (60W)		Y – 25Gbps (SFP28)
H – UPoE+ (90W)		Q – 40G (QSFP)
G - 1Gbps		L – 50G (SFP56)
X – 10Gbps		C – 100G (QSFP28)
N – mGIG (up to 5Gbps)		DD – 400G (QSFP-DD)
M - mGIG		

Catalyst 9000 Family PID decoder

Product ID: C9500X-60L4D

Product Family
C9200
C9300
C9400 (C9404, C9407, C9410)
C9500
C9600 (C9606)

Product sub-family
L – Fixed uplink
C – Compact
X – Expansion
M – Mini (Shallow Depth)
R – Modular Chassis with Redundant Supervisor

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G – 1Gbps		L – 50G (SFP56)
X – 10Gbps		C – 100G (QSFP28)
N – mGIG (up to 5Gbps)		DD – 400G (QSFP-DD)
M – mGIG		

Quiz

C9300LM-48UX-4Y has feature mentioned below:-

IOSd process talks to FMAN-RP process via:-

Name the database having connection to all components of the Doppler/UADP ASIC?

Doppler
L2 & L3 Forwarding

Commands Logic (Doppler)

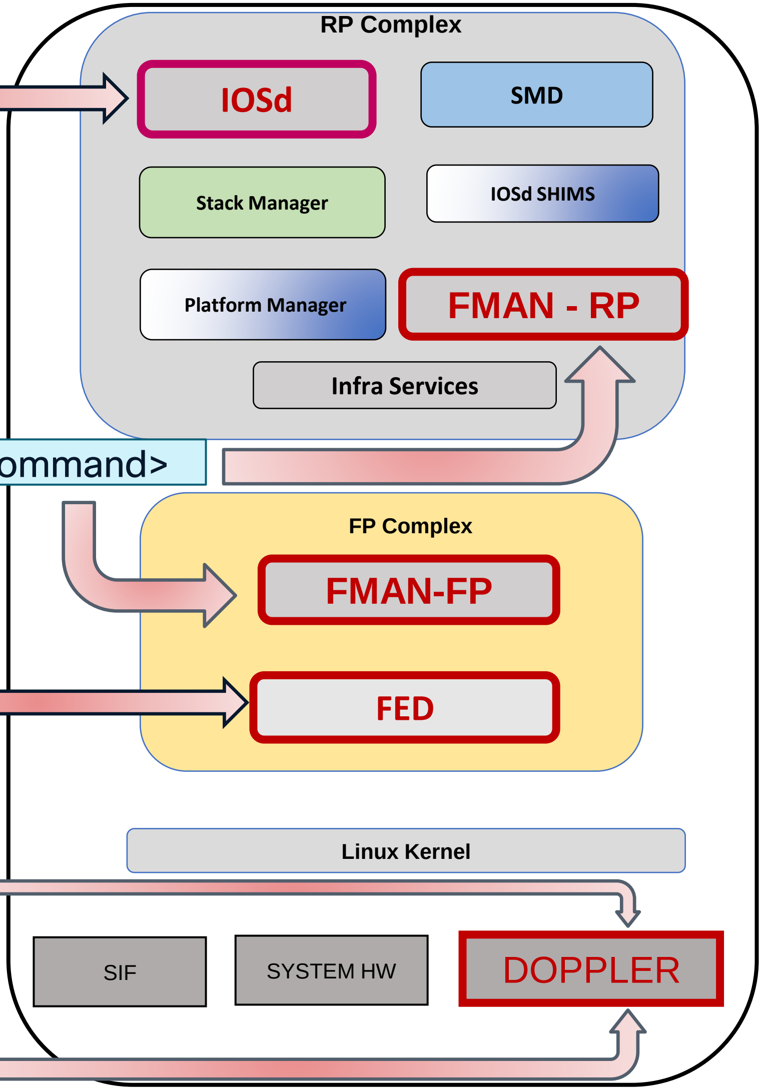
show <Process/Component> <Object> <Variable>

show platform software <Process/Component> switch <id | active | standby> <RP|FP|R0|F0> <command>

show platform software fed switch <id | active | standby> <process> <command>

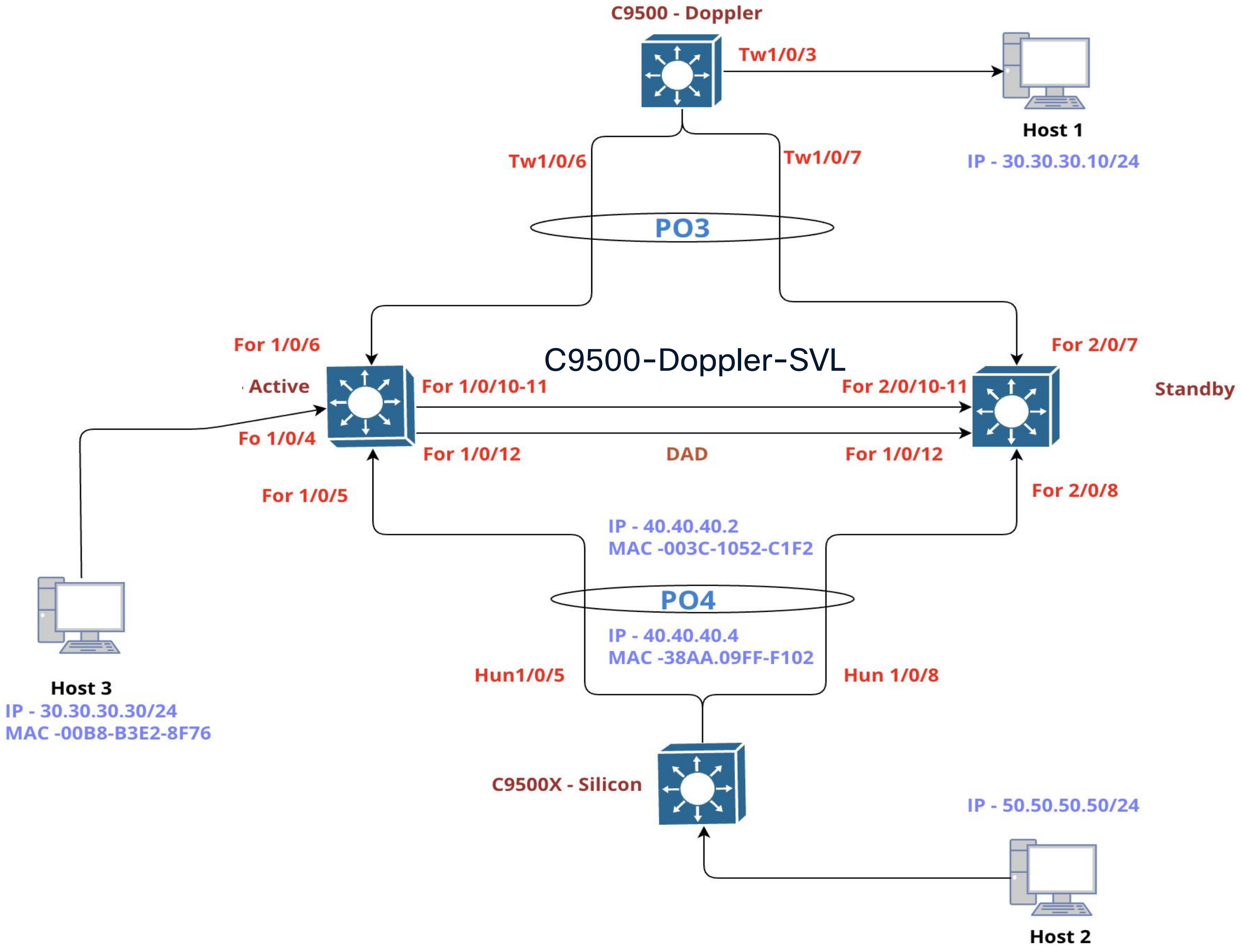
show platform hardware fed switch active fwd-asic abstraction print-resource-handle <>

show platform hardware fed switch active fwd-asic resource asic all <index> range <> <>



Doppler ASIC

Lab Topology



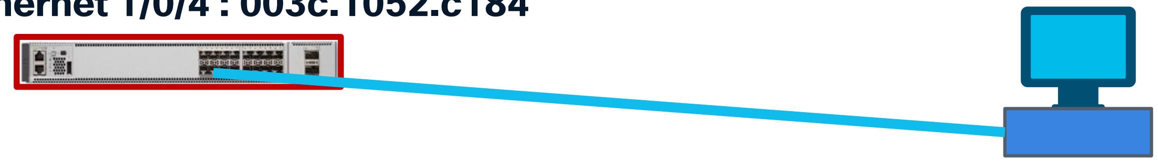
Layer 2 Forwarding

Port Programming

UADP/Doppler L2 Forwarding

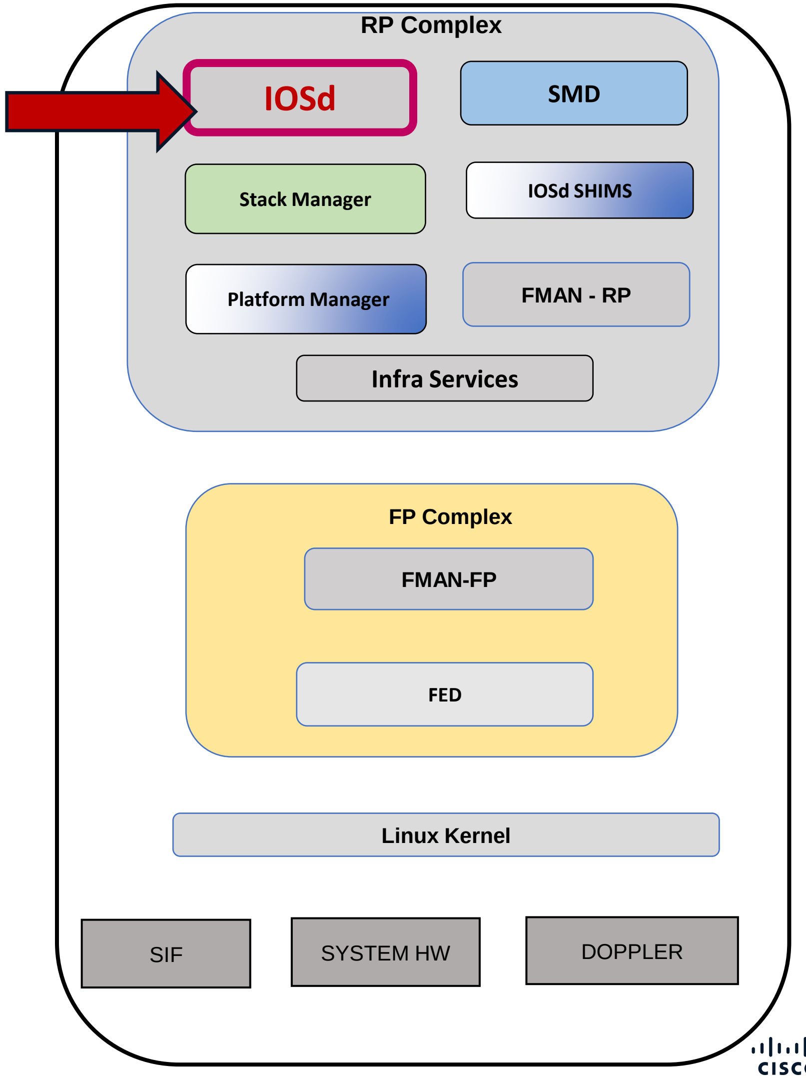
Port Programming
IOSd

FortyGigabitEthernet 1/0/4 : 003c.1052.c184



```
C9500-Doppler-SVL#show  run interface FortyGigabitEthernet 1/0/4
Building configuration...

Current configuration : 126 bytes
!
interface FortyGigabitEthernet1/0/4
 description To-Host3-Gig 1/0/4
 switchport access vlan 30
 switchport mode access
```

```
C9500-Doppler-SVL#show interfaces fortyGigabitEthernet 1/0/4
FortyGigabitEthernet1/0/4 is up, line protocol is up (connected)
  Hardware is Forty Gigabit Ethernet, address is 003c.1052.c184 (bia 003c.1052.c184)
  Description: To-Host3-Gig 1/0/4
  MTU 1500 bytes, BW 1000000 Kbit/sec, DLY 10 usec,
```

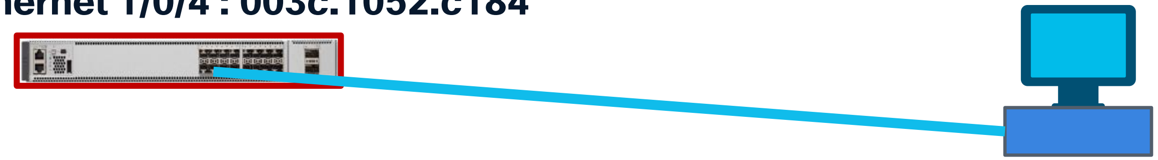


UADP/Doppler L2 Forwarding

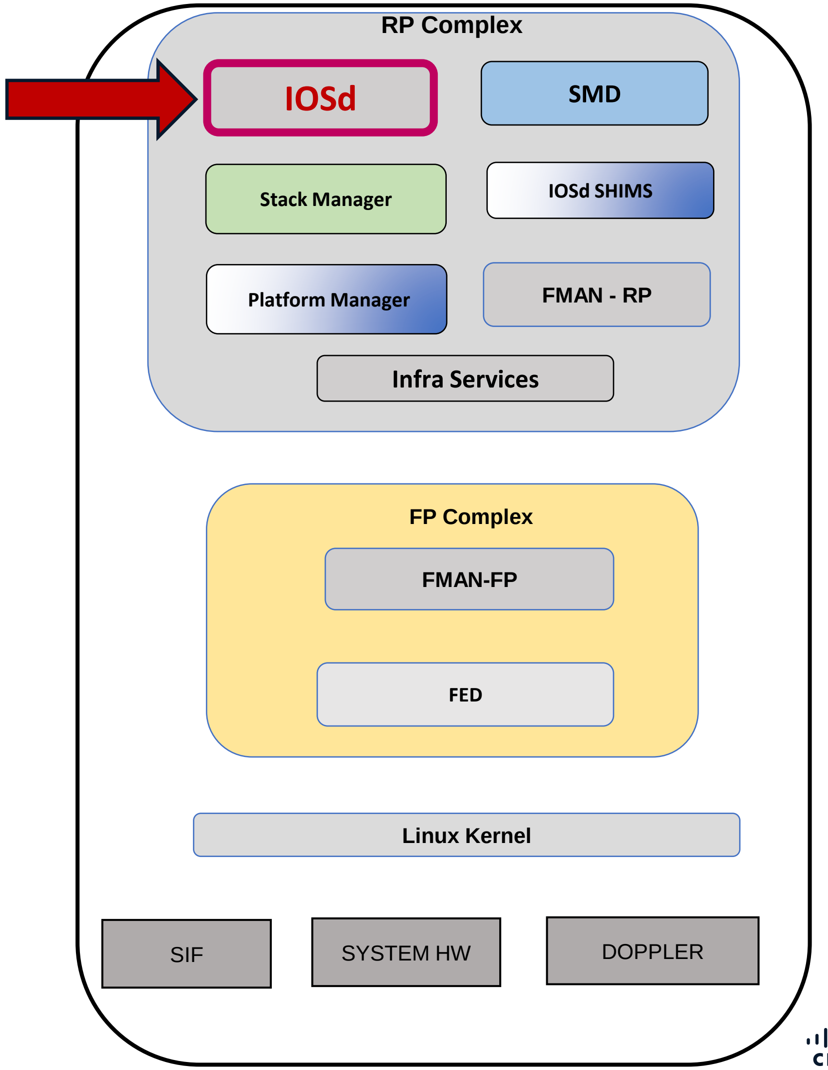
Port Programming

IOSD

FortyGigabitEthernet 1/0/4 : 003c.1052.c184

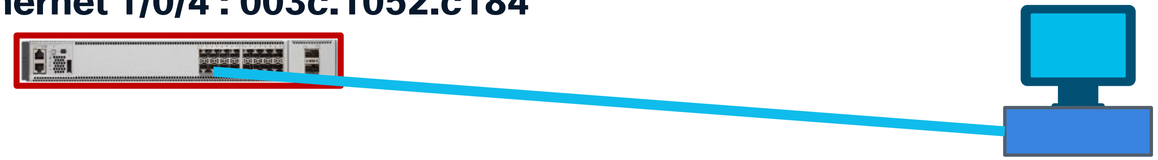


```
C9500-Doppler-SVL#show  idprom interface fortyGigabitEthernet 1/0/4 detail
General SFP Information
-----------------------------------------------
Identifier          :  SFP/SFP+
Ext.Identifier        :  SFP function is defined by two-wire interface ID only
Connector             :  Unknown
Transceiver
 10/40GE Comp code      :  Unknown
 SONET Comp code       :  Unknown
 GE Comp code         :  1000BASE-T
 Link length           :  Unknown
 ~
Length(Copper)           :  1000 m
Vendor Name          :  CISCO-AVAGO
~
Other Information
Chk for link status   : 00
Flow control Receive  : ON
Flow control Send     : Off
Administrative Speed  : auto
Administrative Duplex : auto
Operational Speed     : 1000
Operational Duplex     : full
~
```



UADP/Doppler L2 Forwarding

FortyGigabitEthernet 1/0/4 : 003c.1052.c184

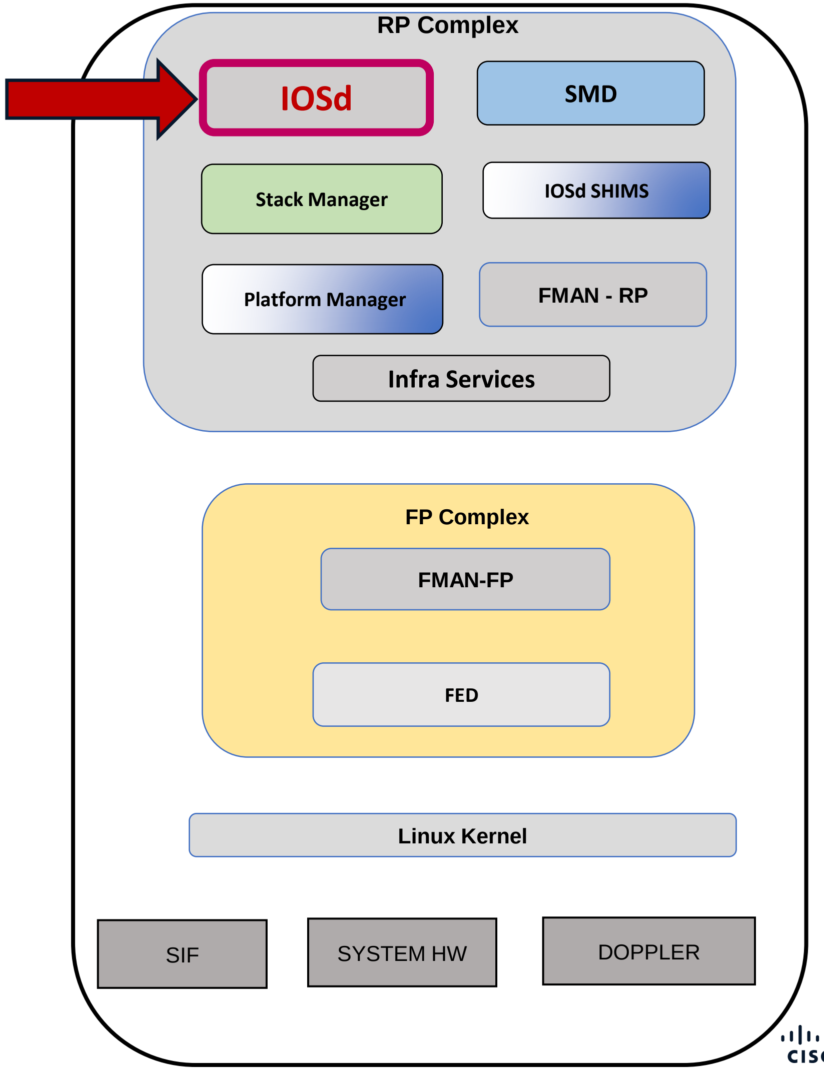


Port Programming

IOSD

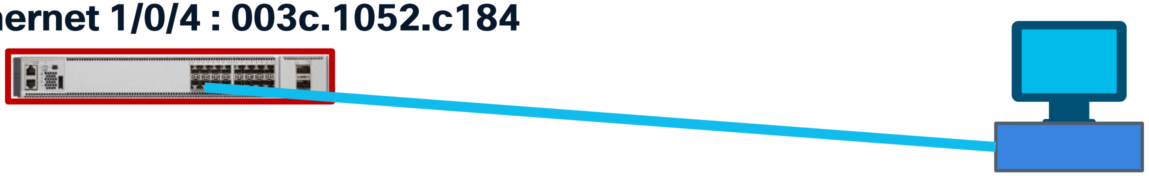
```
C9500-Doppler-SVL#show interfaces fortyGigabitEthernet 1/0/4 switchport
Name: Fo1/0/4
Switchport: Enabled
Administrative Mode: static access
Operational Mode: static access
Administrative Trunking Encapsulation: dot1q
Operational Trunking Encapsulation: native
Negotiation of Trunking: Off
Access Mode VLAN: 30 (VLAN0030)
Trunking Native Mode VLAN: 1 (default)
Administrative Native VLAN tagging: enabled
Voice VLAN: none
~
Unknown unicast blocked: disabled
Unknown multicast blocked: disabled
~
```

```
C9500-Doppler-SVL#show interfaces fortyGigabitEthernet 1/0/4 trunk
Port        Mode          Encapsulation  Status        Native vlan
Fo1/0/4       off         802.1q         not-trunking  1
Port        Vlans allowed on trunk
Fo1/0/4       30
Port        Vlans allowed and active in management domain
Fo1/0/4       30
Port        Vlans in spanning tree forwarding state and not pruned
Fo1/0/4       30
```



UADP/Doppler L2 Forwarding

FortyGigabitEthernet 1/0/4 : 003c.1052.c184



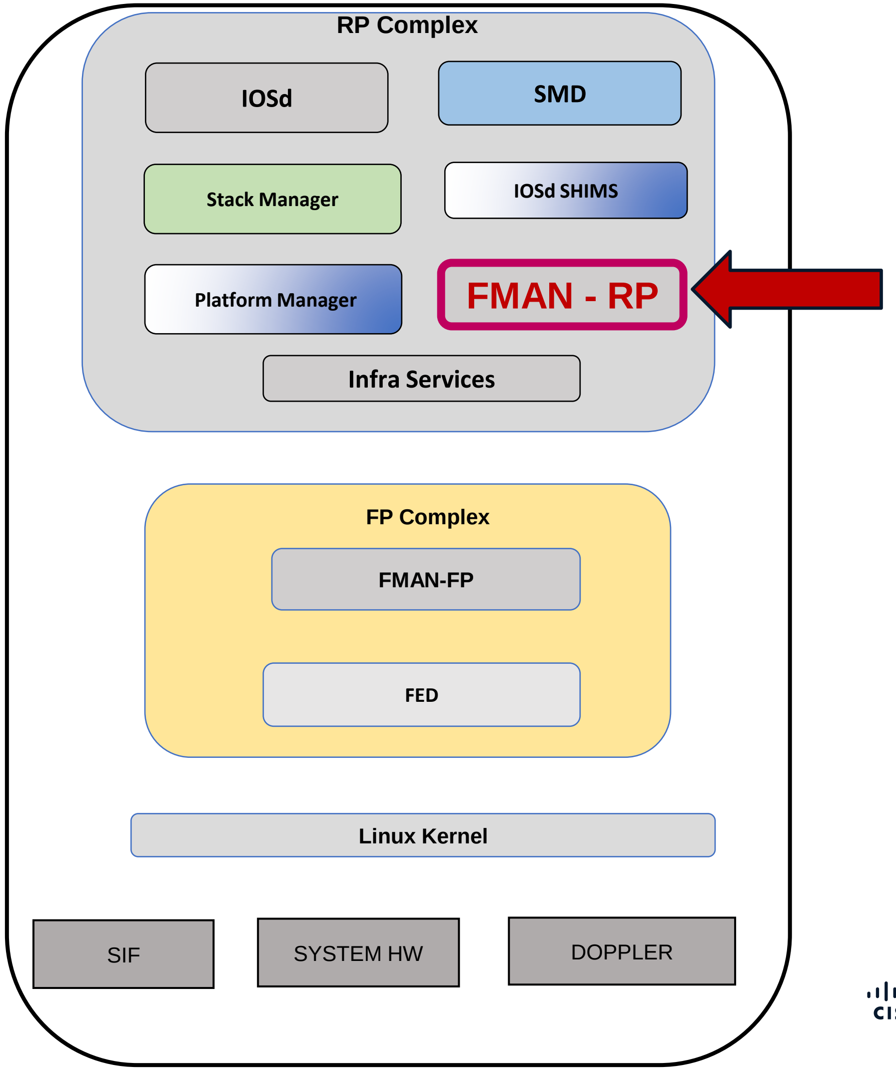
Port Programming

FMAN-RP

Hex(0x0C)=Dec(12)

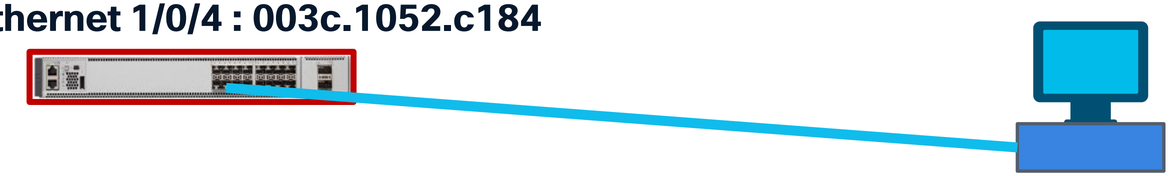
```
C9500-Doppler-SVL#show  platform software interface switch active R0 brief
Forwarding Manager Interfaces Information

Name                          ID          QFP ID
-------------------------------------------------------------
Null0                         1            0
FortyGigabitEthernet1/0/1     9            0
FortyGigabitEthernet1/0/2     10           0
FortyGigabitEthernet1/0/3     11           0
FortyGigabitEthernet1/0/4     12           0
~
```

```
C9500-Doppler-SVL#show  platform software interface switch active R0 name fortyGigabitEthernet 1/0/4
Name: FortyGigabitEthernet1/0/4, ID: 12, QFP ID: 0, Schedules: 4096
Type: PORT, State: enabled, SNMP ID: 12, MTU: 1500
Flow control ID: 65535
bandwidth: 1000000, encap: ARPA
IP Address:
~
QOS trust type: Trust DSCP
```



UADP/Doppler L2 Forwarding

FortyGigabitEthernet 1/0/4 : 003c.1052.c184



Port Programming

FMAN-RP

C9500-Doppler-SVL#show platform software interface switch active R0 statistics

Forwarding Manager Interfaces Statistics

SIP OIR: 0, SPA OIR: 0
DPIDB create: 132, DPIDB update: 1350, DPIDB delete: 2
Vlan update: 0, Vlan delete: 0
EVSI update: 0, EVSI delete: 0
Etherchannel member add: 0, delete: 0, error: 0
Etherchannel bucket add: 0, update: 0, delete: 0, error: 0

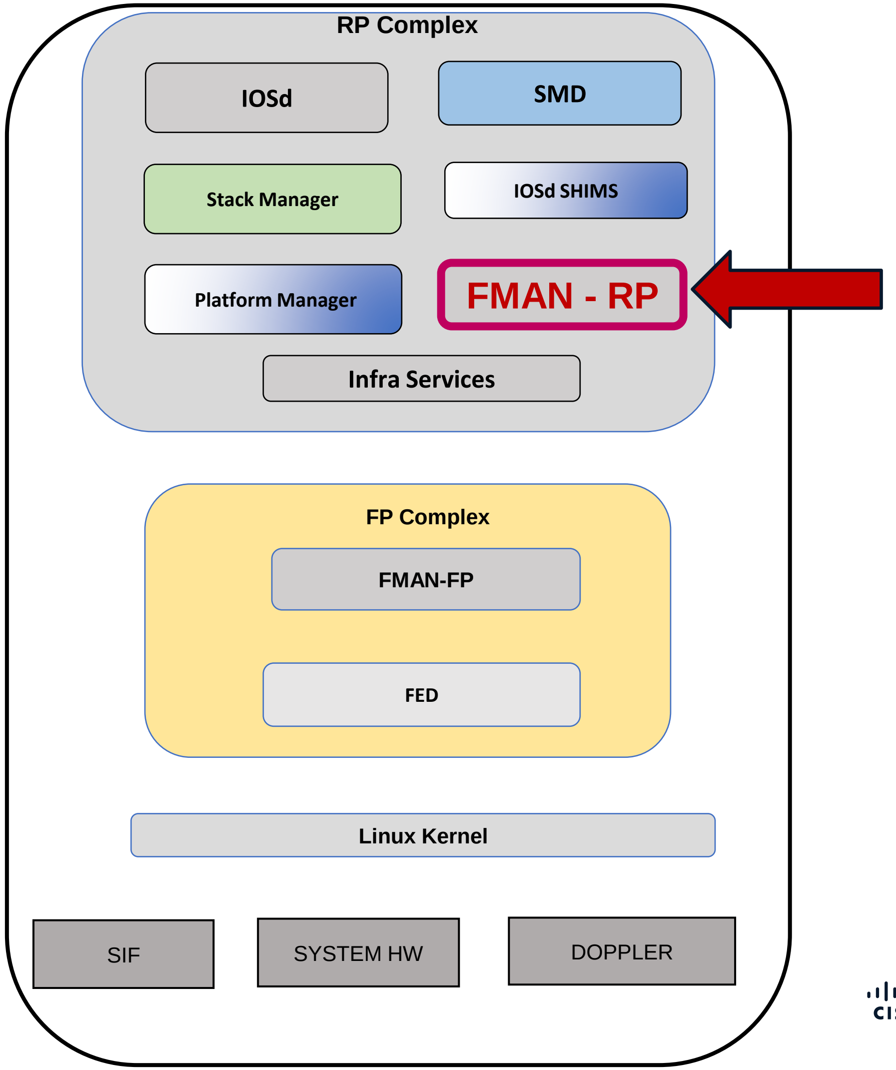
Fman RP-FP Statistics

DPIDB Total: , Sane:
~

AOM errors new: , create: , modify: , delete:
~

GIC API called: , Callback: , Callback error:
~

FR if_config_a: , rsp_cb: , req_err: , rsp_err:
~

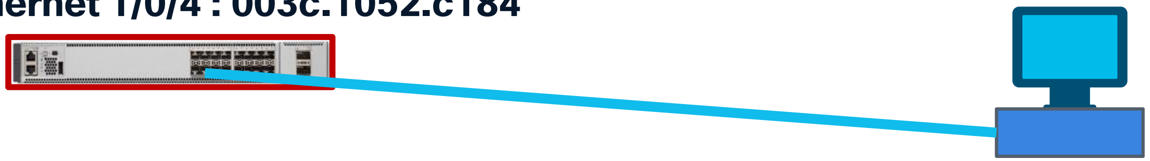


UADP/Doppler L2 Forwarding

Port Programming

FMAN-FP

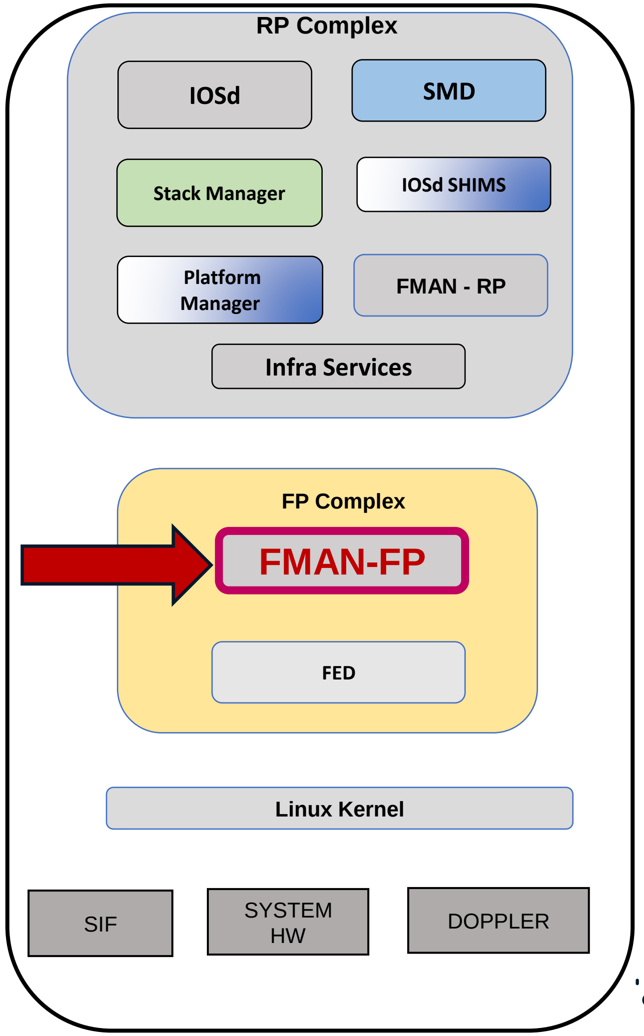
FortyGigabitEthernet 1/0/4 : 003c.1052.c184



Hex(0x0C)=Dec(12)

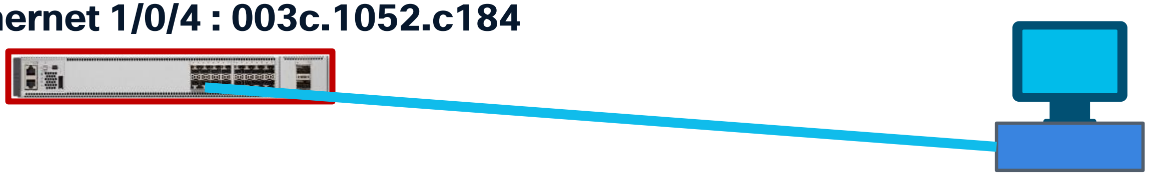
```
C9500-Doppler-SVL#show  platform software interface switch active F0
brief
Forwarding Manager Interfaces Information

Name                          ID            QFP ID
------------------------------------------------------------------
FortyGigabitEthernet1/0/1      9              9
FortyGigabitEthernet1/0/2      10             10
FortyGigabitEthernet1/0/3      11             11
FortyGigabitEthernet1/0/4      12             12
~
```

```
C9500-Doppler-SVL#show  platform software interface switch active F0 name
fortyGigabitEthernet 1/0/4
Name: FortyGigabitEthernet1/0/4, ID: 12, QFP ID: 12, Schedules: 4096
Type: PORT, State: enabled, SNMP ID: 12, MTU: 1500
TX channel ID: 0, RX channel ID: 0, AOM state: created
Flow control ID: 65535
bandwidth: 1000000, encap: ARPA
~
```



UADP/Doppler L2 Forwarding

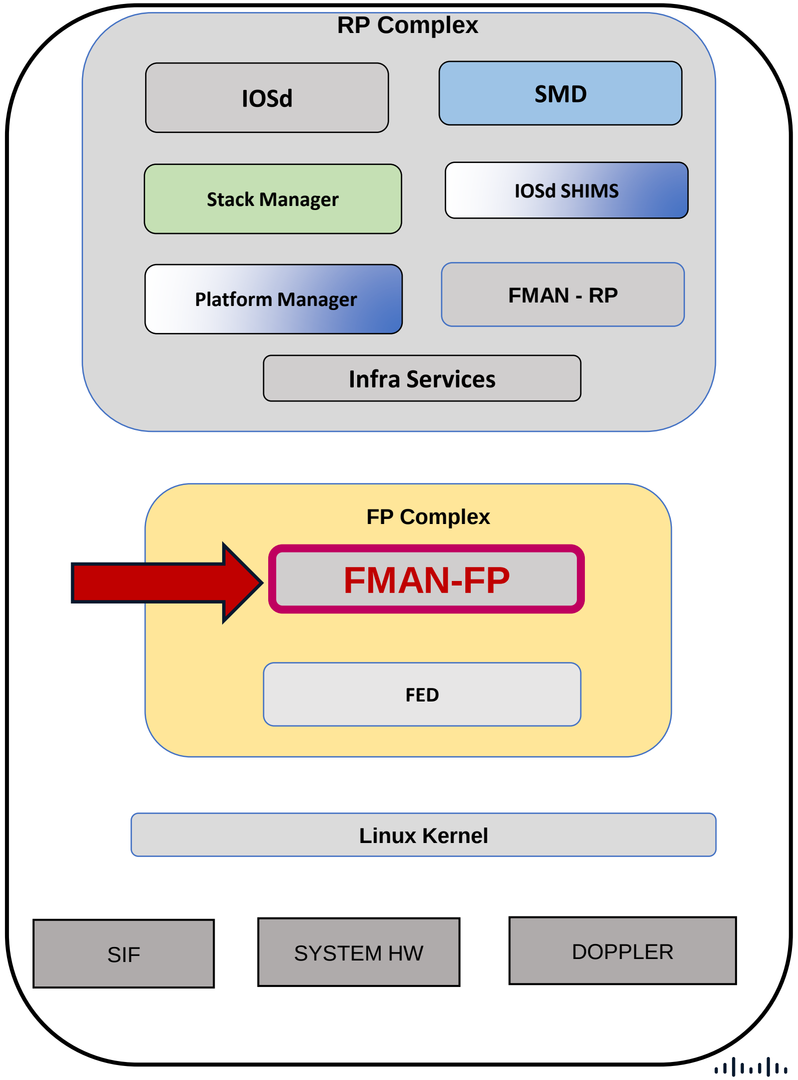
FortyGigabitEthernet 1/0/4 : 003c.1052.c184



Port Programming

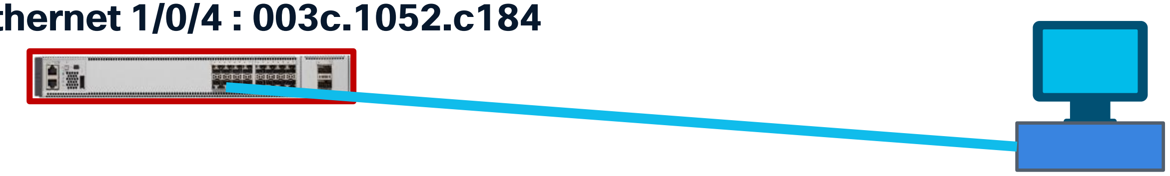
FMAN-FP

```
C9500-Doppler-SVL#show  platform software interface switch active F0 statistics
Forwarding Manager Interfaces Statistics

SIP OIR: 0, SPA OIR: 0
DPIDB create: 534, DPIDB update: 0, DPIDB delete: 2
Vlan update: 0, Vlan delete: 0
EVSI update: 0, EVSI delete: 0
Etherchannel member add: 0, delete: 0, error: 0
Etherchannel bucket add: 0, update: 0, delete: 0, error: 0
CM update: 0, CM delete: 0
DPIDB disable: 14
SG create: 0, SG delete: 0

Fman RP-FP Statistics
DPIDB Total: 130, Sane: 0
Port: 120, Channelized: 0
Virtual: 0, Tunnel: 0, Ether-channel: 2
Sub-interface: 0, EVSI: 0
~
AOM errors new: 0, create: 0, modify: 0, delete: 0
~
Adjacency reparent to Null errors: 0
```



UADP/Doppler L2 Forwarding

FortyGigabitEthernet 1/0/4 : 003c.1052.c184



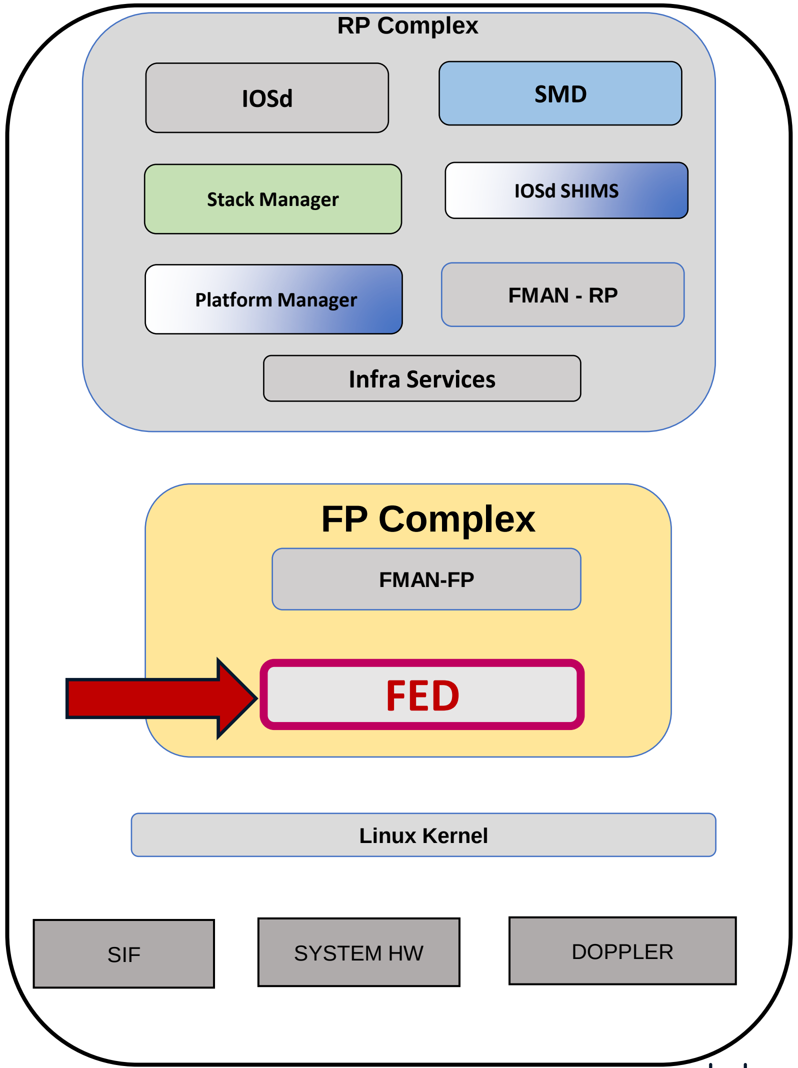
Port Programming

FED – Forwarding Engine Driver

Hex(0x0C)=Dec(12)

```
C9500-Doppler-SVL#show  platform software fed switch active ifm mappings
Interface                    IF_ID   Inst Asic Core Port SubPort Mac  Cntx LPN  GPN  Type Active
FortyGigabitEthernet1/0/1 0x9     3    1    1    0    0       8    0    1    1    NIF  Y
FortyGigabitEthernet1/0/2 0xa     3    1    1    1    0       4    4    2    2    NIF  Y
FortyGigabitEthernet1/0/3 0xb     3    1    1    2    0       0    8    3    3    NIF  Y
FortyGigabitEthernet1/0/4 0xc     2    1    0    3    0       8    0    4    4    NIF  Y
FortyGigabitEthernet1/0/5 0x85    2    1    0    4    0       4    4    5    1370 NIF  Y
FortyGigabitEthernet1/0/6 0xe     2    1    0    5    0       0    8    6    1363 NIF  Y
~
~
```

```
C9500-Doppler-SVL#show platform software fed switch active ifm interfaces ethernet
Interface                    IF_ID           State
---------------------------------------------------------------
FortyGigabitEthernet1/0/1    0x00000009      READY
FortyGigabitEthernet1/0/2    0x0000000a      READY
FortyGigabitEthernet1/0/3    0x0000000b      READY
FortyGigabitEthernet1/0/4    0x0000000c      READY
FortyGigabitEthernet1/0/5    0x00000085      READY
~
~
```

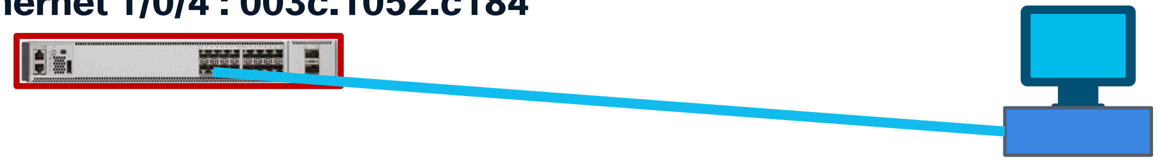


UADP/Doppler L2 Forwarding

Port Programming

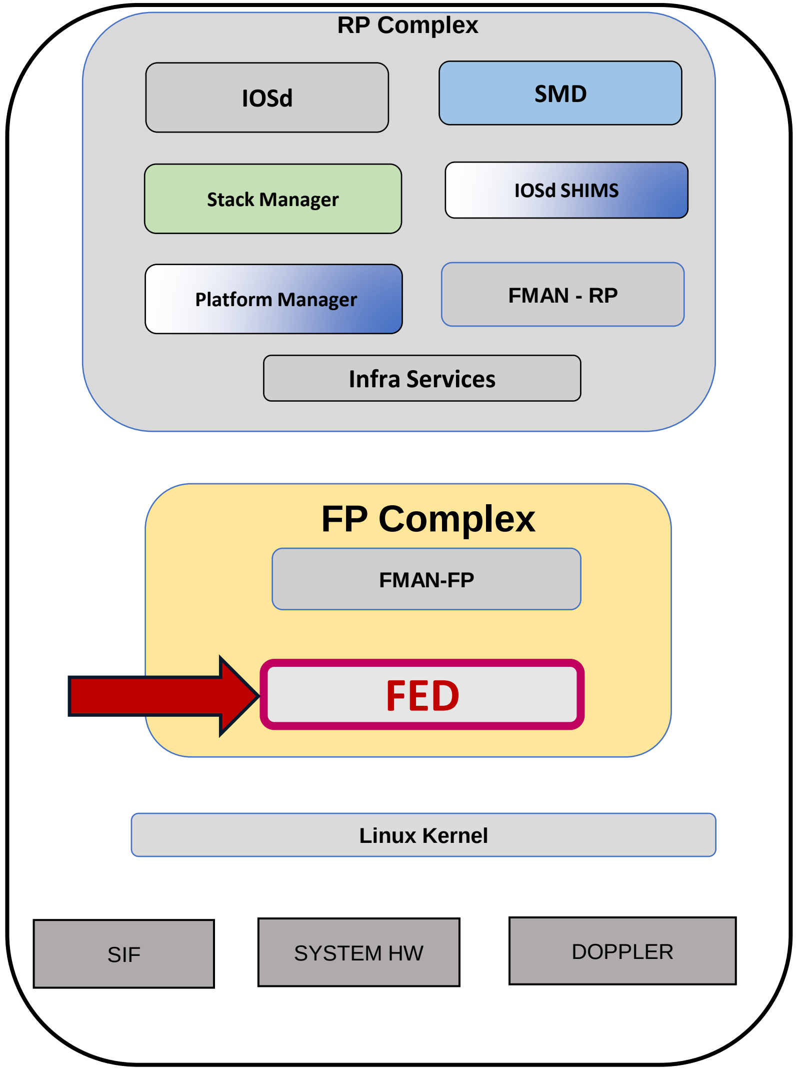
FED – Forwarding Engine Driver

FortyGigabitEthernet 1/0/4 : 003c.1052.c184



```
C9500-Doppler-SVL#show platform software fed switch active ifm if-id 0xc
C9500-Doppler-SVL#show platform software fed switch active ifm if-id 12

Interface IF_ID        : 0x000000000000000c
Interface Name         : FortyGigabitEthernet1/0/4
Interface Block Pointer : 0x7d91341d2648
Interface Block State   : READY
Interface State        : Enabled
Interface Status       : ADD, UPD
Interface Ref-Cnt      : 7
Interface Type         : ETHER
Created Time           : 2026/01/07 07:24:23.010
Last Modified Time     : 2026/01/07 19:05:20.747
Current Time           : 2026/01/08 16:08:40.791
    Port Type          : SWITCH PORT
      Port Location    : LOCAL
      Slot             : 1
~
      Port Information
      Handle ............ [0xac00002d]
    Type .............. [Layer2]
    Identifier ........ [0xc]
      Slot .............. [1]
      Unit .............. [4]
```

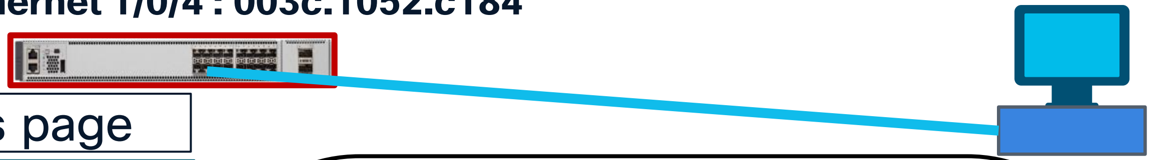
Hex(0x0C)=Dec(12)



Continued →

UADP/Doppler L2 Forwarding

FortyGigabitEthernet 1/0/4 : 003c.1052.c184



Port Programming

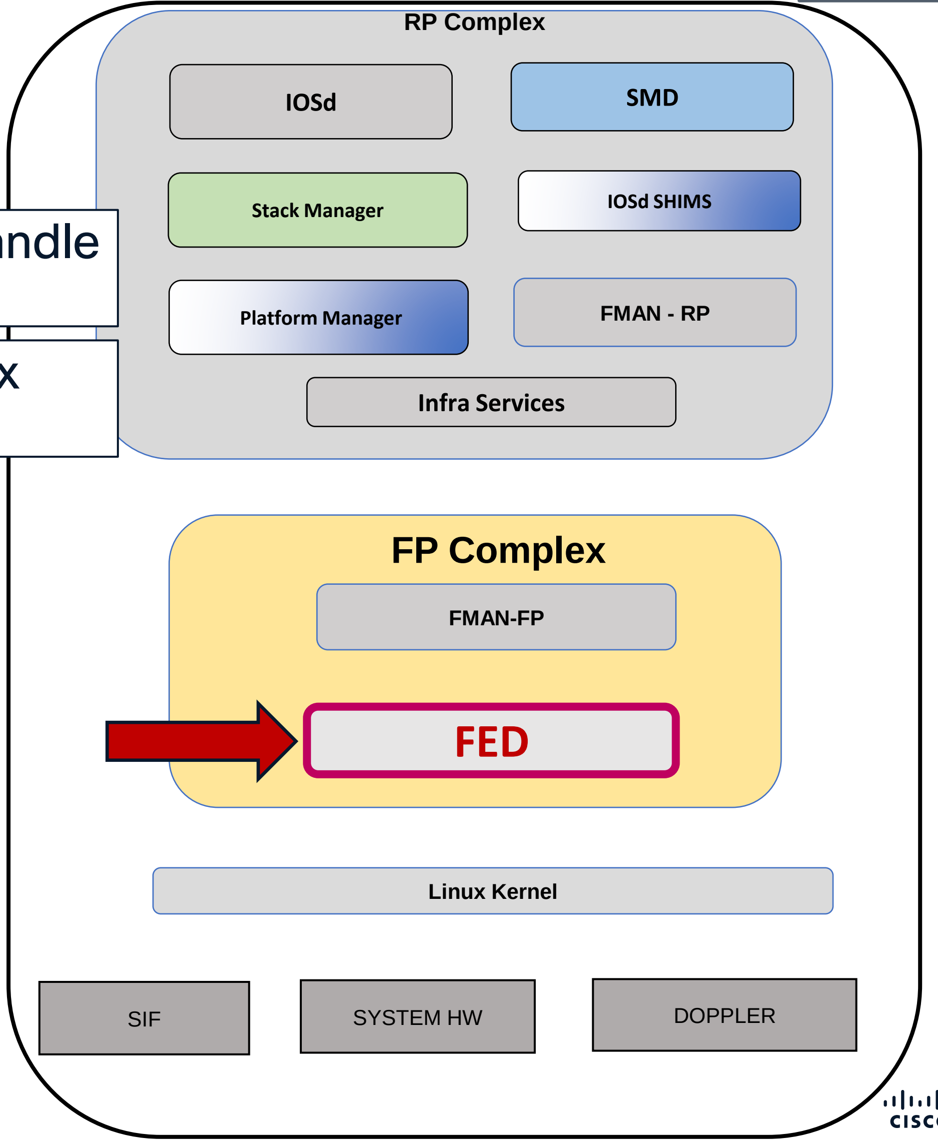
FED

→ Continued from previous page

```
Port Physical Subblock
            Affinity .......... [local]
            Asic Instance ..... [2 (A:1,C:0)]
~
            Speed ............. [40GB]
            type .............. [NIF]
            PORT_LE ........... [0x7d91341d2a78] → →
            L3IF_LE ........... [0x0]
            DI ................ [0x7d91341d96f8]  → →
            SubIf count ....... [0]
      Port L2 Subblock
            Enabled ............. [Yes]
            Allow dot1q ......... [No]
            Allow native ........ [Yes]
            Default VLAN ........ [30]
            Allow priority tag ... [No]
            Allow unknown unicast  [Yes]
            Allow unknown multicast[Yes]
~
            App Hosting........... [Disabled]
      Port QoS Subblock
            Trust Type ................... [0x2]
            Default Value ................ [0]
~
```

Port Logical Entity Handle for Doppler

Port Destination Index Handle for Doppler

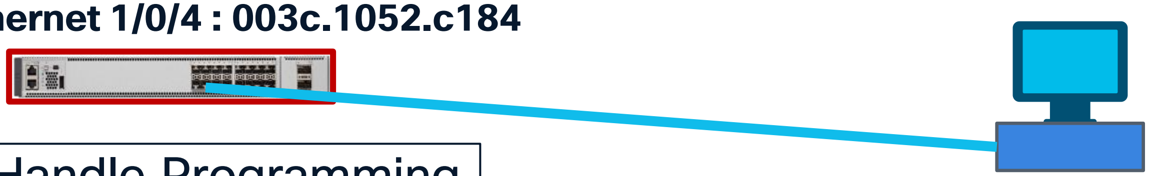


UADP/Doppler L2 Forwarding

Port Programming

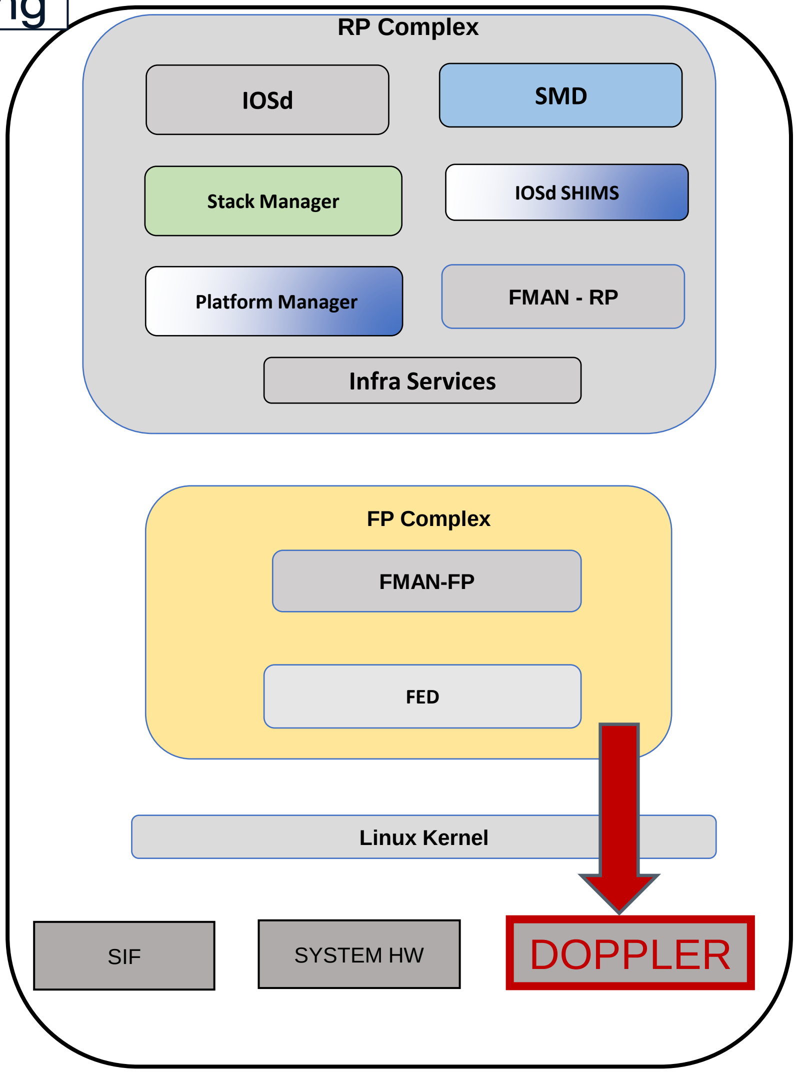
UADP/DOPPLER – Forward ASIC

FortyGigabitEthernet 1/0/4 : 003c.1052.c184



Port Logical Entity (LE) Handle Programming

```
C9500-Doppler-SVL#show  platform hardware fed switch active fwd-asic abstraction print-resource-handle 0x7d91341d2a78 0
Handle:0x7d91341d2a78 Res-Type:ASIC_RSC_PORT_LE Res-Switch-Num:0 Asic-Num:2
Feature-ID:AL_FID_IFM Lkp-ftr-id:LKP_FEAT_INGRESS_PRECLASS1_IPV4 ref_count:1
priv_ri/priv_si Handle: (nil)Hardware Indices/Handles: index2:0x3  mtu_index/l3u_ri_index2:0x2
sm handle [ASIC 2]: 0x7d91341de798

Brief Resource Information (ASIC_INSTANCE# 2)
----------------------------------------
LEAD_PORT_ALLOW_BROADCAST value 1 Pass
LEAD_PORT_ALLOW_CAPWAP value 0 Pass
LEAD_PORT_ALLOW_CTS value 0 Pass
LEAD_PORT_ALLOW_DOT1Q_TAGGED value 0 Pass
LEAD_PORT_ALLOW_MULTICAST value 1 Pass
LEAD_PORT_ALLOW_NATIVE value 1 Pass
LEAD_PORT_ALLOW_NON_CTS value 0 Pass
LEAD_PORT_ALLOW_PRIORITY_TAGGED value 0 Pass
LEAD_PORT_ALLOW_UNICAST value 1 Pass
LEAD_PORT_ALLOW_UNKNOWN_ETHER_TYPE value 0 Pass
LEAD_PORT_ALLOW_UNKNOWN_UNICAST value 1 Pass
LEAD_PORT_ALLOW_VLAN_LOAD_BALANCE_GROUP value 15 Pass
LEAD_PORT_ALLOW_VRF value 0 Pass
~
```



UADP/Doppler L2 Forwarding

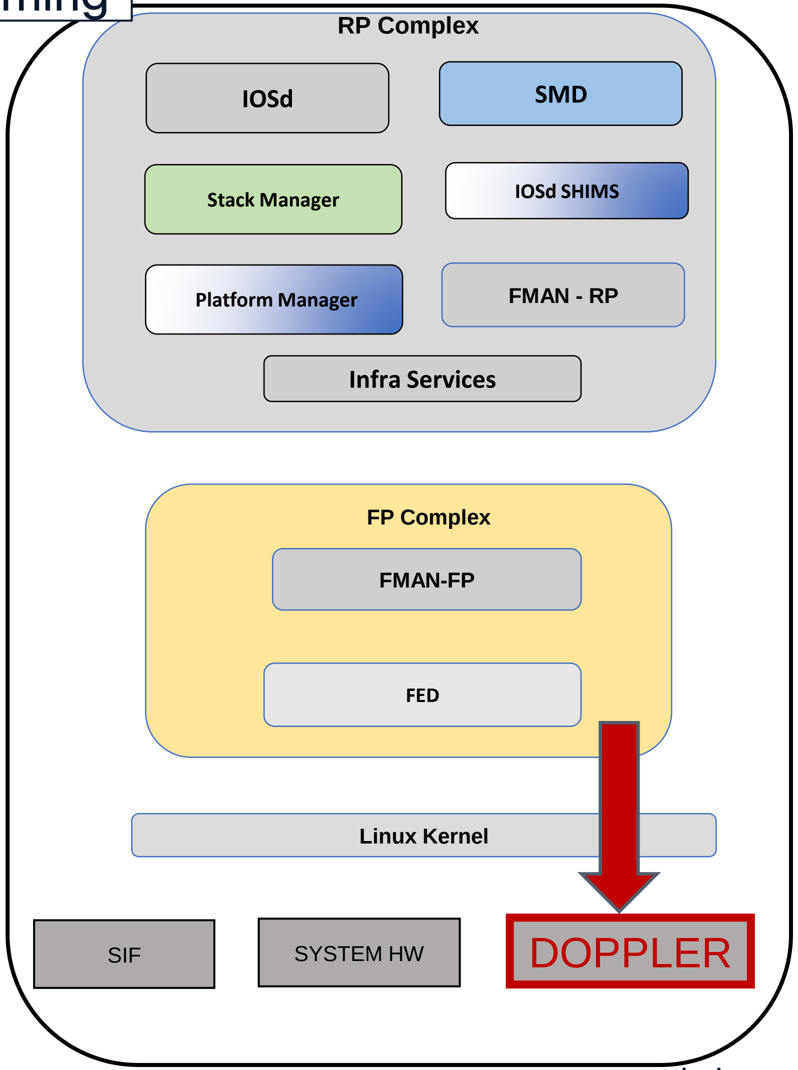
FortyGigabitEthernet 1/0/4 : 003c.1052.c184

Port Programming

UADP/DOPPLER – Forward ASIC

Port Destination Index (DI) Handle Programming

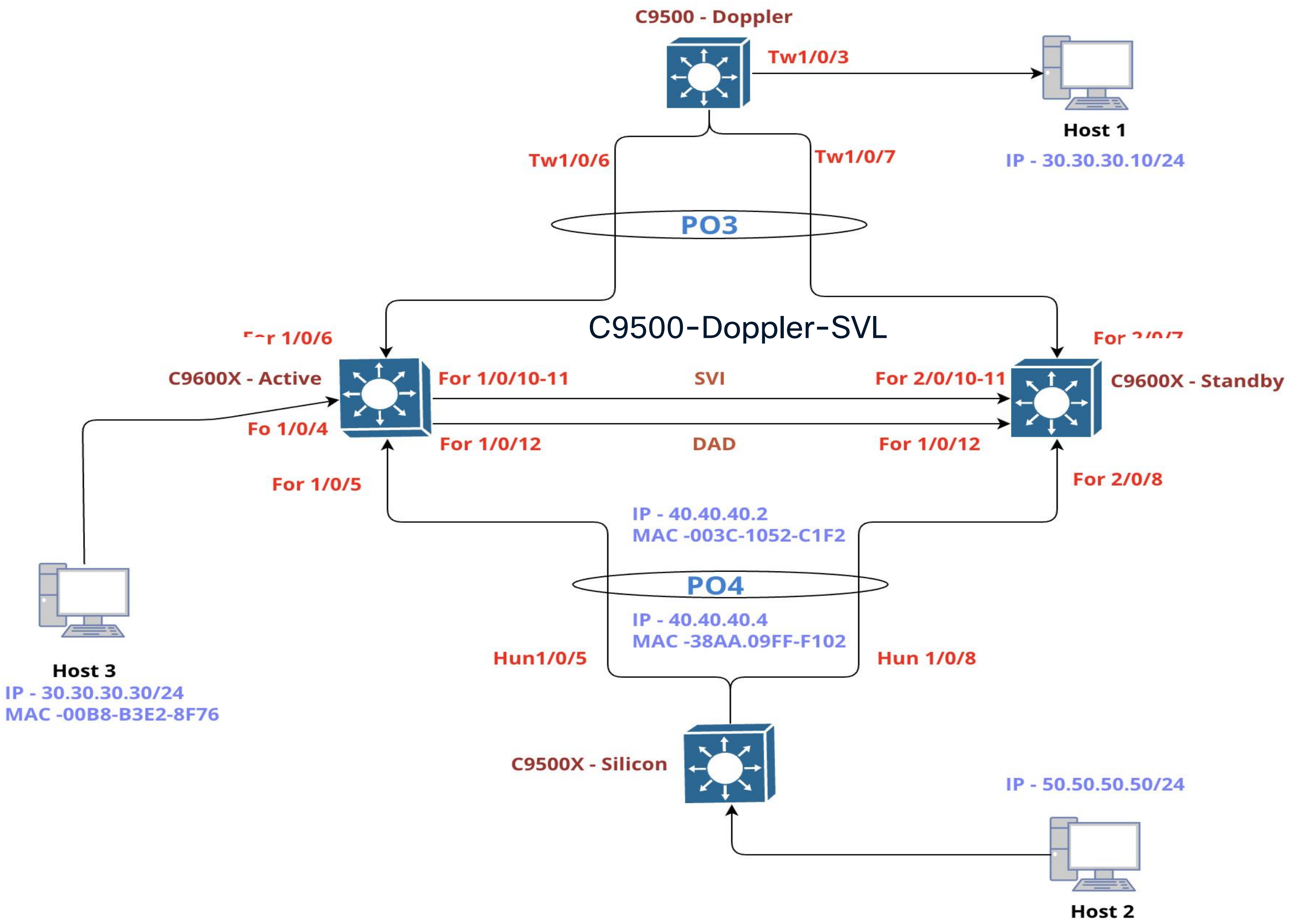
```
C9500-Doppler-SVL#show  platform hardware fed switch active fwd-asic abstraction print-resource-handle 0x7d91341d96f8 0
Handle:0x7d91341d96f8 Res-Type:ASIC_RSC_DI Res-Switch-Num:255 Asic-Num:255
Feature-ID:AL_FID_IFM Lkp-ftr-id:LKP_FEAT_INVALID ref_count:1
priv_ri/priv_si Handle: (nil)Hardware Indices/Handles: index0:0x4  mtu_index/l3u_ri_index0:0x0
index1:0x4  mtu_index/l3u_ri_index1:0x0  index2:0x4  mtu_index/l3u_ri_index2:0x0
index3:0x4  mtu_index/l3u_ri_index3:0x0

Brief Resource Information (ASIC_INSTANCE# 0)
----------------------------------------
Destination index   = 0x4 DI_PORT_GPN
pmap                = 0x00000000 0x00000000

Brief Resource Information (ASIC_INSTANCE# 1)
~
~
Brief Resource Information (ASIC_INSTANCE# 2)
----------------------------------------
Destination index   = 0x4 DI_PORT_GPN
pmap                = 0x00000000 0x00000008
pmap_intf : [FortyGigabitEthernet1/0/4]
~
~
```



VLAN and Spanning-Tree Programming

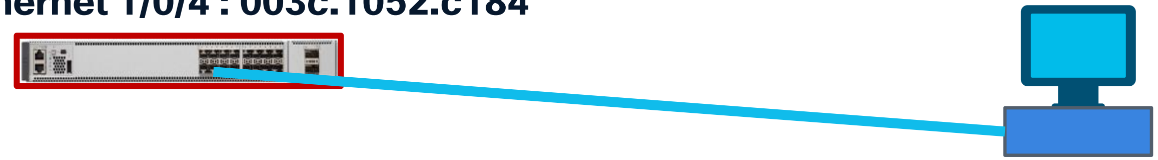
Doppler ASIC

Lab Topology



UADP/Doppler L2 Forwarding

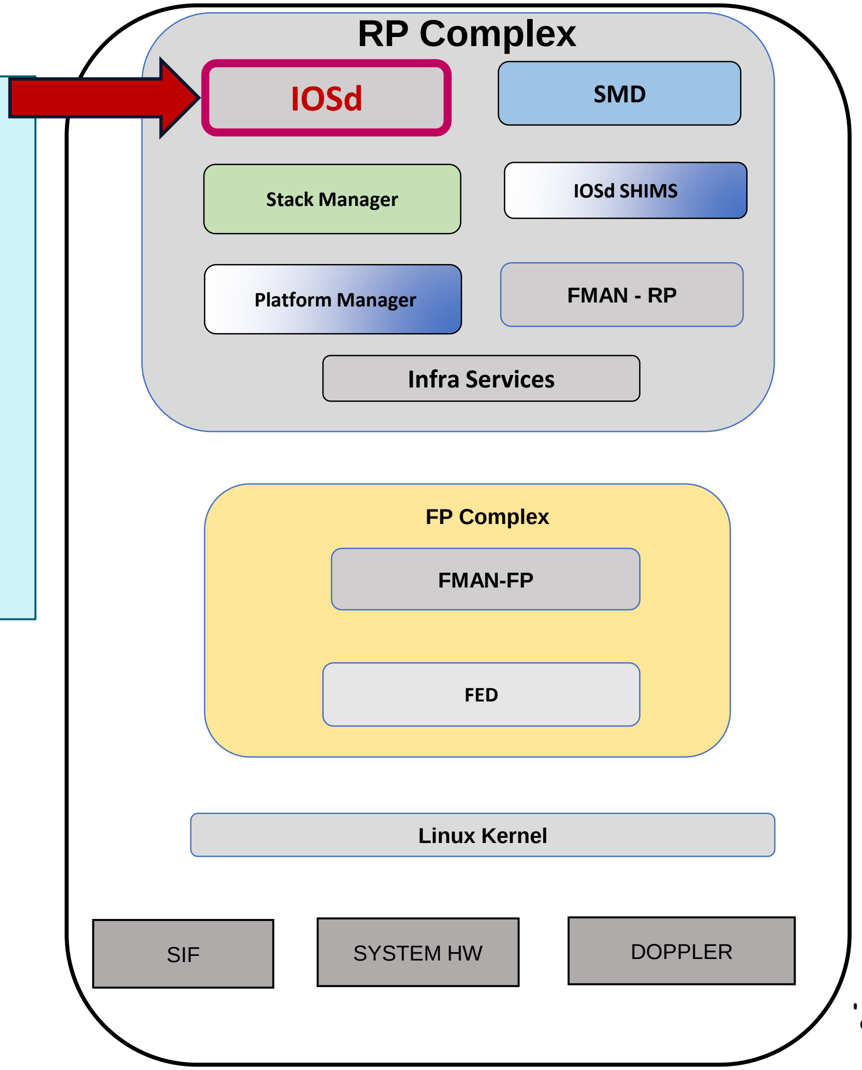
FortyGigabitEthernet 1/0/4 : 003c.1052.c184



VLAN and Spanning-Tree Programming

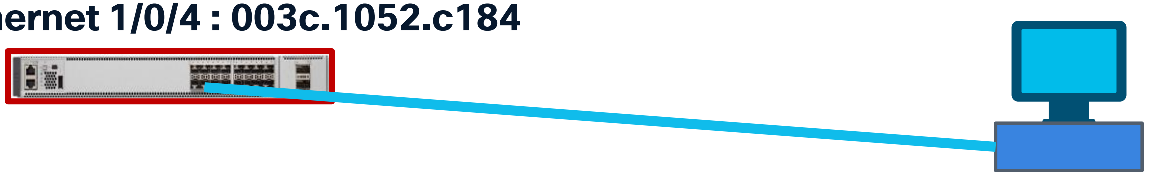
IOSd

```
C9500-Doppler-SVL#show  vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fo1/0/1, Fo1/0/2, Fo1/0/3, Fo1/0/7,
                                                Fo1/0/8, Fo1/0/9, Fo2/0/1, Fo2/0/2,
                                                Fo2/0/4, Fo2/0/5, Fo2/0/6, Fo2/0/9
30   VLAN0030                         active    Fo1/0/4, Fo2/0/3
1002 fddi-default                     act/unsup
1003 token-ring-default               act/unsup
1004 fddinet-default                  act/unsup
1005 trnet-default                    act/unsup
```



UADP/Doppler L2 Forwarding

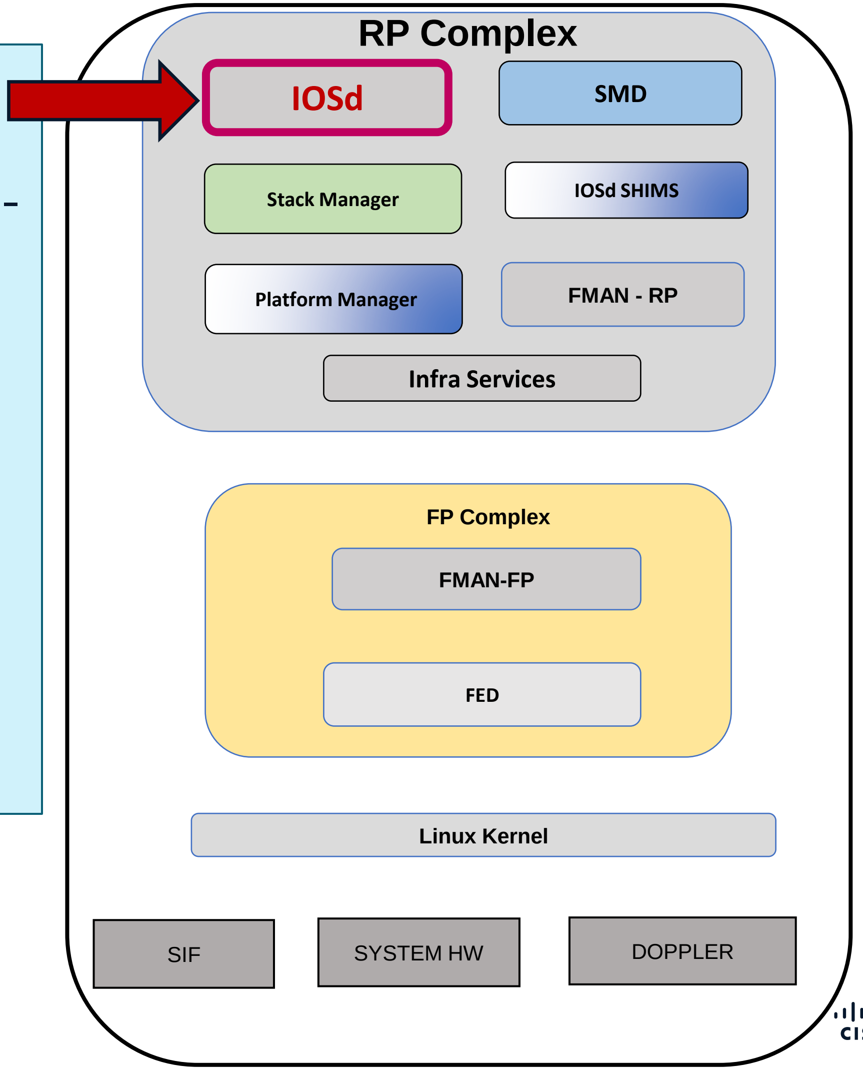
FortyGigabitEthernet 1/0/4 : 003c.1052.c184



VLAN and Spanning-Tree Programming

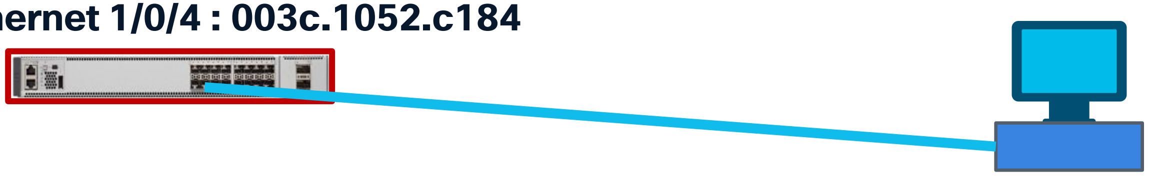
IOSd

```
C9500-Doppler-SVL#show  vlan id 30

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
30   VLAN0030                         active    Fo1/0/4, Fo2/0/3, Po3

VLAN Type  SAID       MTU   Parent RingNo BridgeNo Stp  BrdgMode Trans1 Trans2
---- ----- ---------- ----- ------ ------ -------- ---- -------- ------ ------
30   enet  100030     1500  -      -      -        -    -        0      0

Remote SPAN VLAN
----------------
Disabled

Primary Secondary Type              Ports
------- --------- ----------------- ------------------------------------------
```



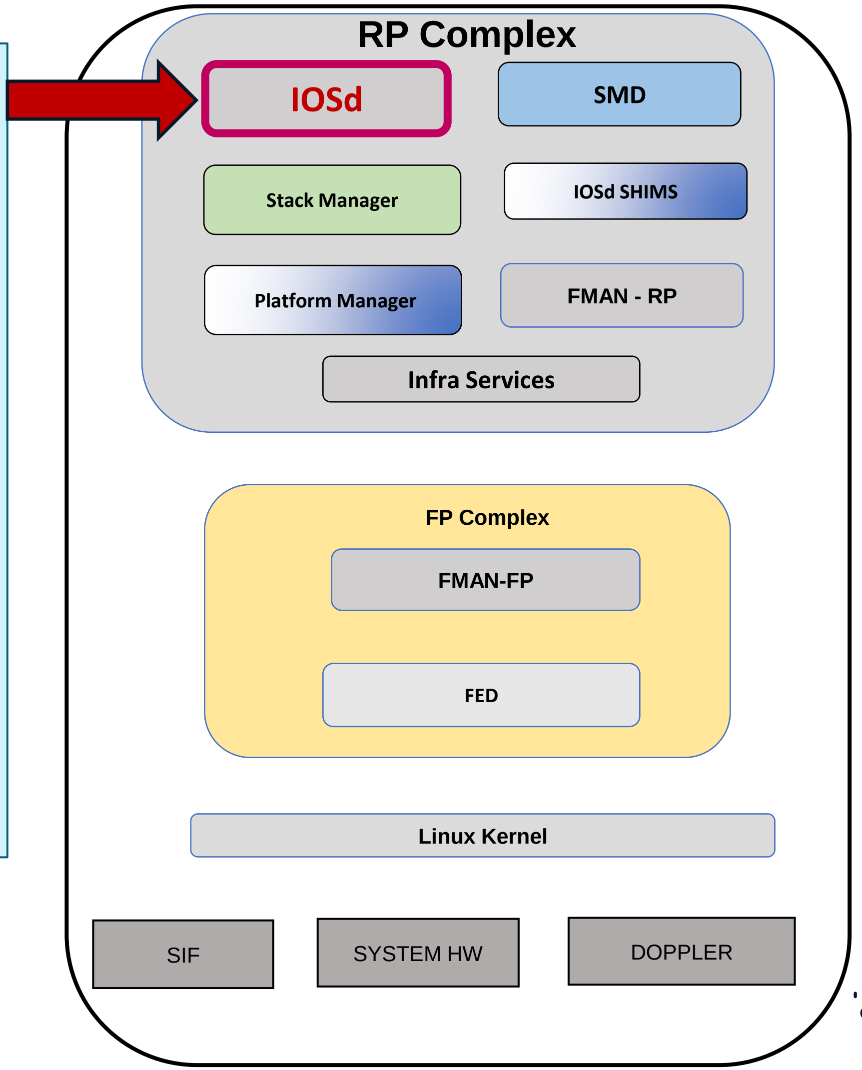
UADP/Doppler L2 Forwarding

FortyGigabitEthernet 1/0/4 : 003c.1052.c184



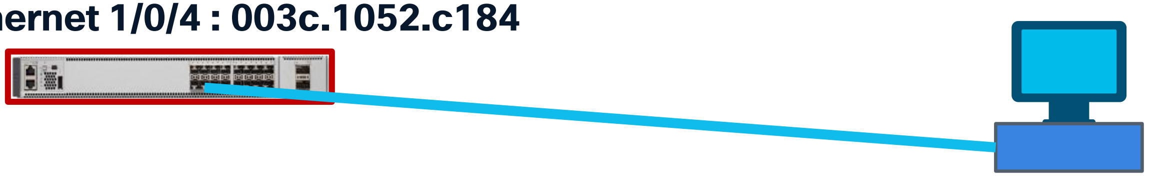
VLAN and Spanning-Tree Programming
IOSd

```
C9500-Doppler-SVL#show  spanning-tree vlan 30

VLAN0030
  Spanning tree enabled protocol rstp
  Root ID    Priority    32798
             Address     003c.1052.c180
             This bridge is the root
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32798  (priority 32768 sys-id-ext 30)
             Address     003c.1052.c180
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  300 sec

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- --------------------------------
Fo1/0/4             Desg FWD 20000     128.4    P2p
Po3                 Desg FWD 1000      128.2283 P2p
```



UADP/Doppler L2 Forwarding

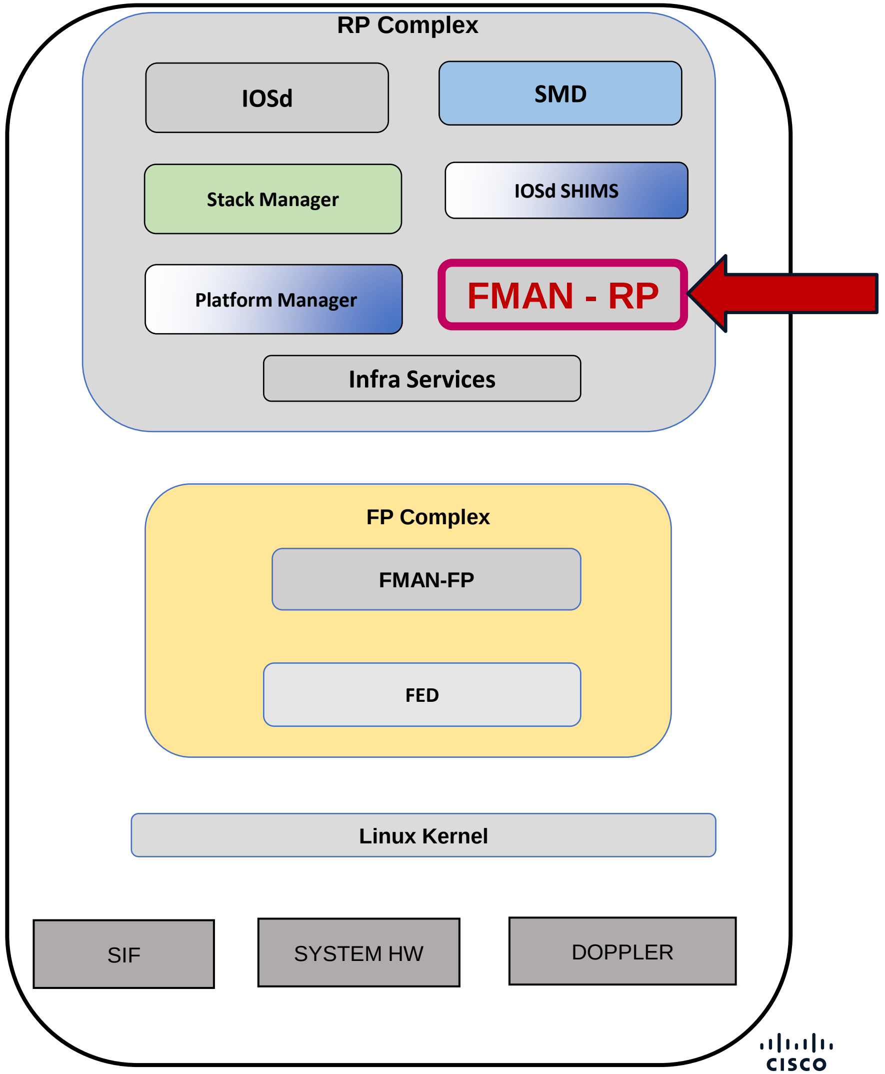
FortyGigabitEthernet 1/0/4 : 003c.1052.c184



VLAN and Spanning-Tree Programming

FMAN-RP

```
C9500-Doppler-SVL#show  platform software vp switch active R0 summary
Forwarding Manager Vlan Port Information

  Vlan    Intf-ID   Stp-state
  --------------------------------------------------------------------
  1       12        Disabled
  30      12        Forwarding          → Corresponding to int Fo 1/0/4
  1       13        Disabled
~
~

Forwarding Manager Vlan Port Information

  Vlan    Intf-ID   Stp-state
  --------------------------------------------------------------------
  1       132       Disabled
  30      132       Forwarding          → Corresponding to int Po 3
  4095    133       Forwarding
  4095    134       Forwarding
```

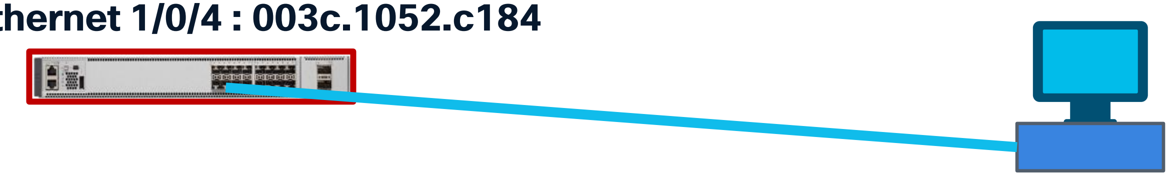


UADP/Doppler L2 Forwarding

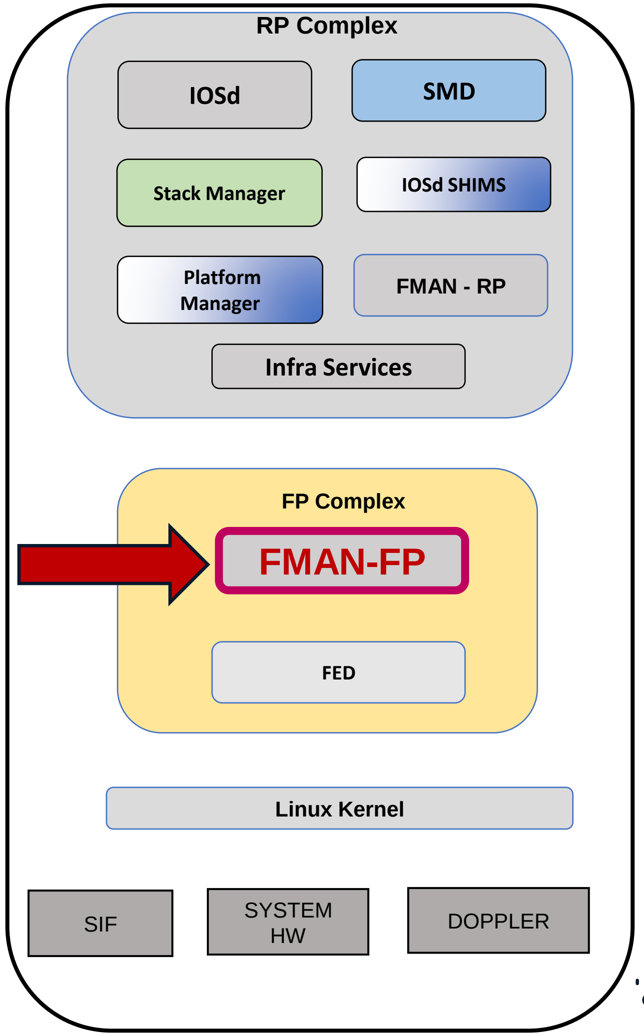
VLAN and Spanning-Tree Programming

FMAN-FP

FortyGigabitEthernet 1/0/4 : 003c.1052.c184

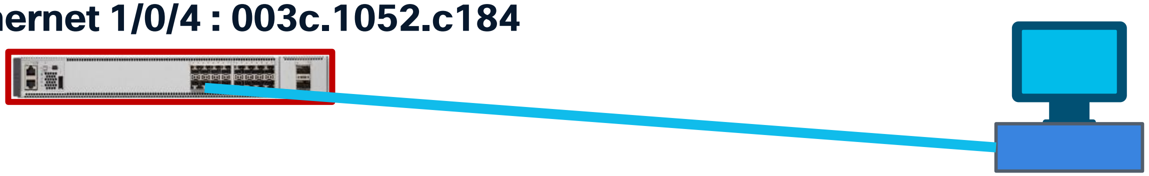


```
C9500-Doppler-SVL#show  platform software vp switch active F0 summary
Forwarding Manager Vlan Port Information

 Vlan        Intf-ID          STP-State         Untagged
 ------------------------------------------------------------------
 0           9              Forwarding           Yes
 0           10             Forwarding           Yes
 0           11             Forwarding           Yes
 0           12             Forwarding           Yes
 1           12             Disabled             Yes
 30          12             Forwarding           Yes
 0           13             Forwarding           Yes
~
~
0            129            Forwarding           Yes
 1           132            Disabled             Yes
 30          132            Forwarding           No
 0           133            Forwarding           Yes
 4095        133            Forwarding           Yes
```



UADP/Doppler L2 Forwarding

FortyGigabitEthernet 1/0/4 : 003c.1052.c184



VLAN and Spanning-Tree Programming

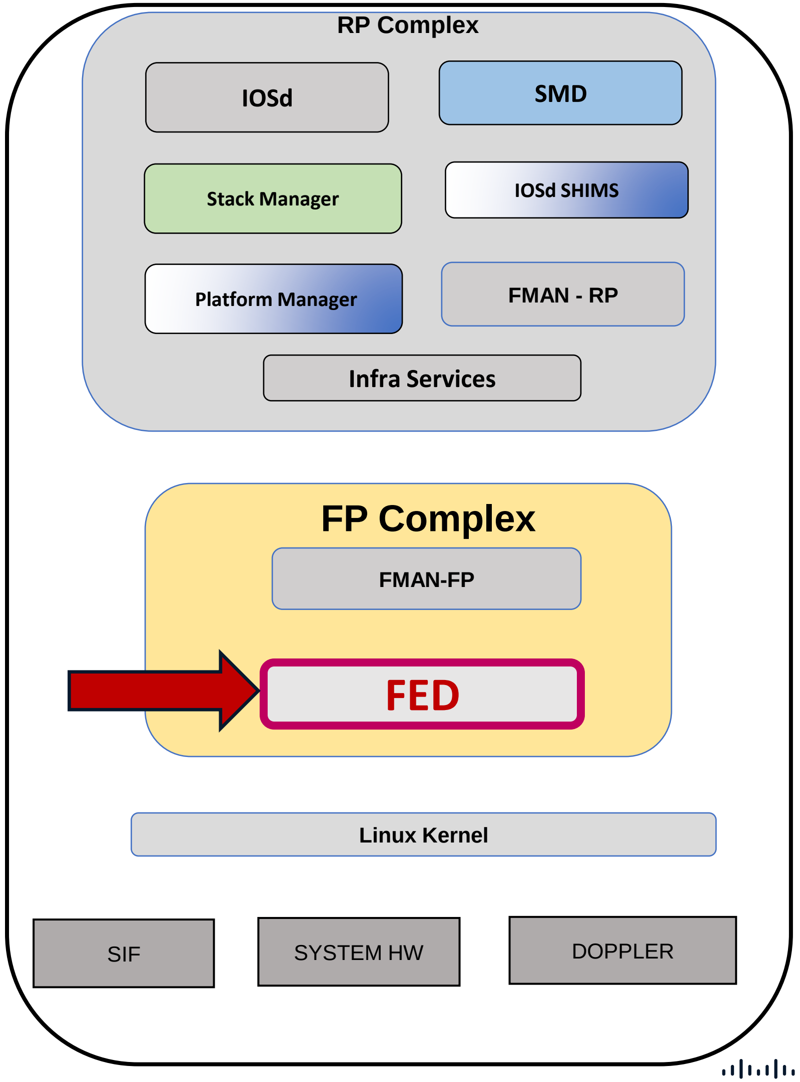
FED – Forwarding Engine Driver

C9500-Doppler-SVL#show platform software fed switch active vp summary interface if_id 12

if_id	vlan_id	pvlan_mode	pvlan_vlan	stp_state	vtp pruned	Untagged
12	**30**	**none**	**30**	**forwarding**	**No**	**Yes**
12	1	none	1	disabled	No	Yes

C9500-Doppler-SVL#show platform software fed switch active vp summary vlan 30

if_id	vlan_id	pvlan_mode	pvlan_vlan	stp_state	vtp pruned	Untagged
132	30	trunk	1	forwarding	No	No
14	30	trunk	1	forwarding	No	No
12	**30**	**none**	**30**	**forwarding**	**No**	**Yes**

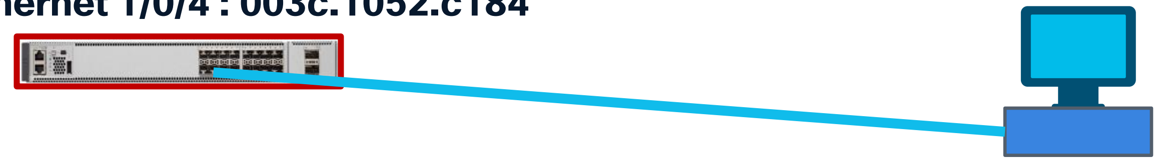


UADP/Doppler L2 Forwarding

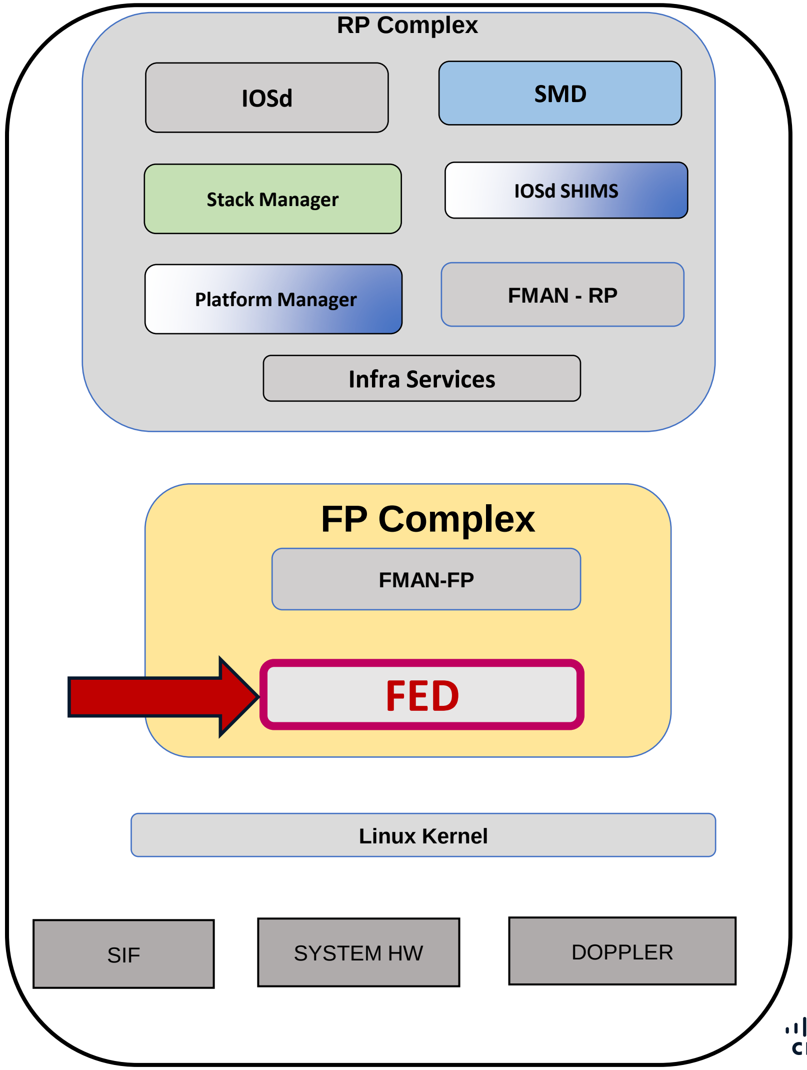
VLAN and Spanning-Tree Programming

FED – Forwarding Engine Driver

FortyGigabitEthernet 1/0/4 : 003c.1052.c184



```
C9500-Doppler-SVL#show platform software fed switch active vp key 12 30
FED PM Virtual Port Data :
 iif_id = 12, vlan_id = 30
  Info:
   stp state: forwarding
   vtp pruned: NO
    Dest Map: TRUE
    limited access: FALSE
    learning cache enable: FALSE
    delete auth behavior: FALSE
    tag native enable: FALSE
    block learn: FALSE
    pvlan port: FALSE
    pvlan mode: none
    pvlan native vlan: 30
```

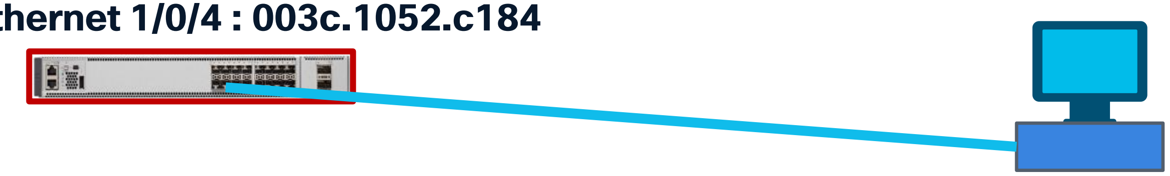


UADP/Doppler L2 Forwarding

VLAN and Spanning-Tree Programming

FED

FortyGigabitEthernet 1/0/4 : 003c.1052.c184



C9500-Doppler-SVL#show platform software fed switch active vlan
VLAN Fed Information

Vlan Id	IF Id	LE Handle	STP Handle	L3 IF Handle	SVI IF ID	MVID
0	0x0000000000000000	0x000055c0aa07f668	0x000055c0a9dc1fa8	0x0000000000000000	0x0000000000000000	2
1	0x000000000005ffff	0x00007d913417bcd8	0x00007d913417c5e8	0x00007d9134600d18	0x0000000000000083	4
30	**0x0000000000420010**	**0x00007d9134798088**	**0x00007d9134817fc8**	**0x00007d913478e648**	**0x0000000000000089**	**5**
4094	0x000000004000117f	0x000055c0aa6c74c8	0x000055c0aa6c7ca8	0x0000000000000000	0x0000000000000000	3
4095	0x000000000005fffa	0x000055c0aa07e8f8	0x000055c0aa07f088	0x0000000000000000	0x0000000000000000	1

C9500-Doppler-SVL#show platform software fed switch active vlan 30
VLAN Fed Information

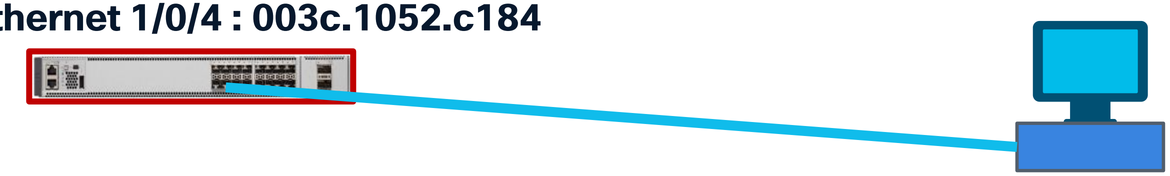
Vlan Id	IF Id	LE Handle	STP Handle	L3 IF Handle	SVI IF ID	MVID
30	0x0000000000420010	**0x00007d9134798088**	0x00007d9134817fc8	0x00007d913478e648	0x0000000000000089	5

UADP/Doppler L2 Forwarding

VLAN and Spanning-Tree Programming

Doppler – Forward ASIC

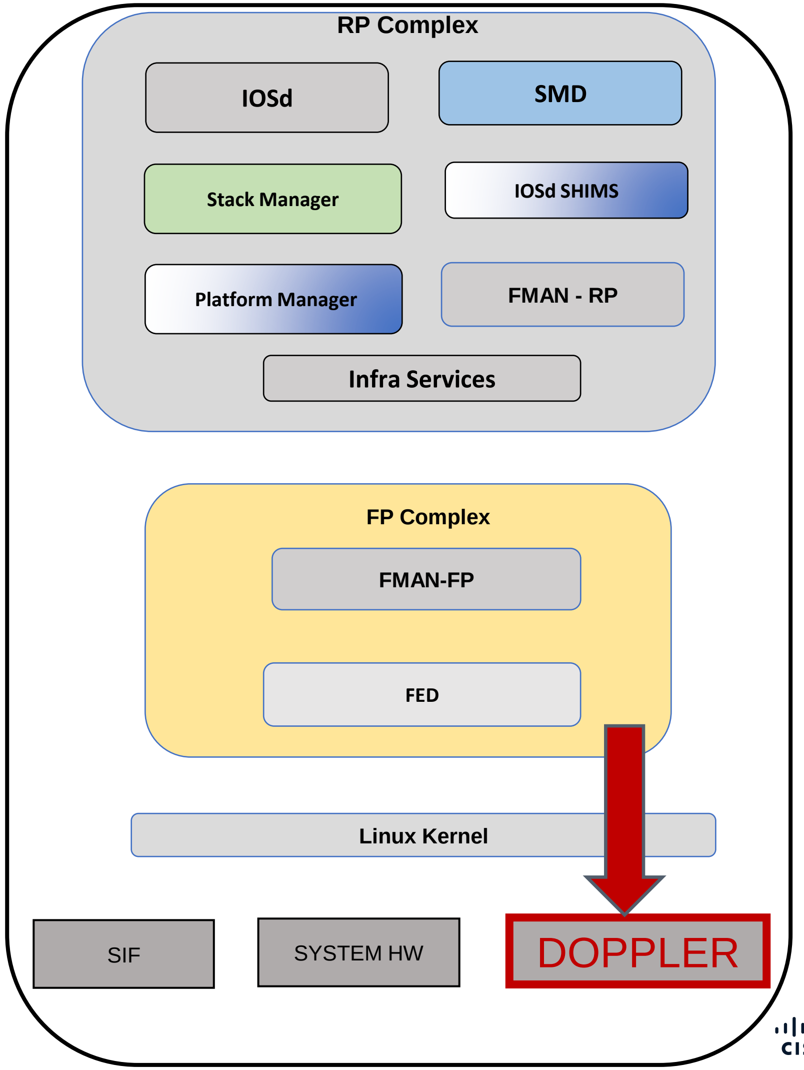
FortyGigabitEthernet 1/0/4 : 003c.1052.c184



Vlan Logical Entity (LE) Handle Programming

C9500-Doppler-SVL#show platform hardware fed switch active fwd-asic abstraction print-resource-handle 0x00007d9134798088 0

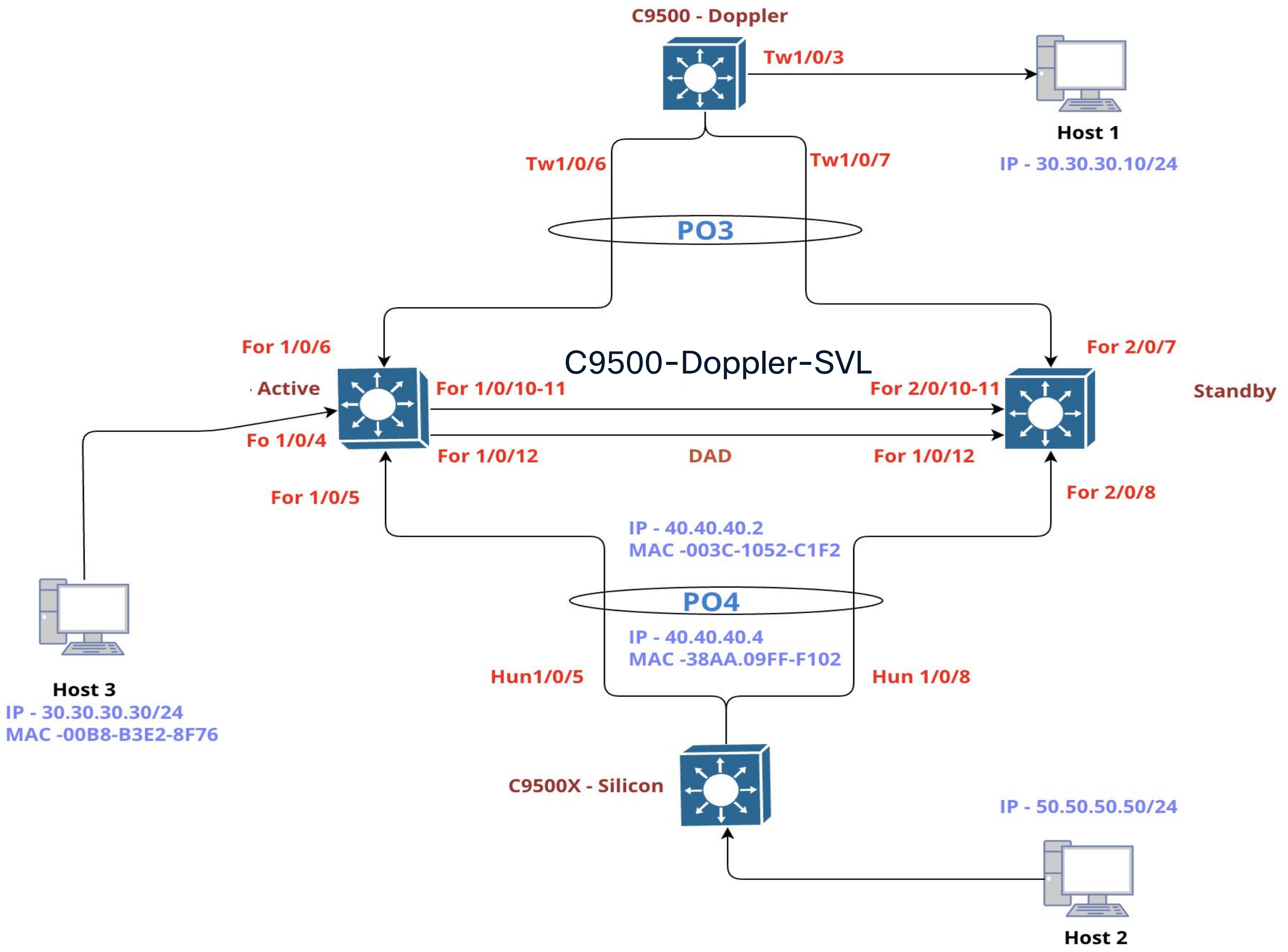
```
Handle:0x7d9134798088 Res-Type:ASIC_RSC_VLAN_LE Res-Switch-Num:255 Asic-
Num:255 Feature-ID:AL_FID_L2 Lkp-ftr-id:LKP_FEAT_INVALID ref_count:1
priv_ri/priv_si Handle: (nil)Hardware Indices/Handles: index0:0x5
mtu_index/l3u_ri_index0:0x0  sm handle [ASIC 0]: 0x7d91347aac48 index1:0x5
mtu_index/l3u_ri_index1:0x0  sm handle [ASIC 1]: 0x7d91347acd38 index2:0x5
mtu_index/l3u_ri_index2:0x0  index3:0x5  mtu_index/l3u_ri_index3:0x0


Brief Resource Information (ASIC_INSTANCE# 0)
----------------------------------------
LEAD_VLAN_IGMP_MLD_SNOOPING_ENABLED_IPV4 value 1 Pass
LEAD_VLAN_IGMP_MLD_SNOOPING_ENABLED_IPV6 value 0 Pass
LEAD_VLAN_ARP_OR_ND_SNOOPING_ENABLED_IPV4 value 0 Pass
LEAD_VLAN_ARP_OR_ND_SNOOPING_ENABLED_IPV6 value 0 Pass
~
~
```



MAC Address Programming

Doppler ASIC

Lab Topology

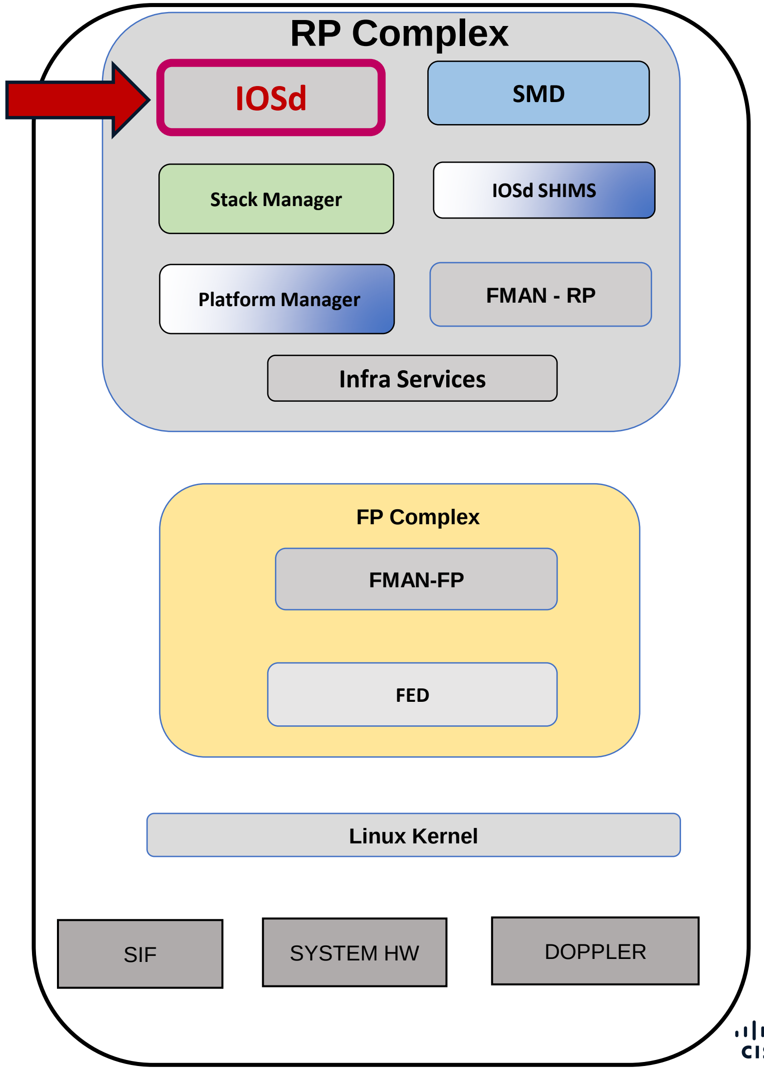


UADP/Doppler L2 Forwarding

Mac Address Programming.
IOSd


```
C9500-Doppler-SVL#show  mac address-table interface fortyGigabitEthernet 1/0/4
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----
 30   00b8.b3e2.8f76   DYNAMIC    Fo1/0/4
Total Mac Addresses for this criterion: 1
```

```
C9500-Doppler-SVL#show mac address-table address 00b8.b3e2.8f76
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----
 30   00b8.b3e2.8f76   DYNAMIC    Fo1/0/4
Total Mac Addresses for this criterion: 1
```



UADP/Doppler L2 Forwarding

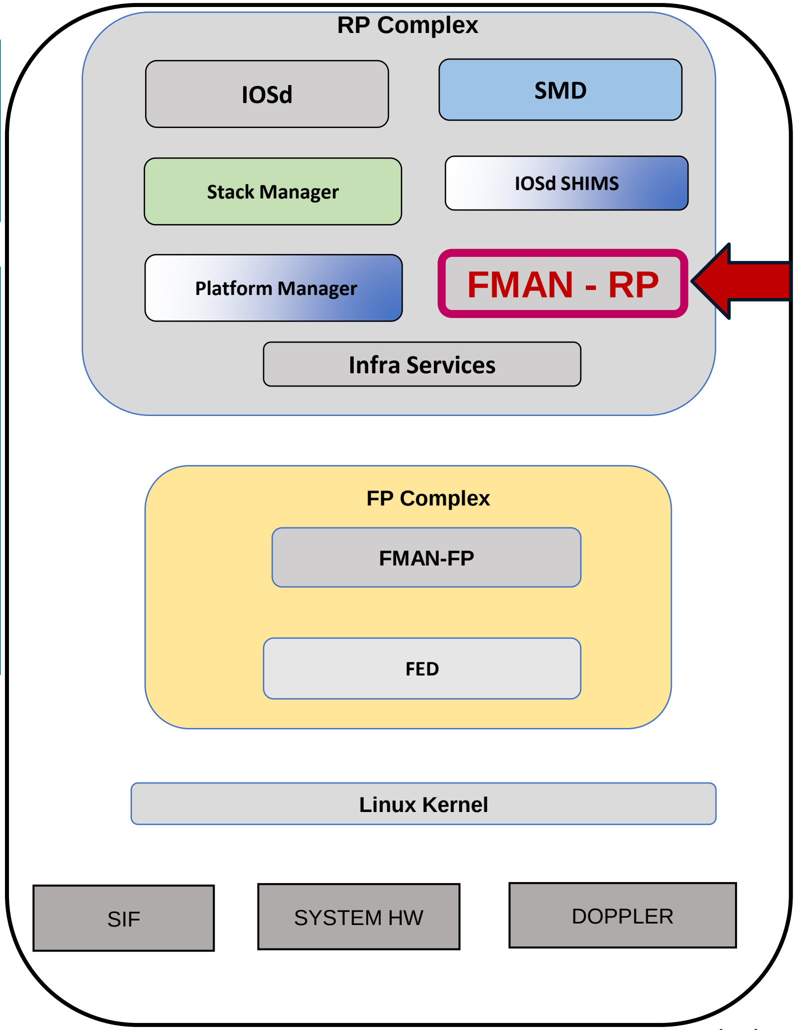
Mac Address Programming.

FMAN-RP


```
C9500-Doppler-SVL#show  platform software matm switch active R0 mac 00b8.b3e2.8f76
Tbl_Type     Tbl_ID    MAC_Address     Type   ECBits   Ports        AOM_ID/OM_PTR
MAT_VLAN      30       00b8.b3e2.8f76   1      0        1           OM: 0x348041ed50
 List of Ports: 12
```

```
C9500-Doppler-SVL#show  platform software matm switch active R0 table
Tbl_Type       Tbl_ID     Num_MAC  Aging       AOM_ID/OM_PTR
MAT_VLAN          1       0          300        OM: 0x34802a6278
MAT_VLAN         30       2          300        OM: 0x348041de10     → 2 mac addresses
MAT_IND           1       21         0          OM: 0x3480249598
MAT_L3IF       4096       0          300        OM: 0x34802a6ea8
MAT_L3IF       4101       1          300        OM: 0x3480405e88
MAT_L3IF       4200       1          300        OM: 0x3480410798
MAT_L3IF       6380       1          300        OM: 0x348040d280
```

```
C9500-Doppler-SVL#show  mac address-table vlan 30
------------------------------------------
Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

 30    003c.1052.c1e5    STATIC      Vl30
 30    00b8.b3e2.8f76    DYNAMIC     Fo1/0/4
Total Mac Addresses for this criterion: 2
```



UADP/Doppler L2 Forwarding

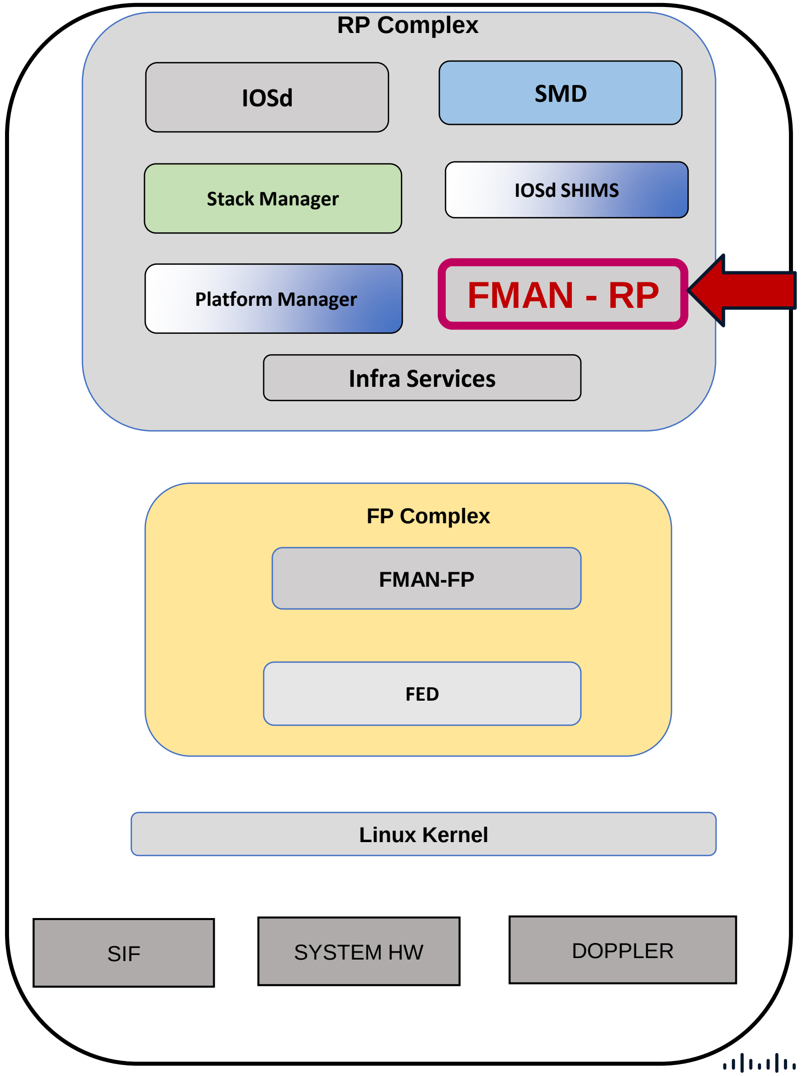
FortyGigabitEthernet 1/0/4
switchport access vlan 30

30.30.30.30 : 00b8.b3e2.8f76

Mac Address Programming.

FMAN-RP

```
C9500-Doppler-SVL#show platform software matm switch active R0 table content
Tbl_Type   Tbl_ID    MAC_Address  Type   ECBits Ports  AOM_ID/OM_PTR
MAT_VLAN        30 003c.1052.c1e5  8002 63700000      1  OM: 0x34802815b8
  List of Ports: 137
MAT_VLAN        30 00b8.b3e2.8f76     1     0      1  OM: 0x348041ed50
  List of Ports: 12
MAT_IND          1 0100.0ccc.cccc     6     0      0  OM: 0x348024c510
~
~

Tbl_Type   Tbl_ID    MAC_Address  Type   ECBits Ports  AOM_ID/OM_PTR
MAT_IND          1 0180.c200.0010     6     0      0  OM: 0x348024bdd8
MAT_IND          1 0180.c200.0021     6     0      0  OM: 0x348024c9e0
MAT_IND          1 ffff.ffff.ffff     6     0      0  OM: 0x348024cc48
MAT_L3IF      4101 003c.1052.c1e4  8002 63700000      1  OM: 0x34803ba8f8
  List of Ports: 133
MAT_L3IF      4200 003c.1052.c1e4  8002 63700000      1  OM: 0x34802554c8
  List of Ports: 134
MAT_L3IF      6380 003c.1052.c1f2  8002 63700000      1  OM: 0x3480409ab0
  List of Ports: 135
```



UADP/Doppler L2 Forwarding

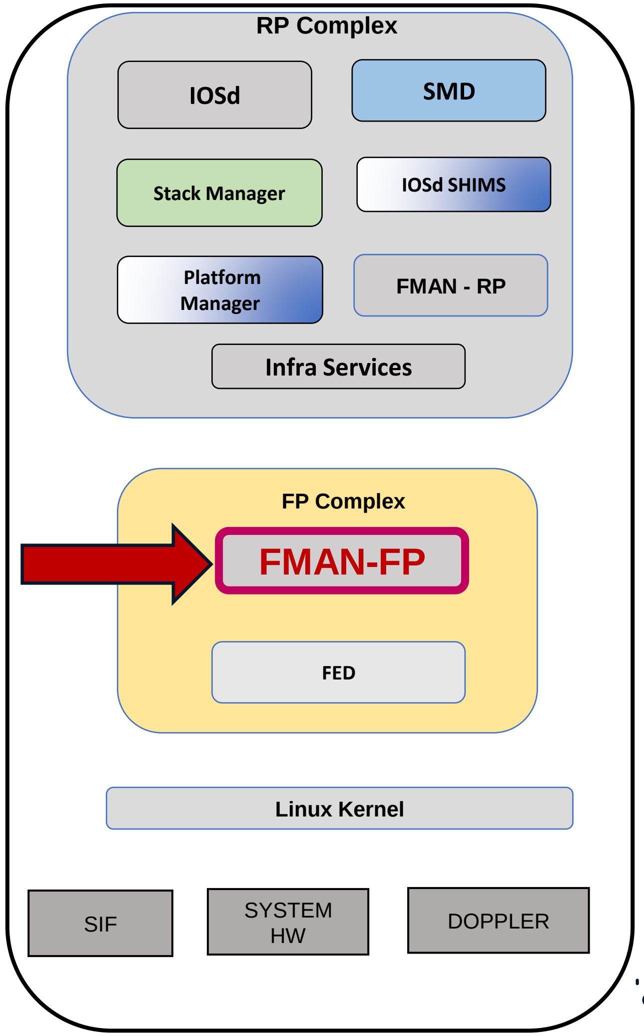
Mac Address Programming.

FMAN-FP

FortyGigabitEthernet 1/0/4
switchport access vlan 30

30.30.30.30 : 00b8.b3e2.8f76

```
C9500-Doppler-SVL#show  platform software matm switch active F0 table content
Tbl_Type   Tbl_ID      MAC_Address  Type   ECBits Ports  AOM_ID/OM_PTR
MAT_VLAN       30      003c.1052.c1e5  8002 63700000      1  3695 created
  List of Ports: 137
MAT_VLAN      30       00b8.b3e2.8f76     1     0     1  3945 created
  List of Ports: 12
MAT_IND         1      0100.0ccc.cccc     6     0     0  460 created
~
MAT_IND         1      0180.c200.000f     6     0     0  456 created

Tbl_Type   Tbl_ID      MAC_Address  Type   ECBits Ports  AOM_ID/OM_PTR
MAT_IND         1      0180.c200.0010     6     0     0  457 created
MAT_IND         1      0180.c200.0021     6     0     0  462 created
MAT_IND         1      ffff.ffff.ffff     6     0     0  463 created
MAT_L3IF     4101      003c.1052.c1e4  8002 63700000      1  3975 created
  List of Ports: 133
MAT_L3IF     4200      003c.1052.c1e4  8002 63700000      1  3417 created
  List of Ports: 134
MAT_L3IF     6380      003c.1052.c1f2  8002 63700000      1  3469 created
  List of Ports: 135
```



UADP/Doppler L2 Forwarding

Mac Address Programming.

FMAN-FP

FortyGigabitEthernet 1/0/4
switchport access vlan 30

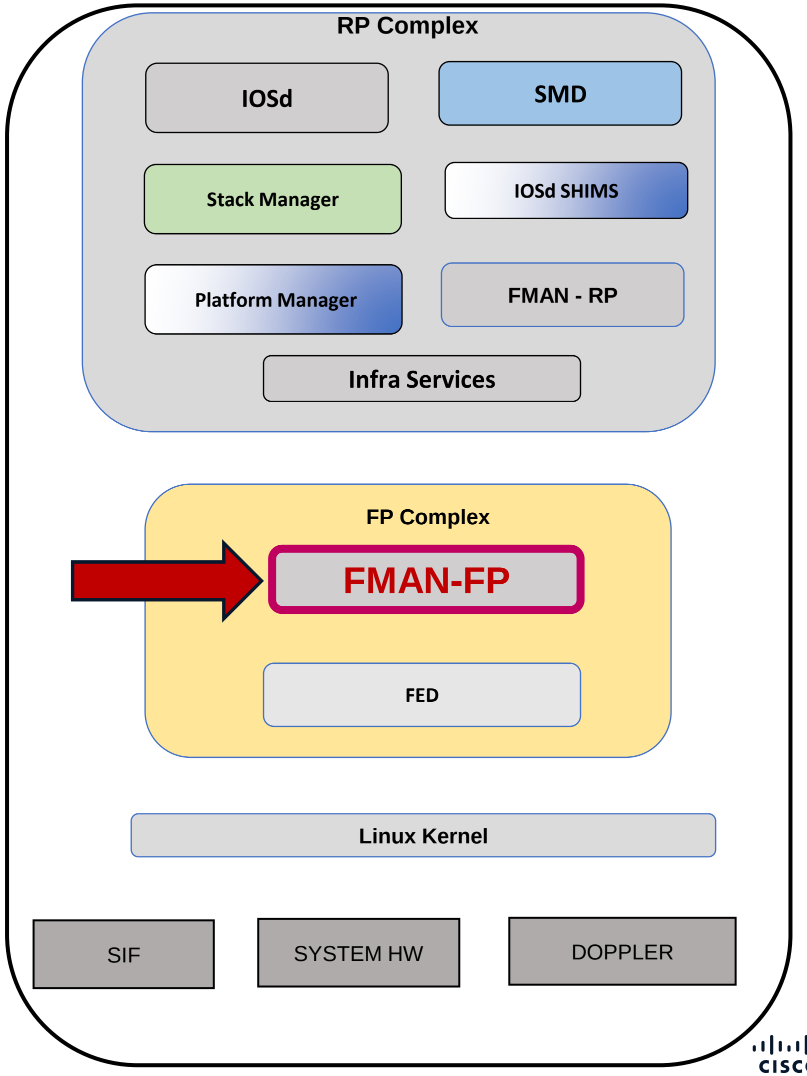
30.30.30.30 : 00b8.b3e2.8f76

C9500-Doppler-SVL#show platform software matm switch active f0 mac 00b8.b3e2.8f76
Tbl_Type Tbl_ID MAC_Address Type ECBits Ports AOM_ID/OM_PTR
MAT_VLAN 30 00b8.b3e2.8f76 1 0 1 3945 created
List of Ports: 12

C9500-Doppler-SVL#show platform software object-manager switch active F0 object 3945
Object identifier: 3945
Description: matm mac entry type **VLAN, id 30, 00b8.b3e2.8f76**
Obj type id: 459
Obj type: MATM mac entry
Status: Done, Epoch: 0, Client data: 0x5b31d48

C9500-Doppler-SVL#show platform software object-manager switch active F0 object 3945 parents
Object identifier: 45
Description: **intf FortyGigabitEthernet1/0/4**, handle 12, hw handle 12, HW dirty: NONE AOM dirty NONE
Status: Done

Object identifier: 3356
Description: **matm table** type VLAN, id 30
Status: Done



UADP/Doppler L2 Forwarding

Mac Address Programming.

FED – Forwarding Engine Driver

FortyGigabitEthernet 1/0/4
switchport access vlan 30

30.30.30.30 : 00b8.b3e2.8f76

```
C9500-Doppler-SVL#show  platform software fed switch active matm macTable
VLAN   MAC                Type  Seq#    EC_Bi  Flags  machandle            siHandle          riHandle       diHandle        *a_time  *e_time
ports                                    Con
------------------------------------------------------------------------------------------------------------------------------------------
0      003c.1052.c1e4     0x8002      0  16682     64  0x7d9134748818    0x7d91347ebb18     0x0        0x5dde               0        0
FortyGigabitEthernet2/0/8                             Yes
 ~
30     00b8.b3e2.8f76      0x1   3120     0     0       0x7d9134634ed8       0x7d913467e2e8     0x0       0x7d91341d96f8    300      15
FortyGigabitEthernet1/0/4                          Yes
 ~
Total Mac number of addresses:: 5
Summary:
Total number of secure addresses:: 0
Total number of drop addresses::
Total number of lisp remote addresses:: 0
*a_time=aging_time(secs)  *e_time=total_elapsed_time(secs)
Type:
~
~
```

Mac address table in FED

UADP/Doppler L2 Forwarding

Mac Address Programming.

FED – Forwarding Engine Driver

FortyGigabitEthernet 1/0/4
switchport access vlan 30

30.30.30.30 : 00b8.b3e2.8f76

```
C9500-Doppler-SVL#show  platform software fed switch active matm macTable vlan 30 mac 00b8.b3e2.8f76
VLAN   MAC               Type  Seq#   EC_Bi  Flags  machandle          siHandle        riHandle     diHandle      *a_time  *e_time
ports                                  Con
------------------------------------------------------------------------------------------------------------------------------------
30     00b8.b3e2.8f76    0x1   3120   0      0      0x7d9134634ed8     0x7d913467e2e8  0x0          0x7d91341d96f8   300      12
FortyGigabitEthernet1/0/4                           Yes

Created Time       : 2026/01/07 19:05:58.453
Last Modified Time : 2026/01/07 19:05:58.453
Current Time       : 2026/01/08 18:39:52.827
Last adj id        : 0
 ======Platform hardware details ======
Asic: 0
 htm-handle = 0x7d91347911a8 MVID = 5 gpn = 997964416
 SI = 0xb2 RI = 0x16 DI = 0x4
 DI = 0x4 pmap = 0x00000000 0x00000000
Asic: 1
~
Asic: 2
SI = 0xb2 RI = 0x16 DI = 0x4
 DI = 0x4 pmap = 0x00000000 0x00000008 pmap_intf : [FortyGigabitEthernet1/0/4]
Asic: 3
~
```

Destination Index Handle (DI handle)

Station Index (SI)	= 0xb2
Destination Index (DI)	= 0x4
Rewrite Index (RI)	= 0x16

UADP/Doppler L2 Forwarding

Mac Address Programming.

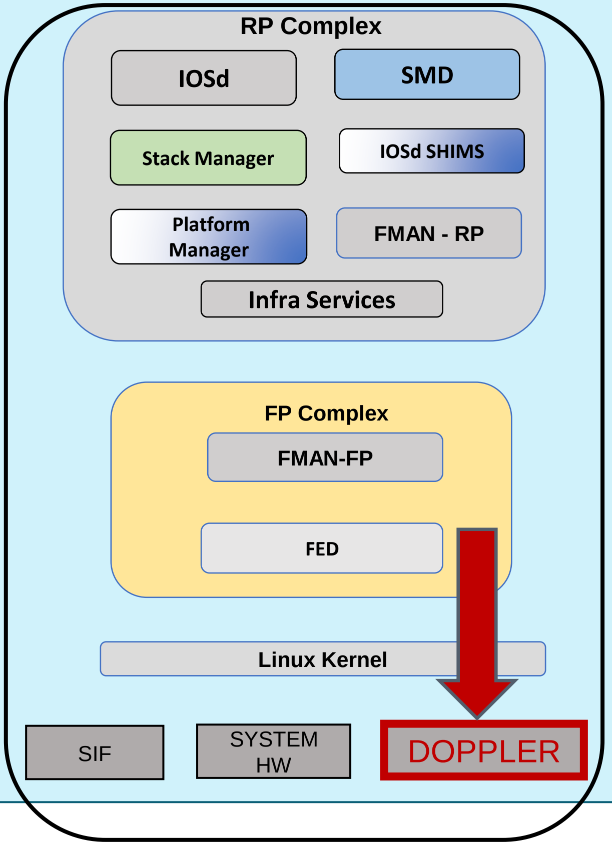
SDK – Forward ASIC

FortyGigabitEthernet 1/0/4
switchport access vlan 30

30.30.30.30 : 00b8.b3e2.8f76

```
C9500-Doppler-SVL#show  platform hardware fed switch active fwd-asic abstraction print-resource-handle 0x7d91341d96f8 0
Handle:0x7d91341d96f8 Res-Type:ASIC_RSC_DI Res-Switch-Num:255 Asic-Num:255 Feature-ID:AL_FID_IFM Lkp-ftr-id:LKP_FEAT_INVALID ref_count:1
priv_ri/priv_si Handle: (nil)Hardware Indices/Handles: index0:0x4  mtu_index/l3u_ri_index0:0x0  index1:0x4  mtu_index/l3u_ri_index1:0x0
index2:0x4  mtu_index/l3u_ri_index2:0x0  index3:0x4  mtu_index/l3u_ri_index3:0x0

Brief Resource Information (ASIC_INSTANCE# 0)
----------------------------------------
Destination index   = 0x4 DI_PORT_GPN
pmap                = 0x00000000 0x00000000
Brief Resource Information (ASIC_INSTANCE# 1)
----------------------------------------
~           Programming same like ASIC_INSTANCE# 0

Brief Resource Information (ASIC_INSTANCE# 2)
----------------------------------------
Destination index   = 0x4 DI_PORT_GPN
pmap                = 0x00000000 0x00000008
pmap_intf : [FortyGigabitEthernet1/0/4]

Brief Resource Information (ASIC_INSTANCE# 3)
----------------------------------------
~           Programming same like ASIC_INSTANCE# 0
```

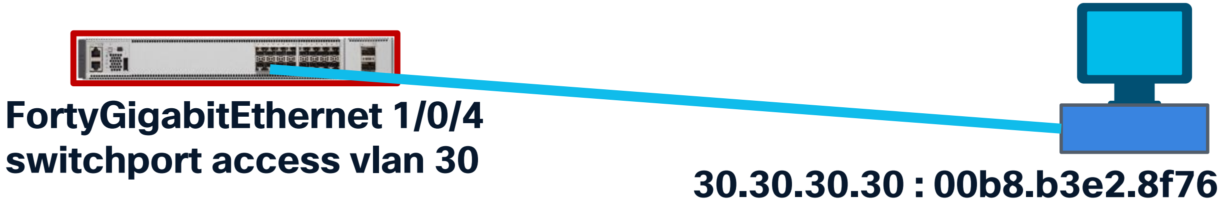
HW programming from DI handle



UADP/Doppler L2 Forwarding

Mac Address Programming.

DOPPLER – HW Programming



C9500-Doppler-SVL#show platform hardware fed switch active fwd-asic resource asic all destination-index range 0x4 0x4

```
ASIC#0:

Destination index    = 0x4 DI_PORT_GPN
pmap                 = 0x00000000 0x00000000
cmi                  = 0x0
rcp_pmap             = 0x0
al_rsc_cmi
CPU Map Index (CMI) [0]
ctiLo0               = 0
ctiLo1               = 0
ctiLo2               = 0
cpuQNum0             = 0
cpuQNum1             = 0
cpuQNum2             = 0
npuIndex             = 0
stripSeg             = 0
copySeg              = 0

ASIC#1:
~ Programming same like ASIC# 0
```

HW programming from DI

```
ASIC#2:

Destination index    = 0x4 DI_PORT_GPN
pmap                 = 0x00000000 0x00000008
pmap_intf : [FortyGigabitEthernet1/0/4]
cmi                  = 0x0
rcp_pmap             = 0x0
al_rsc_cmi
CPU Map Index (CMI) [0]
ctiLo0               = 0
ctiLo1               = 0
ctiLo2               = 0
cpuQNum0             = 0
cpuQNum1             = 0
cpuQNum2             = 0
npuIndex             = 0
stripSeg             = 0
copySeg              = 0

ASIC#3:
~ Programming same like ASIC# 0
```

UADP/Doppler L3 Forwarding

Indirect Route : via Routing Protocol

Doppler – Forward ASIC

Port-map (pmap) decode logic

```
C9500-Doppler-SVL#show platform hardware fed switch active fwd-asic resource asic all destination-index range 0x4 0x4
~
ASIC#2:

Destination index    = 0x4 DI_PORT_GPN
pmap               = 0x00000000 0x00000008
pmap_intf : [FortyGigabitEthernet1/0/4]
~
```

On ASIC Instance 2

Port number 3

$(00000008)_{16}$ ➔ $(0000\ 0000\ 0000\ 0000\ 0000\ 0000\ 0000\ 1000)_2$

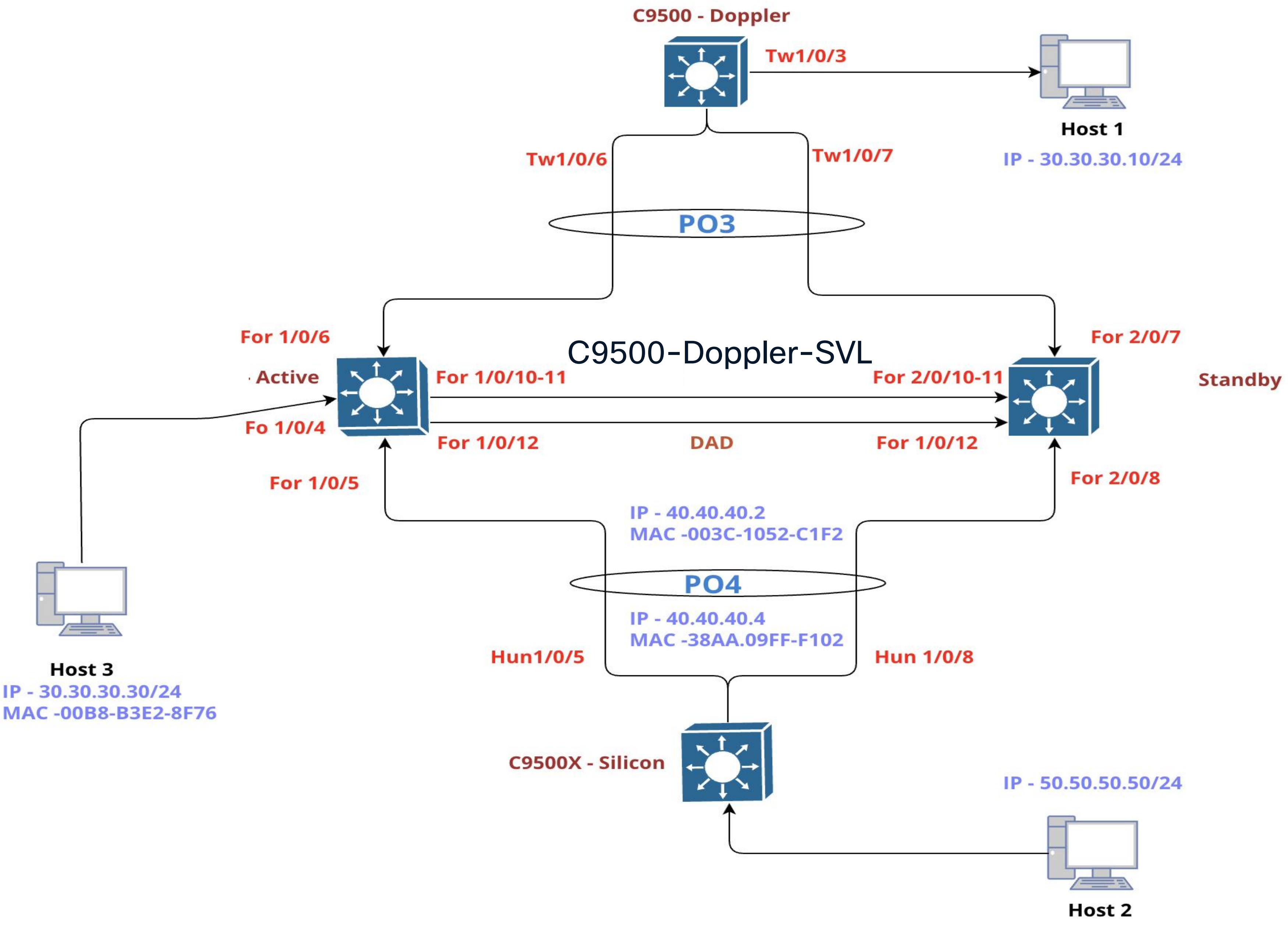
23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0

```
C9500-Doppler-SVL#show  platform software fed switch active ifm mappings
Interface                      IF_ID  Inst Asic Core Port SubPort Mac  Cntx LPN  GPN  Type Active
~
FortyGigabitEthernet1/0/4       0xc    2    1    0    3    0       8    0    4    4    NIF   Y
~
```

Layer 3 Forwarding

Doppler ASIC

Lab Topology

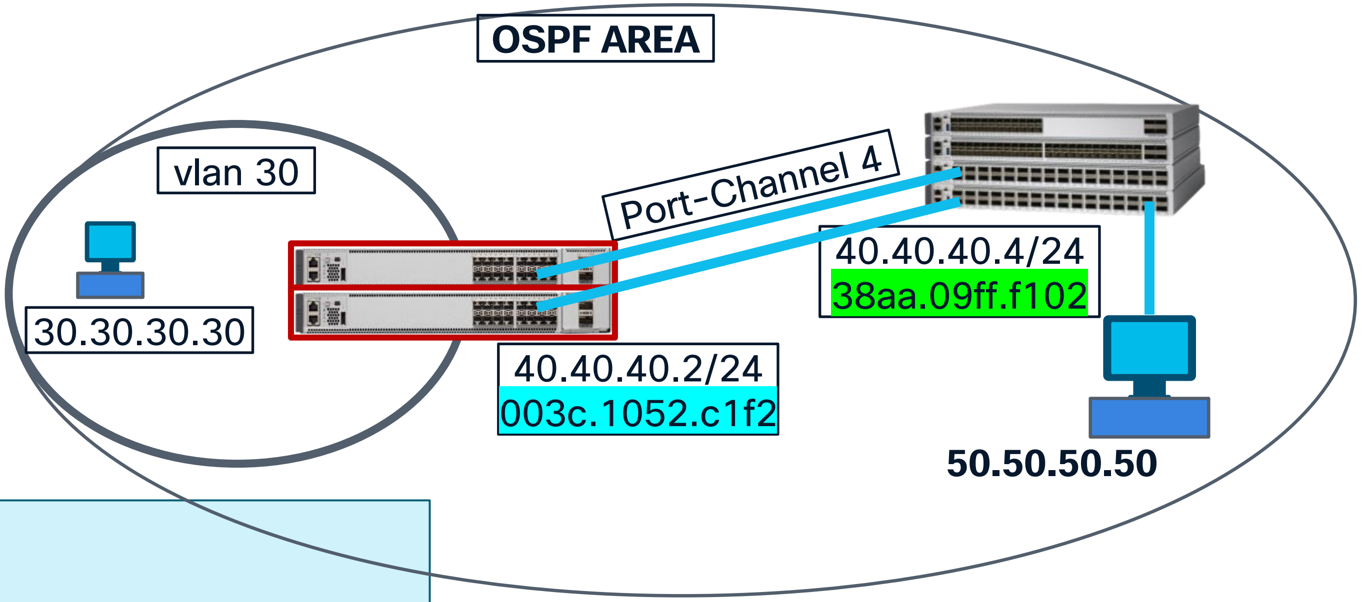


Indirect Route via Routing Protocol

UADP/Doppler L3 Forwarding

Indirect Route : via Routing Protocol

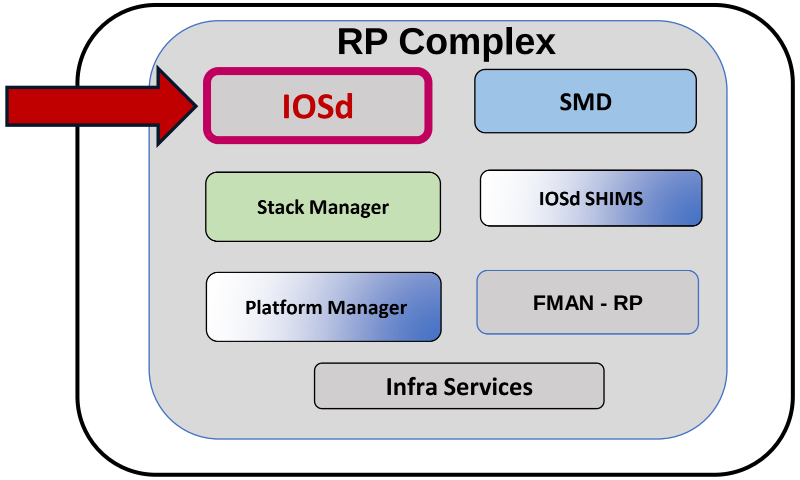
IOSd




```
C9500-Doppler-SVL#show  ip route 50.50.50.50
Routing entry for 50.50.50.0/24
  Known via "ospf 1", distance 110, metric 2, type intra area
  Last update from 40.40.40.4 on Port-channel4, 02:39:37 ago
  Routing Descriptor Blocks:
  * 40.40.40.4, from 4.4.4.4, 02:39:37 ago, via Port-channel4
      Route metric is 2, traffic share count is 1
```

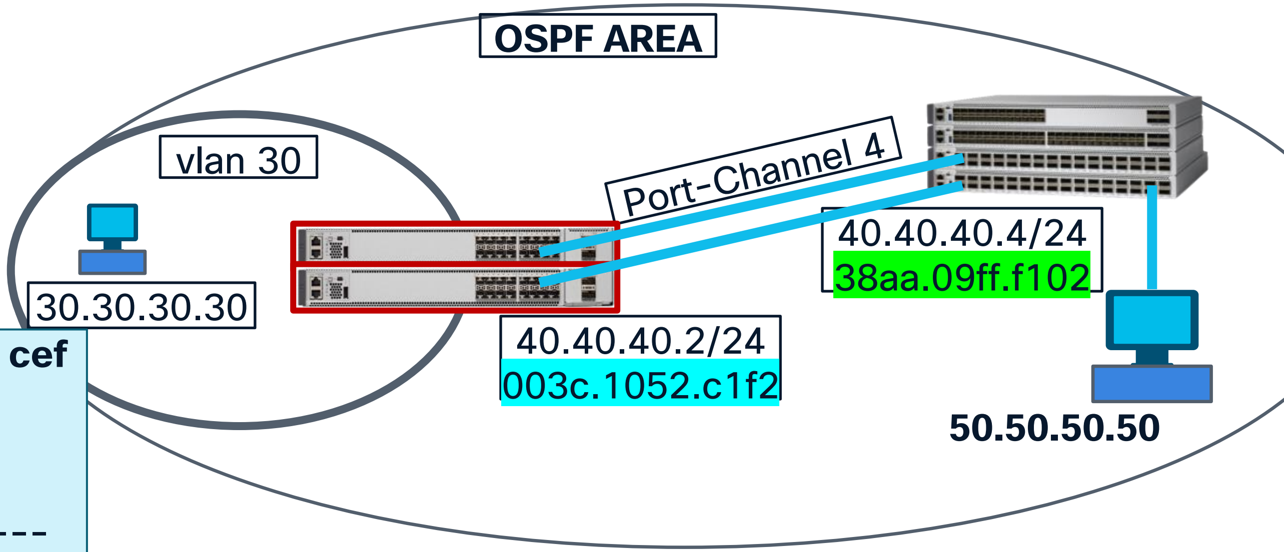
```
C9500-Doppler-SVL#show ip cef 50.50.50.50
50.50.50.0/24
  nexthop 40.40.40.4 Port-channel4
```

```
C9500-Doppler-SVL#show  ip cef exact-route 30.30.30.10 50.50.50.50
30.30.30.30 -> 50.50.50.50 =>IP adj out of Port-channel4, addr 40.40.40.4
```



UADP/Doppler L3 Forwarding

Indirect Route : via Routing Protocol
FMAN-RP



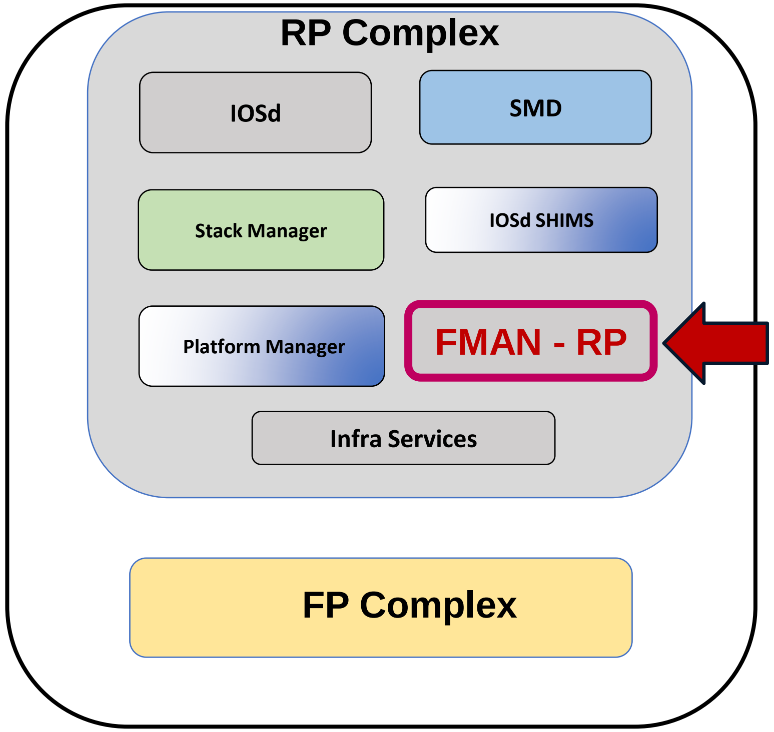

```
C9500-Doppler-SVL#show  platform software ip switch active R0 cef prefix 50.50.50.0/24
Forwarding Table
Prefix/Len               Next Object              Index
-----------------------------------------------------------
50.50.50.0/24            OBJ_ADJACENCY            0x1b
```

```
C9500-Doppler-SVL#show  platform software adjacency switch active R0 index 0x1b
Number of adjacency objects: 5

Adjacency id: 0x1b (27)
  Interface: Port-channel4, IF index: 135, Link Type: MCP_LINK_IP
  Encap: 38:aa:9:ff:f1:2:0:3c:10:52:c1:f2:8:0
  Encap Length: 14, Encap Type: MCP_ET_ARPA, MTU: 1500
  Flags: no-l3-inject
  Incomplete behavior type: None
  Fixup: unknown
  Fixup_Flags_2: unknown
 Nexthop addr: 40.40.40.4
  IP FRR MCP_ADJ_IPFRR_NONE 0
  OM handle: 0x3480417460
```

Source ip : 30.30.30.30
Destination ip : 50.50.50.50

Source mac

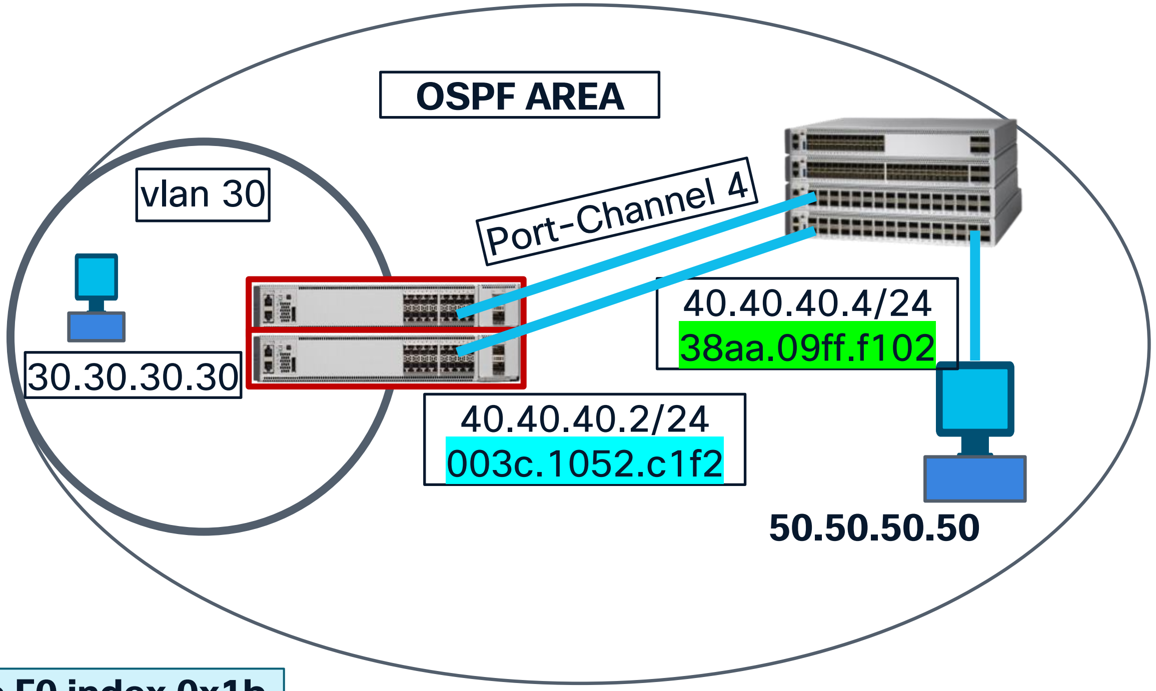
Destination mac



UADP/Doppler L3 Forwarding

Indirect Route : via Routing Protocol

FMAN-FP

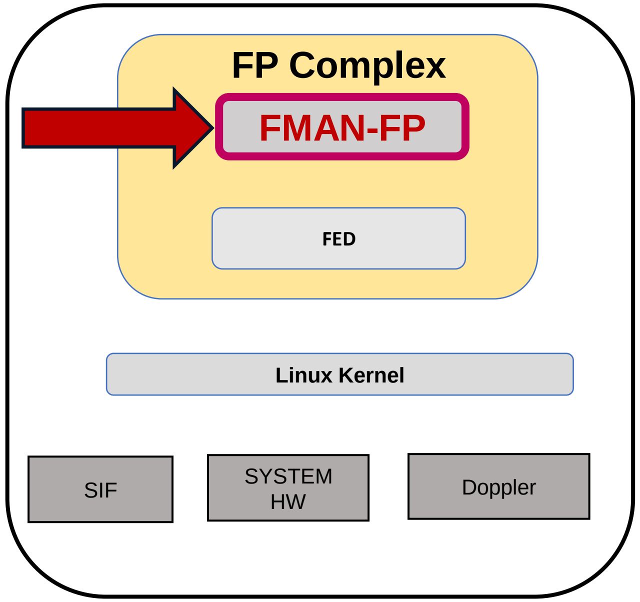


C9500-Doppler-SVL#show platform software ip switch active F0 cef prefix 50.50.50.0/24
Forwarding Table

Prefix/Len	Next Object	Index
50.50.50.0/24	**OBJ_ADJACENCY**	**0x1b**

C9500-Doppler-SVL#show platform software adjacency switch active F0 index 0x1b
Number of adjacency objects: 5

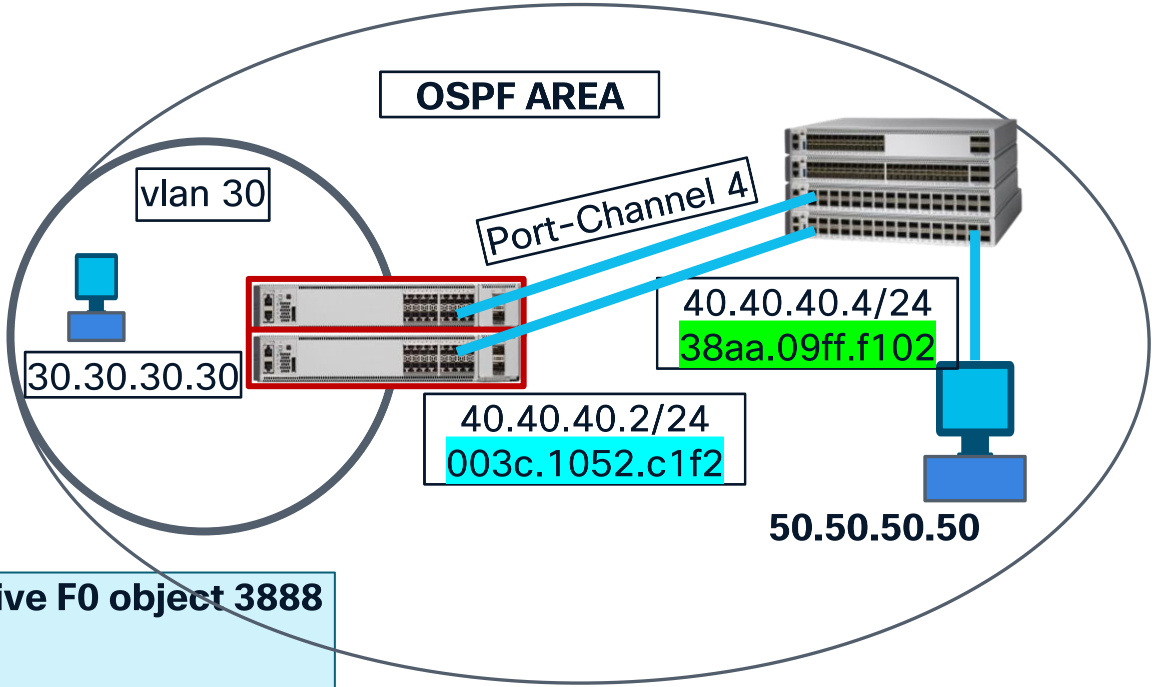
Adjacency id: 0x1b (27)
  Interface: **Port-channel4, IF index: 135**, Link Type: MCP_LINK_IP
  Encap: 38:aa:9:ff:f1:2:0:3c:10:52:c1:f2:8:0
  Encap Length: 14, Encap Type: MCP_ET_ARPA, MTU: 1500
  Flags: no-l3-inject
  Incomplete behavior type: None
  Fixup: unknown
  Fixup_Flags_2: unknown
  Nexthop addr: 40.40.40.4
  IP FRR MCP_ADJ_IPFRR_NONE 0
  aom id: 3511, HW handle: (nil) (created)



UADP/Doppler L3 Forwarding

Indirect Route : via Routing Protocol

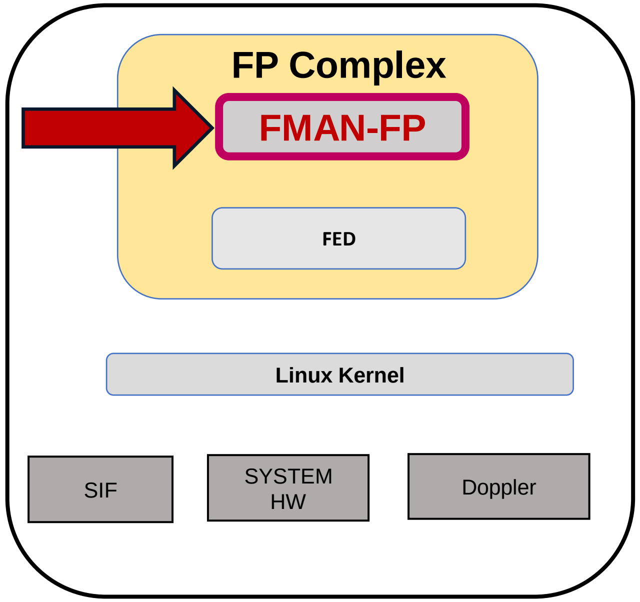
FMAN-FP



C9500-Doppler-SVL#show platform software ip switch active F0 cef prefix 50.50.50.0/24 detail
Forwarding Table
50.50.50.0/24 -> OBJ_ADJACENCY (0x1b), urpf: 39
Prefix Flags: unknown
aom id: 3888, HW handle: (nil) (created)

C9500-Doppler-SVL#show platform software object-manager switch active F0 object 3888
Object identifier: 3888
  Description: **PREFIX 50.50.50.0/24** (Table id 0)
  Obj type id: 72
  Obj type: route-pfx
  Status: Done, Epoch: 0, Client data: 0x5b5a618

C9500-Doppler-SVL#show platform software object-manager switch active F0 object 3888 parents
Object identifier: 22
  Description: ipv4 table 0 (Default), vrf id 0
  Status: **Done**
Object identifier: 3511
  Description: adj 0x1b, Flags None
  Status: **Done**
Object identifier: 3885
  Description: uRPF-list(hdl=0x00000027)
  Status: **Done**

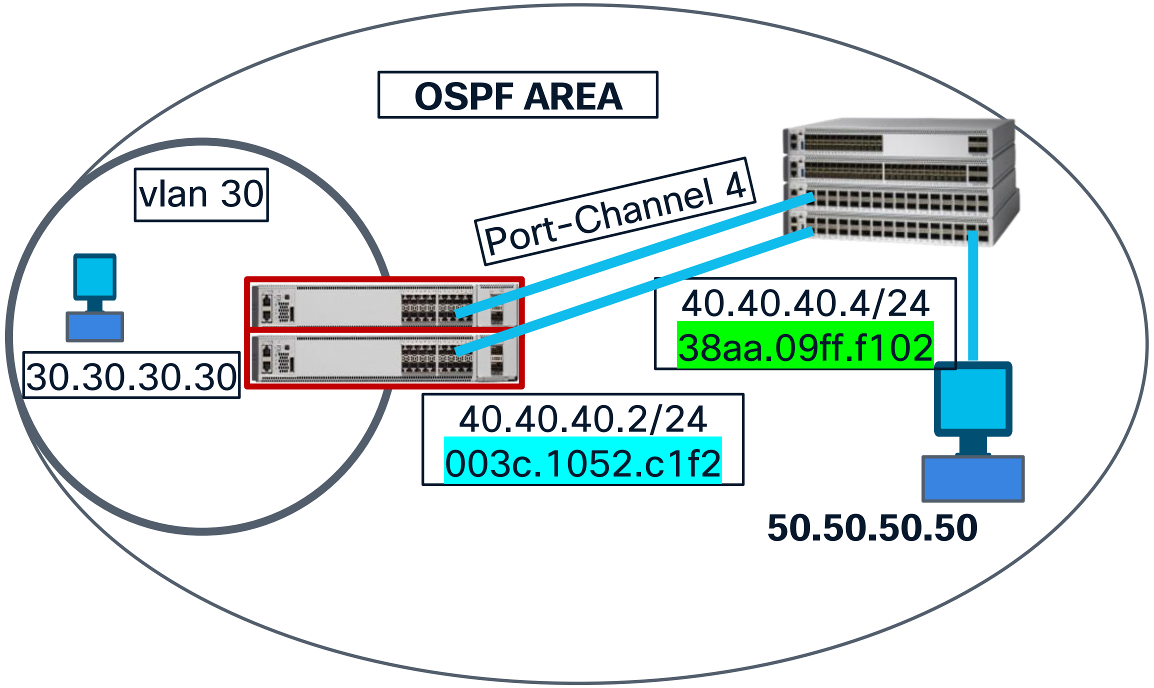


UADP/Doppler L3 Forwarding

Indirect Route : via Routing Protocol

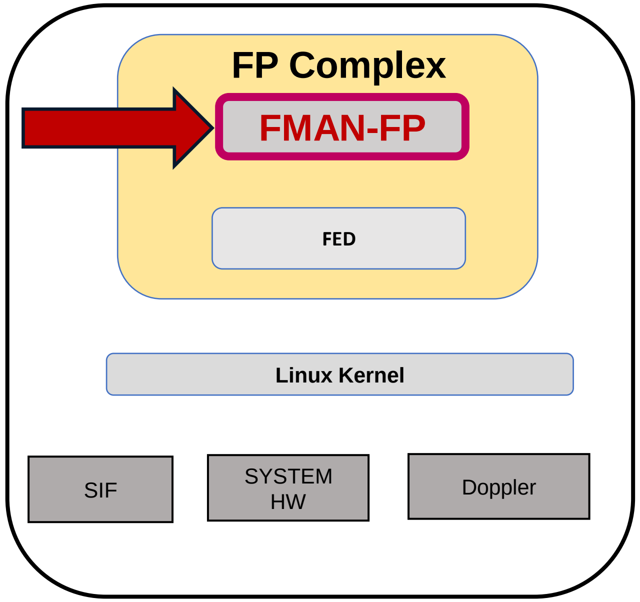
FMAN-FP

```
C9500-Doppler-SVL#show  platform software object-manager
switch active F0 statistics
Forwarding Manager Asynchronous Object Manager Statistics
Object update: Pending-issue: 0, Pending-acknowledgement: 0
Batch begin:   Pending-issue: 0, Pending-acknowledgement: 0
Batch end:     Pending-issue: 0, Pending-acknowledgement: 0
Command:       Pending-acknowledgement: 0
Total-objects: 767
Stale-objects: 0
Resolve-objects: 0
Childless-delete-objects: 0
Backplane-objects: 0
Error-objects: 0
Number of bundles: 0
Paused-types: 8
```




```
C9500-Doppler-SVL#show  platform software object-manager switch active F0 ?
  error-object                 Show object in error state
  pending-ack-batch            Show pending-acknowledgement batch operations
  pending-ack-commands         Show pending-acknowledgement commands
  pending-ack-update      Show pending-acknowledgement object update operations
  pending-issue-batch          Show pending-issue batch operations
  pending-issue-commands       Show pending-issue commands
  pending-issue-update         Show pending-issue object update operations
  stale-object                 Show stale objects
```

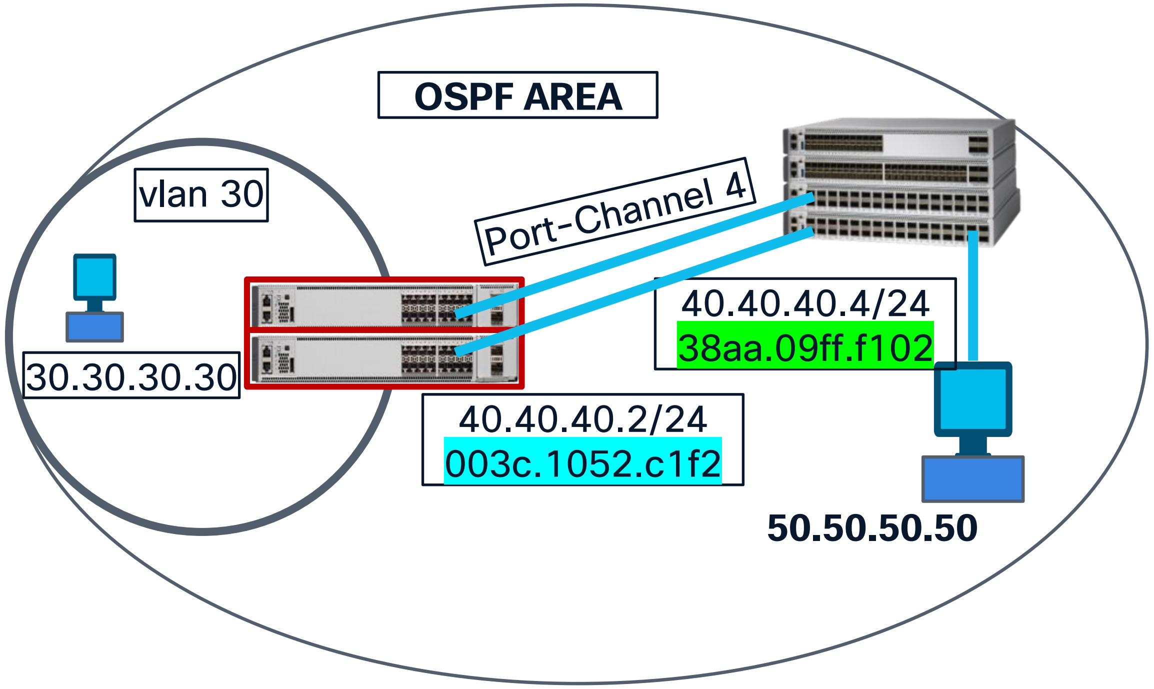
Important Options



UADP/Doppler L3 Forwarding

Indirect Route : via Routing Protocol

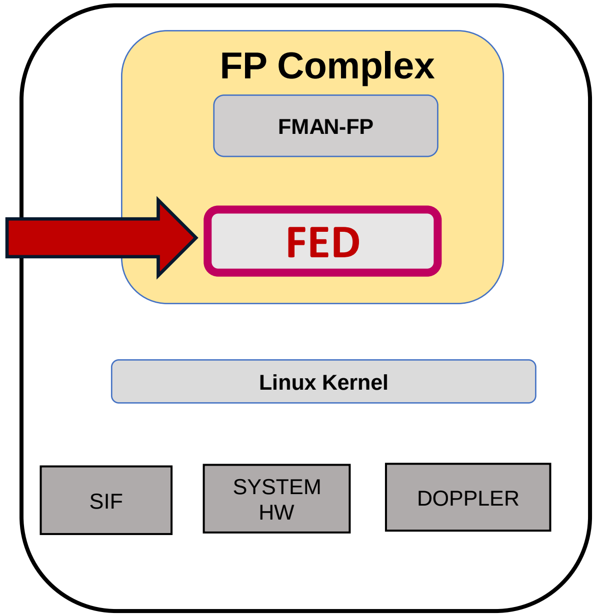
FED – Forwarding Engine Driver



Hash TCAM Handle

```
C9500-Doppler-SVL#show  platform software fed switch active ip route 50.50.50.0/24
vrf   dest                  htm                    flags   SGT   DGID MPLS Last-modified            SecsSinceHit
---   ----                                       ---           -----   ---   ---- ---- ------------------------ ----
0     50.50.50.0/24      0x7d9134a5ab78 0x0      0     0        2026/01/07 18:00:49.504           1
  FIB: prefix_hdl:0x45000017, mpls_ecr_prefix_hdl:0, sgtOverWrite: 0
  ========== OCE chain =====
  ADJ:objid:27 {link_type:IP ifnum:0x87, adj:0x64000027, si: 0x7d91347ef658  IPv4:     40.40.40.4 }
  ===============
  MPLS info: mpls_ecr_scale_prefix_adj:0, mpls_lspa_hdl:0
  ===============
```

Station Index Handle



UADP/Doppler L3 Forwarding

Indirect Route : via Routing Protocol

FED – Forwarding Engine Driver

```
C9500-Doppler-SVL#show  platform software fed switch active ip route 50.50.50.0/24 detail
vrf   dest                                    htm           flags   SGT   DGID MPLS Last-modified          SecsSinceHit
---   ----                                    ---           -----   ---   ---- ---- ----------------------- -----------
0     50.50.50.0/24              0x7d9134a5ab78 0x0     0     0        2026/01/07 18:00:49.504        61935
  FIB: prefix_hdl:0x45000017, mpls_ecr_prefix_hdl:0, sgtOverWrite: 0
  ========== OCE chain =====
  ADJ:objid:27 {link_type:IP ifnum:0x87, adj:0x64000027, si: 0x7d91347ef658  IPv4:      40.40.40.4 }
  ===============
  MPLS info: mpls_ecr_scale_prefix_adj:0, mpls_lspa_hdl:0
  ===============
~
Brief Resource Information (ASIC_INSTANCE# 0)
----------------------------------------
Number of HTM Entries: 1
Entry #0: (handle 0x7d9134a5aad8)
KEY - vrf:0 mtr:0 prefix:50.50.50.0 rcp_redirect_index:0x0
MASK - vrf:4095 mtr:0 prefix:255.255.255.0 rcp_redirect_index:0x0
~
=====================================================================
Asic       SI-Index              DI-Index
----       ---------             --------
0          174                   0x4c9f
```

Continued to next Page.......

UADP/Doppler L3 Forwarding

Indirect Route : via Routing Protocol

FED – Forwarding Engine Driver

```
Detailed Resource Information (ASIC# 0)
----------------------------------------
Station Index (SI) [0xae]
RI = 0x14
DI = 0x4c9f
Replication Bitmap: LD RD CD

Destination index   = 0x4c9f DI_ETHER_CHANNEL
pmap                = 0x00000000 0x00000000
Asic     SI-Index        DI-Index
----     ---------       --------
1        174             0x4c9f
Detailed Resource Information (ASIC# 1)
----------------------------------------
Station Index (SI) [0xae]
RI = 0x14
DI = 0x4c9f
Replication Bitmap: LD RD CD

Destination index   = 0x4c9f DI_ETHER_CHANNEL
pmap                = 0x00000000 0x00000000
Asic     SI-Index        DI-Index
----     ---------       --------
2        174             0x4c9f
```

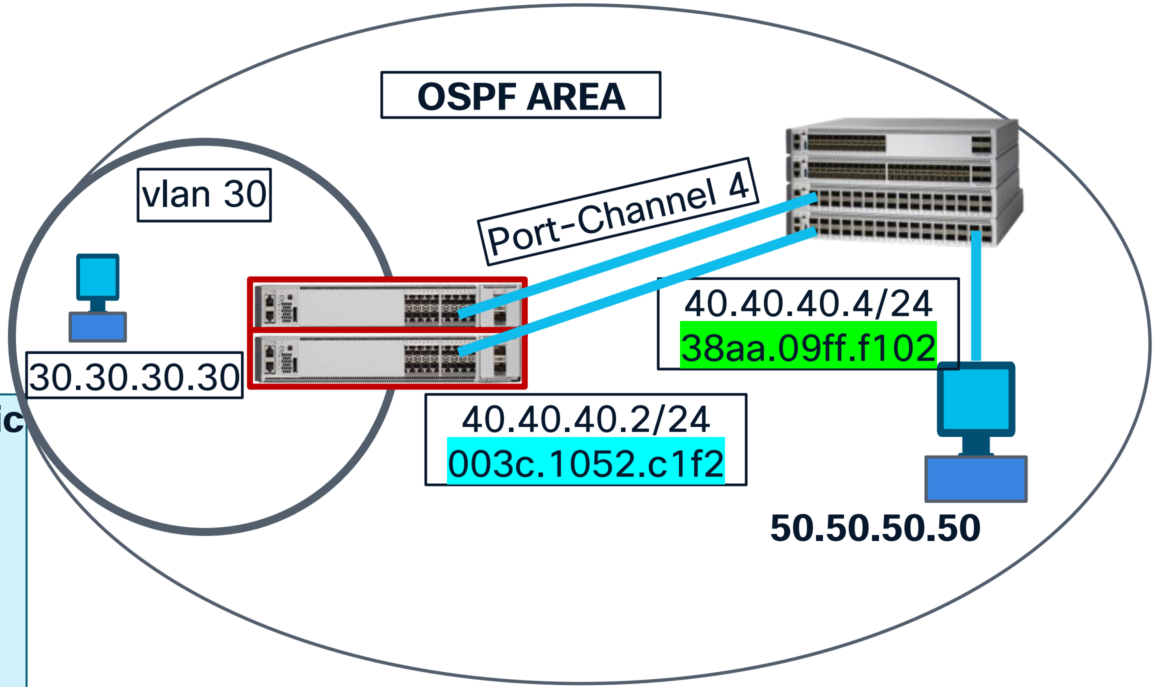
```
Detailed Resource Information (ASIC# 2)
----------------------------------------
Station Index (SI) [0xae]
RI = 0x14
DI = 0x4c9f
Replication Bitmap: LD RD CD

Destination index   = 0x4c9f DI_ETHER_CHANNEL
pmap                = 0x00000000 0x00000010
pmap_intf : [FortyGigabitEthernet1/0/5]
Asic     SI-Index        DI-Index
----     ---------       --------
3        174             0x4c9f
Detailed Resource Information (ASIC# 3)
----------------------------------------
Station Index (SI) [0xae]
RI = 0x14
DI = 0x4c9f
Replication Bitmap: LD RD CD

Destination index   = 0x4c9f DI_ETHER_CHANNEL
pmap                = 0x00000000 0x00000000
```

UADP/Doppler L3 Forwarding

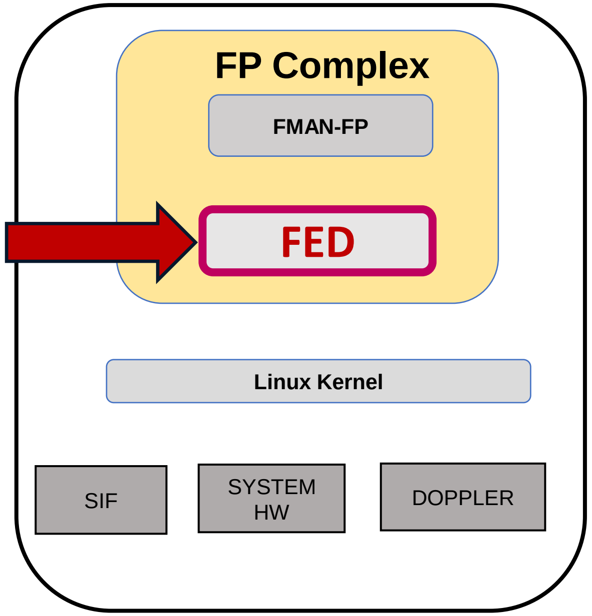
Indirect Route : via Routing Protocol

UADP/Doppler

Station index programming in Hardware




```
C9500-Doppler-SVL#show platform hardware fed switch active fwd-asic abstraction print-resource-handle 0x7d91347ef658 1
Handle:0x7d91347ef658 Res-Type:ASIC_RSC_SI Res-Switch-Num:255
~
Detailed Resource Information (ASIC_INSTANCE# 0)
----------------------------------------
Station Index (SI) [0xae]
RI = 0x14
DI = 0x4c9f
stationTableGenericLabel = 0
stationFdConstructionLabel = 0x7
lookupSkipIdIndex = 0
rcpServiceId = 0
dejaVuPreCheckEn = 0x1
Replication Bitmap: LD RD CD

Detailed Resource Information (ASIC_INSTANCE# 1)
>>>>> (SAME STATION INDEX PROGRAMMING LIKE ASIC_INSTANCE# 0)
Detailed Resource Information (ASIC_INSTANCE# 2)
>>>>> (SAME STATION INDEX PROGRAMMING LIKE ASIC_INSTANCE# 0)
Detailed Resource Information (ASIC_INSTANCE# 3)
>>>>> (SAME STATION INDEX PROGRAMMING LIKE ASIC_INSTANCE# 0)
```

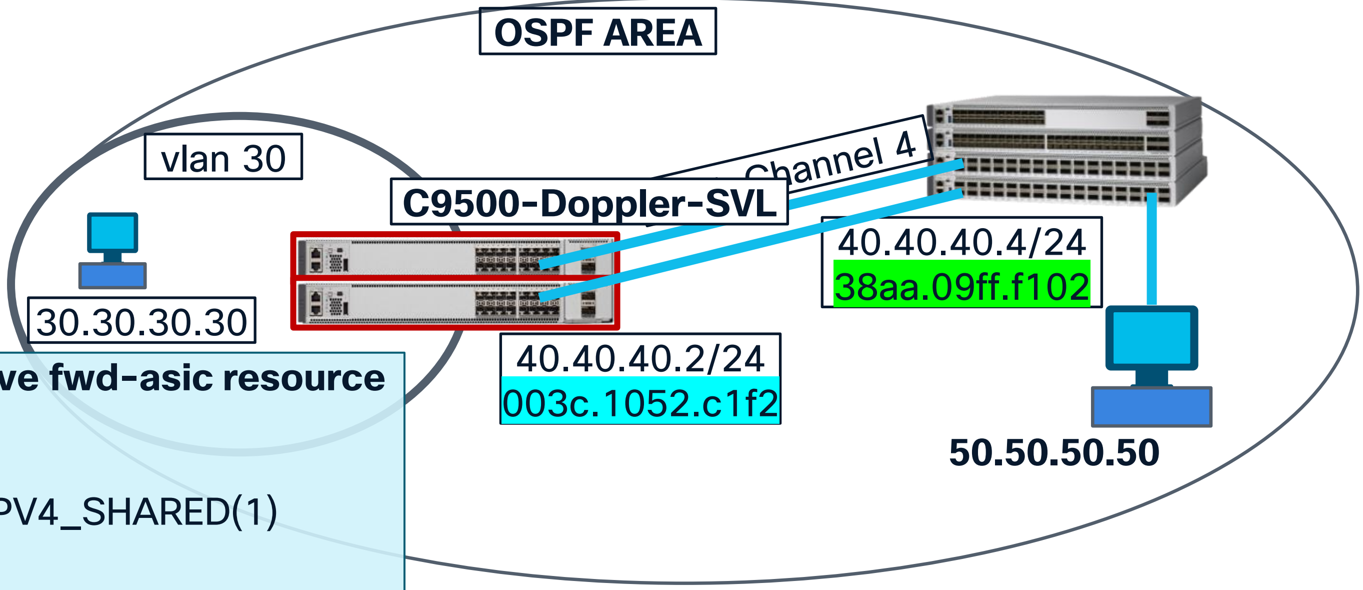


UADP/Doppler L3 Forwarding

Indirect Route : via Routing Protocol

Doppler – Forward ASIC

Rewrite index programming in Doppler



C9500-Doppler-SVL#show platform hardware fed switch active fwd-asic resource asic all rewrite-index range 0x14 0x14

ASIC#:0 RI:20 Rewrite_type:AL_RRM_REWRITE_L3_UNICAST_IPV4_SHARED(1)
Mapped_rii:L3_UNICAST_IPV4(9)
MAC Addr: MAC Addr: 38:aa:09:ff:f1:02,
L3IF LE Index 40

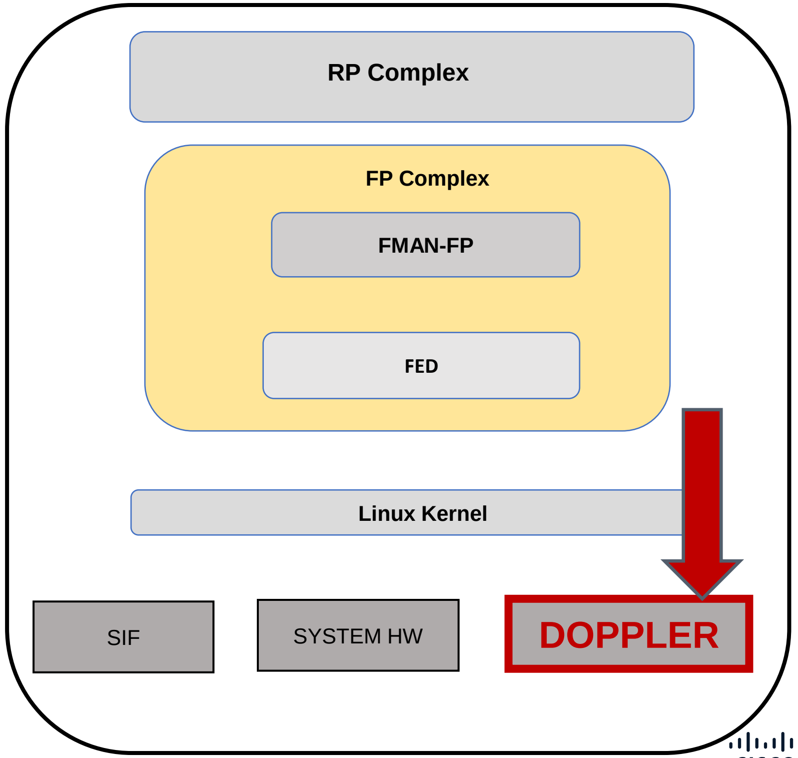
ASIC#:1 RI:20 Rewrite_type:AL_RRM_REWRITE_L3_UNICAST_IPV4_SHARED(1)
Mapped_rii:L3_UNICAST_IPV4(9)
MAC Addr: MAC Addr: 38:aa:09:ff:f1:02,
L3IF LE Index 40

Rewrite mac address

Next hop mac address : 38aa.09ff.f102

ASIC#:2 RI:20 Rewrite_type:AL_RRM_REWRITE_L3_UNICAST_IPV4_SHARED(1)
Mapped_rii:L3_UNICAST_IPV4(9)
MAC Addr: MAC Addr: 38:aa:09:ff:f1:02,
L3IF LE Index 40

ASIC#:3 RI:20 Rewrite_type:AL_RRM_REWRITE_L3_UNICAST_IPV4_SHARED(1)
Mapped_rii:L3_UNICAST_IPV4(9)
MAC Addr: MAC Addr: 38:aa:09:ff:f1:02,
L3IF LE Index 40

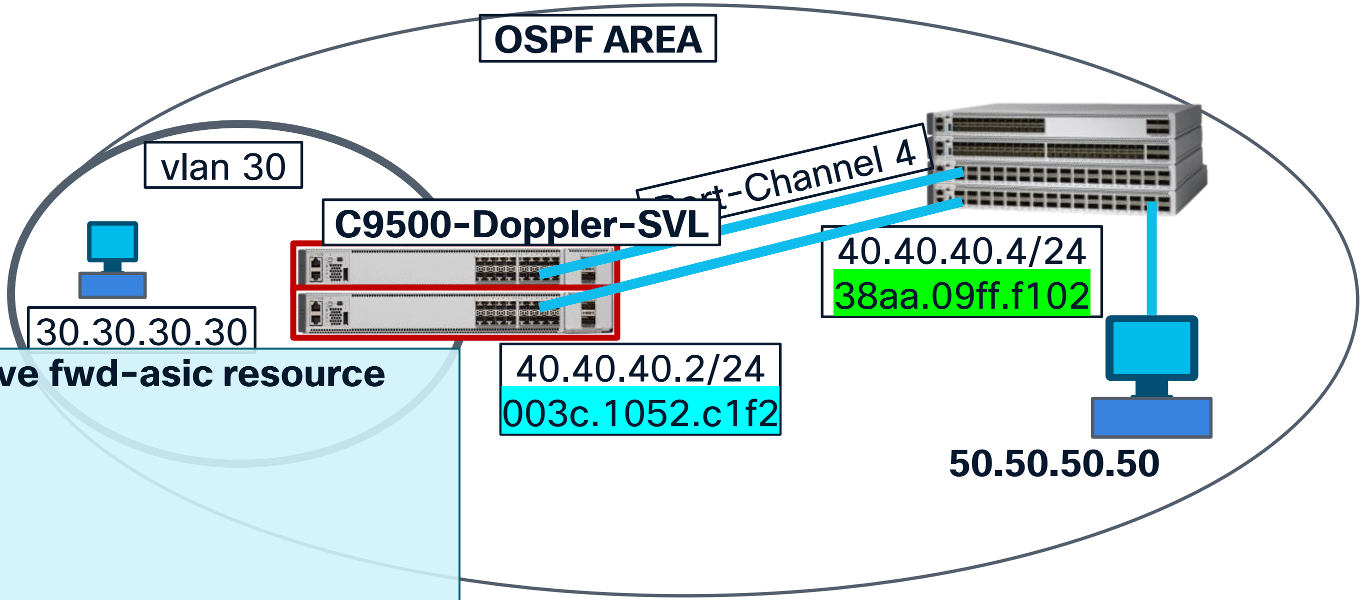


UADP/Doppler L3 Forwarding

Indirect Route : via Routing Protocol

Doppler – Forward ASIC

Destination index programming in Doppler



C9500-Doppler-SVL#show platform hardware fed switch active fwd-asic resource asic all destination-index range 0x4c9f 0x4c9f

ASIC#0:
Destination index = 0x4c9f DI_ETHER_CHANNEL
pmap = 0x00000000 0x00000000
~

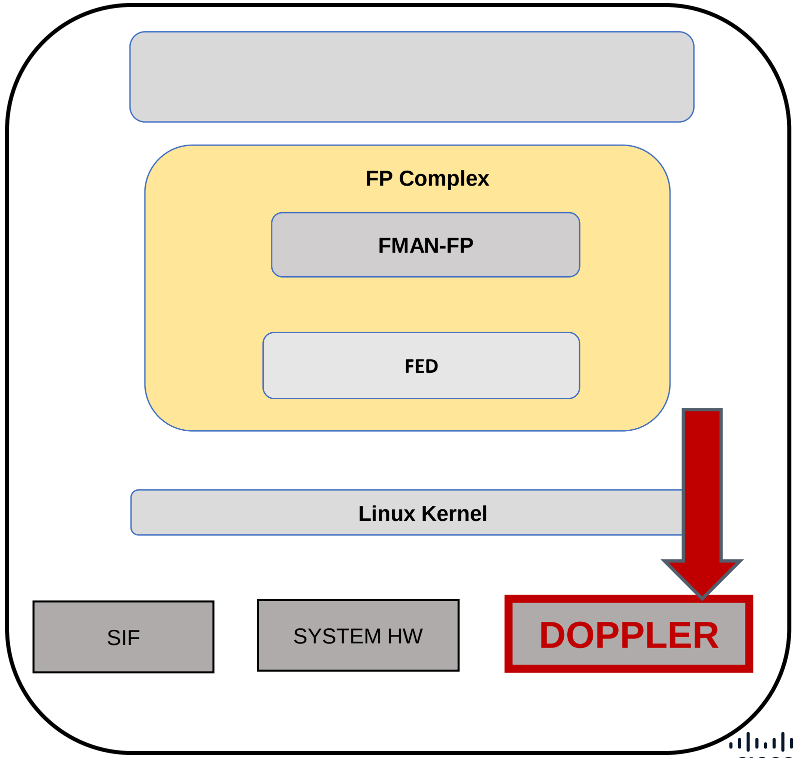
ASIC#1:
Destination index = 0x4c9f DI_ETHER_CHANNEL
pmap = 0x00000000 0x00000000
~

Where to egress out the packet?

ASIC#2:
Destination index = 0x4c9f DI_ETHER_CHANNEL
pmap = 0x00000000 0x00000010
pmap_intf : [FortyGigabitEthernet1/0/5]
cmi = 0x0
~
~

Non ZERO interface port map

ASIC#3:
Destination index = 0x4c9f DI_ETHER_CHANNEL
pmap = 0x00000000 0x00000000
~



UADP/Doppler L3 Forwarding

Indirect Route : via Routing Protocol

Doppler – Forward ASIC

Port-map (pmap) decode logic

```
C9500-Doppler-SVL#show platform hardware fed switch active fwd-asic resource asic all destination-index range 0x4c9f 0x4c9f
~
ASIC#2:
Destination index   = 0x4c9f DI_ETHER_CHANNEL
pmap               = 0x00000000 0x00000010
pmap_intf : [FortyGigabitEthernet1/0/5]
~
```

On ASIC Instance 2

Port number 4

$(00000010)_{16}$ ➔ $(0000\ 0000\ 0000\ 0000\ 0000\ 0000\ 0001\ 0000)_2$

23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0

```
C9500-Doppler-SVL#show  platform software fed switch active ifm mappings
Interface                    IF_ID  Inst Asic Core Port SubPort Mac  Cntx LPN  GPN  Type Active
~
FortyGigabitEthernet1/0/5    0x85   2    1    0    4    0       4    4    5    1368 NIF  Y
~
```

UADP/Doppler L3 Forwarding

Indirect Route : via Routing Protocol

FED – Forwarding Engine Driver

```
C9500-Doppler-SVL#show  platform software fed switch active ip route 50.50.50.0/24 detail
vrf   dest                              htm           flags   SGT   DGID MPLS Last-modified           SecsSinceHit
---   ----                              ---           -----   ---   ---- ---- ----------------------- ------------
0     50.50.50.0/24      0x7d9134a5ab78 0x0     0     0        2026/01/07 18:00:49.504       61935
  FIB: prefix_hdl:0x45000017, mpls_ecr_prefix_hdl:0, sgtOverWrite: 0
  ========== OCE chain =====
  ADJ:objid:27 {link_type:IP ifnum:0x87, adj:0x64000027, si: 0x7d91347ef658  IPv4:      40.40.40.4 }
 ~
Hardware entry details
---------------------
 ~
Detailed Resource Information (ASIC# 1)
----------------------------------------
Station Index (SI) [0xae]
RI = 0x14
DI = 0x4c9f
Replication Bitmap: LD RD CD

Destination index    = 0x4c9f DI_ETHER_CHANNEL
pmap              = 0x00000000 0x00000010
pmap_intf : [FortyGigabitEthernet1/0/5]
~
```

Quiz

Ideal troubleshooting sequence on Doppler/UADP ASIC based platform is:-

Which command is used to verify the state of an SVI interface in the FED on a Cisco Catalyst 9300 switch?

Which command(s) is/are used to verify the consistency of a route between IOSd, FMAN_FP, FED?

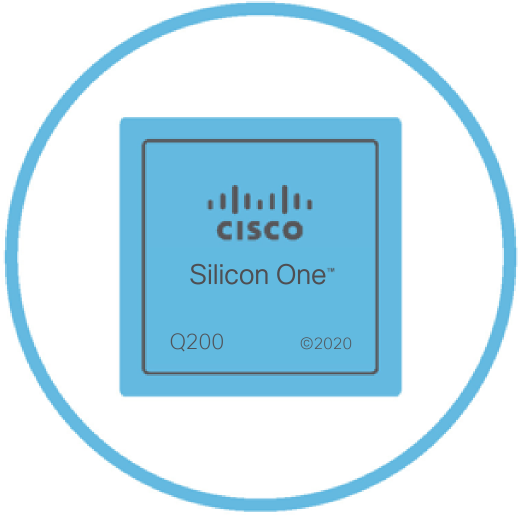
Silicon 1 is The FUTURE

Enterprise
Campus

This session will make you FUTURE aware.

Silicon ONE
Introduction & Architecture

Catalyst 9000 Series – Common Building Blocks



Programmable x86 Multi-Core CPU

Application Hosting
Secure Containers

Open IOS XE® Polaris

Model-Driven APIs
Modular Patching

Cisco Silicon One™

Programmable Pipeline
Flexible Tables

Same binary image for both UADP and Silicon One C9K platforms*

* C9200 uses IOS XE Lite

Cisco Silicon One

Switching Silicon

- **High Throughput**
  extremely fast hardware-based L2-L4 forwarding and services (measured in Terabits per second)
- **Optimized Scale**
  optimized for Campus LAN environments with moderate IP & MAC scale (10s-100s of thousands)
- **Low Latency**
  extremely low hardware-based system latency (measured in Nanoseconds & Microseconds)
- **Streamlined Buffering**
  shallow buffering systems to reduce latency, with very high throughput

Routing Silicon

- **Flexible Features**
  complex, stateful L3-L7 forwarding and services (measured in Gigabits per second)
- **Massive Scale**
  optimized for WAN/SP environments with very high IP scale (100s of thousands – millions)
- **Mixed Interfaces**
  support for Ethernet, Serial, Cellular and other types and speeds in a single system
- **Deeper Buffering**
  deep buffers to accommodate different speeds, bursts and different flow patterns

Cisco Silicon One Bringing Switching and Routing convergence

Silicon One
Q200

400G Support

Deep Buffers 8GB

Routes 10M

ASIC Performance 12.8 Tbps, 8.1 Bpps

Multi Slice Architecture

P4 NPL

56G PAM4

VOQ Architecture

7 nm Technology

Cisco Silicon One™ Q200

Industry leading Switching and Routing Silicon

12.8T BW

8.1 Bpps

8G HBM for deep buffers

10M IPv4 or 5M IPv6 route scale

Fully P4 Programmable Pipeline

50G PAM4 Serdes

Cisco Silicon ONE Q200

Industry Leading
12.8T System on Chip

First 7nm ASIC providing lowest watts/GE power consumption

Fully P4 programmable enabling feature velocity

Multi slice architecture for flexibility and scale

Routing Capabilities with Switching Power and Performance

ONE Architecture Multiple Devices – Silicon ONE

9500X, 9600X

S1- Q200
12.8 Tbps
7 nm
8.1 Bpps

CS-C9610

S1- K100
51.2 Tbps
7 nm
15.6 Bpps

CS-C9350

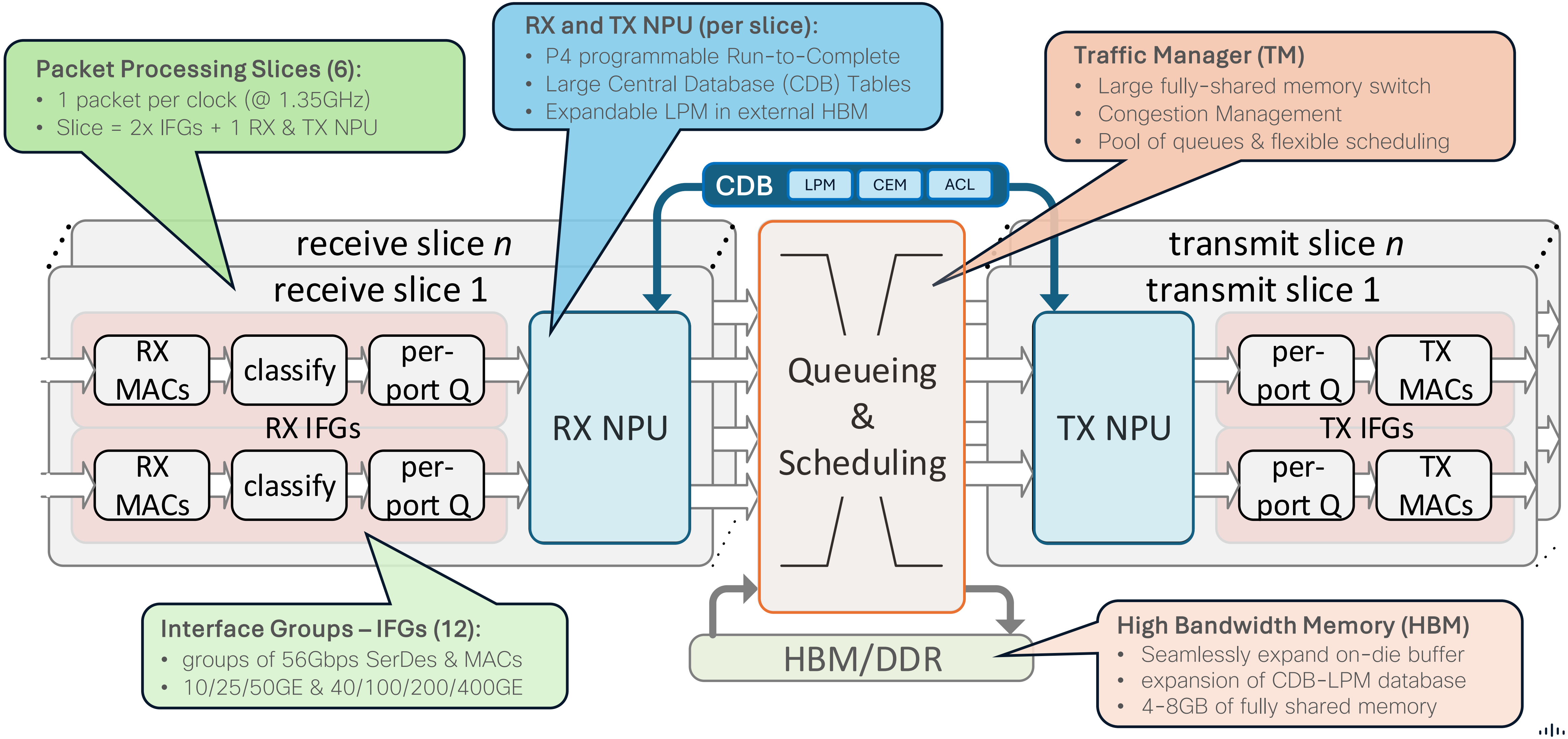
S1- A100
1.4 Tbps
16nm
1.5 Bpps

- Multiple generations and formats, same architecture
- Rich flexible forwarding & services memories
- First fully programmable microcode network

- Multiple functions: system-on-chip or line-card
- Multiple form factors: fixed or modular
- Multiple places: Access, Distribution and Core

Cisco Silicon One™ Q200

ASIC Architecture & Block Diagram

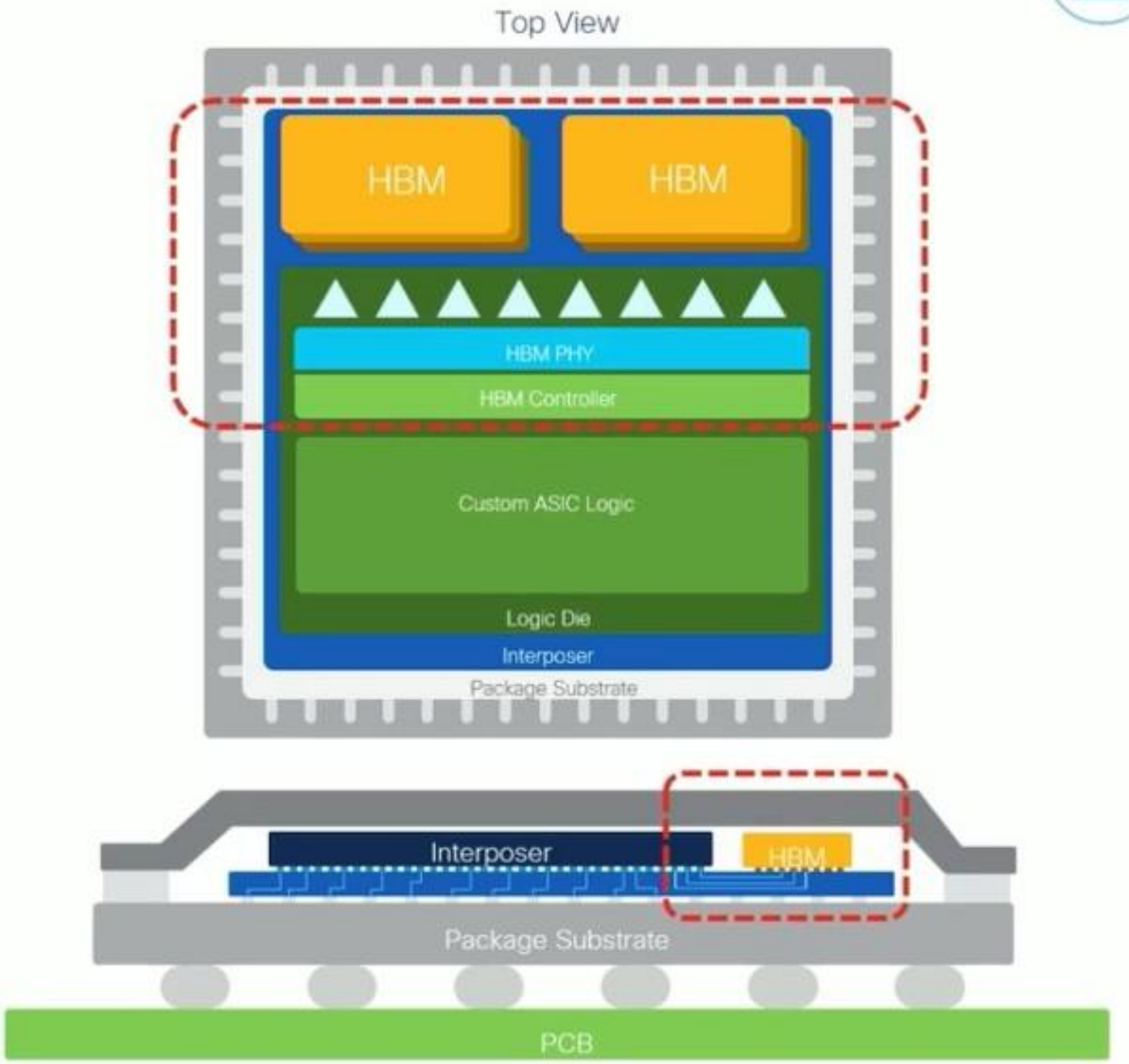


Cisco Silicon One - HBM

High Bandwidth Memory

- Augments local on-die memory
  - use local (SMS) buffers until full
  - use HBM for bursts or congestion
- Deep buffering + FIB expansion on-package, at high-speed
- 2 x stacks of 2.5D memory with wide-bus interposer = ~2.4 Tbps full duplex
- Interposer connects ASIC die to on-package HBM memory

en.wikipedia.org/wiki/High_Bandwidth_Memory



Cisco Silicon One™ Q200 – Central Databases

Onboard LPM, CEM & ACL memory

Q200 CDB includes the Central L2/L3 Forwarding and ACL databases:

LPM – SRAM database for IP/mask routing implemented by Longest Prefix Match algorithm

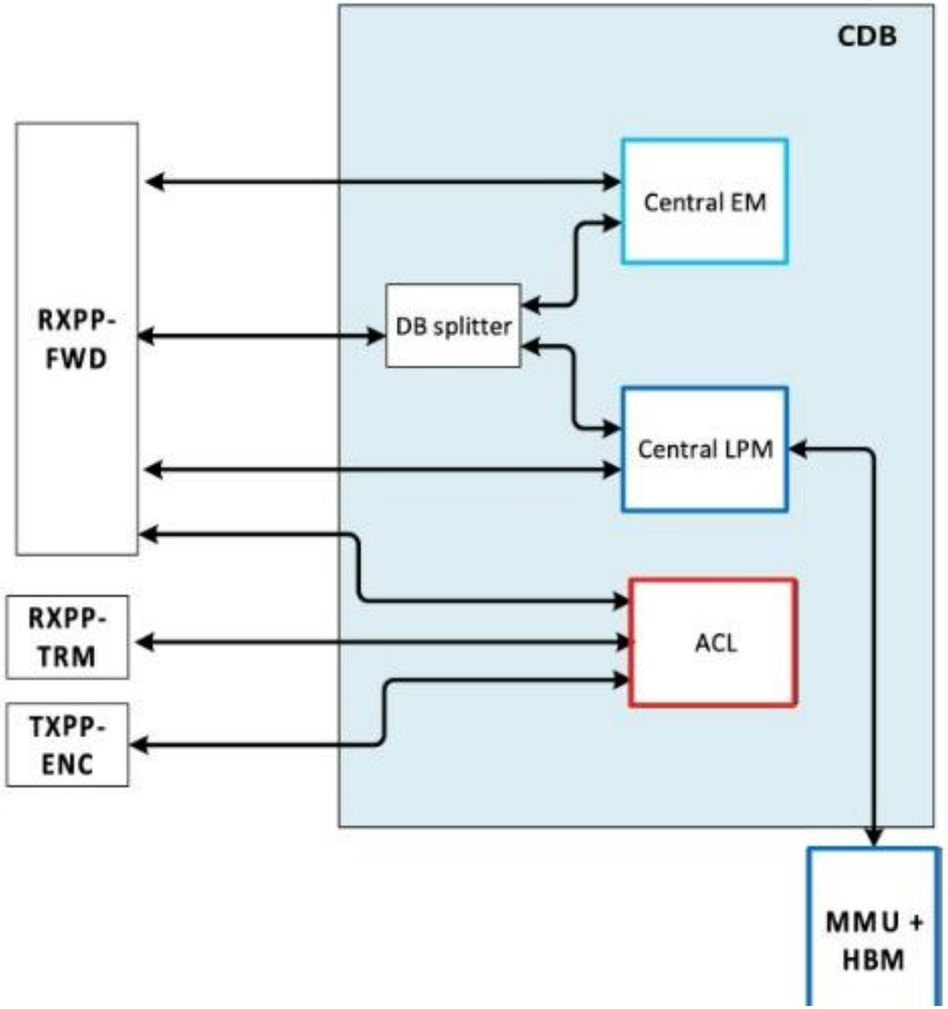
- Primarily used by IPv4 and IPv6 unicast routing
- Up to* 2M IPv4 route entries, or 1M IPv6 route entries
- LPM can be extended (from CDB) to HBM

CEM – SRAM database for MAC & Host (/48, /32 or /128), Multicast & Labels implemented by Exact Match algorithm

- For features using an exact match (every bit, no mask)
- Up to 608K IPv4 entries, or 304K IPv6 entries
- CEM can be flexibly reallocated for different tables

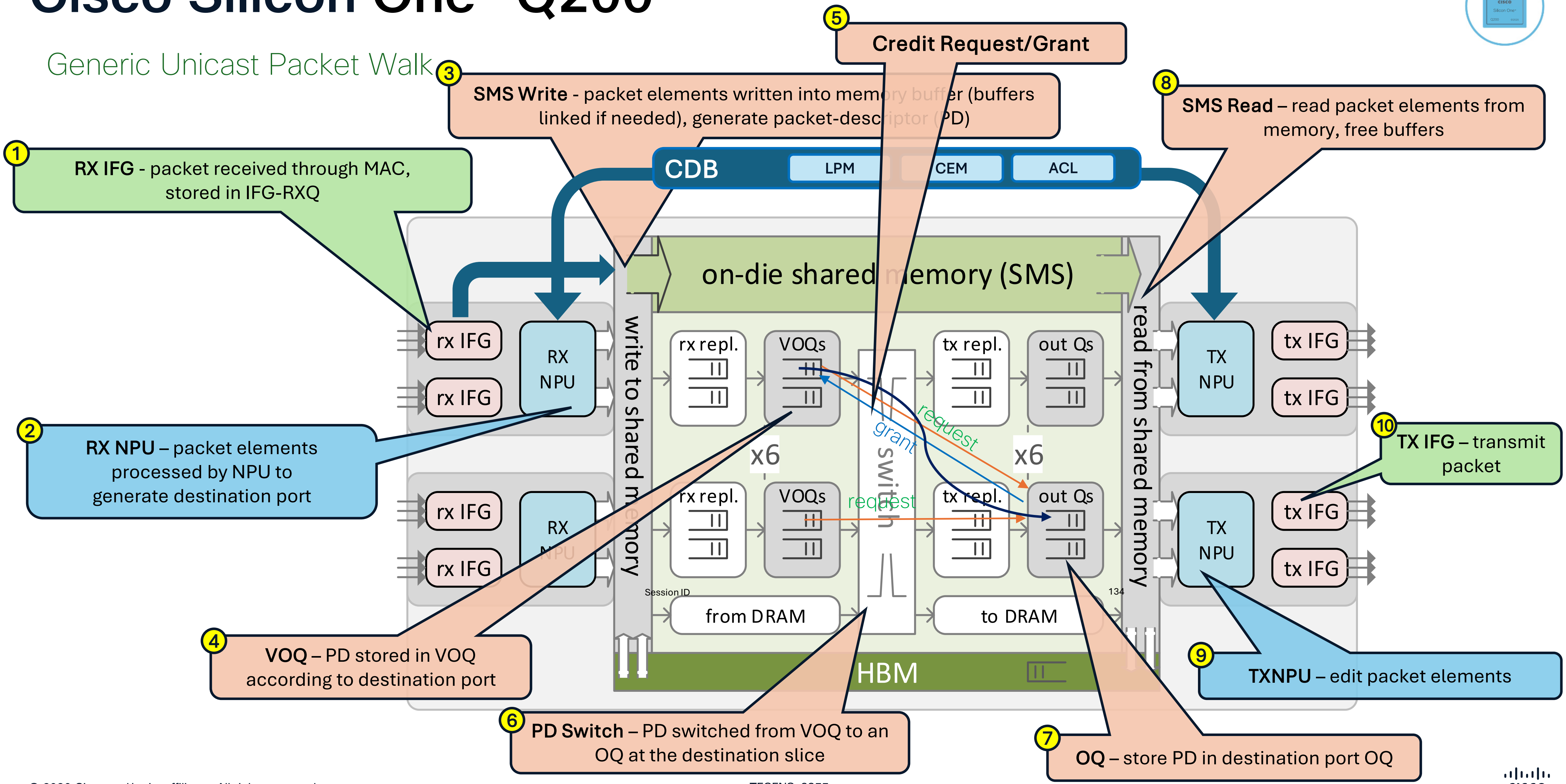
ACL – TCAM classification database, contains Security, QoS and Services Access Control List entries

- For features that use (match criteria + action) policies
- Up to 8K IPv4 ACL entries, or 4K IPv6 ACL entries
- OG/SGACLs use CEM, with only action ACEs in TCAM

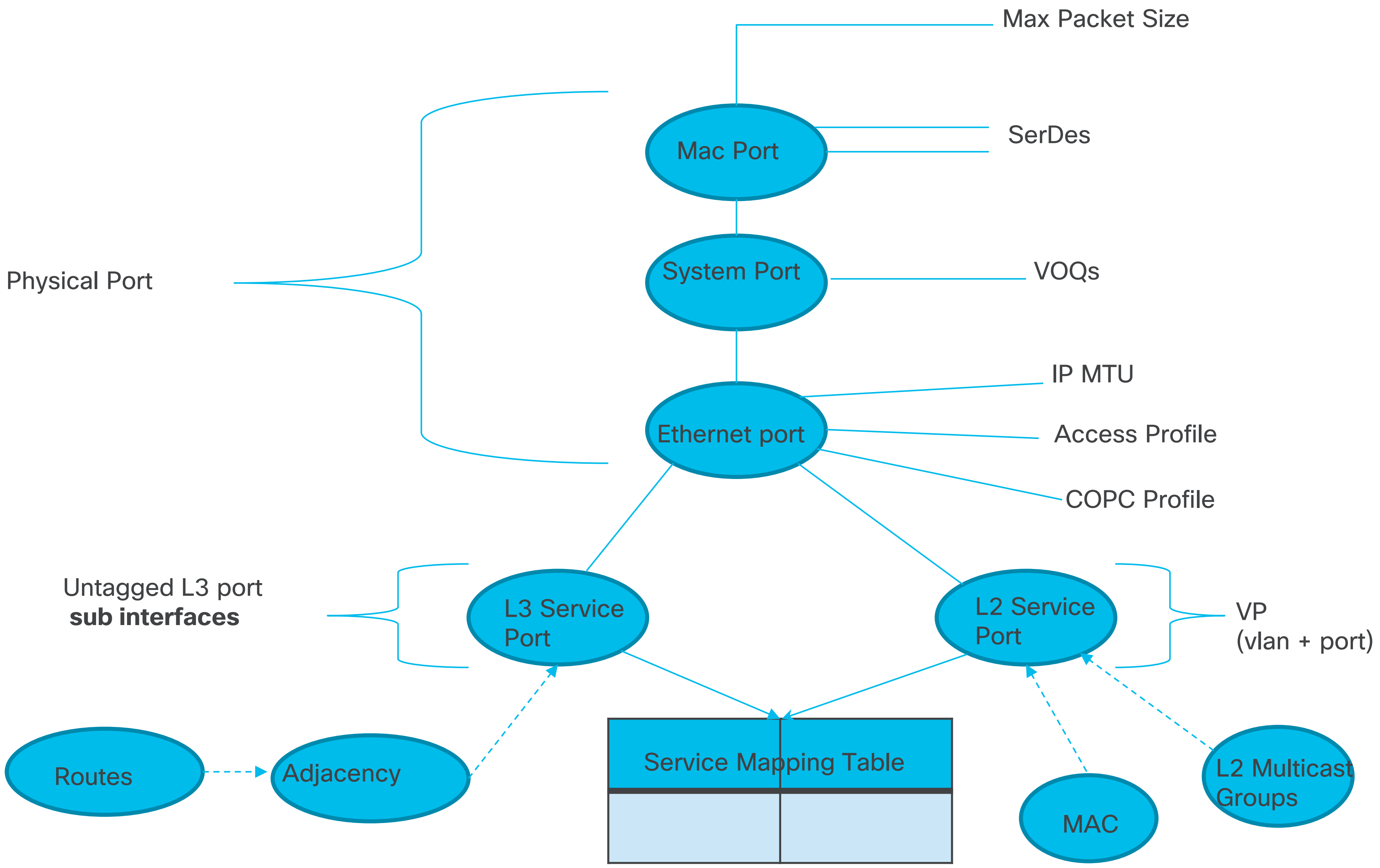


Cisco Silicon One™ Q200

Generic Unicast Packet Walk



Object Model



Quiz

What is the maximum possible throughput of UADP & Silicon One ASIC?

Silicon One Q200 based on:-

What is the name of the interface between FED 3.0 and SiliconOne ASIC?

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Silicon ONE
L2 and L3 Forwarding

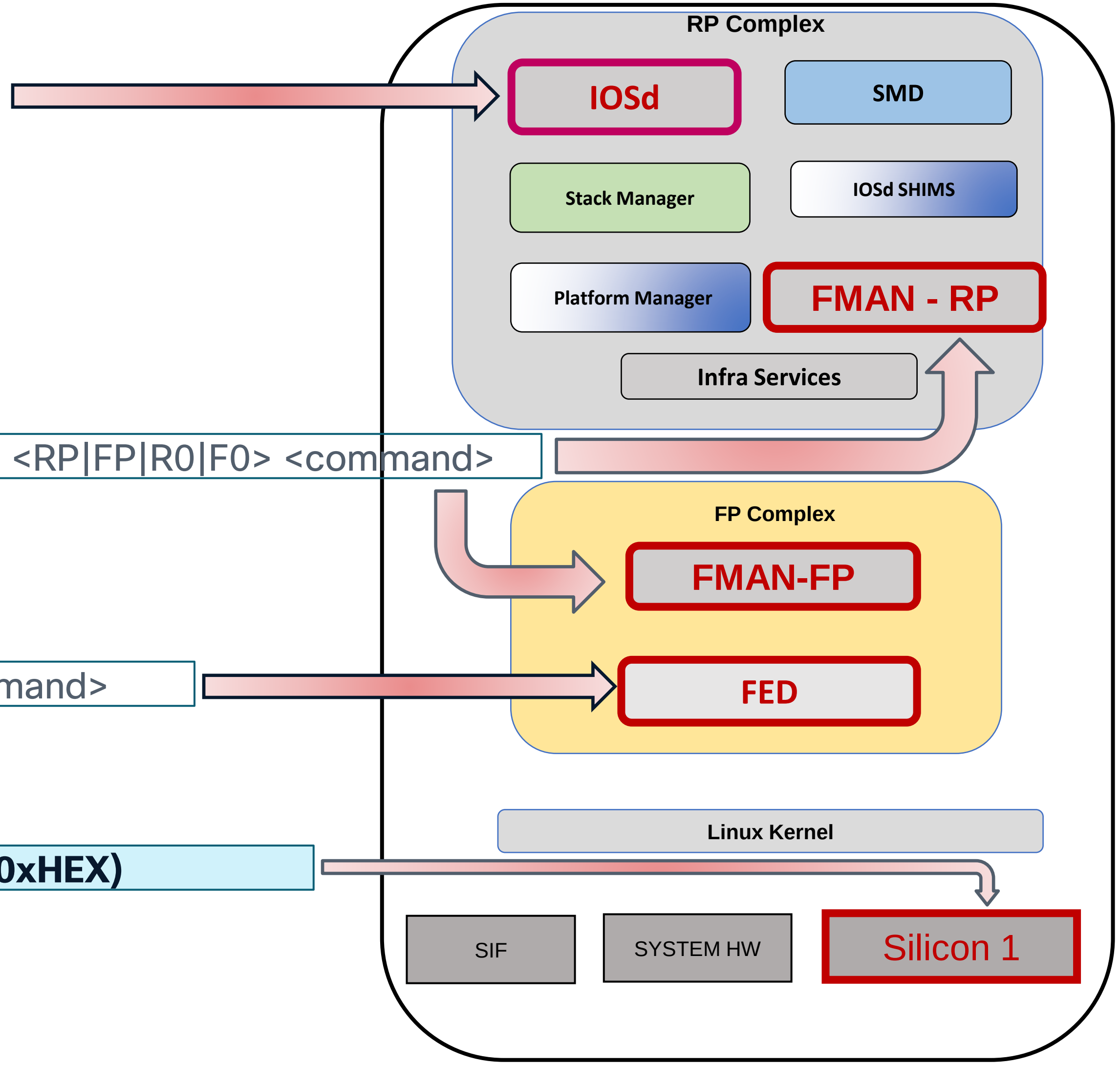
Commands Logic (Silicon ONE)

show <Process/Components> <Objects> <Variables>

show platform software <software_process> switch <id | active | standby> <RP|FP|R0|F0> <command>

show platform software fed switch <id | active | standby> <process> <command>

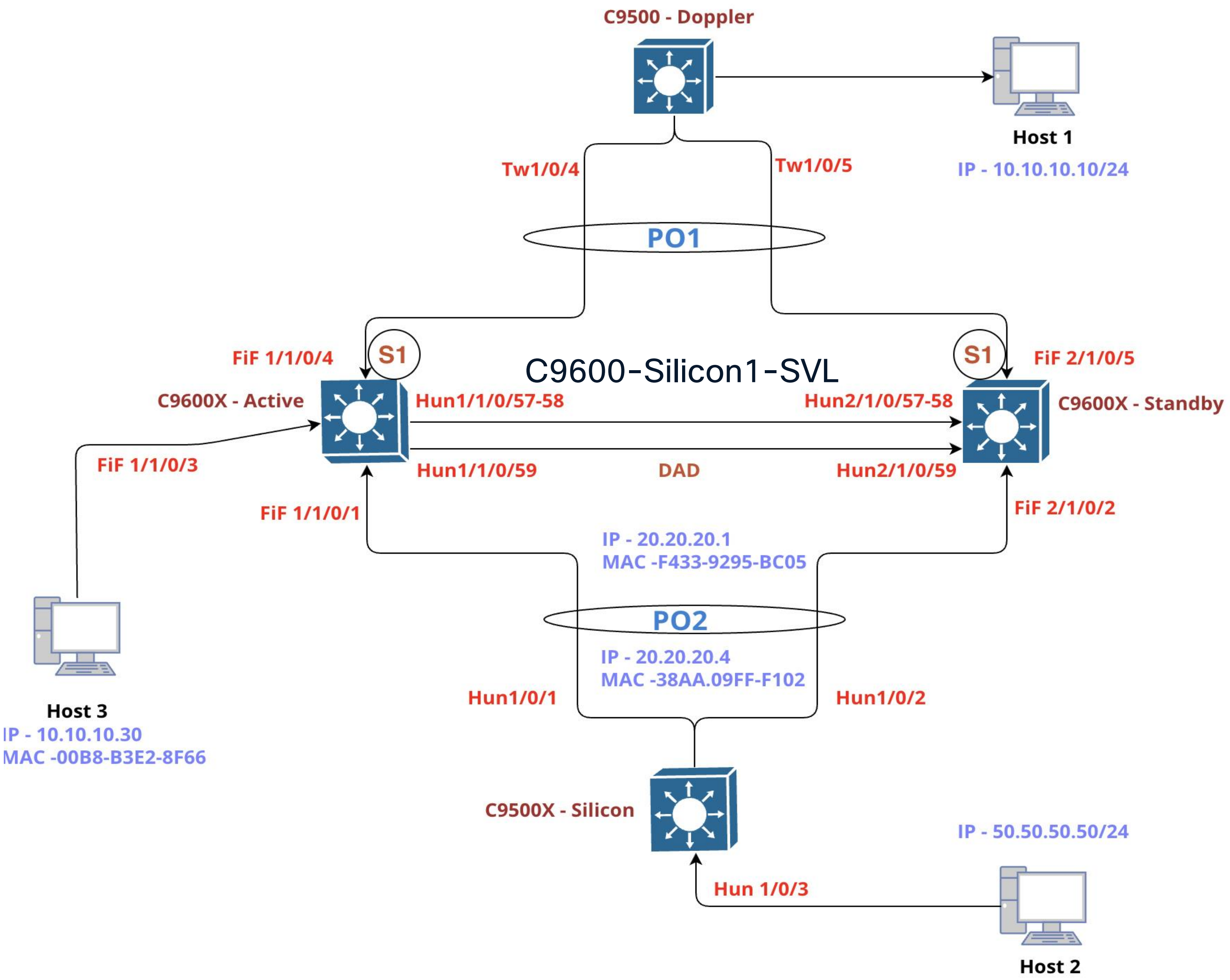
show platform hardware fed switch active fwd-asic insight sdk_object(0xHEX)



S1 - L2 Forwarding

Silicon ASIC

Lab Topology



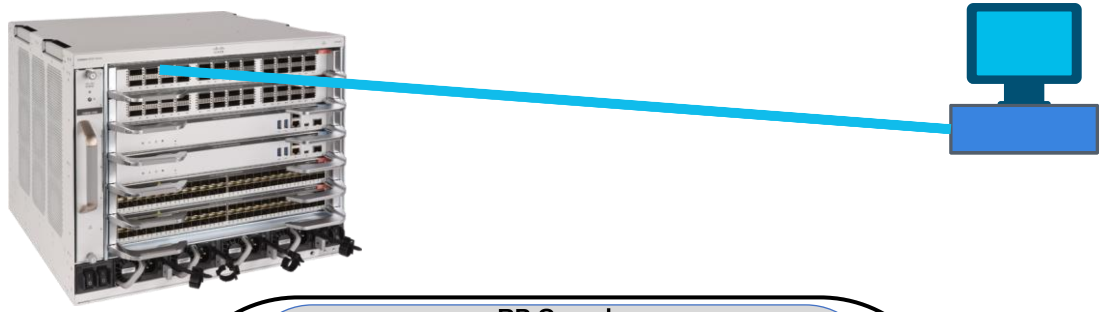
Port Programming

Silicon ONE L2 Forwarding

Port Programming

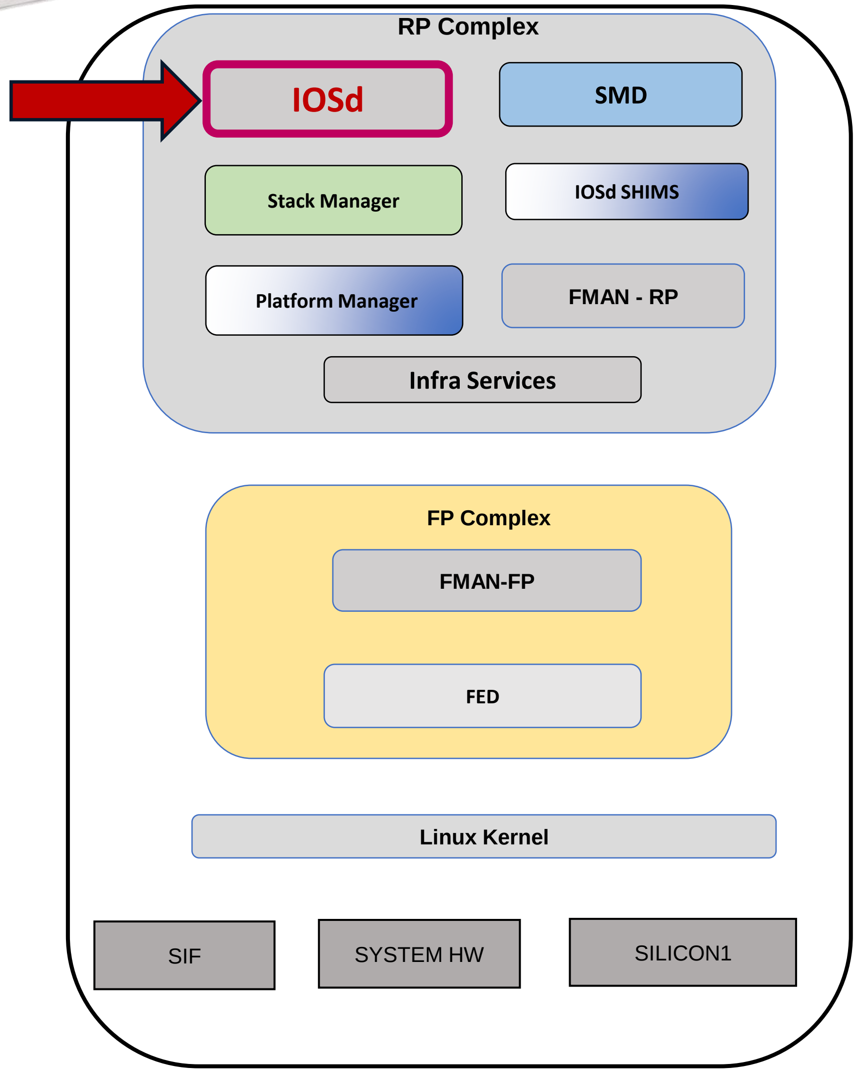
IOSd

FiftyGigE 1/1/0/3 : f003.bce9.3402



```
C9600-Silicon1-SVL#show  run interface fiftyGigE 1/1/0/3
Building configuration...

Current configuration : 117 bytes
!
interface FiftyGigE1/1/0/3
 description To-Host3-Ten 1/1/3
 switchport access vlan 10
 switchport mode access
end
```

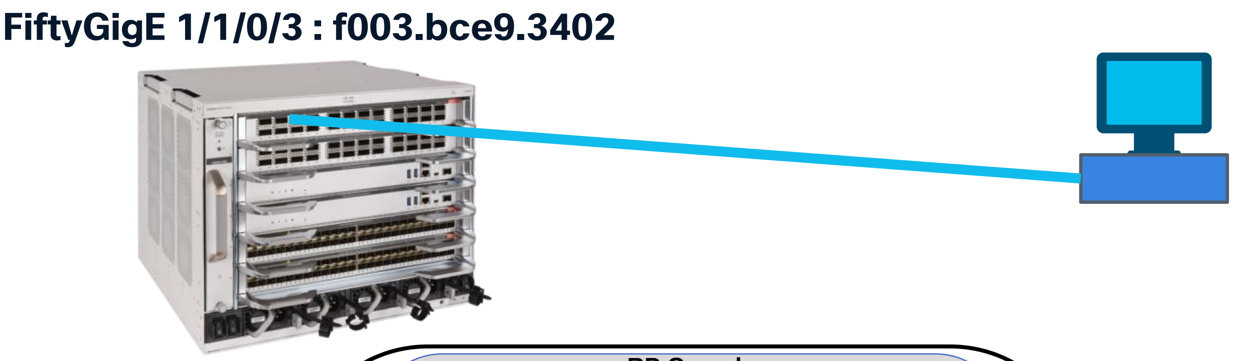
```
C9600-Silicon1-SVL#show  interfaces fiftyGigE 1/1/0/3
FiftyGigE1/1/0/3 is up, line protocol is up (connected)
  Hardware is Fifty Gigabit Ethernet, address is f003.bce9.3402 (bia
f003.bce9.3402)
  Description: To-Host3-Ten 1/1/3
  MTU 1500 bytes, BW 10000000 Kbit/sec, DLY 10 usec,
 ~
 ~
```



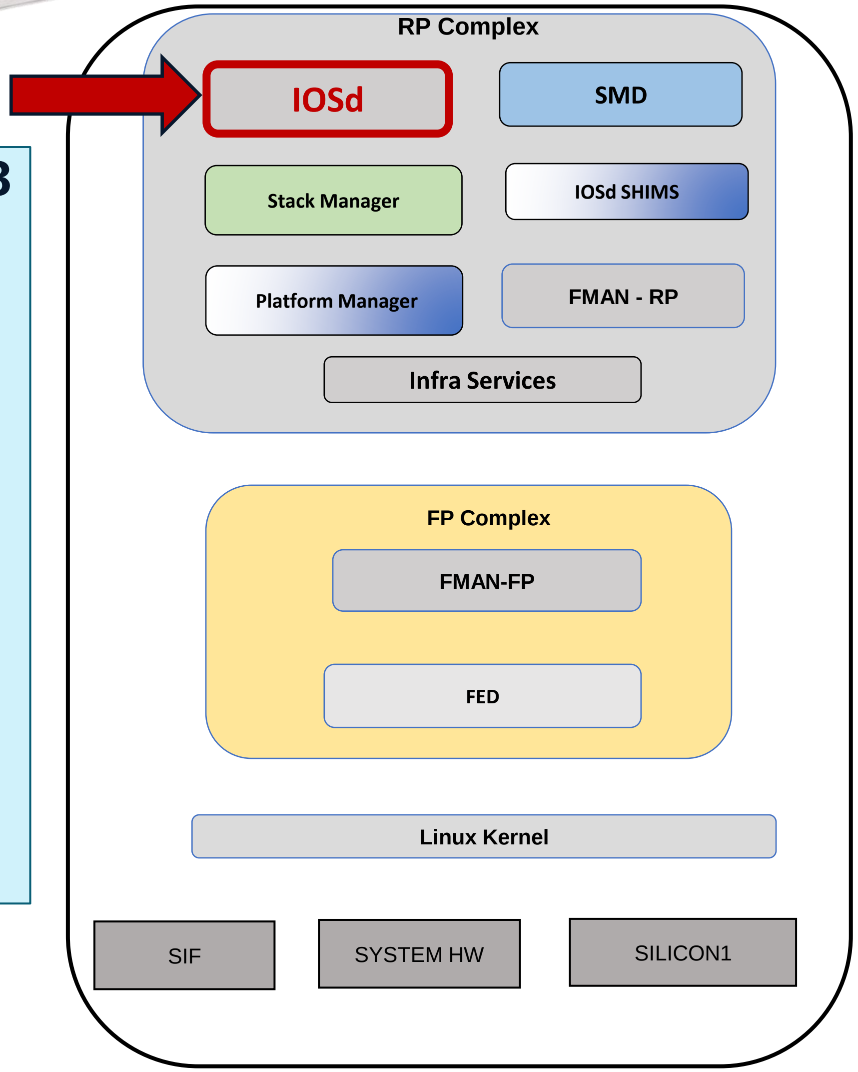
Silicon ONE L2 Forwarding

Port Programming

IOSD




```
C9600-Silicon1-SVL#show  idprom interface fiftyGigE 1/1/0/3
IDPROM for transceiver :
  Description                    = SFP or SFP+ optics (type 3)
  Transceiver Type:              = 10GE CU2M (522)
  Product Identifier (PID)       = SFP-H10GB-CU2M
  Vendor Revision                = U
  Serial Number (SN)             = TED2305H2P1
  Vendor Name                    = CISCO-TYCO
  ~
  Device State                   = Enabled.
 Connector type                  = Copper pigtail
 ~
!
```



Silicon ONE L2 Forwarding

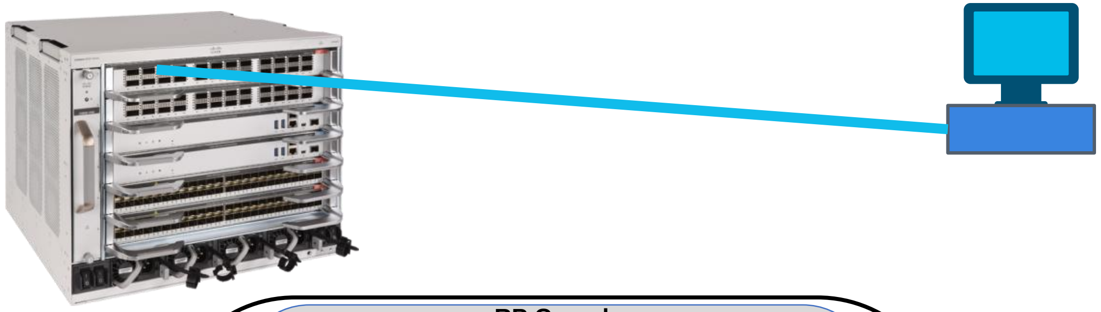
Port Programming

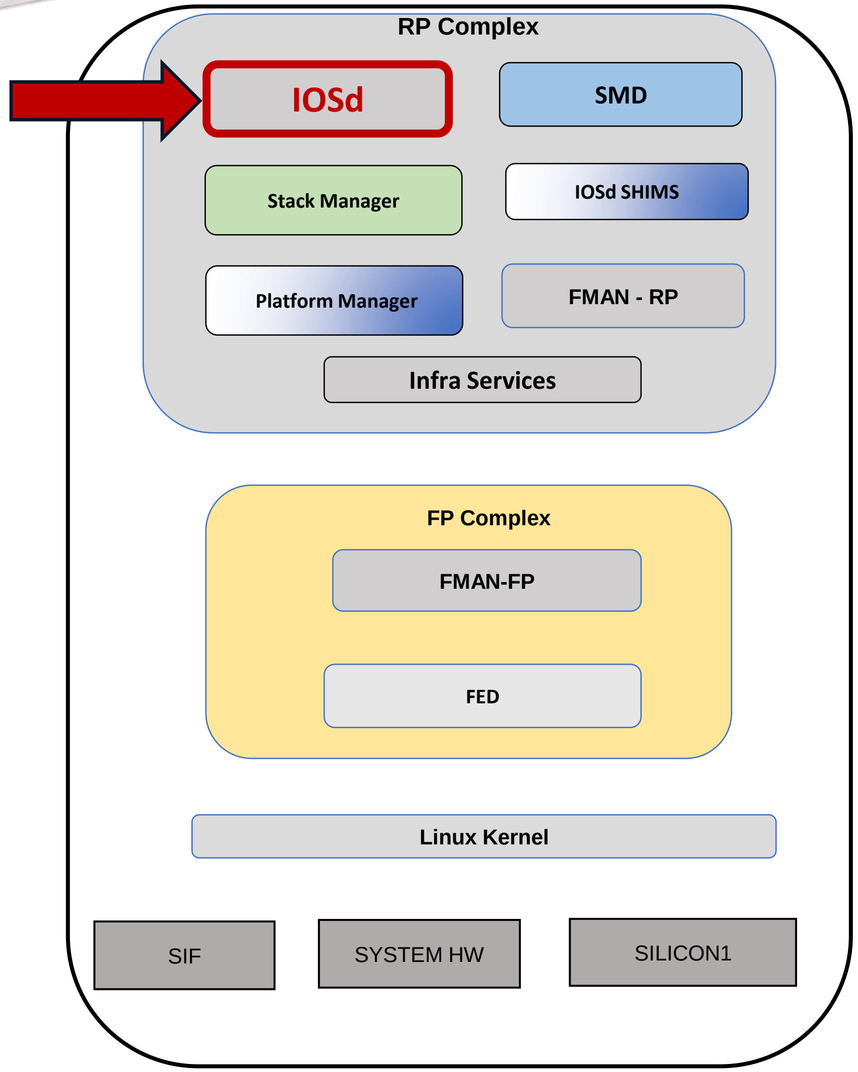
IOSD

```
C9600-Silicon1-SVL#show  interfaces fiftyGigE 1/1/0/3 switchport
Name: Fif1/1/0/3
Switchport: Enabled
Administrative Mode: static access
Operational Mode: static access
Administrative Trunking Encapsulation: dot1q
Operational Trunking Encapsulation: native
Negotiation of Trunking: Off
Access Mode VLAN: 10 (VLAN0010)
Trunking Native Mode VLAN: 1 (default)
Administrative Native VLAN tagging: enabled
Voice VLAN: none
```

```
C9600-Silicon1-SVL#show  interfaces fiftyGigE 1/1/0/3 trunk
Port         Mode          Encapsulation   Status          Native vlan
Fif1/1/0/3    off            802.1q          not-trunking        1
Port       Vlans allowed on trunk
Fif1/1/0/3   10
Port       Vlans allowed and active in management domain
Fif1/1/0/3   10
Port       Vlans in spanning tree forwarding state and not pruned
Fif1/1/0/3   10
```

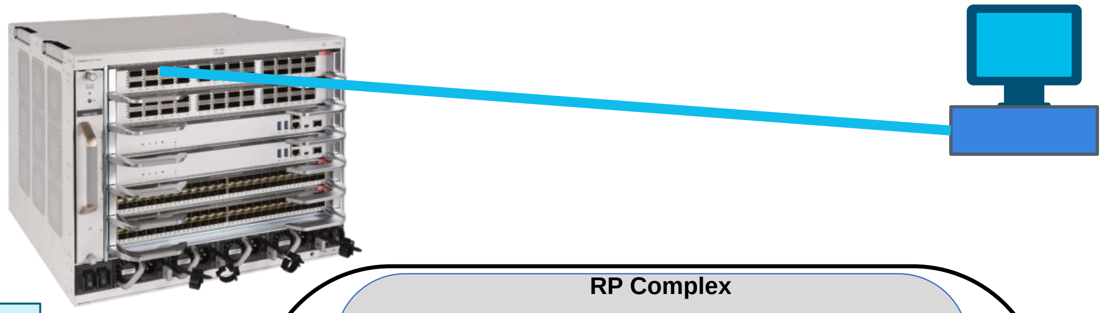
FiftyGigE 1/1/0/3 : f003.bce9.3402





Silicon ONE L2 Forwarding

FiftyGigE 1/1/0/3 : f003.bce9.3402



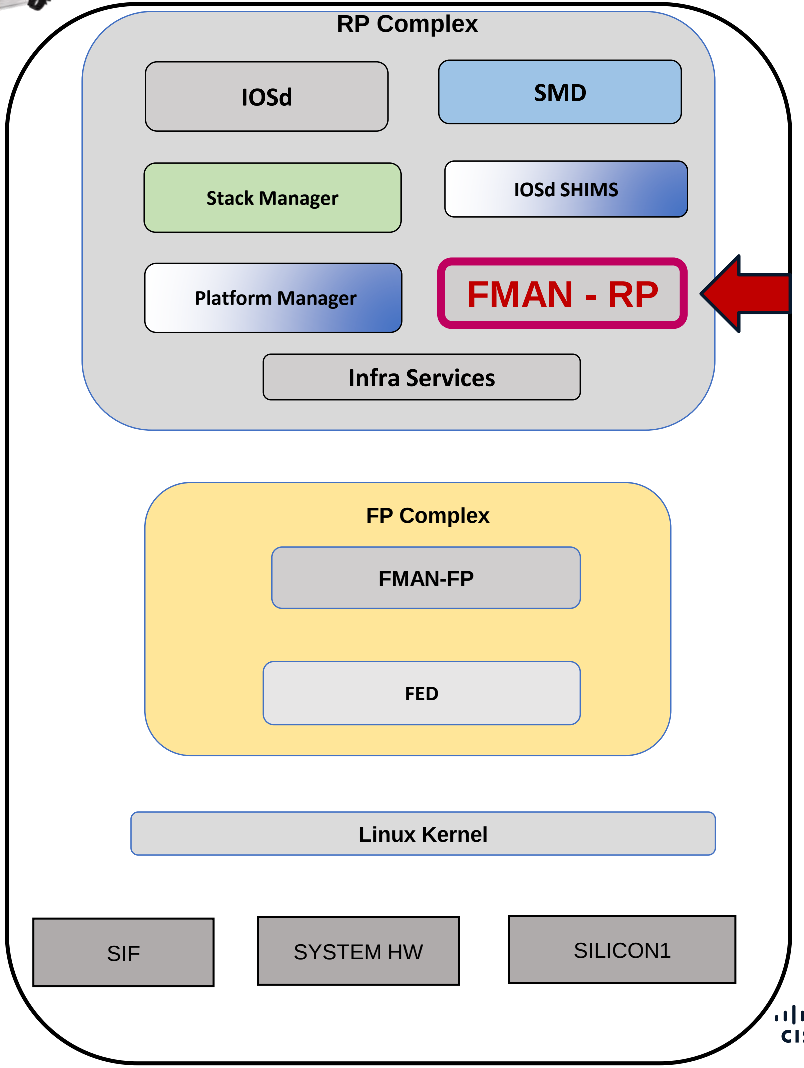
Port Programming

FMAN-RP

Hex(0x40B)=Dec(1035)

```
C9600-Silicon1-SVL#show  platform software interface switch active R0 brief
 Forwarding Manager Interfaces Information

 Name                          ID          QFP ID
 --------------------------------------------------------------
 Null0                        1024          0
 FiftyGigE1/1/0/2             1034          0
 FiftyGigE1/1/0/3             1035          0
 FiftyGigE1/1/0/4             1036          0
 ~
 ~
```

```
C9600-Silicon1-SVL#show  platform software interface switch active R0 name fiftyGigE 1/1/0/3
Name: FiftyGigE1/1/0/3, ID: 1035, QFP ID: 0, Schedules: 4096
Type: PORT, State: enabled, SNMP ID: 7, MTU: 1500
Flow control ID: 65535
bandwidth: 10000000, encap: ARPA
~
~
QOS trust type: Trust DSCP
```

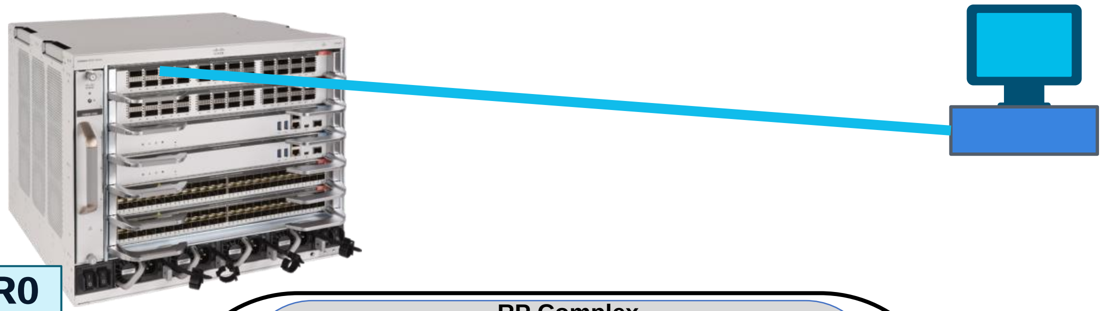


Silicon ONE L2 Forwarding

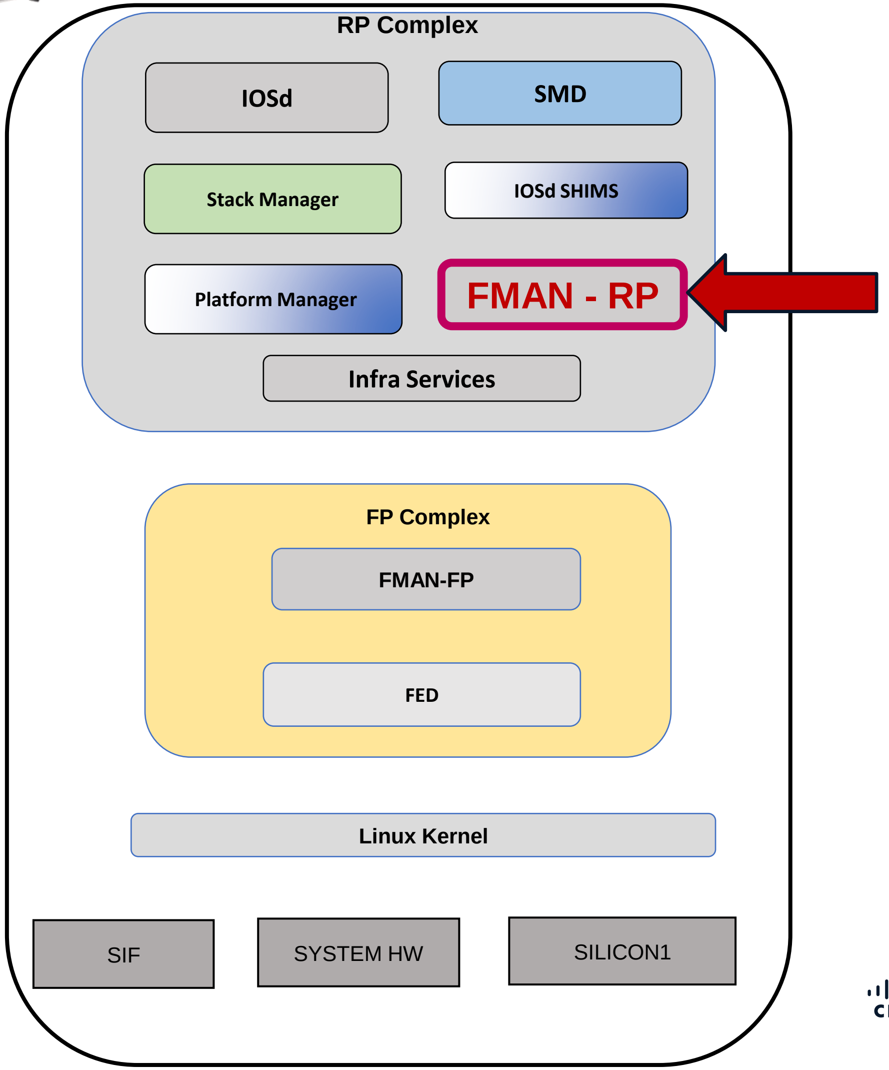
Port Programming

FMAN-RP

FiftyGigE 1/1/0/3 : f003.bce9.3402



```
C9600-Silicon1-SVL#show  platform software interface switch active R0 statistics
 Forwarding Manager Interfaces Statistics

SIP OIR: 6, SPA OIR: 0
~

Fman RP-FP Statistics

~

AOM errors new: , create: , modify: , delete:
Database addition errors:
Adjacency reparent to Null errors:

EVC API called: , Callback: , Callback error:
~
```



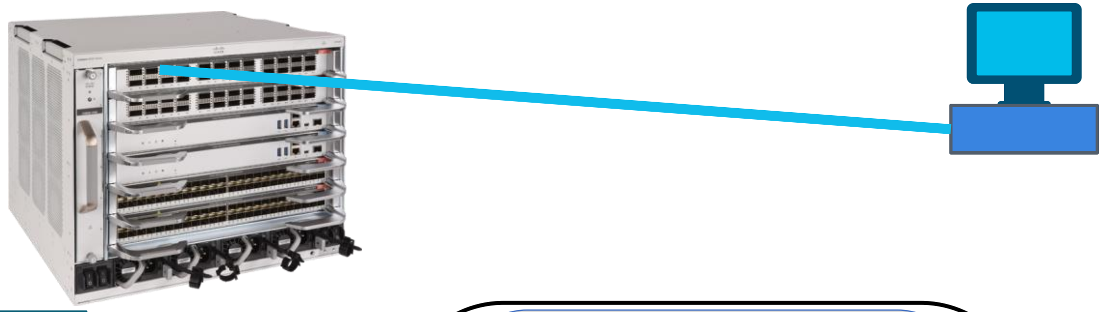
Silicon ONE L2 Forwarding

Port Programming

FMAN-FP

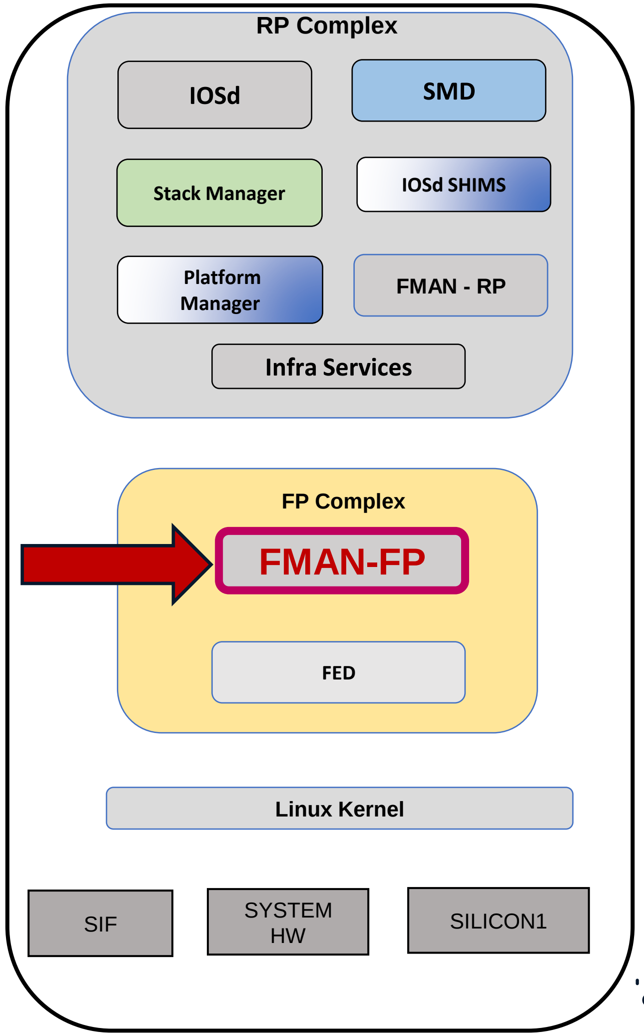
Hex(0x40B)=Dec(1035)

FiftyGigE 1/1/0/3 : f003.bce9.3402



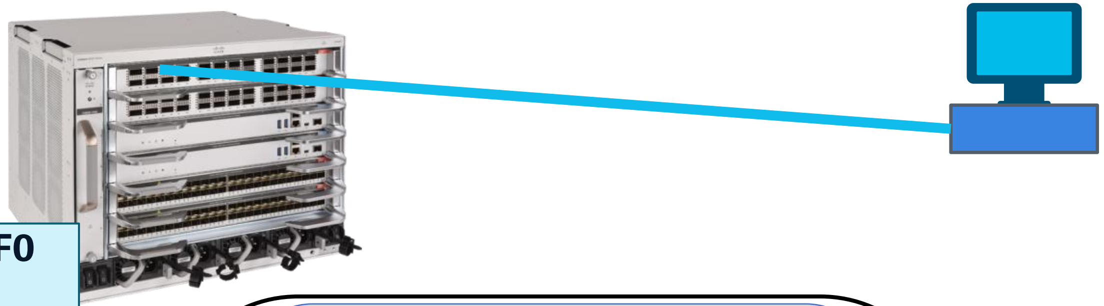
```
C9600-Silicon1-SVL#show  platform software interface switch active F0 brief
Forwarding Manager Interfaces Information
Name                          ID            QFP ID
-----------------------------------------------------
~
HundredGigE2/1/0/59/4          1168          1168
FiftyGigE1/1/0/2               1034          1034
FiftyGigE1/1/0/3               1035          1035
FiftyGigE1/1/0/4               1036          1036
FiftyGigE1/1/0/5               1037          1037
~
```

```
C9600-Silicon1-SVL#show  platform software interface switch active F0 name fiftyGigE 1/1/0/3
Name: FiftyGigE1/1/0/3, ID: 1035, QFP ID: 1035, Schedules: 4096
Type: PORT, State: enabled, SNMP ID: 7, MTU: 1500
~
```



Silicon ONE L2 Forwarding

FiftyGigE 1/1/0/3 : f003.bce9.3402

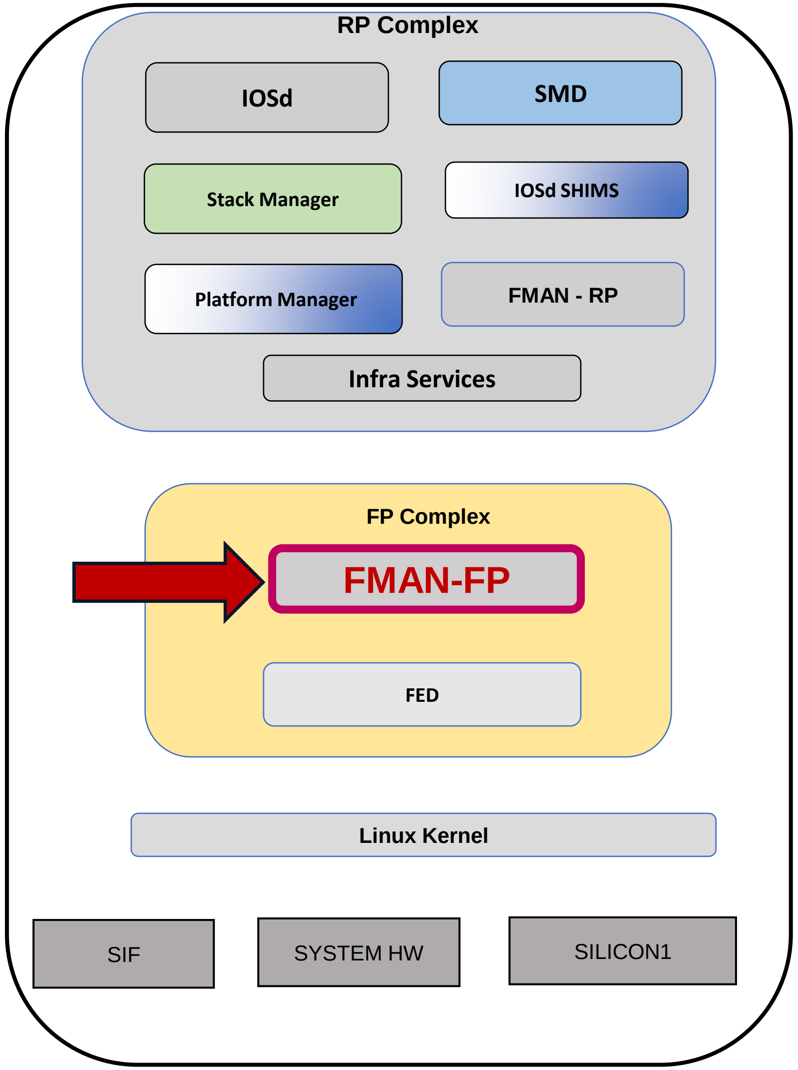


Port Programming

FMAN-FP

C9600-Silicon1-SVL#show platform software interface switch active F0 statistics

```
Forwarding Manager Interfaces Statistics

SIP OIR: 5, SPA OIR: 0
DPIDB create: 306, DPIDB update: 115, DPIDB delete: 1
Vlan update: 0, Vlan delete: 0
EVSI update: 0, EVSI delete: 0
Etherchannel member add: 0, delete: 0, error: 0
Etherchannel bucket add: 0, update: 0, delete: 0, error: 0

Fman RP-FP Statistics
~
Port: 133, Channelized: 0
Virtual: 1, Tunnel: 2, Ether-channel: 3
~

AOM errors new: 0, create: 0, modify: 0, delete: 0
~
FR if_config_a: , rsp_cb: , req_err: , rsp_err:
FR if_bind_a: , rsp_cb: , req_err: , rsp_err:
FR if_unbind_a: , rsp_cb: , req_err: , rsp_err:
FR dlci_create_a: , rsp_cb: , req_err: , rsp_err:
```

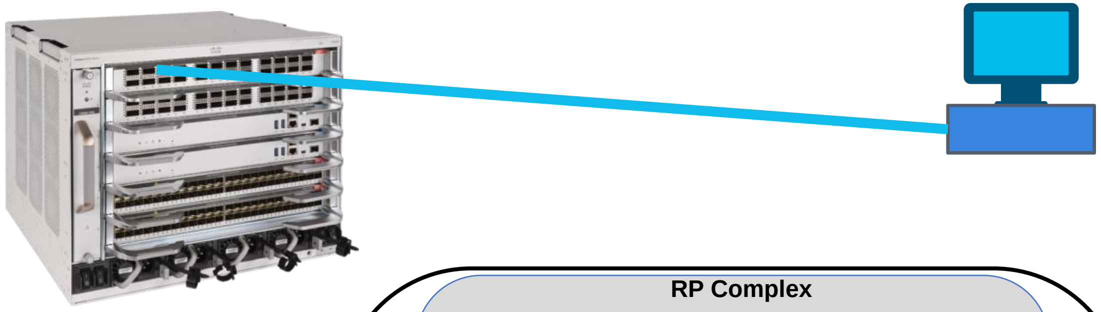


Silicon ONE L2 Forwarding

Port Programming

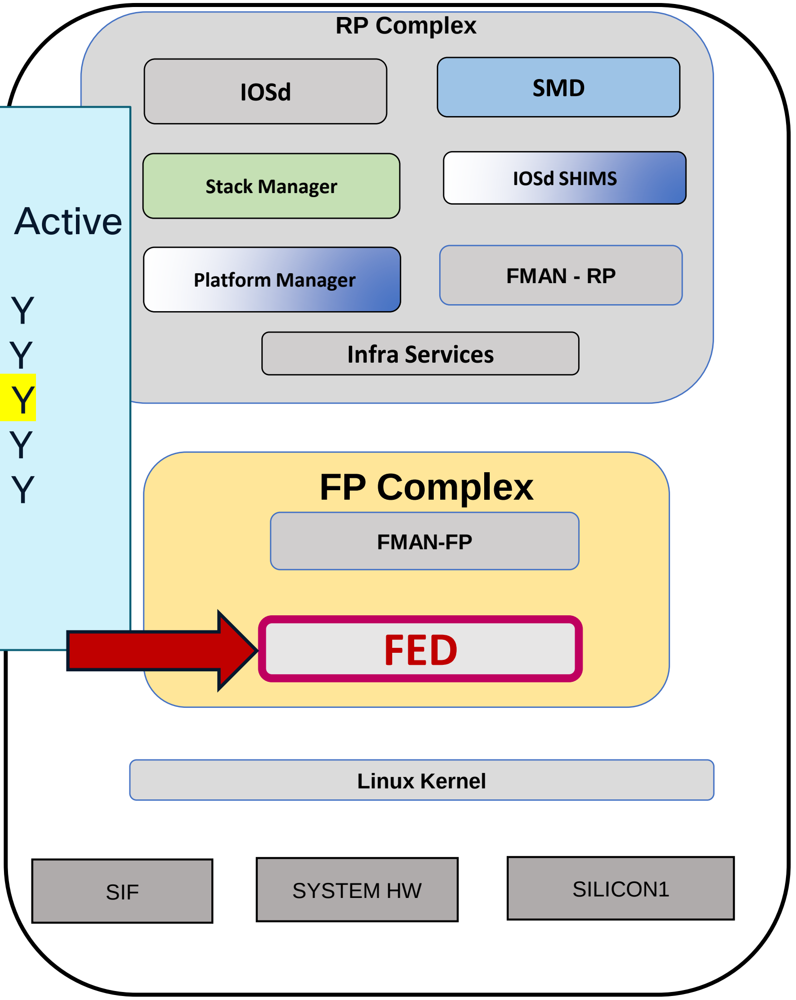
FED – Forwarding Engine Driver

FiftyGigE 1/1/0/3 : f003.bce9.3402



Hex(0x40B)=Dec(1035)

```
C9600-Silicon1-SVL#show  platform software fed switch active ifm mappings
Interface          IF_ID  Inst Asic Core     Port       Mac        Last Serdes    LPN  GPN        Active
                                    IFG_ID.  SubPort  First Serdes        Cntx              Type
FiftyGigE1/1/0/1   0x497   0   0    0    0   0    0   1   0        0      0   1   1   NIF   Y
FiftyGigE1/1/0/2   0x40a   0   0    0    0   0    0   2   1        1      0   2   2   NIF   Y
FiftyGigE1/1/0/3   0x40b   0   0    0    0   0    0   3   2        2      0   3   3   NIF   Y
FiftyGigE1/1/0/4   0x40c   0   0    0    0   0    0   4   3        3      0   4   4   NIF   Y
FiftyGigE1/1/0/5   0x40d   0   0    0    0   0    0   5   4        4      0   5   5   NIF   Y
~
~
```

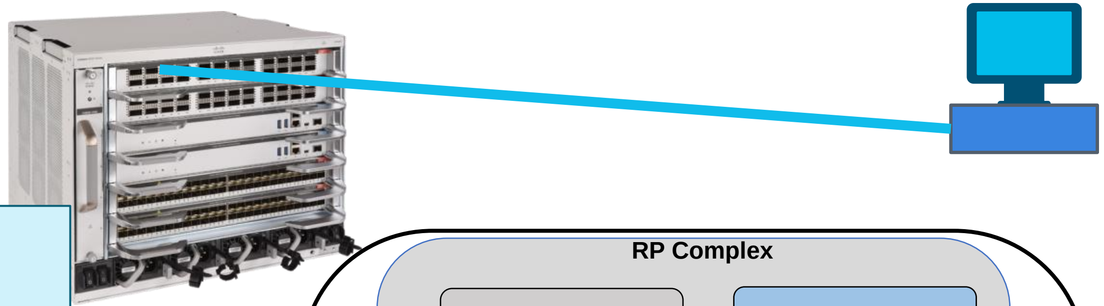


Silicon ONE L2 Forwarding

Port Programming

FED – Forwarding Engine Driver

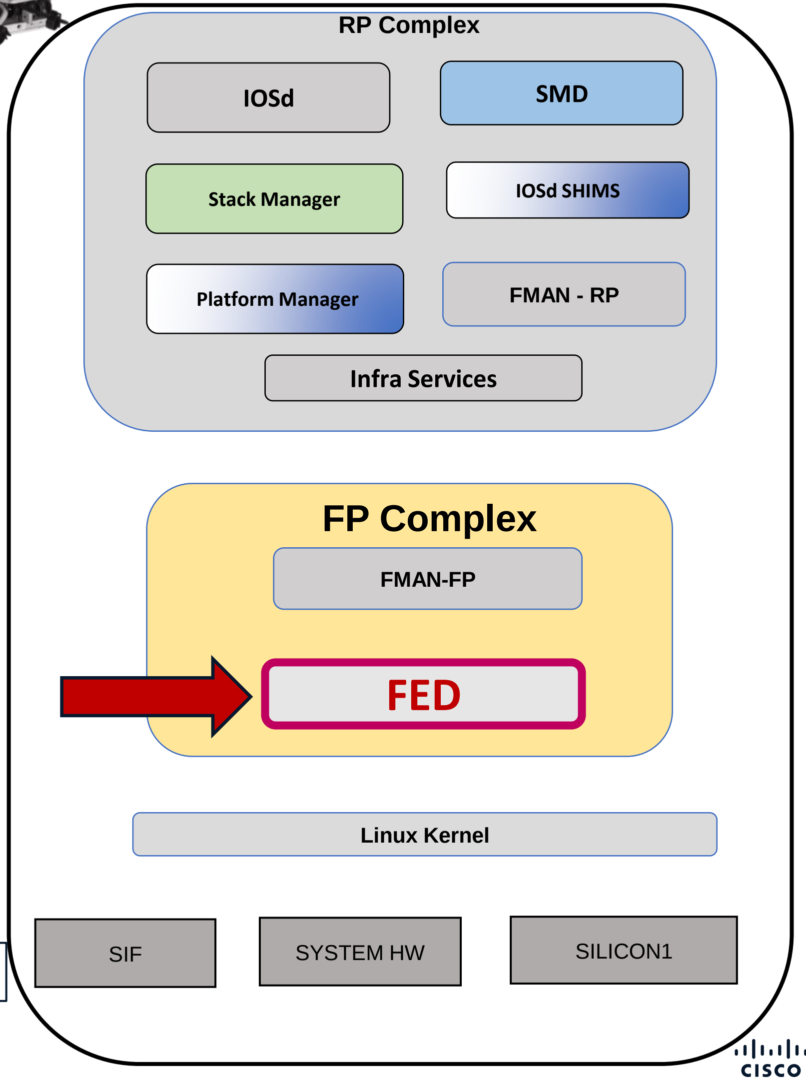
FiftyGigE 1/1/0/3 : f003.bce9.3402



```
C9600-Silicon1-SVL#show  platform software fed switch active ifm
interface_name fif1/1/0/3
Interface IF_ID                 : 0x000000000000040b
Interface Name                  : FiftyGigE1/1/0/3
Interface Block Pointer         : 0x7b1407244738
Interface Block State           : Ready
Interface State                 : Enabled
Interface Admin mode            : Admin Up
Interface Status                : NPD
Interface Ref-Cnt               : 1
Interface Type                  : ETHER
        Port Type               : SWITCH PORT
        Port Location           : LOCAL
~
~
IFM Feature Sub block information
        Port Physical Subblock
                Affinity ..........     [local]
                LPN ...............     [3]
                GPN ...............     [3]
                Speed .............     [10GB]
                type ..............     [IFM_PORT_TYPE_L2]
                MTU ...............     [1522]
                ac profile ........     [IFM_AC_PROFILE_L2_DEFAULT]
```

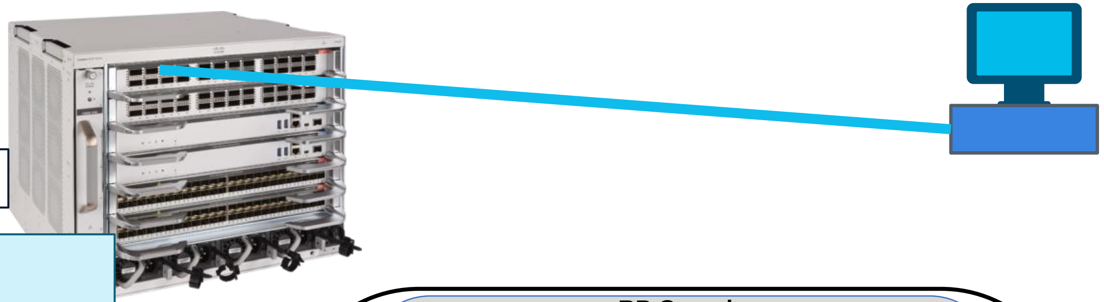
Hex(0x40B)=Dec(1035)

Continued........



Silicon ONE L2 Forwarding

FiftyGigE 1/1/0/3 : f003.bce9.3402

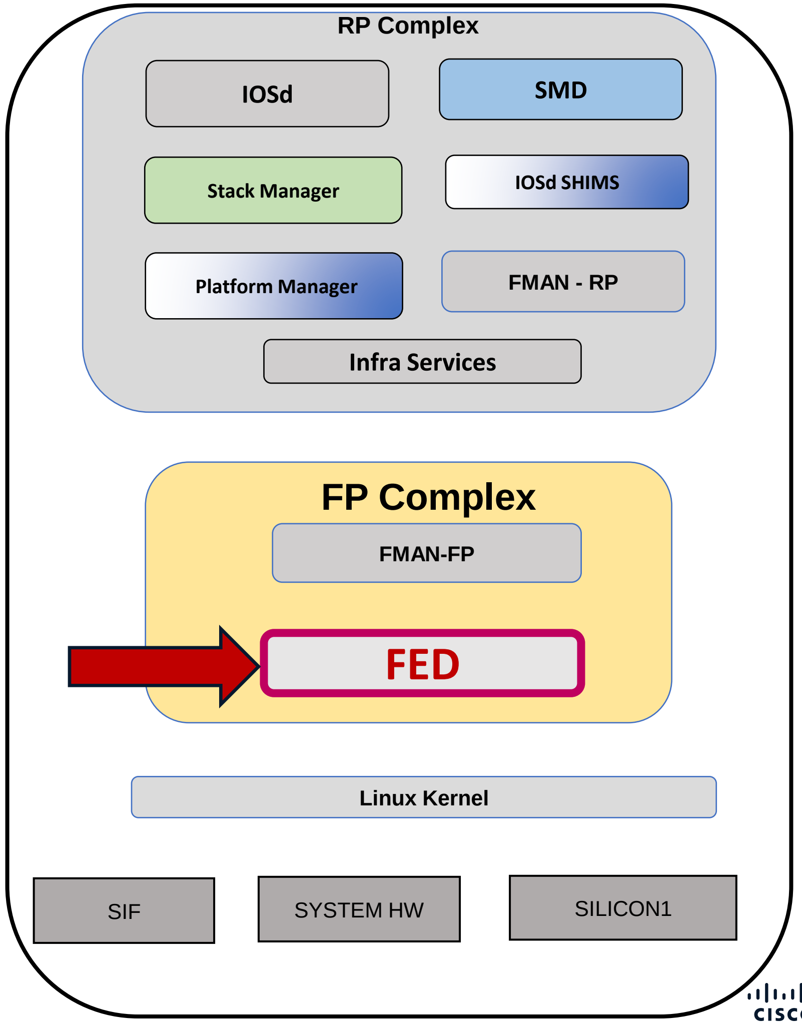


Port Programming

FED

........Continued from previous page

```
C9600-Silicon1-SVL#show  platform software fed switch active ifm
interface_name fif1/1/0/3

Port Subblock [0]
        Mac port oid..................          [0x2e0(736)]
        System port oid...............          [0x2e4(740)]
        System port gid...............          [44]
        Ethernet port oid.............          [0x2ef(751)]     → SDK
        Voq oid.......................          [0x2e2(738)]     → SDK
    Platform Subblock
        Asic.............          [0]
        Core.............          [0]
~
    Port L2 Subblock
        L2 Port Mode ................           [port_mode_access]
        L2 Port Mode set.............           [Yes]
        Ethertype....................           [8100]
        Port vlan  ..................           [10]
            Native vlan (trunk) ............. [0]
~
Events Log
[2026/01/07 19:11:05.917] link state down
~
```

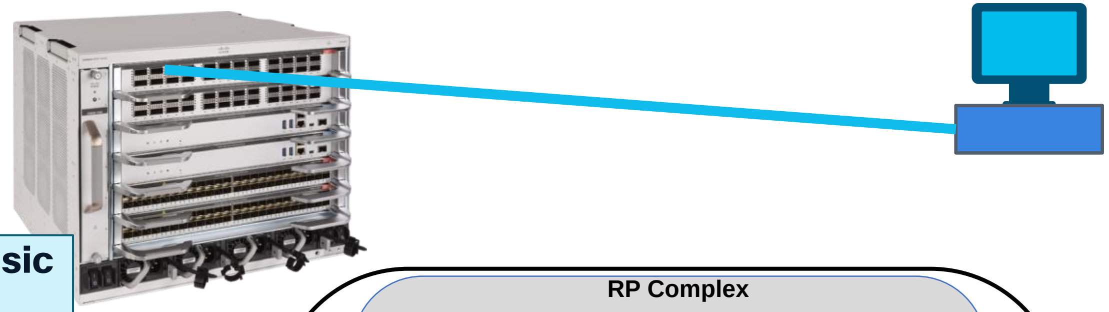


Silicon ONE L2 Forwarding

Port Programming

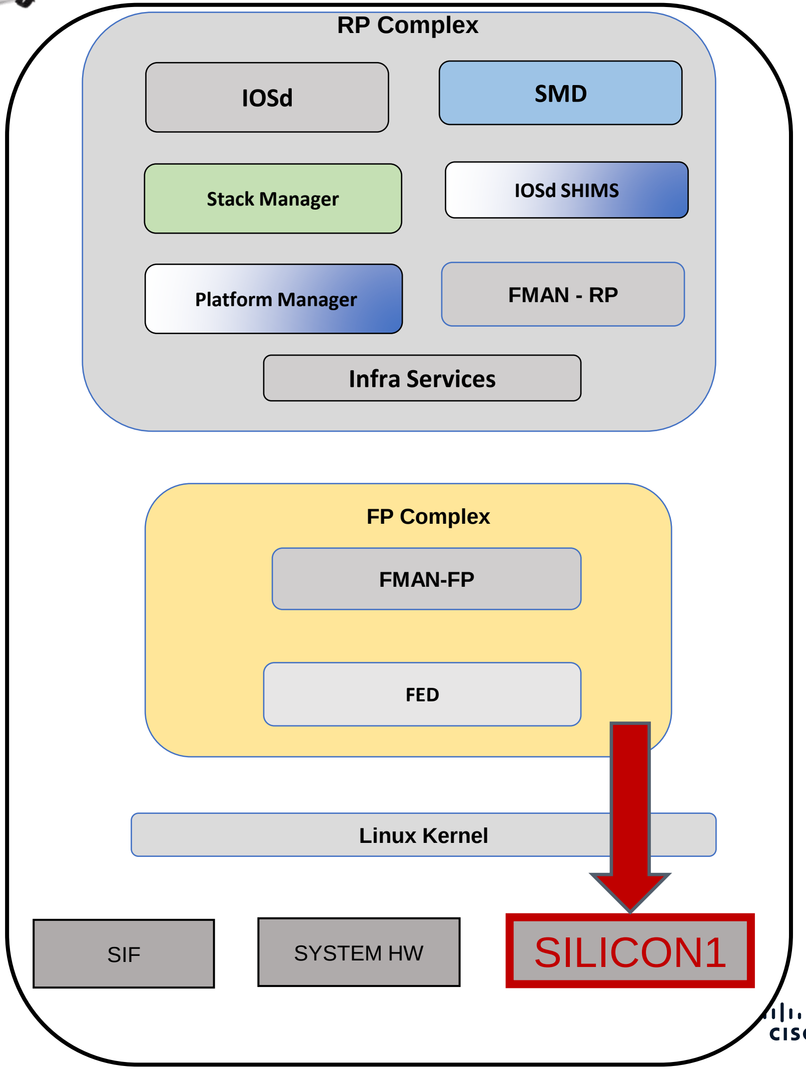
SDK – Forward ASIC

FiftyGigE 1/1/0/3 : f003.bce9.3402



C9600-Silicon1-SVL#show platform hardware fed switch active fwd-asic insight sdk_object(751)
ethernet_port object
ac_profile: =oref('ac_profile', oid=0xf0)
cookie: Fif1/1/0/3
copc_profile: =0x9
decrement_ttl: true

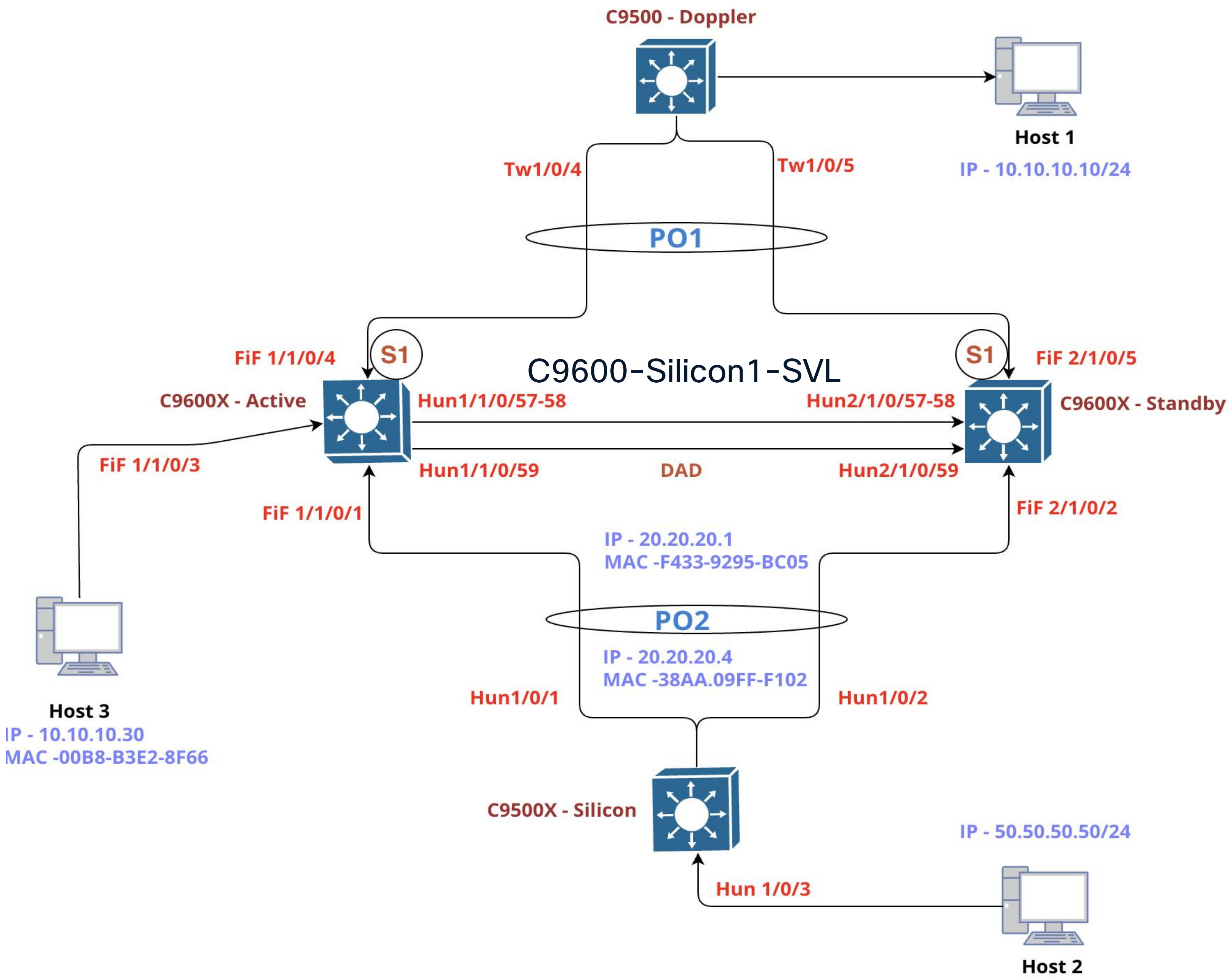
C9600-Silicon1-SVL#show platform hardware fed switch active fwd-asic insight sdk_object(738)
voq_set object
base_voq_id: =0x1c8
base_vsc_vec: =(328, 392, 456, 520, 584, 648)
cookie: ''
destination_device: =0x0
destination_ifg: =0x0
destination_slice: =0x0
device: =oref('device', oid=0x0)
local_fabric_mode: =0x0
oid: =0x2e2
set_size: =0x8
source_slice_vec: =(0, 1, 2, 3, 4, 5)



VLAN and Spanning-Tree Programming

Silicon ASIC

Lab Topology

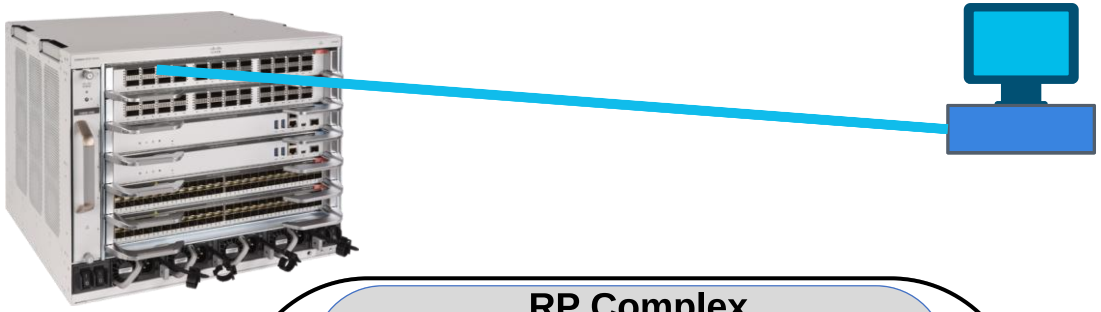


Silicon ONE L2 Forwarding

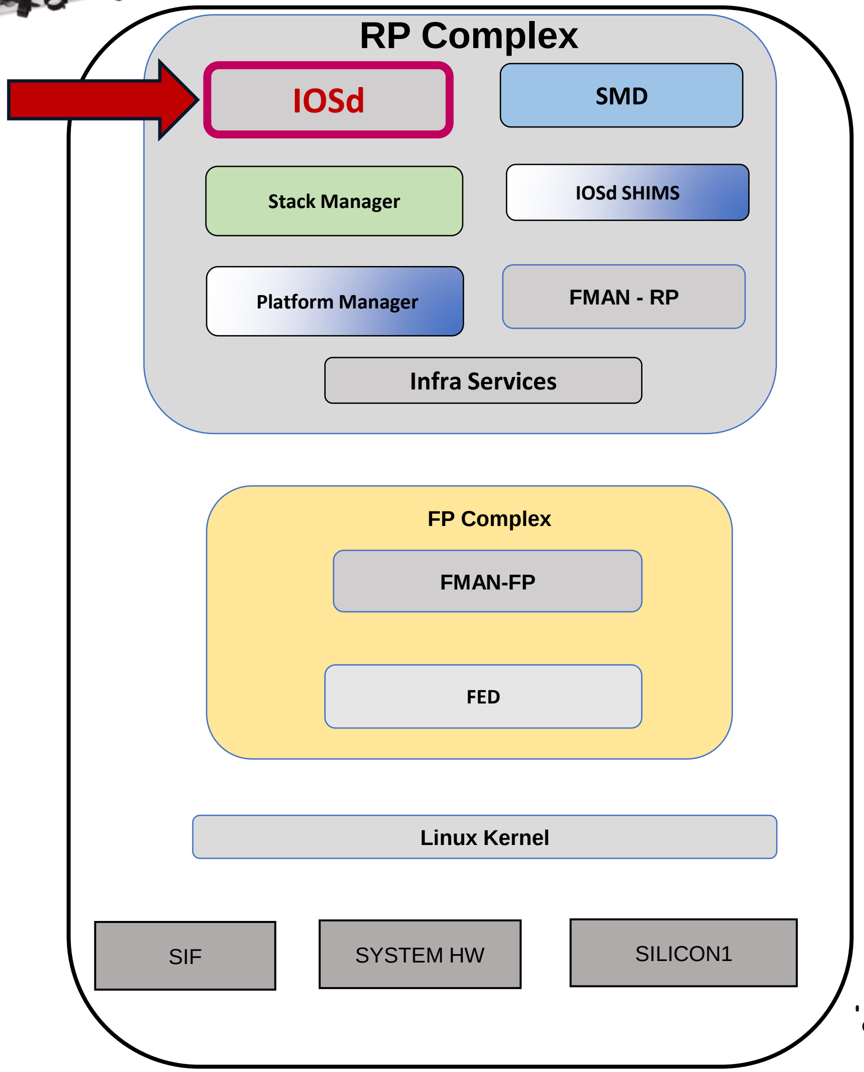
VLAN and Spanning-Tree Programming

IOSd

FiftyGigE 1/1/0/3 : f003.bce9.3402



```
C9600-Silicon1-SVL#show  vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
-----
1    default                          active       Fif1/1/0/2, Fif1/1/0/5, Fif1/1/0/6,
Fif1/1/0/7, Fif1/1/0/8, Fif1/1/0/9
~
~
                                      Fif2/1/0/56, Hu2/1/0/60

10   VLAN0010                         active       Fif1/1/0/3, Fif2/1/0/4
1002 fddi-default                     act/unsup
1003 token-ring-default               act/unsup
1004 fddinet-default                  act/unsup
1005 trnet-default                    act/unsup
```

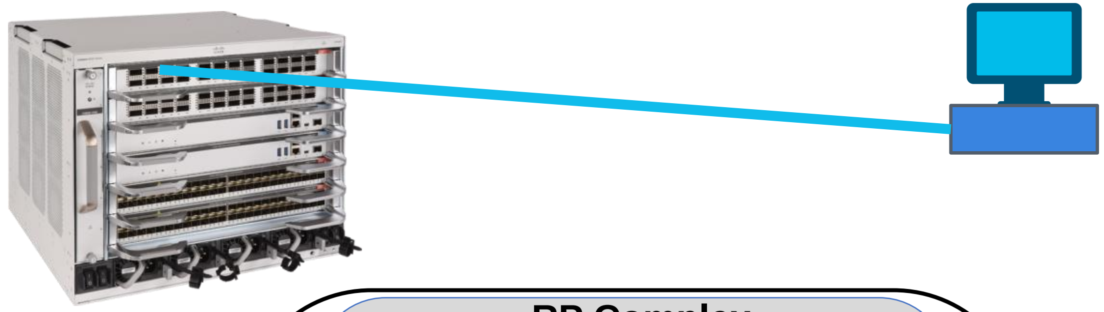


Silicon ONE L2 Forwarding

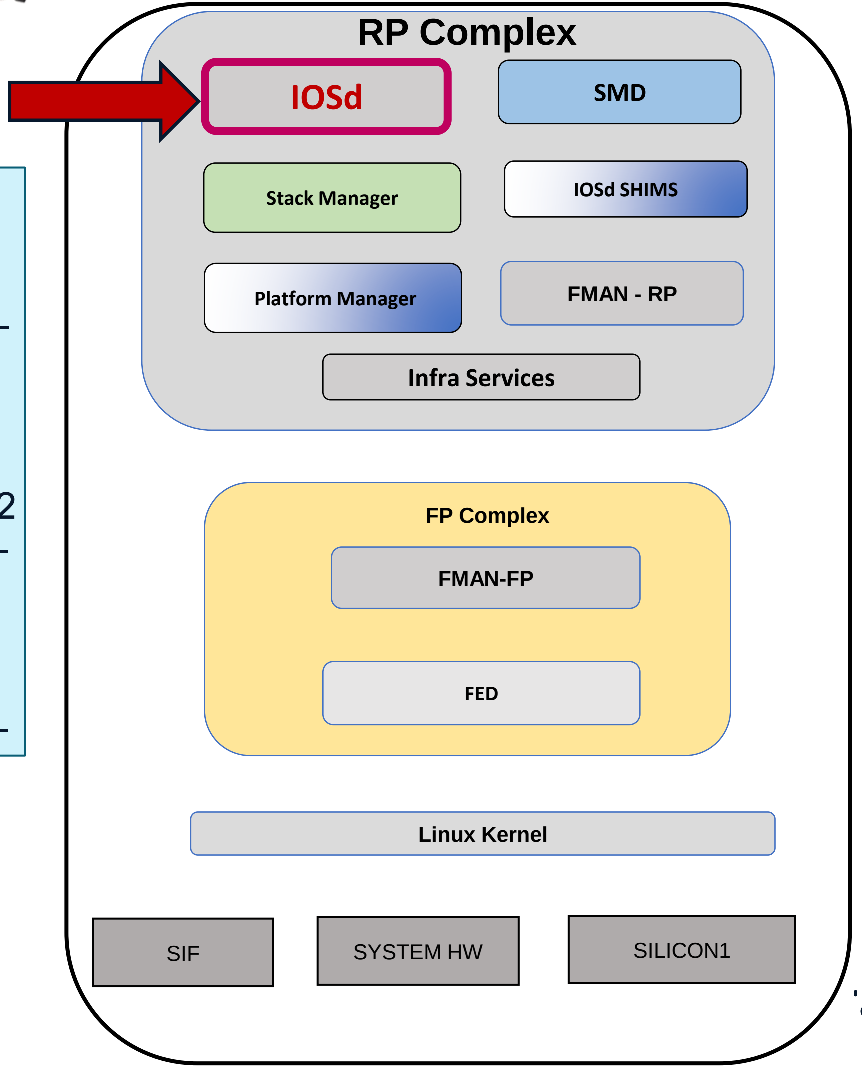
VLAN and Spanning-Tree Programming

IOSd

FiftyGigE 1/1/0/3 : f003.bce9.3402



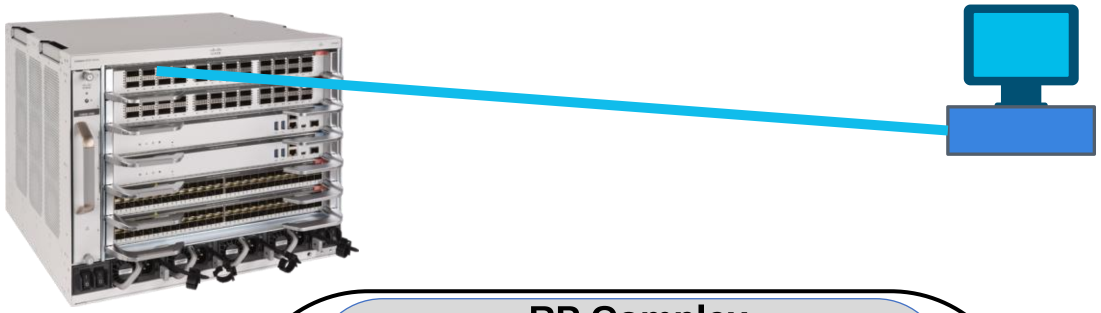
```
C9600-Silicon1-SVL#show  vlan id 10

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
-----
10   VLAN0010                         active    Fif1/1/0/3, Fif2/1/0/4, Po1

VLAN Type  SAID       MTU   Parent RingNo BridgeNo Stp  BrdgMode Trans1 Trans2
---- ----- ---------- ----- ------ ------ -------- ---- -------- ------ ------
10   enet  100010     1500  -      -      -        -    -        0      0

Primary Secondary Type              Ports
------- --------- ----------------- ------------------------------------------
```



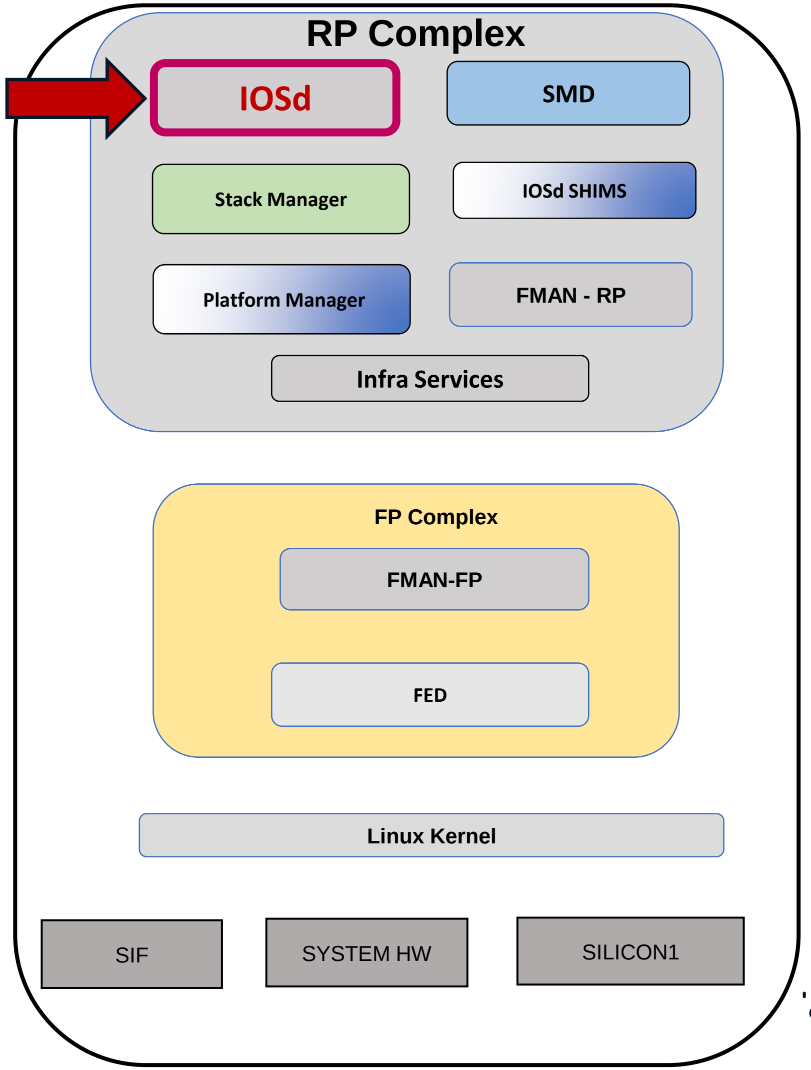
Silicon ONE L2 Forwarding

VLAN and Spanning-Tree Programming
IOSd

FiftyGigE 1/1/0/3 : f003.bce9.3402



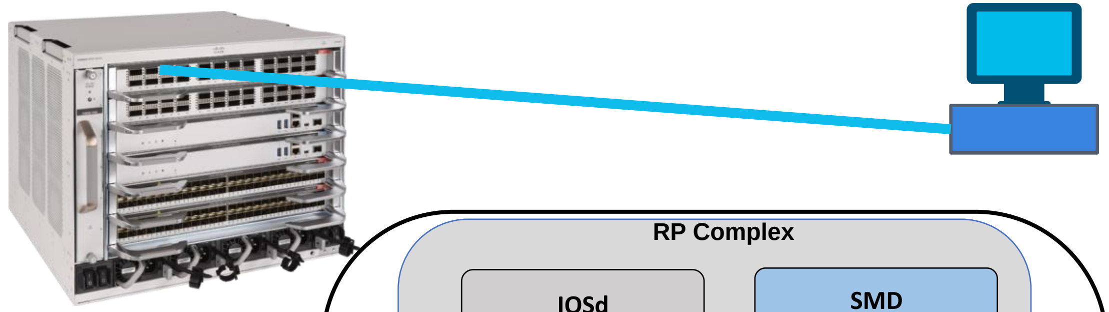
```
C9600-Silicon1-SVL#show  spanning-tree vlan 10

VLAN0010
  Spanning tree enabled protocol rstp
  Root ID    Priority    32778
             Address     4c5d.3ccd.dda0
             Cost        1000
             Port        4873 (Port-channel1)
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32778  (priority 32768 sys-id-ext 10)
             Address     f433.9295.bc00
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  300 sec

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- --------------------------------
Fif1/1/0/3          Desg FWD 2000      128.1179 P2p
Po1                 Root FWD 1000      128.777  P2p
```



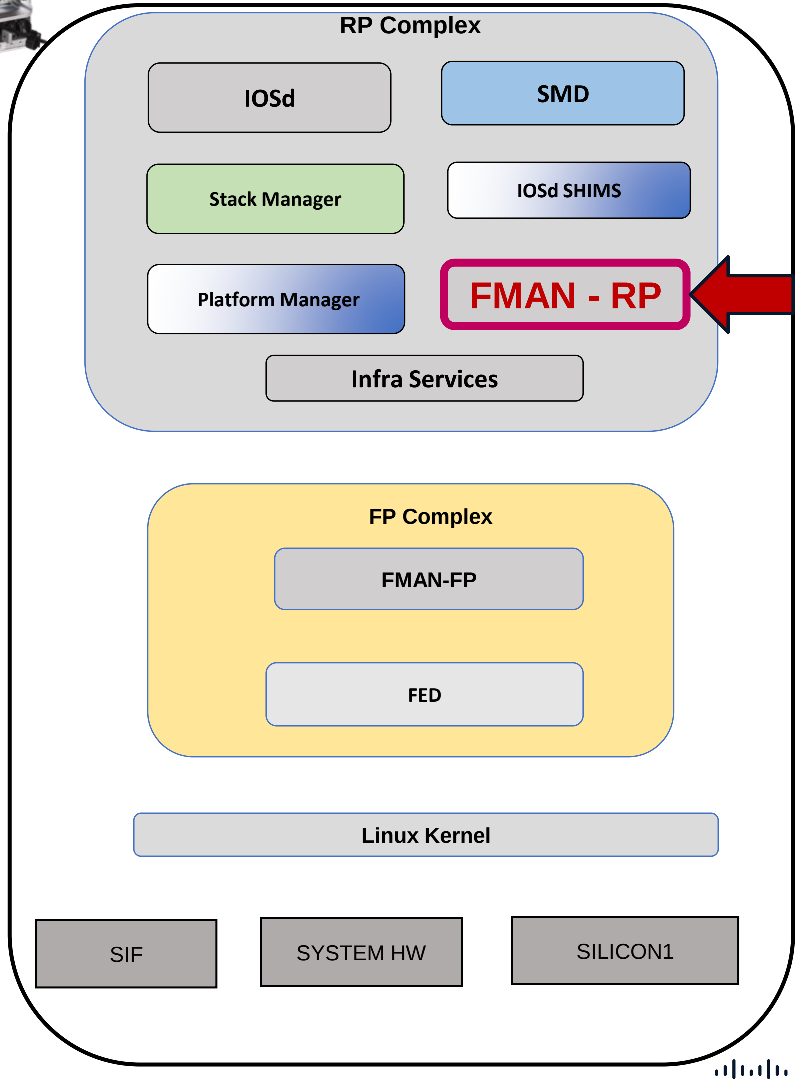
Silicon ONE L2 Forwarding

VLAN and Spanning-Tree Programming
FMAN-RP

FiftyGigE 1/1/0/3 : f003.bce9.3402



```
C9600-Silicon1-SVL#show  platform software vp switch  active R0 summary
Forwarding Manager Vlan Port Information

 Vlan       Intf-ID      Stp-state
 -------------------------------------------------------------------------
 1          1035         Disabled
 10         1035         Forwarding
 1          1036         Blocking
 10         1036         Blocking
~
~
~
 Vlan       Intf-ID      Stp-state
 -------------------------------------------------------------------------
 1          1172         Forwarding
 10         1172         Forwarding
 4095       1175         Forwarding
 4095       1176         Forwarding
```

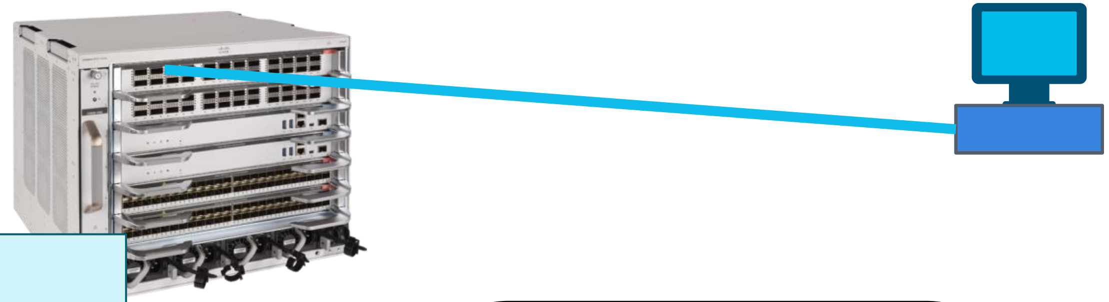


Silicon ONE L2 Forwarding

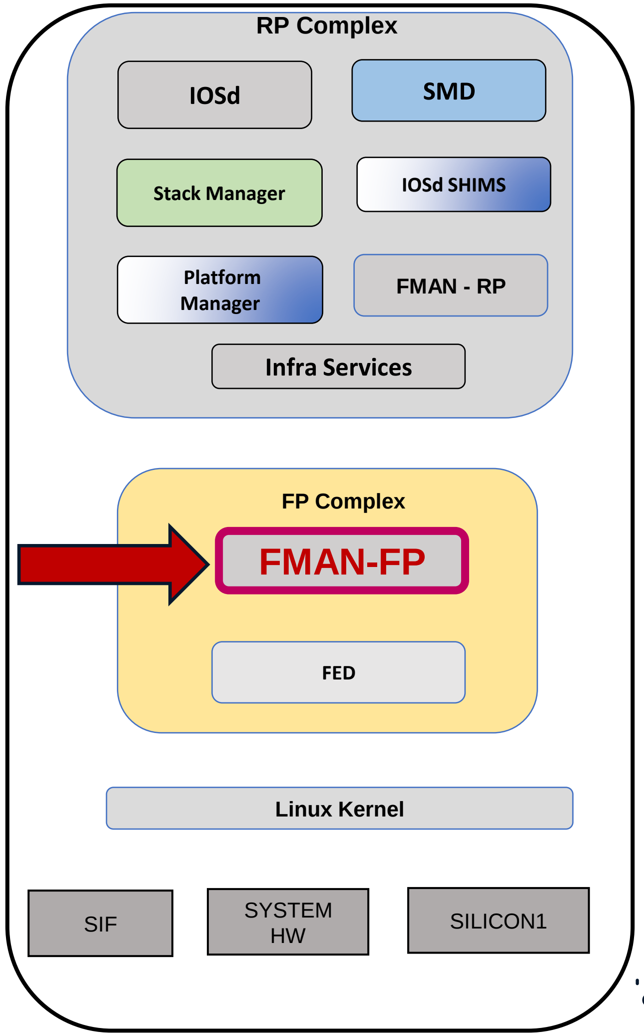
VLAN and Spanning-Tree Programming

FMAN-FP

FiftyGigE 1/1/0/3 : f003.bce9.3402



```
C9600-Silicon1-SVL#show  platform software vp switch active F0
summary
Forwarding Manager Vlan Port Information

 Vlan              Intf-ID     STP-State         Untagged
 -----------------------------------------------------------
 0                 59          Forwarding          Yes
 4095              59          Forwarding          Yes
~
 0                 1035        Forwarding          Yes
 1                 1035        Disabled            Yes
 10                1035        Forwarding          Yes
 0                 1036        Forwarding          Yes
 1                 1036        Blocking            Yes
 10                1036        Blocking            No
~
 0                 1168        Forwarding          Yes
 1                 1172        Forwarding          Yes
 10                1172        Forwarding          No
 0                 1175        Forwarding          Yes
 4095              1175        Forwarding          Yes
~
```

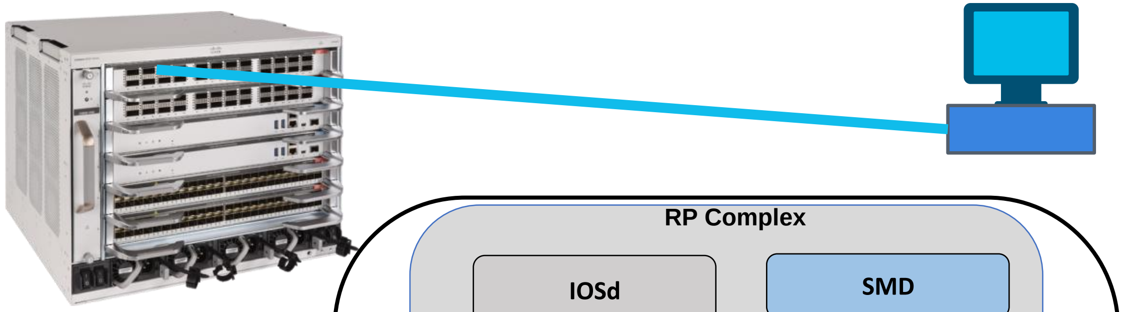


Silicon ONE L2 Forwarding

VLAN and Spanning-Tree Programming

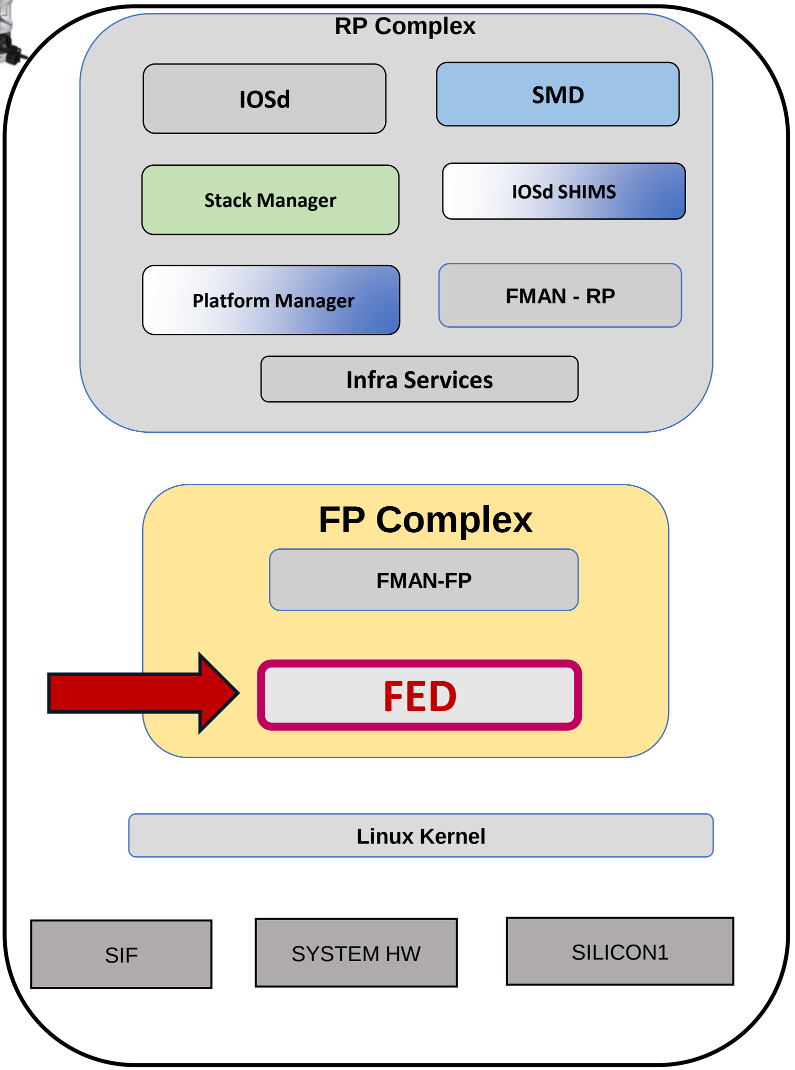
FED – Forwarding Engine Driver

FiftyGigE 1/1/0/3 : f003.bce9.3402



C9600-Silicon1-SVL#show platform software fed switch active vp summary interface if_id 1035

if_id	vlan_id	pvlan_mode	pvlan_vlan	stp_state	vtp pruned	Untagged
1035	**10**	**none**	**10**	**forwarding**	**No**	**Yes**
1035	1	none	1	disabled	No	No

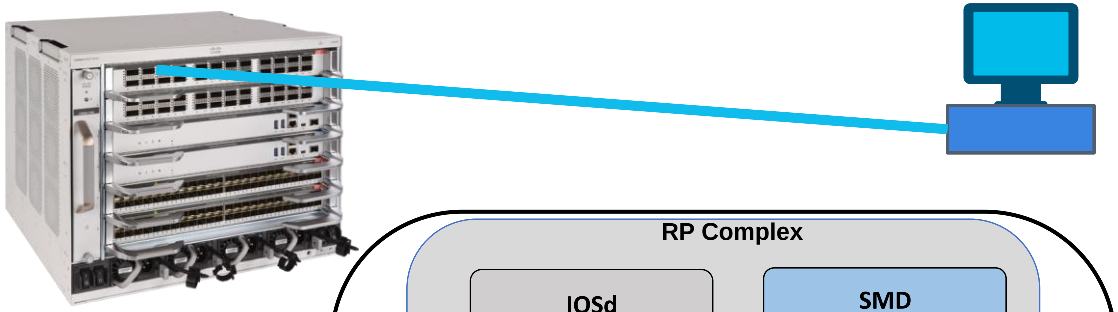


Silicon ONE L2 Forwarding

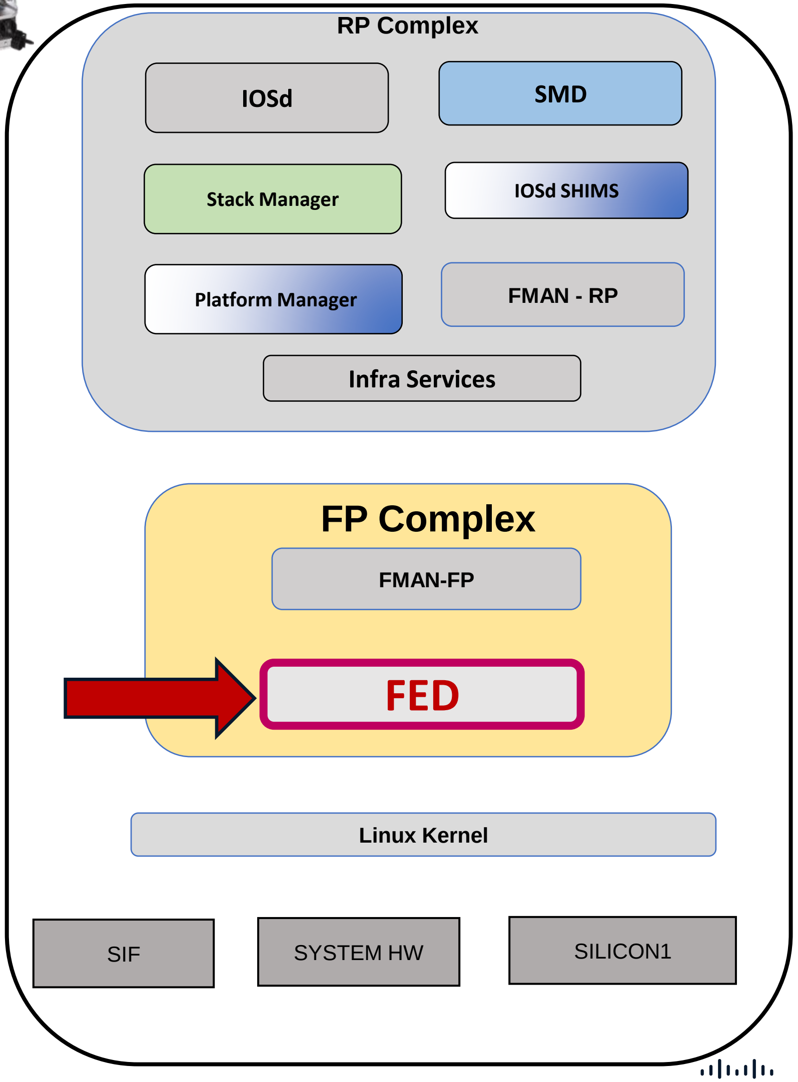
VLAN and Spanning-Tree Programming

FED – Forwarding Engine Driver

FiftyGigE 1/1/0/3 : f003.bce9.3402



```
C9600-Silicon1-SVL#show platform software fed switch active vp key 1035 10
FED PM Virtual Port Data :
 iif_id = 1035, vlan_id = 10
  Info:
   stp state: forwarding
    vtp pruned: NO
    Dest Map: TRUE
    limited access: FALSE
    learning cache enable: FALSE
    delete auth behavior: FALSE
   tag native enable: FALSE
    block learn: FALSE
    pvlan port: FALSE
    pvlan mode: none
    pvlan native vlan: 10
```

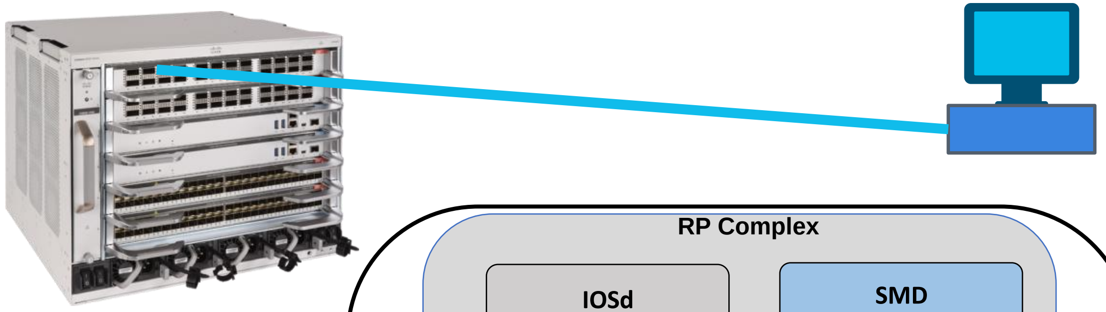


Silicon ONE L2 Forwarding

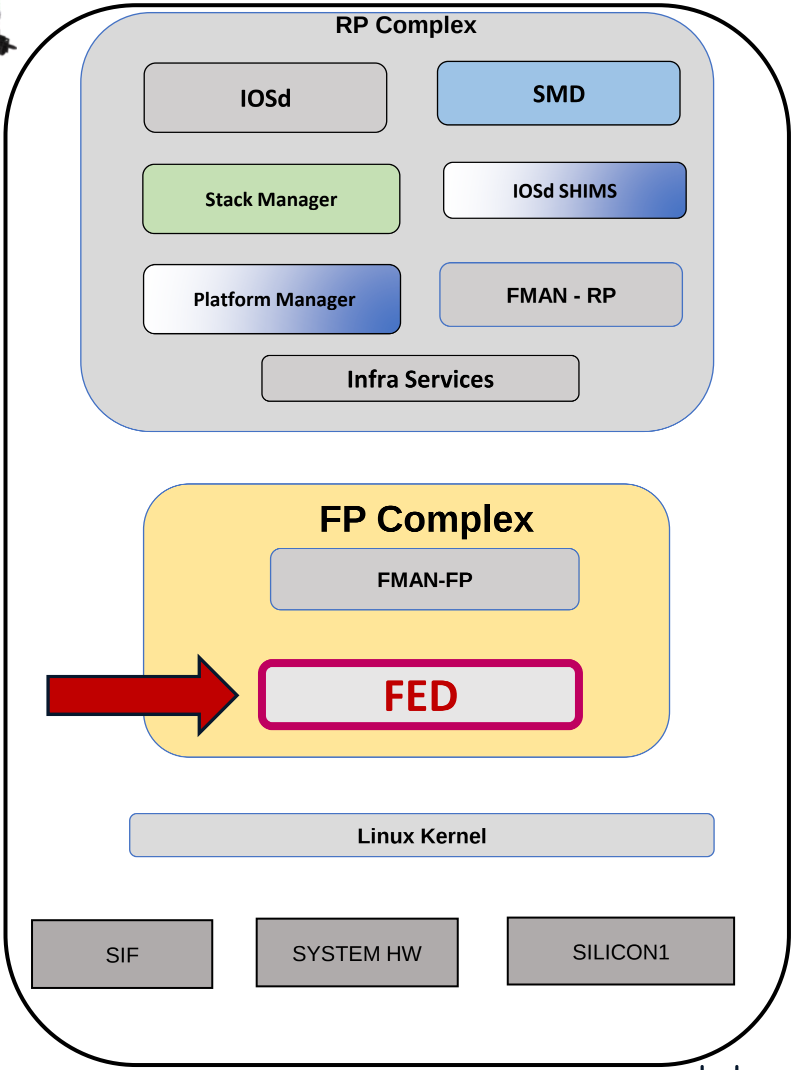
VLAN and Spanning-Tree Programming

FED

FiftyGigE 1/1/0/3 : f003.bce9.3402



```
C9600-Silicon1-SVL#show platform software fed switch active vp key 1035 10 detail
FED PM Virtual Port Data :
 iif_id = 1035, vlan_id = 10,
 vp_if_id = 0x3000a,
 l2_srv_port_oid = 1938, l2_srv_port_gid = 133        ← SDK object.
 match_vlan = 0,
 default_vlan = 10,
 Info:
   STP State : forwarding
   port  mode : Access
   port tagging : Native
```

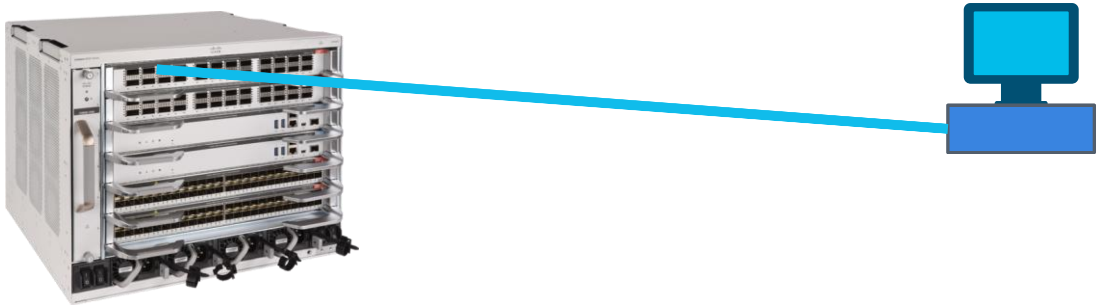


Silicon ONE L2 Forwarding

VLAN and Spanning-Tree Programming

FED

FiftyGigE 1/1/0/3 : f003.bce9.3402



C9600-Silicon1-SVL#show platform software fed switch active stp-vlan 10

Interface	pvlan_mode	stp_state	stp_state(HW)	vtp pruned	Untagged	Ingress(HW)	Egress(HW)	GID	Mac learn
Port-channel1	trunk	forwarding	Forwarding	No	No	Forwarding	Forwarding	132	Enable
FiftyGigE1/1/0/3	none	forwarding	Forwarding	No	Yes	Forwarding	Forwarding	33	Enable

HW flood list: : Port-channel1, FiftyGigE1/1/0/3

C9600-Silicon1-SVL#show platform software fed switch active vlan 10

VLAN Fed Information

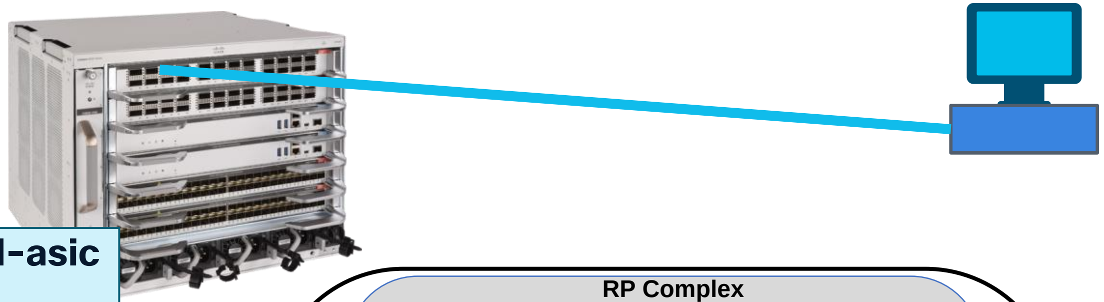
Vlan Id	IF Id	LE Handle	STP Handle	L3 IF Handle	SVI IF ID	MVID
10	0x0000000000420010	0x0000000000000813	0x0000000000000000	0x0000000000000000	0x0000000000000000	11

Silicon ONE L2 Forwarding

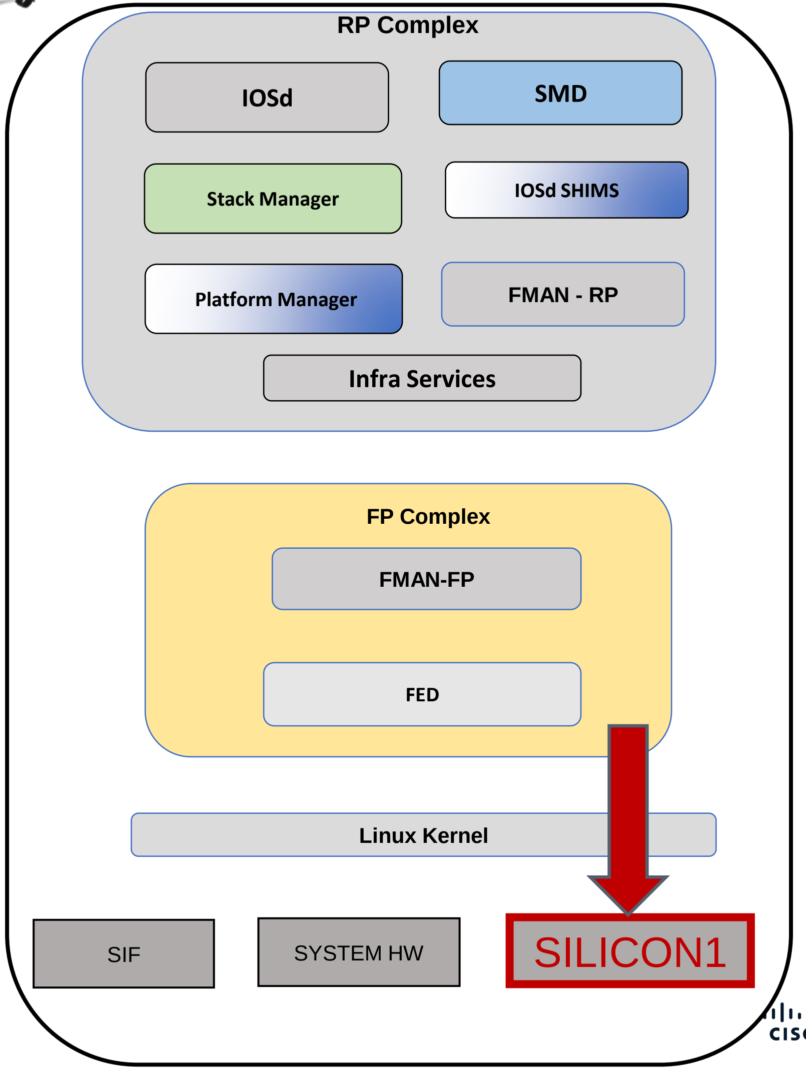
VLAN and Spanning-Tree Programming

SDK – Forward ASIC

FiftyGigE 1/1/0/3 : f003.bce9.3402



```
C9600-Silicon1-SVL#show  platform hardware fed switch active fwd-asic insight sdk_object(1938)
# l2_service_port object
ac_port_vid_group_mode: =0x0
aci_dest_tunnel_type: =0x0
aci_ip_tunnel_mode: =0x0
~
~
l3_destination: null
mac_learning_mode: =0x2
meter: null
oid: =0x792
port_subtype: =0x0
port_type: =0x1
security_group_policy_enforcement: false
service_mapping_vid_list: []
smac_miss_enabled: false
stp_state: =0x3
svl_transport_enabled: false
tunnel_mode: =0x0
vrf: null
vxlan_tunnel_mode: =0x0
```

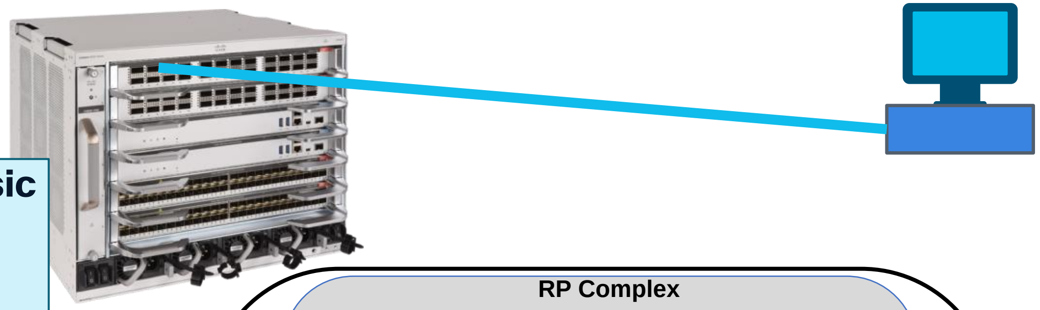


Silicon ONE L2 Forwarding

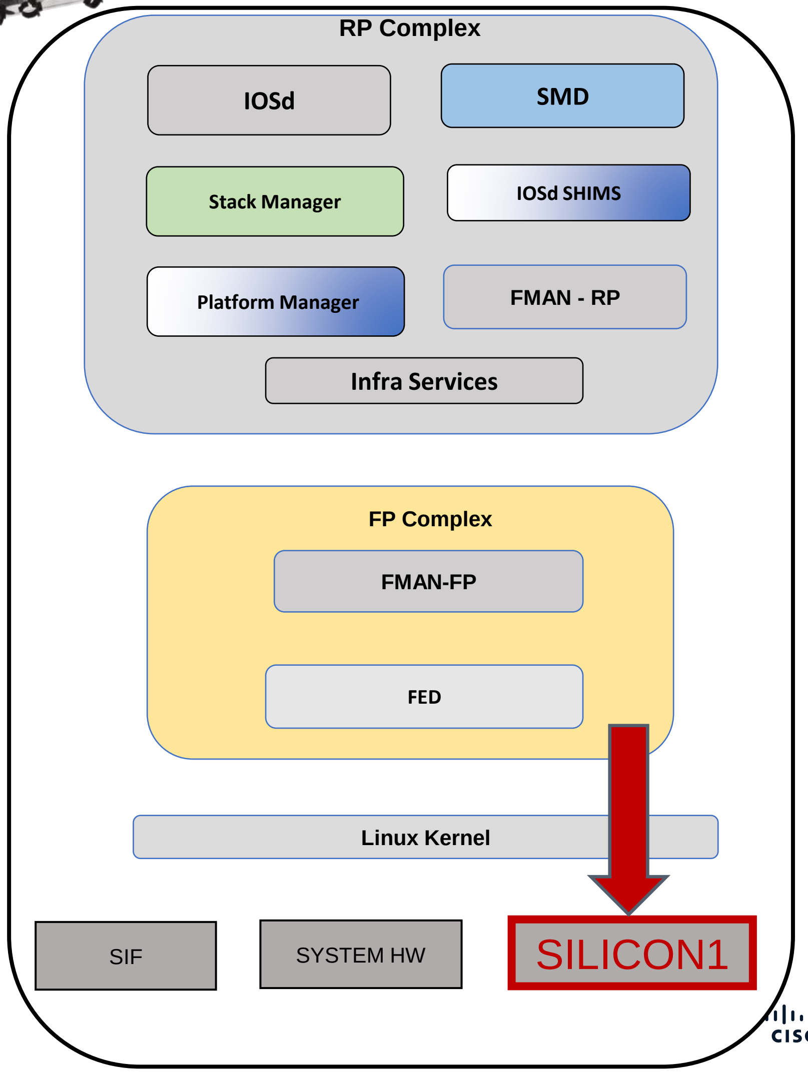
VLAN and Spanning-Tree Programming

SDK – Forward ASIC

FiftyGigE 1/1/0/3 : f003.bce9.3402



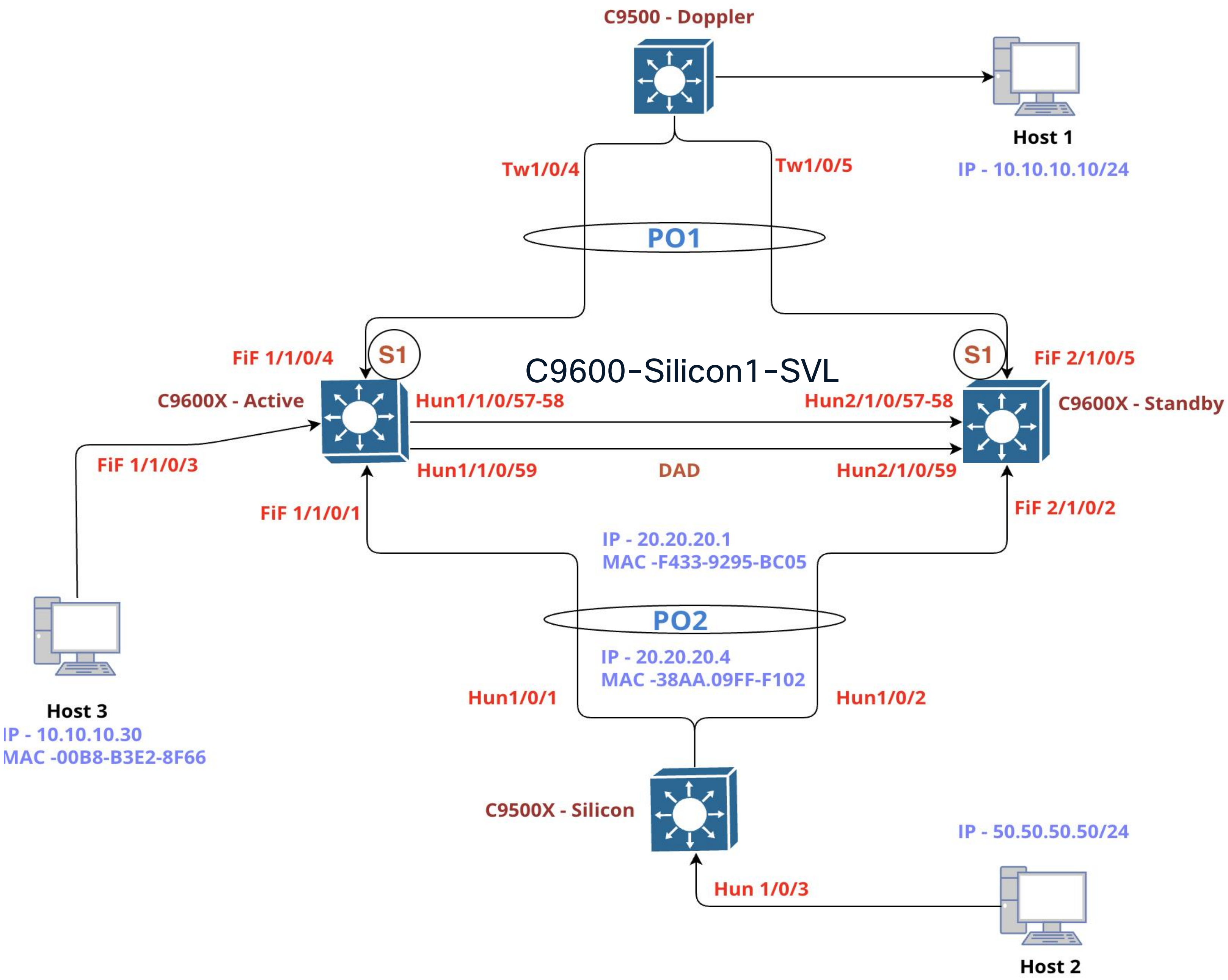
```
C9600-Silicon1-SVL#show  platform hardware fed switch active fwd-asic
insight sdk_object(0x813)
# switch object
cookie: '10'
~
mac_entries:
- addr:
    bytes:
    - =0x66
    - =0x8f
    - =0xe2
    - =0xb3
    - =0xb8
    - =0x0
   flat: =0xb8b3e28f66
~
- addr:
    bytes:
    - =0xa4
    - =0xdd
    - =0xcd
    - =0x3c
    - =0x5d
    - =0x4c
   flat: =0x4c5d3ccddda4
~
```



MAC Address Programming

Silicon ASIC

Lab Topology

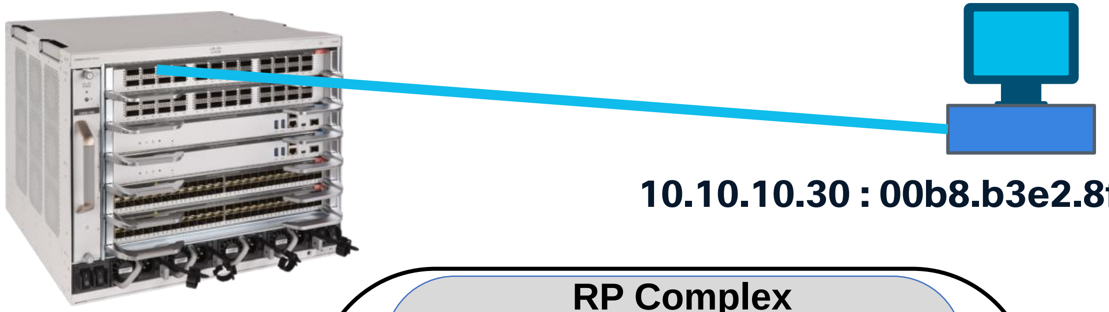


Silicon ONE L2 Forwarding

Mac Address Programming.

IOSd

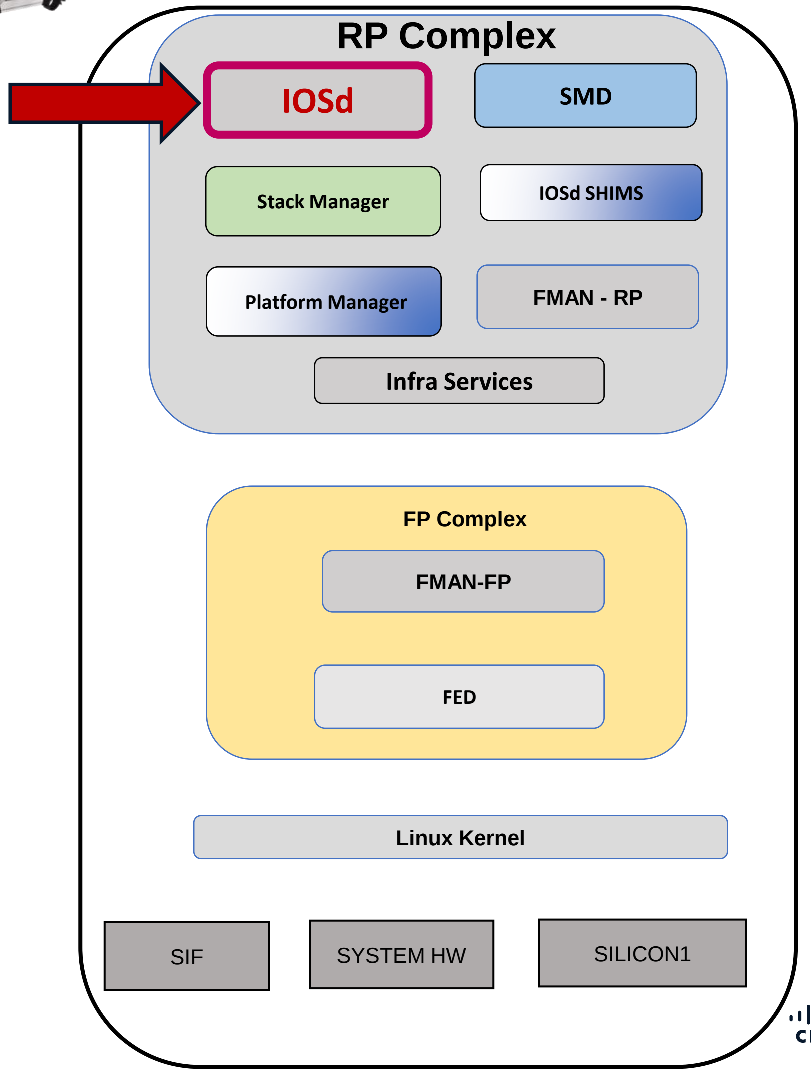
FiftyGigE 1/1/0/3 : f003.bce9.3402



10.10.10.30 : 00b8.b3e2.8f66

```
C9600-Silicon1-SVL#show  mac address-table interface fiftyGigE 1/1/0/3
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----
  10    00b8.b3e2.8f66    DYNAMIC     Fif1/1/0/3
Total Mac Addresses for this criterion: 1
```

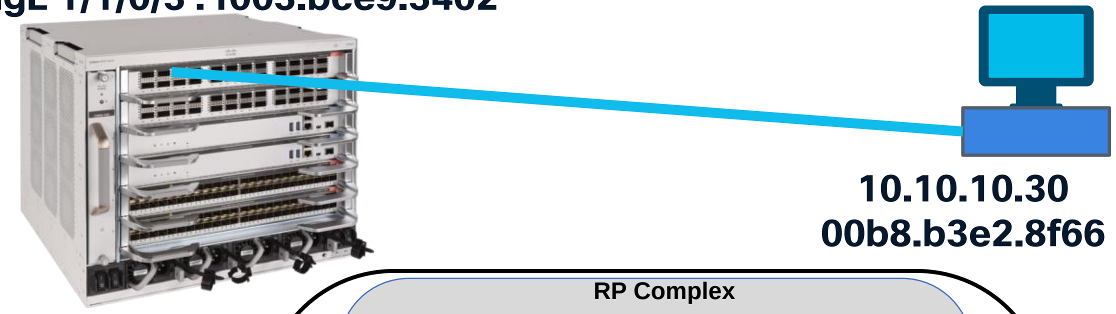
```
C9600-Silicon1-SVL#show  mac address-table address 00b8.b3e2.8f66
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----
  10    00b8.b3e2.8f66    DYNAMIC     Fif1/1/0/3
Total Mac Addresses for this criterion: 1
```



Silicon ONE L2 Forwarding

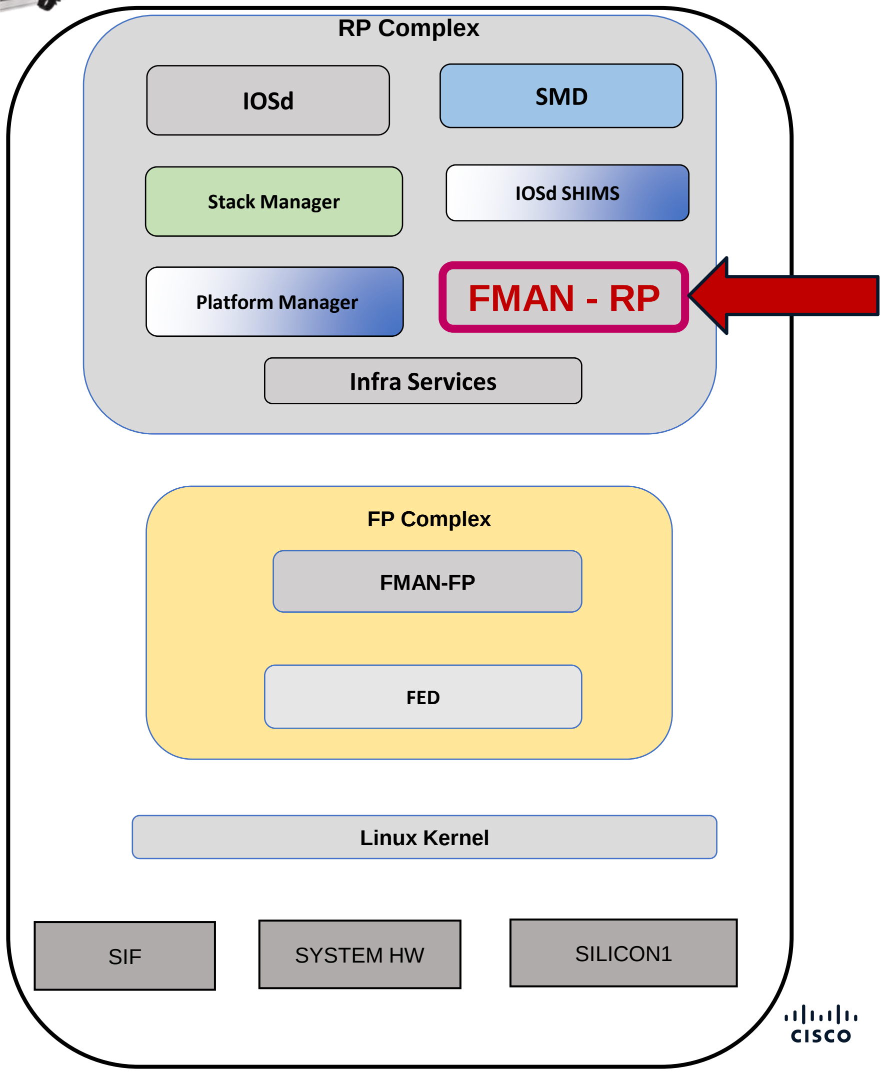
Mac Address Programming.

FMAN-RP




```
C9600-Silicon1-SVL#show  platform software matm switch active R0 mac
00b8.b3e2.8f66
Tbl_Type   Tbl_ID    MAC_Address  Type    ECBits Ports  AOM_ID/OM_PTR
MAT_VLAN     10     00b8.b3e2.8f66    1      0           1       OM: 0x34804b0da0
  List of Ports: 1035
```

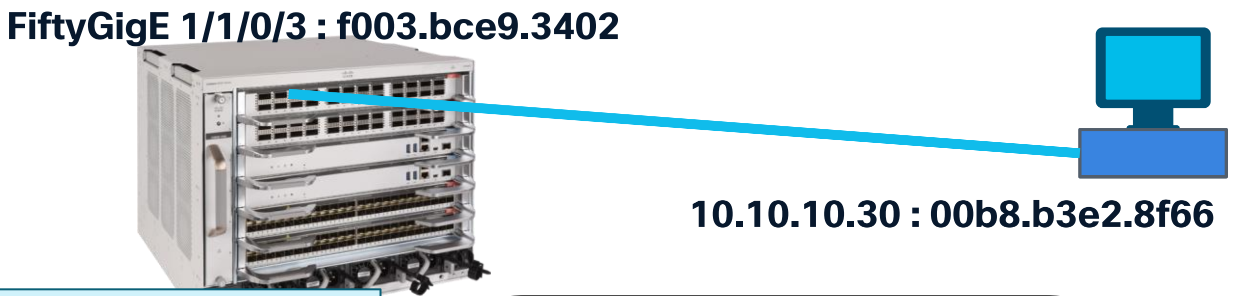
```
C9600-Silicon1-SVL#show platform software matm switch active R0 Table
Tbl_Type       Tbl_ID  Num_MAC  Aging  AOM_ID/OM_PTR
MAT_VLA           1          3       300          OM: 0x3480323098
MAT_VLAN       10          3       300          OM: 0x34802edbd8
MAT_IND           1        21       0              OM: 0x34802b33a0
MAT_L3IF     4096          0       300          OM: 0x34804be058
MAT_L3IF     5273          1       30            OM: 0x3480208e28
MAT_L3IF     6450          1       300          OM: 0x3480301570
MAT_L3IF     8970          1       300          OM: 0x3480327858
```



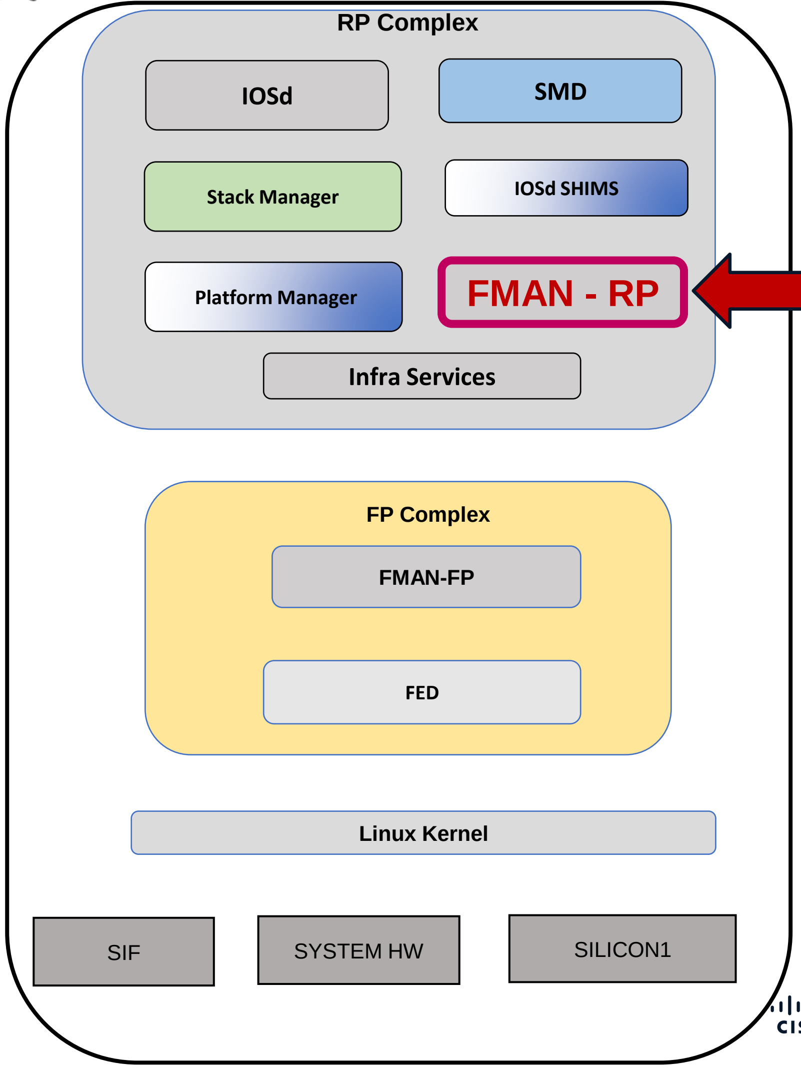
Silicon ONE L2 Forwarding

Mac Address Programming

FMAN-RP



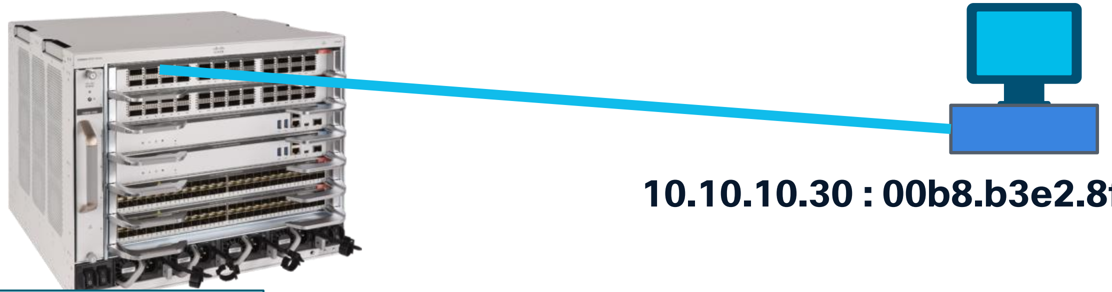

```
C9600-Silicon1-SVL#show platform software matm switch active R0 table content
Tbl_Type   Tbl_ID    MAC_Address  Type    ECBits Ports  AOM_ID/OM_PTR
MAT_VLAN        1 4c5d.3ccd.dda4     1      4      1  OM: 0x34804b3330
  List of Ports: 1172
MAT_VLAN        1 4c5d.3ccd.dda5     1      4      1  OM: 0x34804c7ad8
  List of Ports: 1172
MAT_VLAN        1 f433.9295.bc05  8002      0      1  OM: 0x3480324250
  List of Ports: 1170
MAT_VLAN       10 00b8.b3e2.8f66     1      0      1  OM: 0x34804b0da0
  List of Ports: 1035
MAT_VLAN       10 4c5d.3ccd.dda4     1      4      1  OM: 0x34804bf720
  List of Ports: 1172
MAT_VLAN       10 f433.9295.bc05  8002      0      1  OM: 0x34802e83b8
  List of Ports: 1177
MAT_IND         1 0100.0ccc.cccc     6      0      0  OM: 0x34802b6318
~
Tbl_Type   Tbl_ID    MAC_Address  Type    ECBits Ports  AOM_ID/OM_PTR
MAT_IND         1 0180.c200.000c     6      0      0  OM: 0x34802b54a8
~
MAT_IND         1 ffff.ffff.ffff     6      0      0  OM: 0x34802b6a50
MAT_L3IF     5273 f433.9295.bc05  8002      0      1  OM: 0x34804c33c0
  List of Ports: 1175
~
```



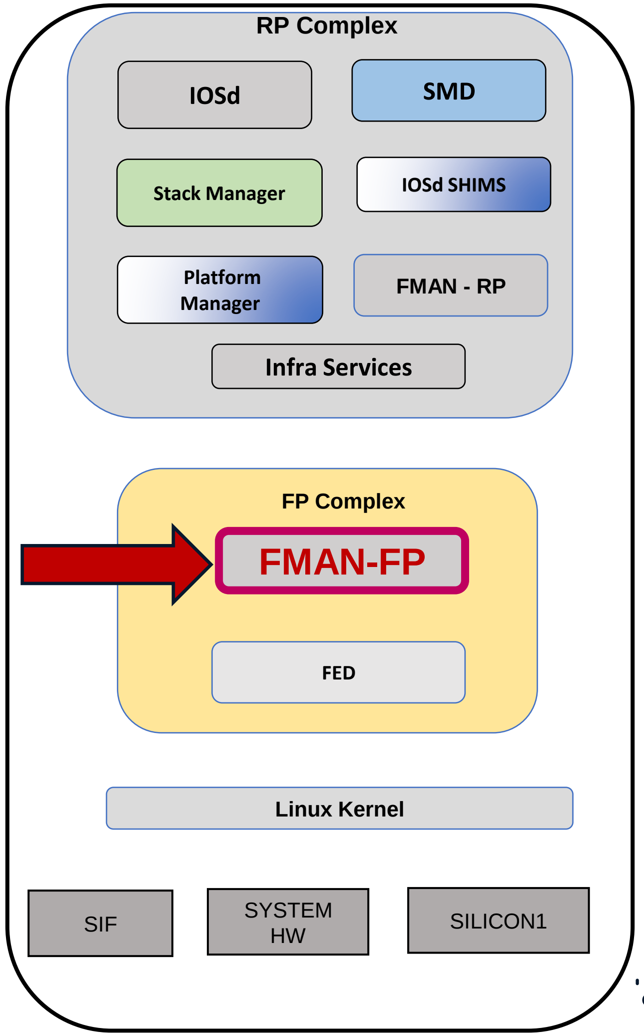
Silicon ONE L2 Forwarding

Mac Address Programming.

FMAN-FP




```
C9600-Silicon1-SVL#show platform software matm switch active F0 table content
Tbl_Type   Tbl_ID    MAC_Address  Type   ECBits Ports  AOM_ID/OM_PTR
~
MAT_VLAN        1   f433.9295.bc05  8002     0     1        727 created
  List of Ports: 1170
MAT_VLAN      10   00b8.b3e2.8f66    1        0     1        6996 created
  List of Ports: 1035
MAT_VLAN       10  4c5d.3ccd.dda4     1       4     1        6884 created
  List of Ports: 1172
MAT_VLAN       10  f433.9295.bc05  8002     0     1        736 created
  List of Ports: 1177
MAT_IND         1     0100.0ccc.cccc     6       0     0         48 created
~

Tbl_Type   Tbl_ID    MAC_Address  Type   ECBits Ports  AOM_ID/OM_PTR
MAT_IND         1     0180.c200.000c     6     0     0           42 created
~
MAT_IND         1     ffff.ffff.ffff              6     0     0           51 created
MAT_L3IF   5273   f433.9295.bc05  8002   0     1           5806 created
  List of Ports: 1175
~
```

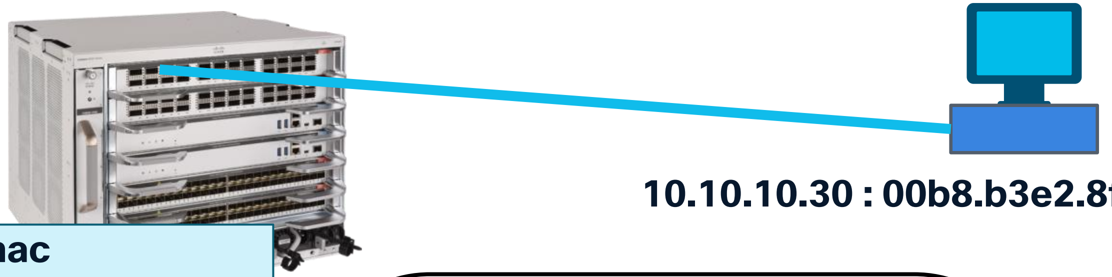


Silicon ONE L2 Forwarding

Mac Address Programming.

FMAN-FP

FiftyGigE 1/1/0/3 : f003.bce9.3402

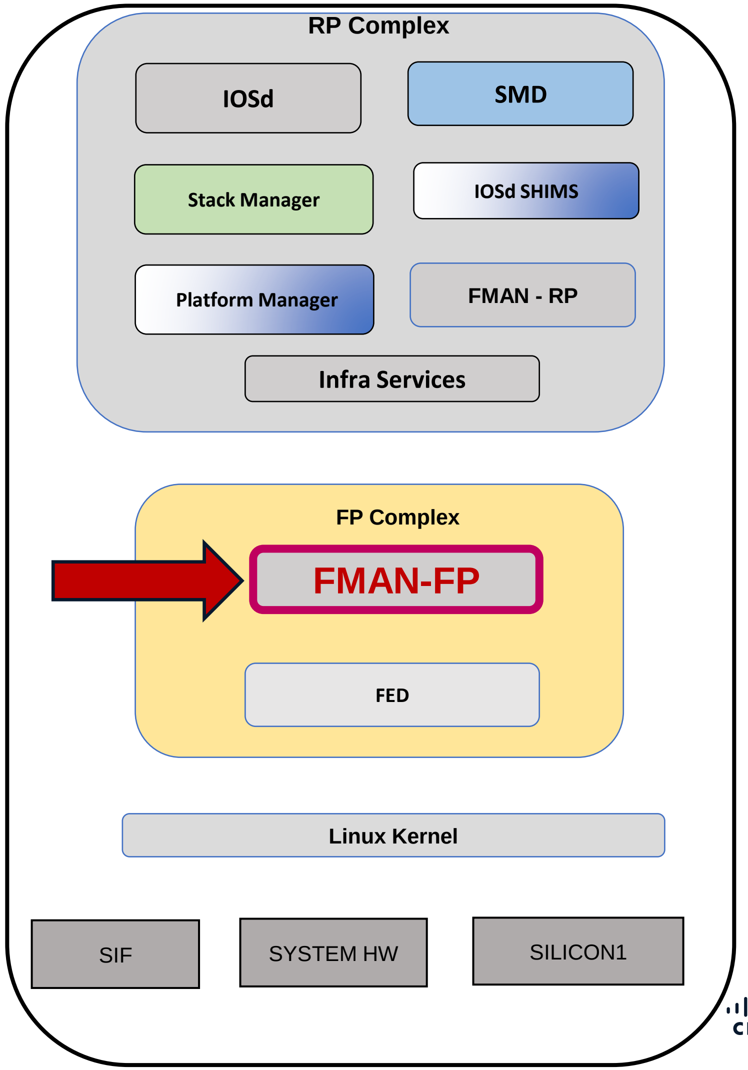


10.10.10.30 : 00b8.b3e2.8f66

```
C9600-Silicon1-SVL#show  platform software matm switch active F0 mac 00b8.b3e2.8f66
Tbl_Type   Tbl_ID    MAC_Address  Type    ECBits Ports  AOM_ID/OM_PTR
MAT_VLAN       10 00b8.b3e2.8f66     1      0      1  6996 created
  List of Ports: 1035
```

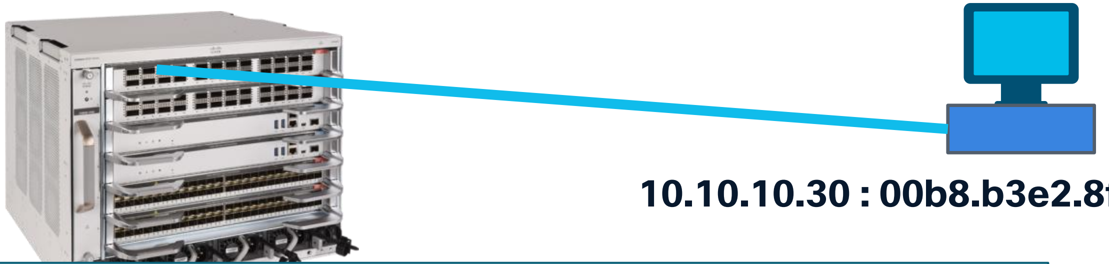
```
C9600-Silicon1-SVL#show platform software object-manager switch active F0 object 6996
Object identifier: 6996
  Description: matm mac entry type VLAN, id 10, 00b8.b3e2.8f66
  Obj type id: 459
  Obj type: MATM mac entry
  Status: Done, Epoch: 0, Client data: 0x10d54ff8
```

```
C9600-Silicon1-SVL#show platform software object-manager switch active F0 object 6996 parents
Object identifier: 176
  Description: intf FiftyGigE1/1/0/3, handle 1035, hw handle 1035, HW dirty: NONE AOM dirty NONE
  Status: Done

Object identifier: 591
  Description: matm table type VLAN, id 10
  Status: Done
```



Silicon ONE L2 Forwarding

FiftyGigE 1/1/0/3 : f003.bce9.3402



10.10.10.30 : 00b8.b3e2.8f66

Mac Address Programming.

FED – Forwarding Engine Driver

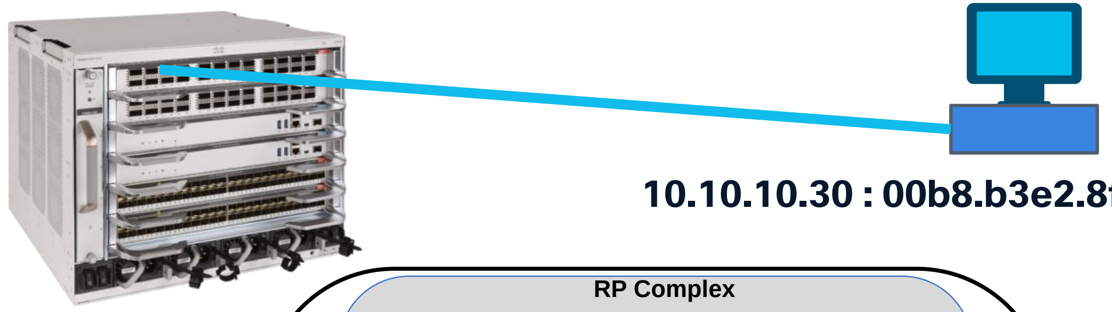
```
C9600-Silicon1-SVL#show  platform software fed switch active matm macTable vlan 10 mac 00b8.b3e2.8f66 detail
VLAN   MAC                Type  Seq#    EC_Bi  Flags  machandle   siHandle  riHandle  diHandle  *a_time  *e_time  ports Con
-----------------------------------------------------------------------------------------------------------------------------
10     00b8.b3e2.8f66     0x1   5699    0      0      0x0         0x0       0x0       0x0       300      60       No

======platform software details ======
ASIC [0]:
        L2 Service Port OID.........[1938]
        L2 Service Port GID.........[133]

======platform hardware details ======
ASIC [0]:
        Age Value...................[12]
        Age Remaining...............[48]
        L2 Service Port OID.........[1938]
        L2 Service Port GID.........[133]
        -------- L2 Service port OID Info --------
        L2 Service Port Object
        Port Type: AC Port
        STP Type: Forwarding
~
~
```

Silicon ONE L2 Forwarding

Mac Address Programming.

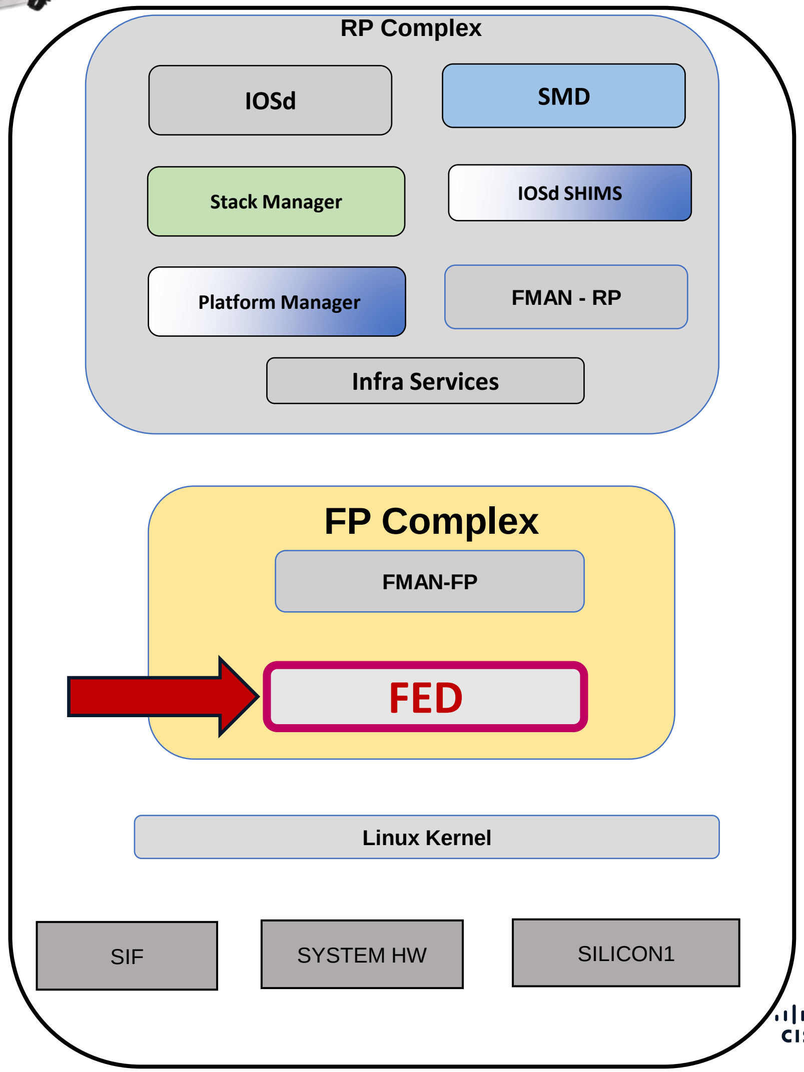
FED – Forwarding Engine Driver

FiftyGigE 1/1/0/3 : f003.bce9.3402



10.10.10.30 : 00b8.b3e2.8f66

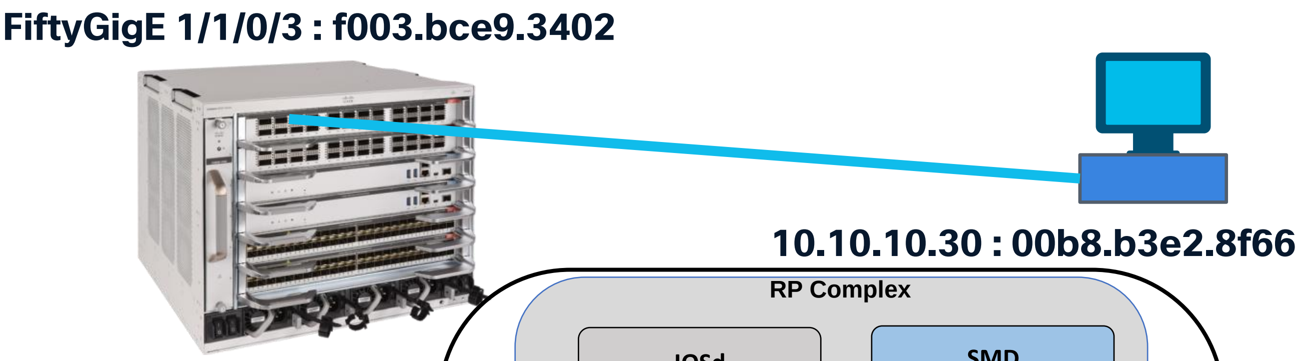
```
C9600-Silicon1-SVL#show platform software fed switch active vp key 1035 10
FED PM Virtual Port Data :
 iif_id = 1035, vlan_id = 10
 Info:
   stp state: forwarding
    vtp pruned: NO
    Dest Map: TRUE
    limited access: FALSE
    learning cache enable: FALSE
    delete auth behavior: FALSE
   tag native enable: FALSE
    block learn: FALSE
    pvlan port: FALSE
    pvlan mode: none
    pvlan native vlan: 10
```



Silicon ONE L2 Forwarding

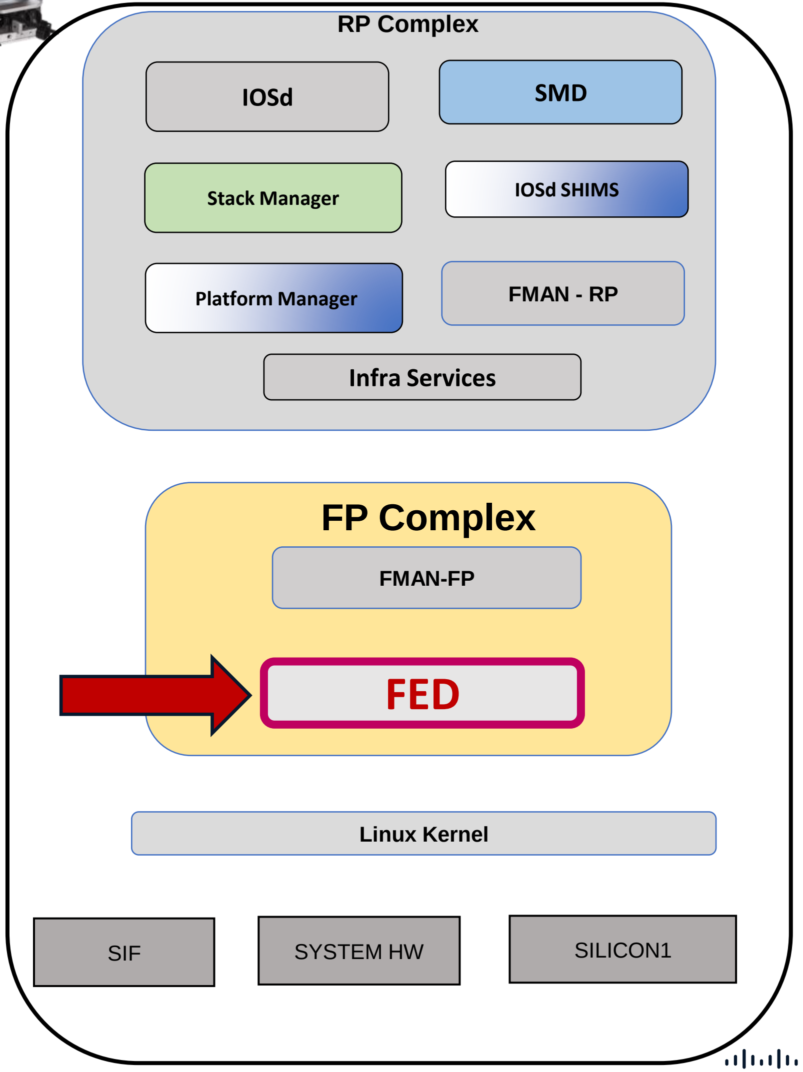
Mac Address Programming.

FED – Forwarding Engine Driver



C9600-Silicon1-SVL#show platform software fed switch active vp key 1035 10 detail

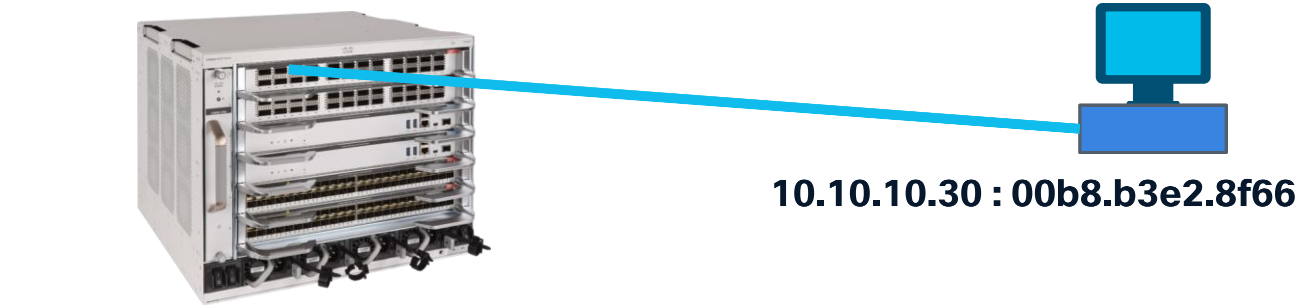
```
FED PM Virtual Port Data :
 iif_id = 1035, vlan_id = 10,
  vp_if_id = 0x3000a,
  l2_srv_port_oid = 1938, l2_srv_port_gid = 133       ← SDK object.
  match_vlan = 0,
  default_vlan = 10,
  Info:
   STP State : forwarding
   port  mode : Access
   port tagging : Native
```



Silicon ONE L2 Forwarding

Mac Address Programming.

FED – Forwarding Engine Driver



C9600-Silicon1-SVL#show platform software fed switch active vlan 10
VLAN Fed Information

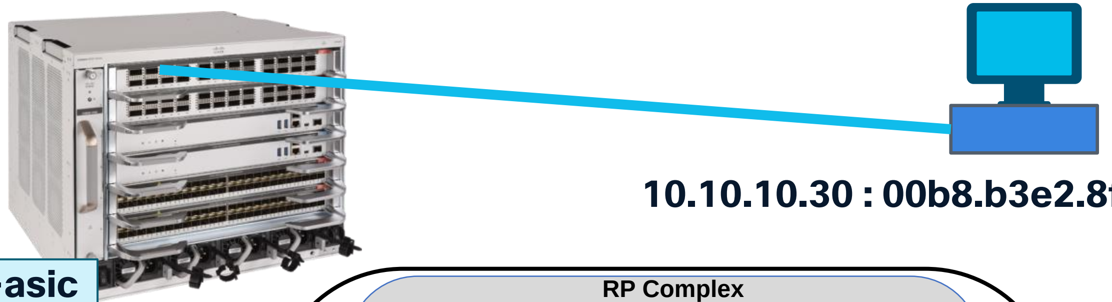
Vlan Id	IF Id	LE Handle	STP Handle	L3 IF Handle	SVI IF ID	MVID
10	0x0000000000420010	**0x0000000000000813**	0x0000000000000000	0x0000000000000000	0x0000000000000000	**11**

C9600-Silicon1-SVL#show platform software fed switch active vlan
VLAN Fed Information

Vlan Id	IF Id	LE Handle	STP Handle	L3 IF Handle	SVI IF ID	MVID
0	0x0004000000000000	0x0000000000000144	0x0000000000000000	0x0000000000000000	0x0000000000000000	4
1	0x000000000005ffff	0x000000000000080d	0x0000000000000000	0x0000000000000000	0x0000000000000000	10
10	**0x0000000000420010**	**0x0000000000000813**	**0x0000000000000000**	**0x0000000000000000**	**0x0000000000000000**	**11**
11	0x000400000000000b	0x0000000000000793	0x0000000000000000	0x0000000000000000	0x0000000000000000	12
4095	0x000000000005fffa	0x0000000000000146	0x0000000000000000	0x0000000000000000	0x0000000000000000	7

Silicon ONE L2 Forwarding

FiftyGigE 1/1/0/3 : f003.bce9.3402

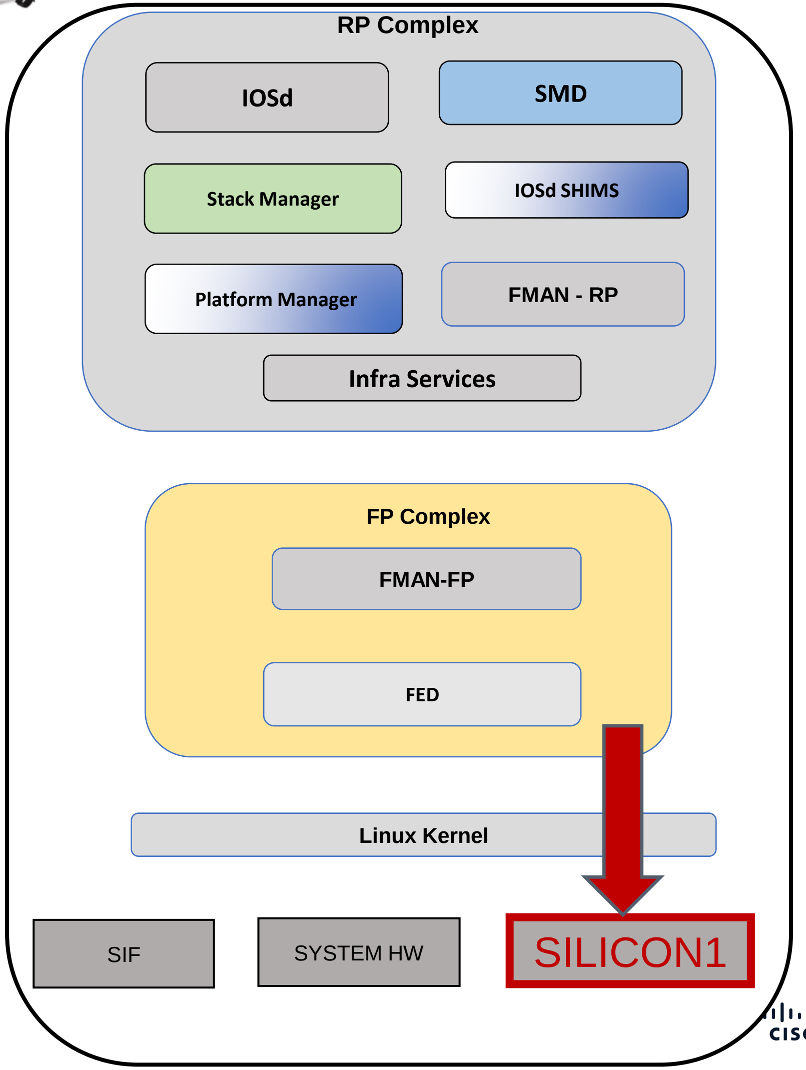


10.10.10.30 : 00b8.b3e2.8f66

Mac Address Programming.

SDK – Forward ASIC

C9600-Silicon1-SVL#show platform hardware fed switch active fwd-asic insight sdk_object(1938)
l2_service_port object
ac_port_vid_group_mode: =0x0
aci_dest_tunnel_type: =0x0
aci_ip_tunnel_mode: =0x0
~
~
l3_destination: null
mac_learning_mode: =0x2
meter: null
oid: =0x792
port_subtype: =0x0
port_type: =0x1
security_group_policy_enforcement: false
service_mapping_vid_list: []
smac_miss_enabled: false
stp_state: =0x3
svl_transport_enabled: false
tunnel_mode: =0x0
vrf: null
vxlan_tunnel_mode: =0x0

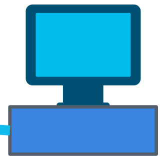


Silicon ONE L2 Forwarding

Mac Address Programming.

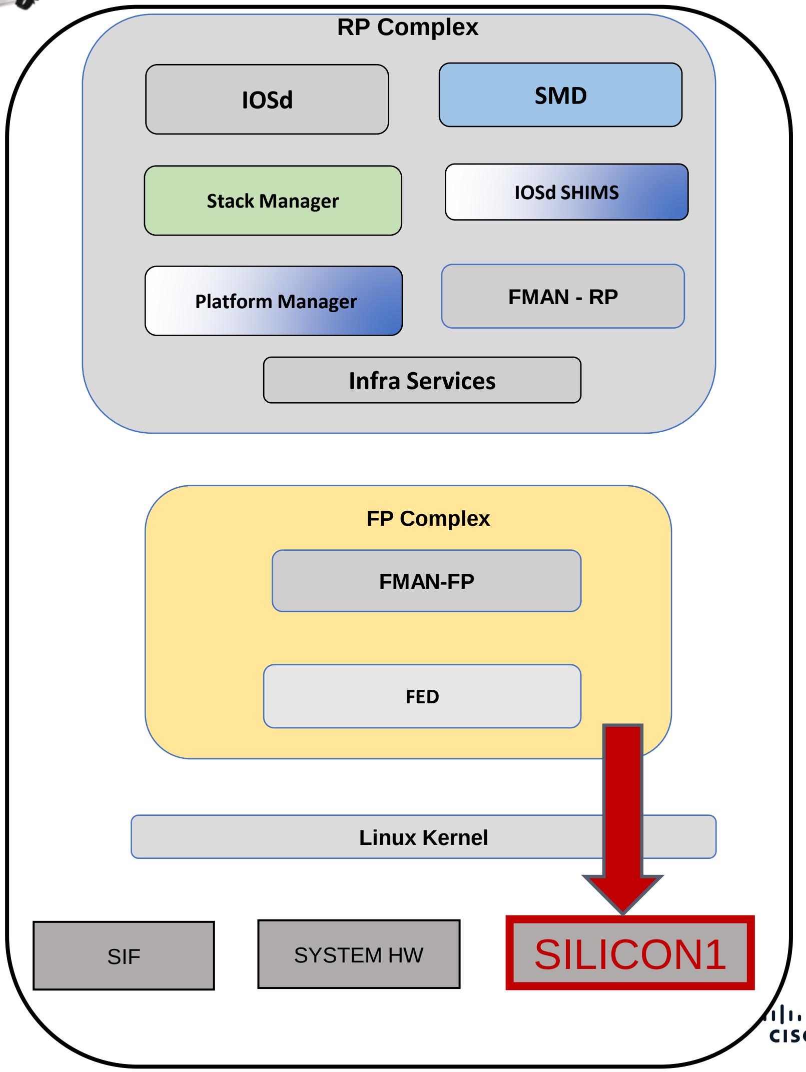
SDK – Forward ASIC

FiftyGigE 1/1/0/3 : f003.bce9.3402



10.10.10.30 : 00b8.b3e2.8f66

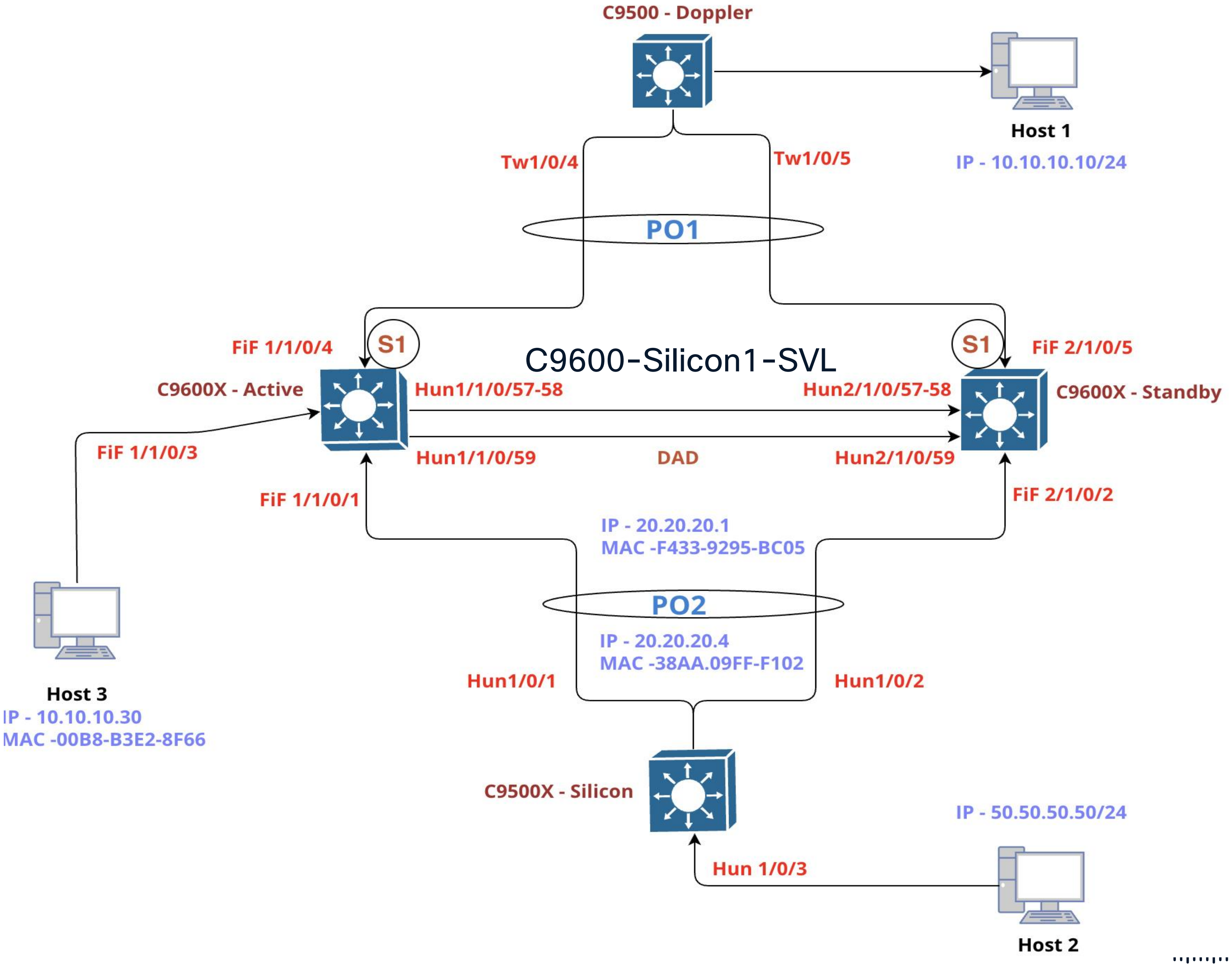
```
C9600-Silicon1-SVL#show  platform hardware fed switch active fwd-asic insight
sdk_object(0x813)
# switch object
cookie: '10'
~
mac_entries:
- addr:
    bytes:
    - =0x66
    - =0x8f
    - =0xe2
    - =0xb3
    - =0xb8
    - =0x0
   flat: =0xb8b3e28f66
~
- addr:
    bytes:
    - =0xa4
    - =0xdd
    - =0xcd
    - =0x3c
    - =0x5d
    - =0x4c
   flat: =0x4c5d3ccddda4
```



S1 – L3 Forwarding

Silicon ASIC

Lab Topology



L3 Interface Programming

Silicon ONE L3 Forwarding

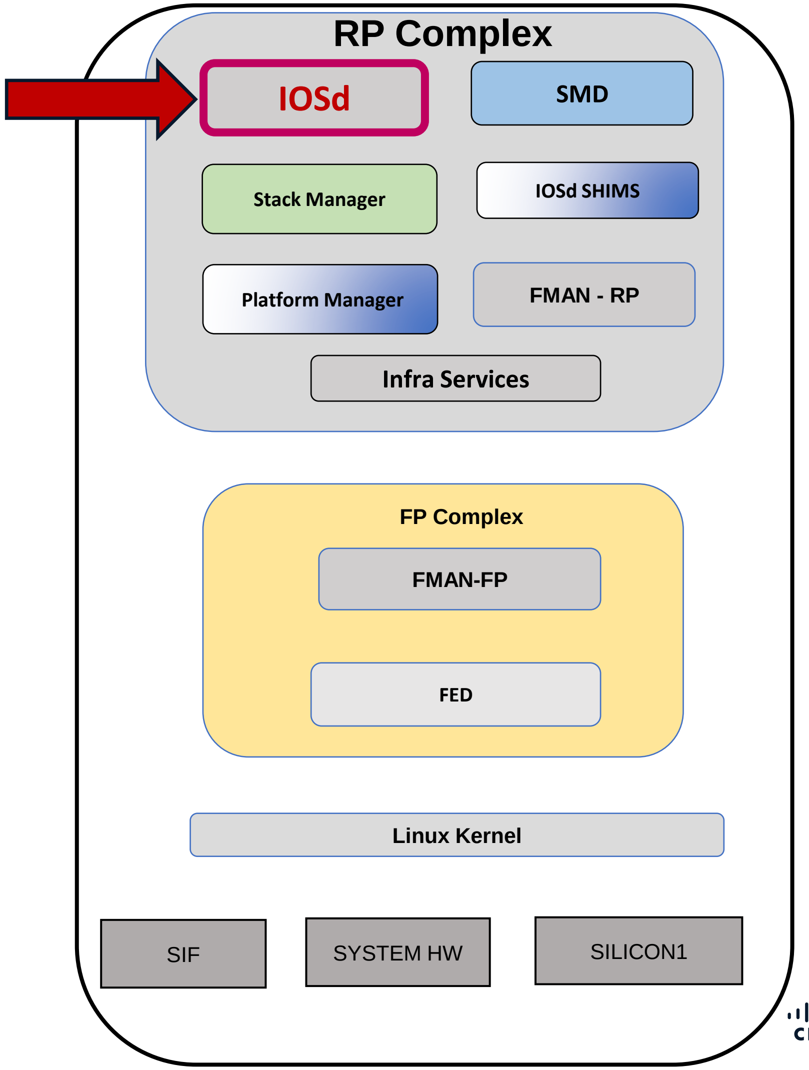
L3 Interface Programming.

IOSd

interface vlan 10
ip address : 10.10.10.1/24
mac address : f433.9295.bc05
interface id : dec (1177) = hex (0x499)

```
C9600-Silicon1-SVL#show  run interface vlan 10
Building configuration...

Current configuration : 131 bytes
!
interface Vlan10
 ip address 10.10.10.1 255.255.255.0
 ip pim sparse-mode
 ip igmp join-group 239.255.0.1
 ip ospf 1 area 0
end
```

```
C9600-Silicon1-SVL#show  interfaces vlan 10
Vlan10 is up, line protocol is up, Autostate Enabled
  Hardware is Ethernet SVI, address is f433.9295.bc05 (bia f433.9295.bc05)
  Internet address is 10.10.10.1/24
  MTU 1500 bytes, BW 1000000 Kbit/sec, DLY 10 usec,
~
~
```



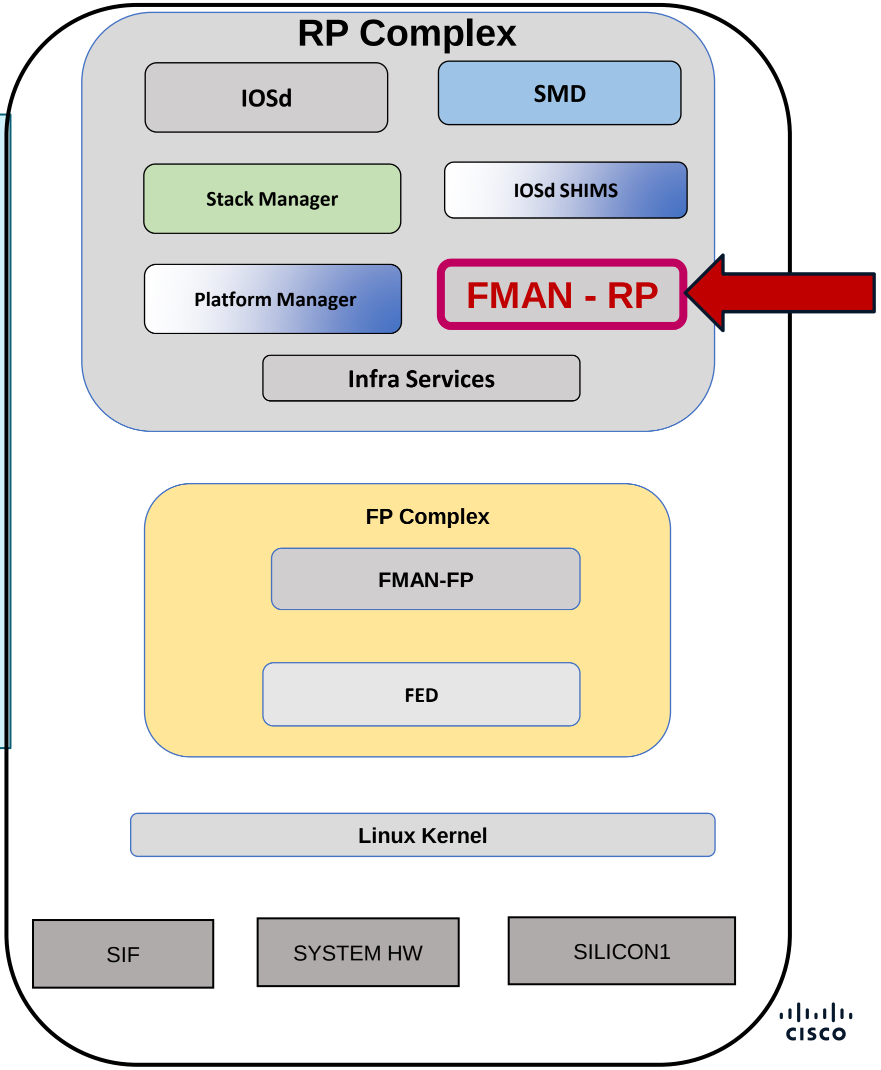
Silicon ONE L3 Forwarding

L3 Interface Programming.

FMAN-RP

interface vlan 10
ip address : 10.10.10.1/24
mac address : f433.9295.bc05
interface id : dec (1177) = hex (0x499)

C9600-Silicon1-SVL#show platform software interface switch active R0 name vlan 10
Name: Vlan10, **ID: 1177**, **QFP ID: 0**, Schedules: 4096
Type: Switch-Virtual-Interface, **State: enabled**, SNMP ID: 151, **MTU: 1500**
IP Address: 10.10.10.1
IPV6 Address:
Flags: ipv4
      mcast-ipv4
ICMP Flags: unreachables, redirects, no-info-reply, no-mask-reply
ICMP6 Flags: unreachables, redirects
SMI enabled on protocol(s): UNKNOWN
Authenticated-user:
FRR linkdown ID:
vNet Name: , vNet Tag: 0, vNet Extra Information: 0
QOS trust type: Unknown



Silicon ONE L3 Forwarding

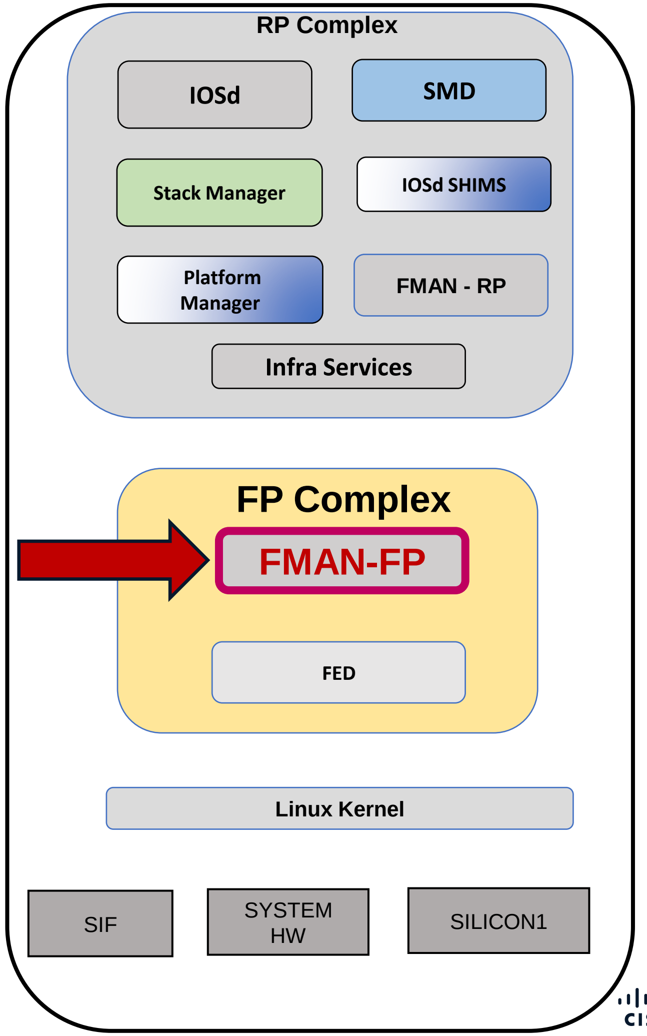
L3 Interface Programming.

FMAN-FP

interface vlan 10
ip address : 10.10.10.1/24
mac address : f433.9295.bc05
interface id : dec (1177) = hex (0x499)

```
C9600-Silicon1-SVL#show  platform software interface switch active F0 name vlan 10
Name: Vlan10, ID: 1177, QFP ID: 1177, Schedules: 4096
Type: Switch-Virtual-Interface, State: enabled, SNMP ID: 151, MTU: 1500
IP Address: 10.10.10.1
IPV6 Address: ::
Flags: ipv4
ICMP Flags: unreachables, redirects, no-info-reply, no-mask-reply
ICMP6 Flags: unreachables, redirects
SMI enabled on protocol(s): UNKNOWN
Authenticated-user:
FRR linkdown ID: 65535
vNet Name: , vNet Tag: 0, vNet Extra Information: 0
Dirty: unknown
AOM dependency sanity check: PASS
AOM Obj ID: 439
```

```
C9600-Silicon1-SVL#show  platform software object-manager switch active F0 Object 439
Object identifier: 439
  Description: intf Vlan10, handle 1177, hw handle 1177,  HW dirty: NONE   AOM dirty NONE
  Obj type id: 32
  Obj type: dpidb-config
  Status: Done, Epoch: 0, Client data: 0x10bae2c8
```



Silicon ONE L3 Forwarding

L3 Interface Programming.

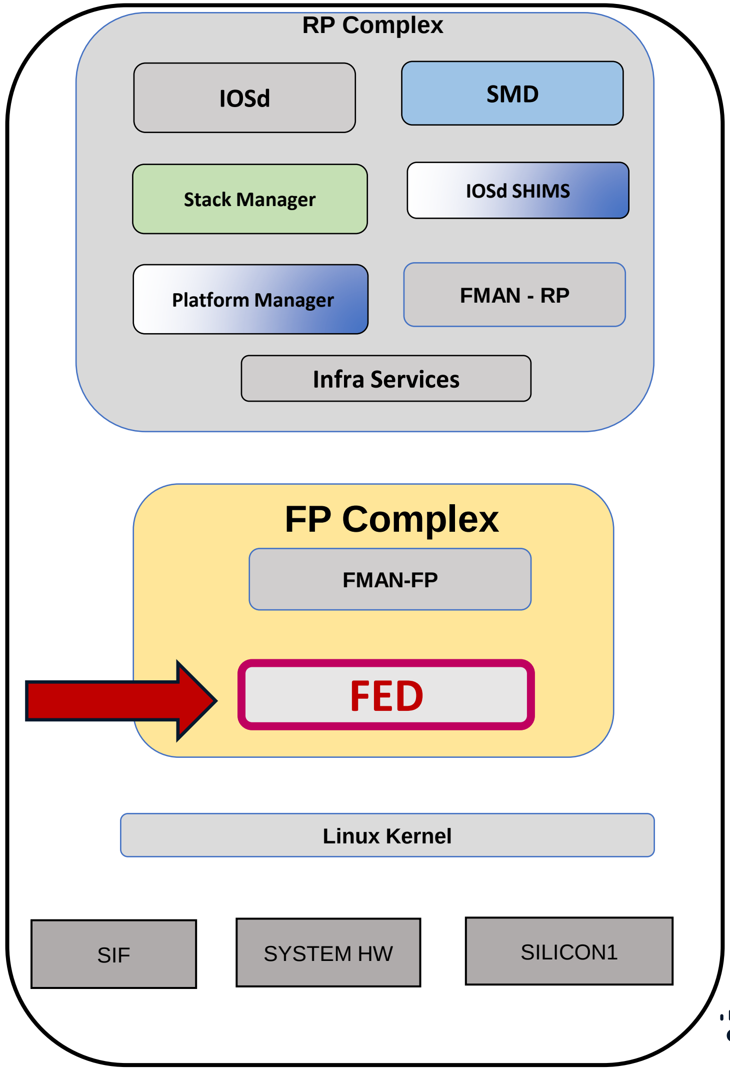
FED – Forwarding Engine Driver

interface vlan 10
ip address : 10.10.10.1/24
mac address : f433.9295.bc05
interface id : dec (1177) = hex (0x499)

```
C9600-Silicon1-SVL#show  platform software fed switch active ifm interfaces svi
Interface                    IF_ID              State
---------------------------------------------------------------------------
Vlan1                       0x00000492         Ready
Vlan10                      0x00000499         Ready
```

Few imp FED commands :-

show platform software fed switch active ifm interfaces **svi**
show platform software fed switch active ifm interfaces **loopback**
show platform software fed switch active ifm interfaces **vlan**
show platform software fed switch active ifm interfaces **sw-subif**
show platform software fed switch active ifm mappings



Silicon ONE L3 Forwarding

L3 Interface Programming.

FED – Forwarding Engine Driver

interface vlan 10
ip address : 10.10.10.1/24
mac address : f433.9295.bc05
interface id : dec (1177) = hex (0x499)

C9600-Silicon1-SVL#show platform software fed switch active ifm interface_name vlan 10

```
Interface IF_ID          : 0x0000000000000499
Interface Name           : Vlan10
Interface Block Pointer : 0x7b12f48a7ef8
Interface Block State : Ready
Interface State          : Enabled
Interface Status         : UPD,
Interface Ref-Cnt        : 1
Interface Type           : SVI
     Vlan id                : 10
     SNMP IF Index          : 151
     IPv4 MTU               : 1500
     IPv6 MTU               : 0
     IPv4 VRF ID            : 0x0
     IPv6 VRF ID            : 0x65535
     Protocol flags         : 0x0005 [ ipv4 pim_ipv4 ]
     Misc flags             : 0x0045 [ ipv4 mcast_v4 --- ]
     ICMPv4 flags           : 0x03 [ unreachable redirect ]
     ICMPv6 flags           : 0x03 [ unreachable redirect ]
     Mac Address            : f4:33:92:95:bc:05
Ref Count                : 1 (feature Ref Counts + 1)
```

```
No Feature Reference count Present
IFM Feature Sub block information
      Port L3 Subblock
          VRF ID ................................ [0]
          IPv4 Routing Enabled ...... [Yes]
            IPv6 Routing Enabled .... [No]
            MPLS Enabled ............. [No]
          Pimv4 Enabled ................ [Yes]
            Pimv6 Enabled ............. [No]
          IPv4 MTU ......................... [1500]
            IPv6 MTU ..................... [0]
            L3 srv port gid ............. [43]
          L3 srv port oid for Asic[0] ......... [0x817(2071)]
       Port CTS Subblock is NULL if_id = 0x499
       Port CTS Subblock                    <<<<< Security block
            Disable SGACL ................... [0x0]
            Trust ................................. [0x0]
            Propagate ........................... [0x0]
            Port SGT ............................ [0xffff]
Events Log     ?
```

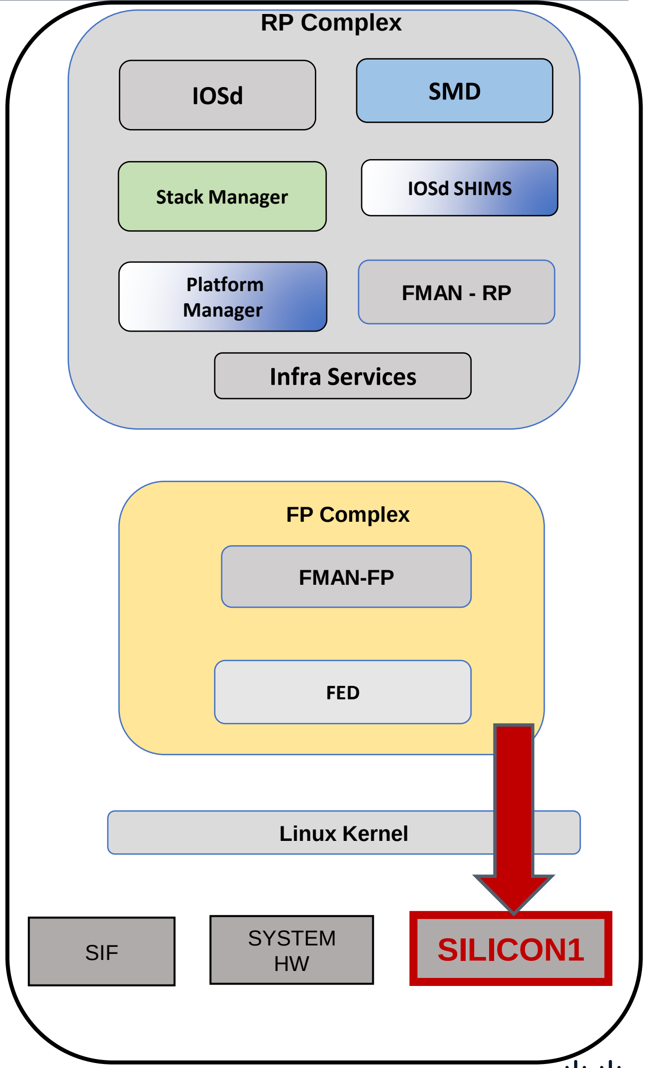
SDK

Silicon ONE L3 Forwarding

L3 Interface Programming.

SDK – Forward ASIC

```
interface vlan 10
ip address : 10.10.10.1/24
mac address : f433.9295.bc05
interface id : dec (1177) = hex (0x499)
```



C9600-Silicon1-SVL#show platform hardware fed switch active fwd-asic insight sdk_object(2071)

```
# svi_port object
active: true
all_vrf_ipv4_hosts:
~
ipv4_subnets:
- addr:
   b_addr:
   - =0x0
   - =0xa
   - =0xa
   - =0xa
  s_addr: =0xa0a0a00    <<<  10.10.10.0
 length: =0x18          <<<  Hex(0x18)=Dec(24)
ipv6_hosts: []
ipv6_subnets: []
load_balancing_profile: =0x1
```

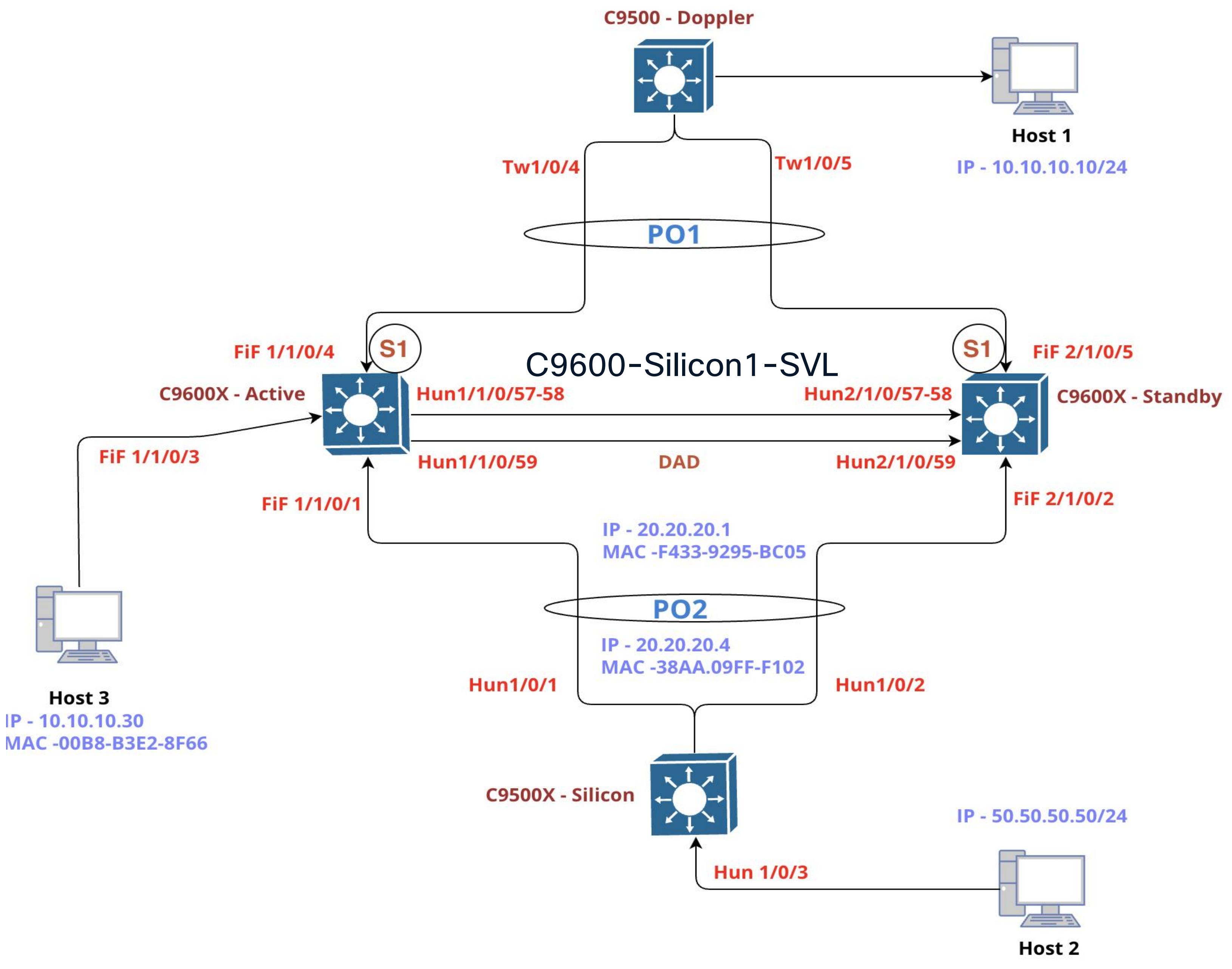
```
mac:
  bytes:
  - =0x5
  - =0xbc
  - =0x95
  - =0x92
  - =0x33
  - =0xf4
  flat: =0xf4339295bc05
  word:
  - =0xbc05
  - =0x9295
  - =0xf433
oid: =0x817
~
~
```

mac address : f433.9295.bc05

Directly Connected Routes Programming

Silicon ASIC

Lab Topology



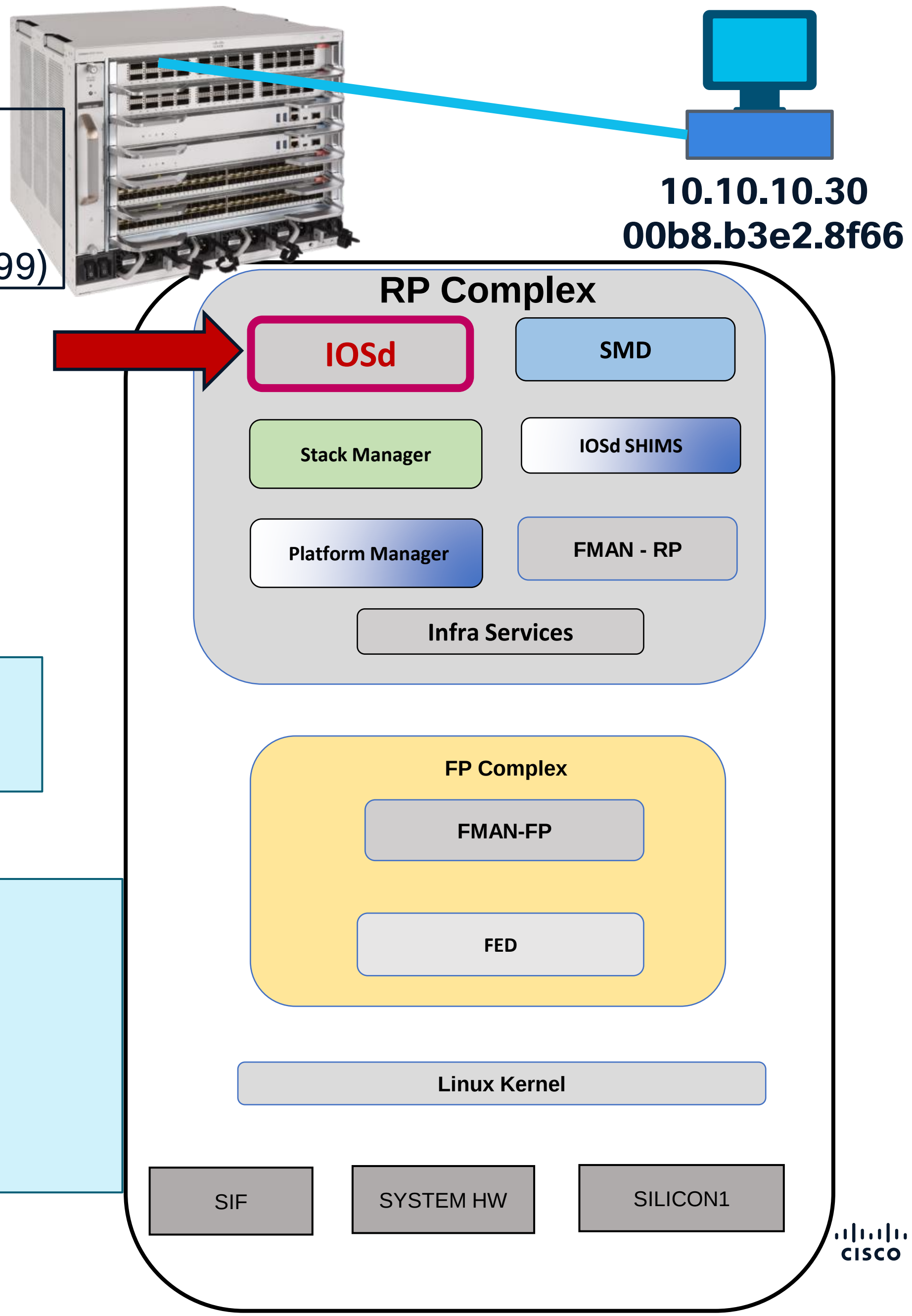
Silicon ONE L3 Forwarding

Directly Connected Routes

IOSd

FiftyGigE 1/1/0/3 : Access Vlan 10

interface vlan 10
ip address : 10.10.10.1/24
mac address : f433.9295.bc05
interface id : dec (1177) = hex (0x499)




```
C9600-Silicon1-SVL#show  ip route 10.10.10.30
Routing entry for 10.10.10.0/24
  Known via "connected", distance 0, metric 0 (connected, via interface)
  Routing Descriptor Blocks:
  * directly connected, via Vlan10
      Route metric is 0, traffic share count is 1
```

```
C9600-Silicon1-SVL#show  ip arp 10.10.10.30
Protocol  Address          Age (min)  Hardware Addr   Type   Interface
Internet  10.10.10.30           74    00b8.b3e2.8f66  ARPA   Vlan10
```

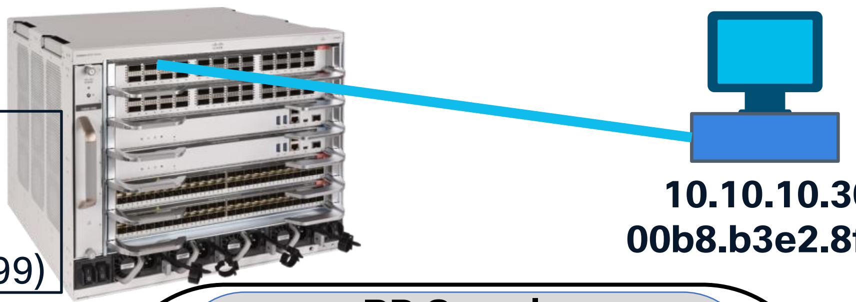
```
C9600-Silicon1-SVL#show  mac address-table address 00b8.b3e2.8f66
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----
  10    00b8.b3e2.8f66    DYNAMIC     Fif1/1/0/3
Total Mac Addresses for this criterion: 1
```

Silicon ONE L3 Forwarding

Directly Connected Routes

IOSd

FiftyGigE 1/1/0/3 : Access Vlan 10



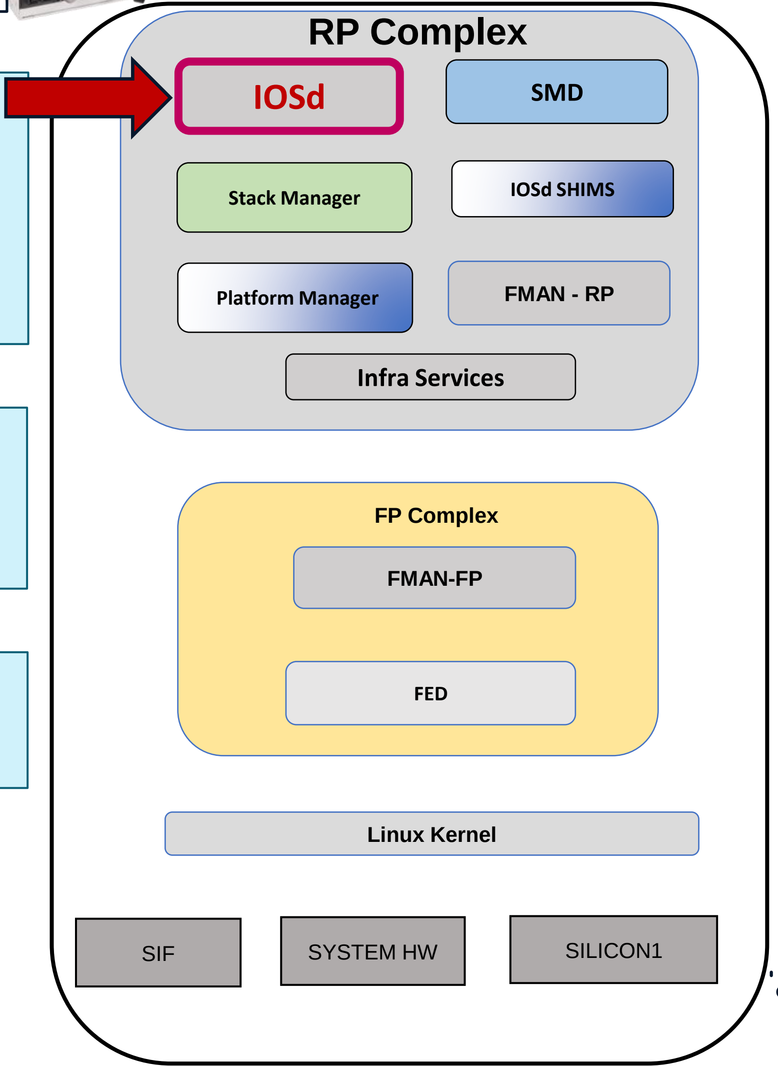
10.10.10.30
00b8.b3e2.8f66

interface vlan 10
ip address : 10.10.10.1/24
mac address : f433.9295.bc05
interface id : dec (1177) = hex (0x499)

```
C9600-Silicon1-SVL#show  interfaces vlan 10
Vlan10 is up, line protocol is up, Autostate Enabled
  Hardware is Ethernet SVI, address is f433.9295.bc05 (bia f433.9295.bc05)
  Internet address is 10.10.10.1/24
~
~
```

```
C9600-Silicon1-SVL#show  ip cef 10.10.10.30
10.10.10.30/32
 attached to Vlan10
```

```
C9600-Silicon1-SVL#show  ip cef exact-route 10.10.10.1 10.10.10.30
10.10.10.1 -> 10.10.10.30 =>IP adj out of Vlan10, addr 10.10.10.30
```



Silicon ONE L3 Forwarding

Directly Connected Routes

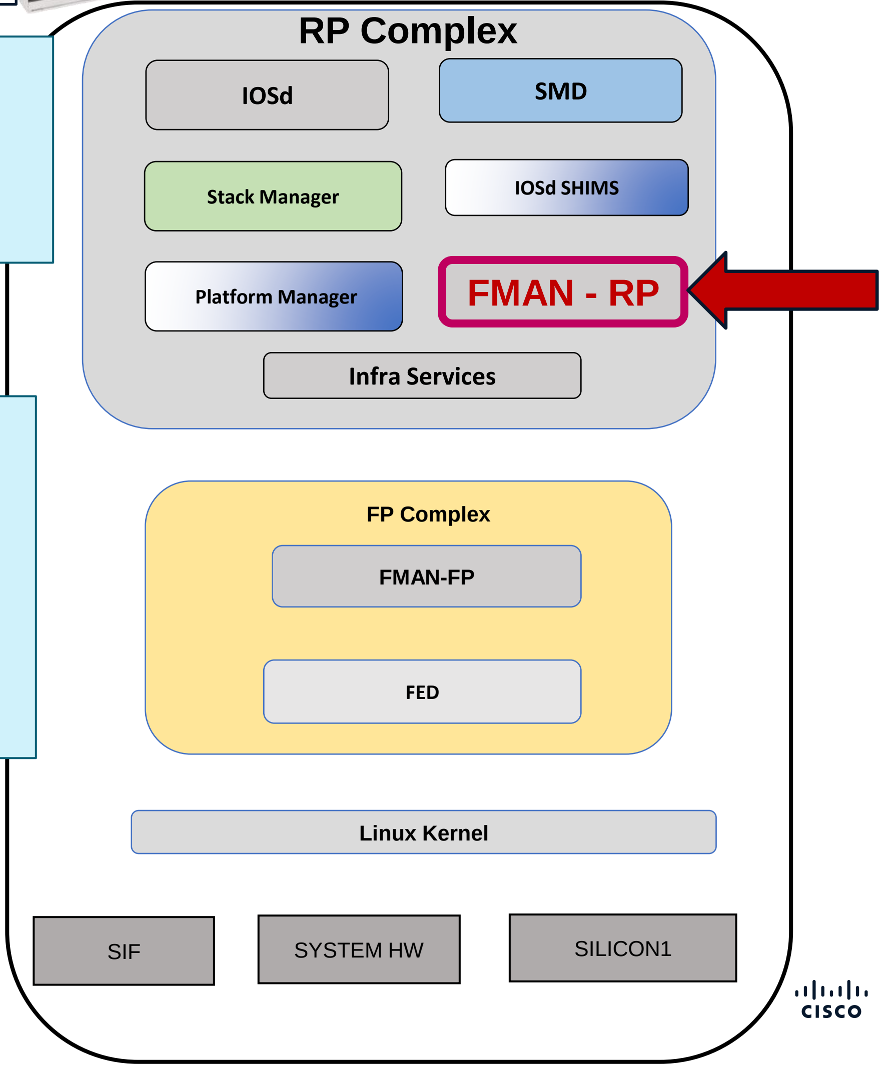
FMAN-RP

FiftyGigE 1/1/0/3 : Access Vlan 10

interface vlan 10
ip address : 10.10.10.1/24
mac address : f433.9295.bc05
interface id : dec (1177) = hex (0x499)

```
C9600-Silicon1-SVL#show  platform software ip switch active R0 cef prefix
10.10.10.30/32
Forwarding Table
Prefix/Len                  Next Object       Index
------------------------------------------------------------------
10.10.10.30/32              OBJ_ADJACENCY    0x12
```

```
C9600-Silicon1-SVL#show  platform software ip switch active R0 cef prefix
10.10.10.30/32 detail
Forwarding Table

10.10.10.30/32 -> OBJ_ADJACENCY (0x12), urpf: 19
Connected Interface: 1177
Prefix Flags: Directly L2 attached
OM handle: 0x34804b9b80
```



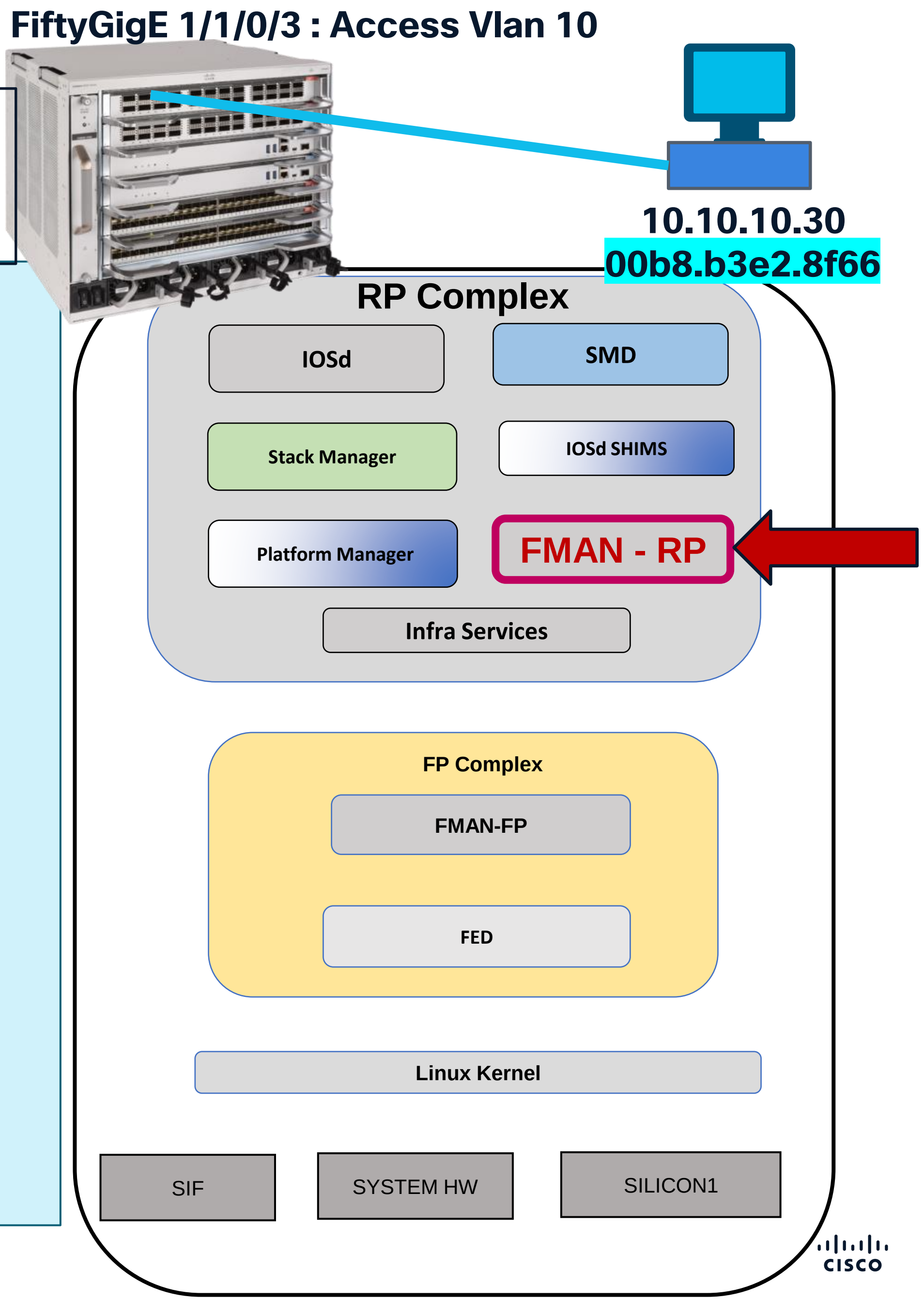
Silicon ONE L3 Forwarding

Directly Connected Routes

FMAN-RP

interface vlan 10
ip address : 10.10.10.1/24
mac address : f433.9295.bc05
interface id : dec (1177) = hex (0x499)

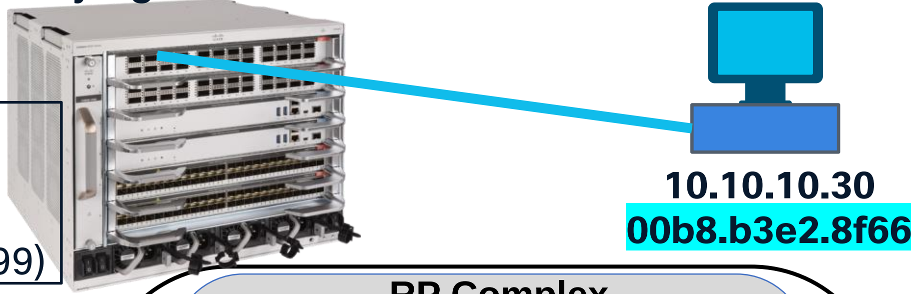
```
C9600-Silicon1-SVL#show  platform software ip switch active R0 cef prefix
10.10.10.30/32 oce
=== Prefix OCE ===
Prefix/Len: 10.10.10.30/32
  Next Obj Type: OBJ_ADJACENCY
  Next Obj Handle: 0x12, urpf: 19
  Connected Interface: 1177
  Prefix Flags: Directly L2 attached
  OM handle: 0x34804b9b80

=== Adjacency OCE ===
Adjacency id: 0x12 (18)
  Interface: Vlan10, IF index: 1177, Link Type: MCP_LINK_IP
  Encap: 0:b8:b3:e2:8f:66:f4:33:92:95:bc:5:8:0
  Encap Length: 14, Encap Type: MCP_ET_ARPA, MTU: 1500
  Flags: no-l3-inject
  Incomplete behavior type: None
  Fixup: unknown
  Fixup_Flags_2: unknown
  Nexthop addr: 10.10.10.30
  IP FRR MCP_ADJ_IPFRR_NONE 0
  OM handle: 0x34804be8e0
```



Silicon ONE L3 Forwarding

Directly Connected Routes
FMAN-RP

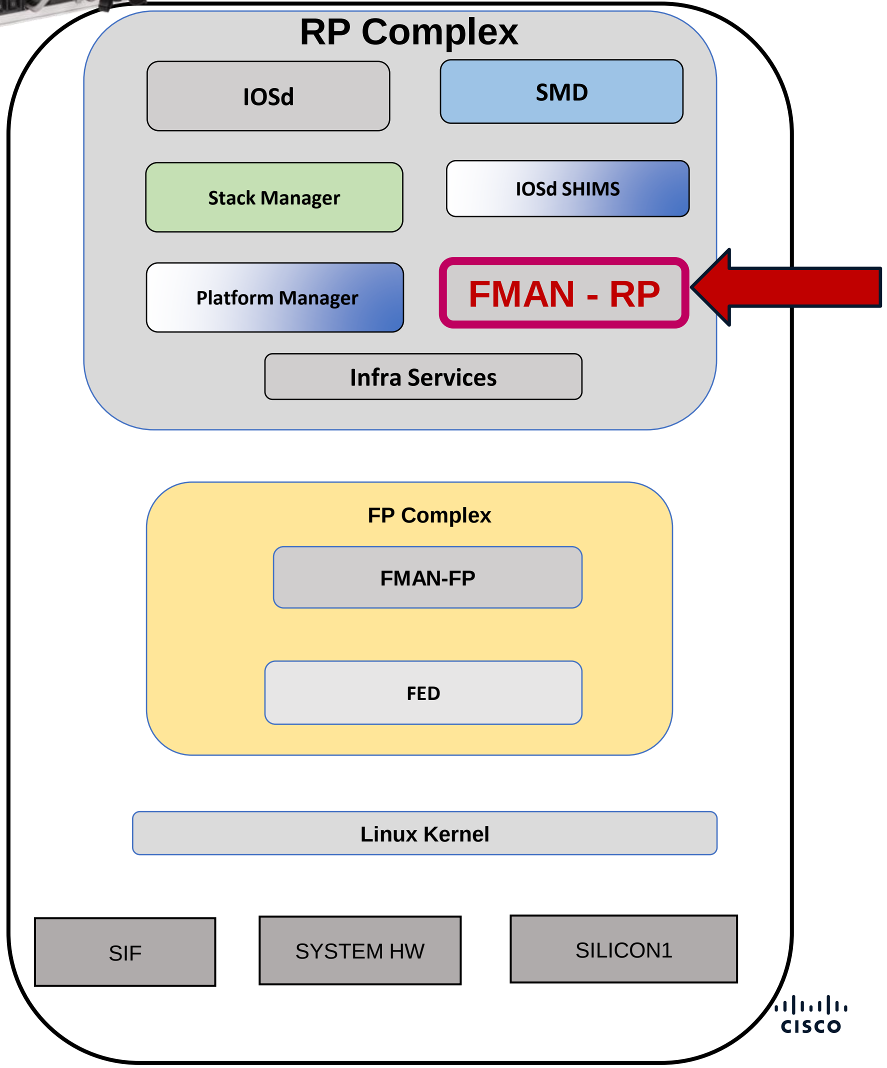
FiftyGigE 1/1/0/3 : Access Vlan 10




```
interface vlan 10
ip address : 10.10.10.1/24
mac address : f433.9295.bc05
interface id : dec (1177) = hex (0x499)
```

```
C9600-Silicon1-SVL#show  platform software adjacency switch active R0 index 18
Number of adjacency objects: 12

Adjacency id: 0x12 (18)
  Interface: Vlan10, IF index: 1177, Link Type: MCP_LINK_IP
  Encap: 0:b8:b3:e2:8f:66:f4:33:92:95:bc:5:8:0
  Encap Length: 14, Encap Type: MCP_ET_ARPA, MTU: 1500
  Flags: no-l3-inject
  Incomplete behavior type: None
  Fixup: unknown
  Fixup_Flags_2: unknown
  Nexthop addr: 10.10.10.30
  IP FRR MCP_ADJ_IPFRR_NONE 0
  OM handle: 0x34804be8e0
```

```
C9600-Silicon1-SVL#show  ip arp vlan 10
Protocol  Address          Age (min)  Hardware Addr   Type   Interface
Internet  10.10.10.1              -   f433.9295.bc05  ARPA   Vlan10
Internet  10.10.10.3             75   4c5d.3ccd.de3f  ARPA   Vlan10
Internet  10.10.10.10            99   00b8.b3d7.1274  ARPA   Vlan10
Internet  10.10.10.30            83   00b8.b3e2.8f66  ARPA   Vlan10
```

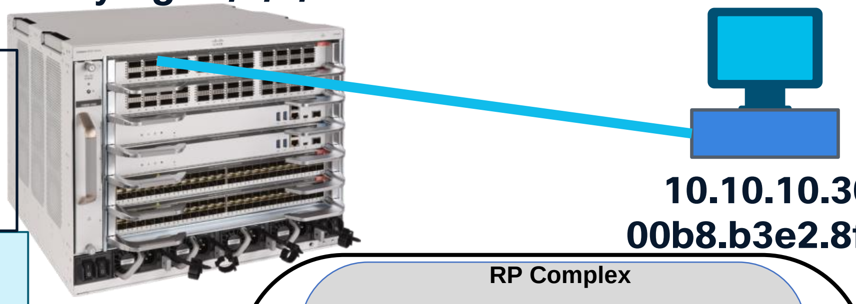


Silicon ONE L3 Forwarding

Directly Connected Routes

FMAN-FP

interface vlan 10
ip address : 10.10.10.1/24
mac address : f433.9295.bc05
interface id : dec (1177) = hex (0x499)

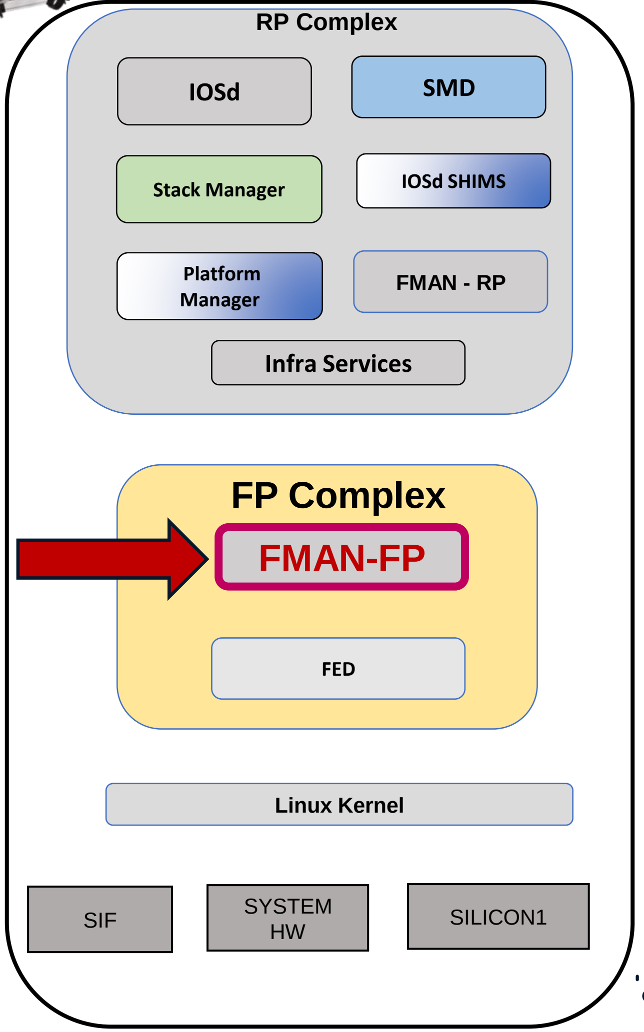


C9600-Silicon1-SVL#show platform software ip switch active F0 cef prefix 10.10.10.30/32
Forwarding Table
Prefix/Len   Next Object   Index

10.10.10.30/32   OBJ_ADJACENCY   0x12

C9600-Silicon1-SVL#show platform software ip switch active F0 cef prefix 10.10.10.30/32 detail
Forwarding Table

10.10.10.30/32 -> OBJ_ADJACENCY (0x12), urpf: 19
Connected Interface: 1177
Prefix Flags: Directly L2 attached
aom id: 6889, HW handle: (nil) (**created**)



Silicon ONE L3 Forwarding

Directly Connected Routes

FMAN-FP

interface vlan 10
ip address : 10.10.10.1/24
mac address : f433.9295.bc05
interface id : dec (1177) = hex (0x499)

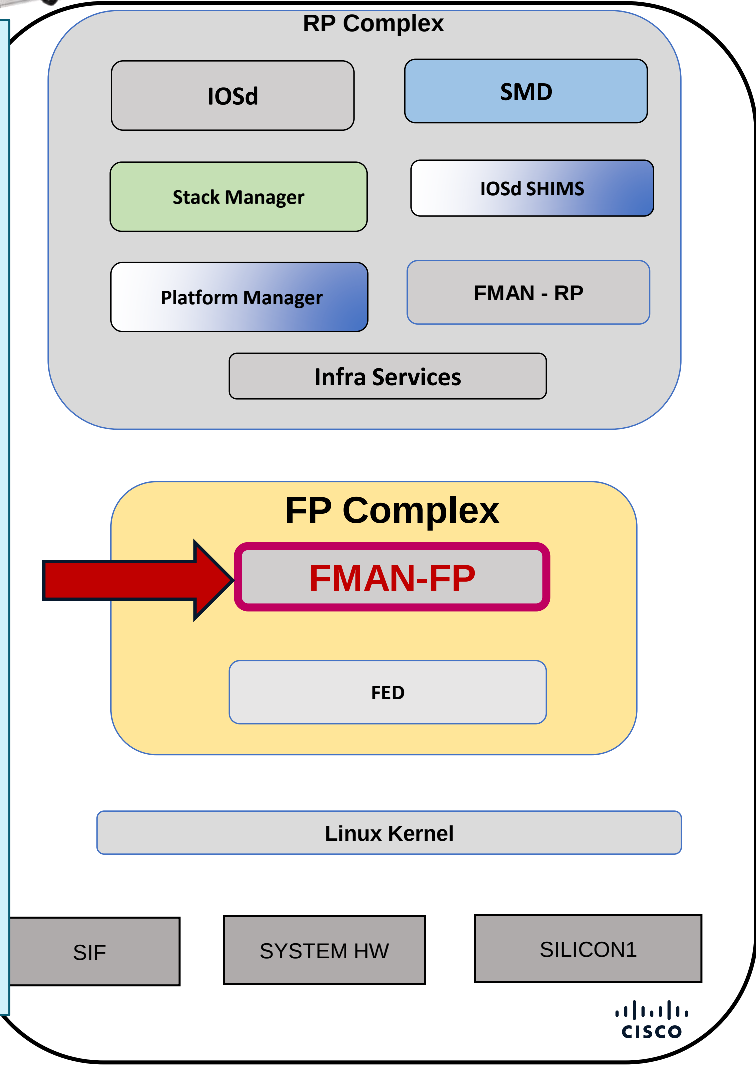
FiftyGigE 1/1/0/3 : Access Vlan 10

10.10.10.30
00b8.b3e2.8f66

C9600-Silicon1-SVL#show platform software ip switch active F0 cef prefix 10.10.10.30/32 oce

=== Prefix OCE ===
Prefix/Len: 10.10.10.30/32
  Next Obj Type: OBJ_ADJACENCY
  Next Obj Handle: 0x12, urpf: 19
  Connected Interface: 1177
  Prefix Flags: Directly L2 attached
  aom id: 6889, **HW handle: (nil) (created)**

=== Adjacency OCE ===
Adjacency id: 0x12 (18)
  Interface: Vlan10, IF index: 1177, Link Type: MCP_LINK_IP
  Encap: 0:b8:b3:e2:8f:66:f4:33:92:95:bc:5:8:0
  Encap Length: 14, Encap Type: MCP_ET_ARPA, MTU: 1500
  Flags: no-l3-inject
  Incomplete behavior type: None
  Fixup: unknown
  Fixup_Flags_2: unknown
  Nexthop addr: 10.10.10.30
  IP FRR MCP_ADJ_IPFRR_NONE 0
  aom id: 6887, **HW handle: (nil) (created)**



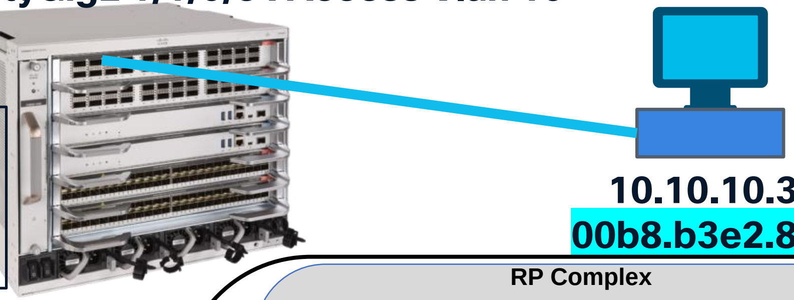
Silicon ONE L3 Forwarding

Directly Connected Routes

FMAN-FP

FiftyGigE 1/1/0/3 : Access Vlan 10

interface vlan 10
ip address : 10.10.10.1/24
mac address : f433.9295.bc05
interface id : dec (1177) = hex (0x499)

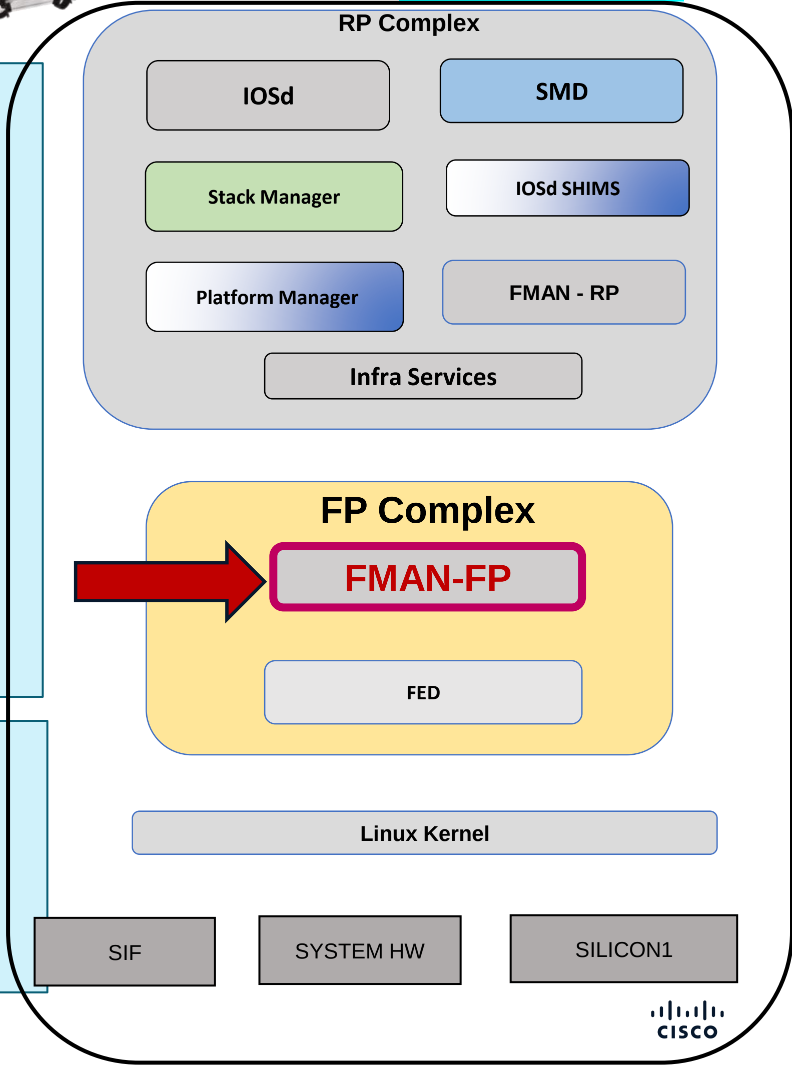


10.10.10.30
00b8.b3e2.8f66

```
C9600-Silicon1-SVL#show  platform software adjacency switch active F0 index 0x12
Number of adjacency objects: 13

Adjacency id: 0x12 (18)
  Interface: Vlan10, IF index: 1177, Link Type: MCP_LINK_IP
  Encap: 0:b8:b3:e2:8f:66:f4:33:92:95:bc:5:8:0
  Encap Length: 14, Encap Type: MCP_ET_ARPA, MTU: 1500
  Flags: no-l3-inject
  Incomplete behavior type: None
  Fixup: unknown
  Fixup_Flags_2: unknown
  Nexthop addr: 10.10.10.30
  IP FRR MCP_ADJ_IPFRR_NONE 0
  aom id: 6887, HW handle: (nil) (created)
```

```
C9600-Silicon1-SVL#show  platform software object-manager switch active F0 object 6889
Object identifier: 6889
  Description: PREFIX 10.10.10.30/32 (Table id 0)
  Obj type id: 72
  Obj type: route-pfx
  Status: Done, Epoch: 0, Client data: 0x10cfcc68
```

<<<<<< Check in FED

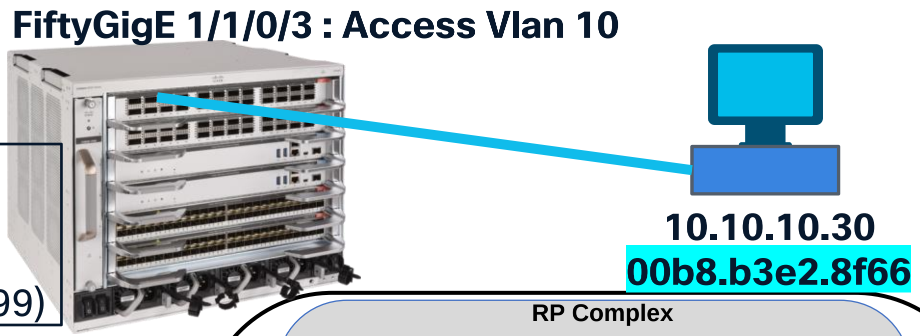


Silicon ONE L3 Forwarding

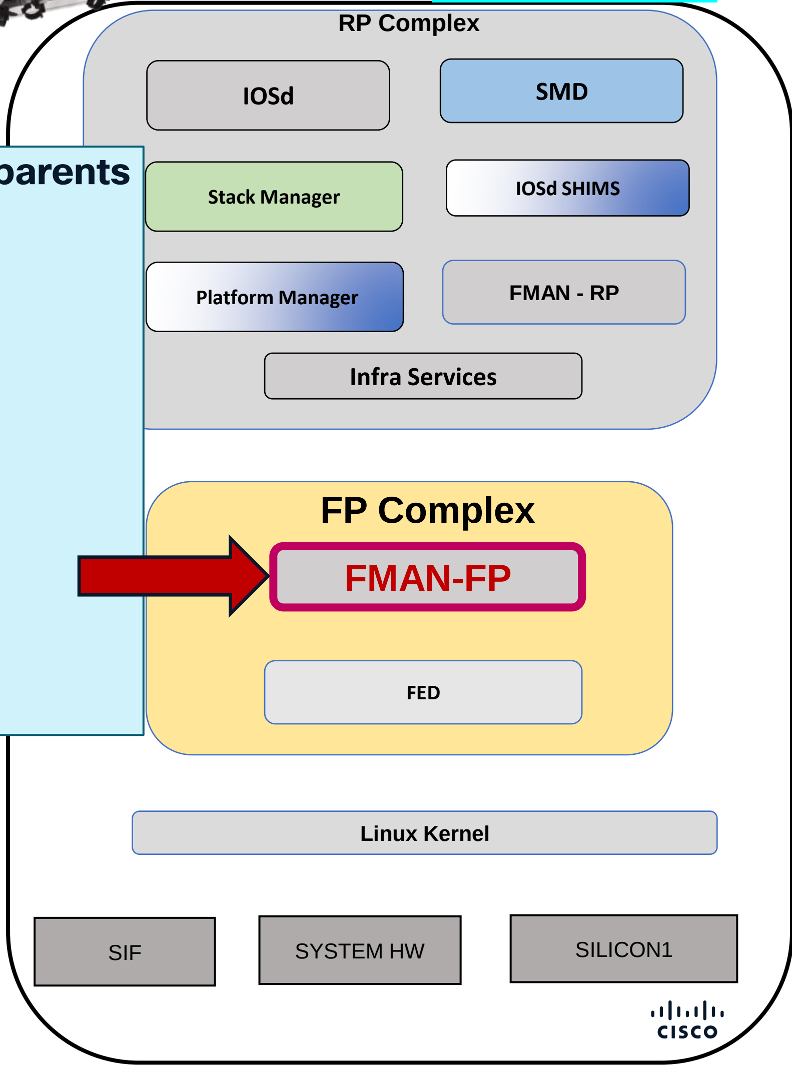
Directly Connected Routes

FMAN-FP

interface vlan 10
ip address : 10.10.10.1/24
mac address : f433.9295.bc05
interface id : dec (1177) = hex (0x499)




```
C9600-Silicon1-SVL#show  platform software object-manager switch active F0 object 6889 parents

Object identifier: 151
  Description: ipv4 table 0 (Default), vrf id 0
  Status: Done

Object identifier: 6887
  Description: adj 0x12, Flags None
  Status: Done

Object identifier: 6888
  Description: uRPF-list(hdl=0x00000013)
  Status: Done
```



Silicon ONE L3 Forwarding

Directly Connected Routes

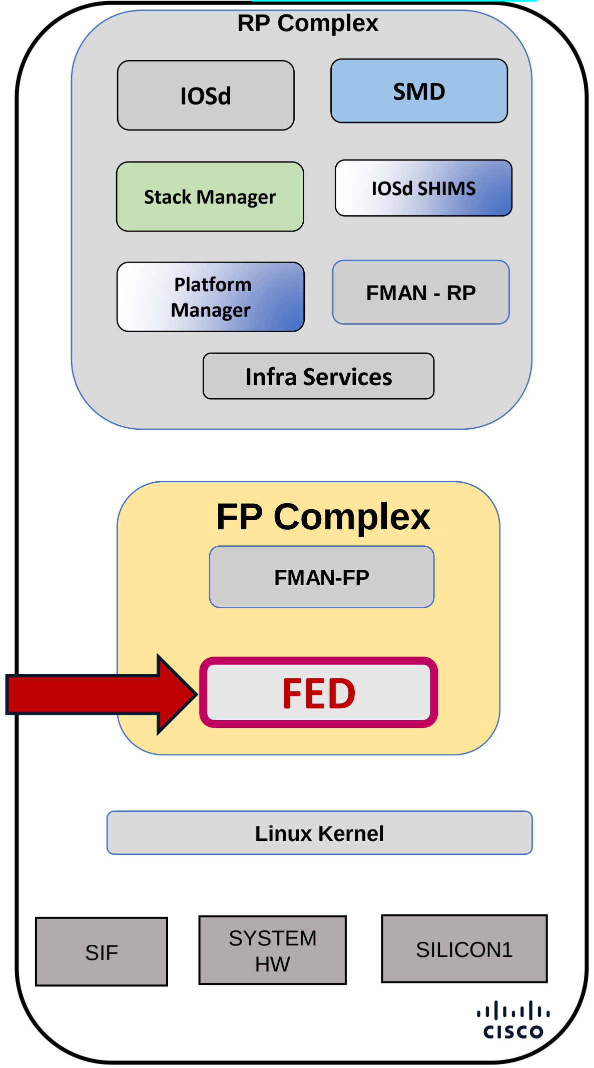
FED – Forwarding Engine Driver

FiftyGigE 1/1/0/3 : Access Vlan 10

interface vlan 10
ip address : 10.10.10.1/24
mac address : f433.9295.bc05
interface id : dec (1177) = hex (0x499)

10.10.10.30
00b8.b3e2.8f66

```
C9600-Silicon1-SVL#show  platform software fed switch active ip route 10.10.10.30/32
IPV4ROUTE_ID:id:0x63a010cfcc68 nobj:(IPNEXTHOP_ID,0x12) 10.10.10.30/32 tblid:0 DA:1
   State:(0) success
   NPD: device:0 lspa_rec:0 api_type:host(1)
   State:(0) success
   Mac address of Host:00b8.b3e2.8f66
  Host L3Port OID:0x817 vrf(gid/oid):0/0x2c1
  ADJ:objid:0x12 (IPv4: 10.10.10.30)  nh_type:NHADJ_NORMAL iif_id:0x499 ether_type:0x8 #child:1
     srcmac:f433.9295.bc05 dstmac:00b8.b3e2.8f66
     State:(0) success
    NPD: fec_oid:0 was_nor_nh:0 cr_def:1 stale:0 l3port_valid:0
    NPD: device:0 nh_gid/oid:0/0x0 old_gid/oid:0/0x0 parent_oid:0
    State:(0) success
```



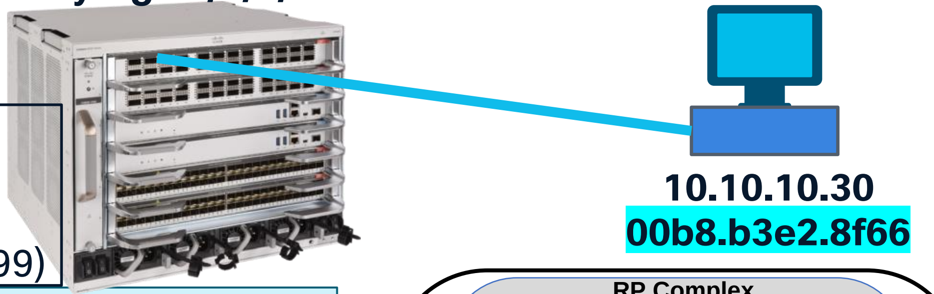
Silicon ONE L3 Forwarding

Directly Connected Routes

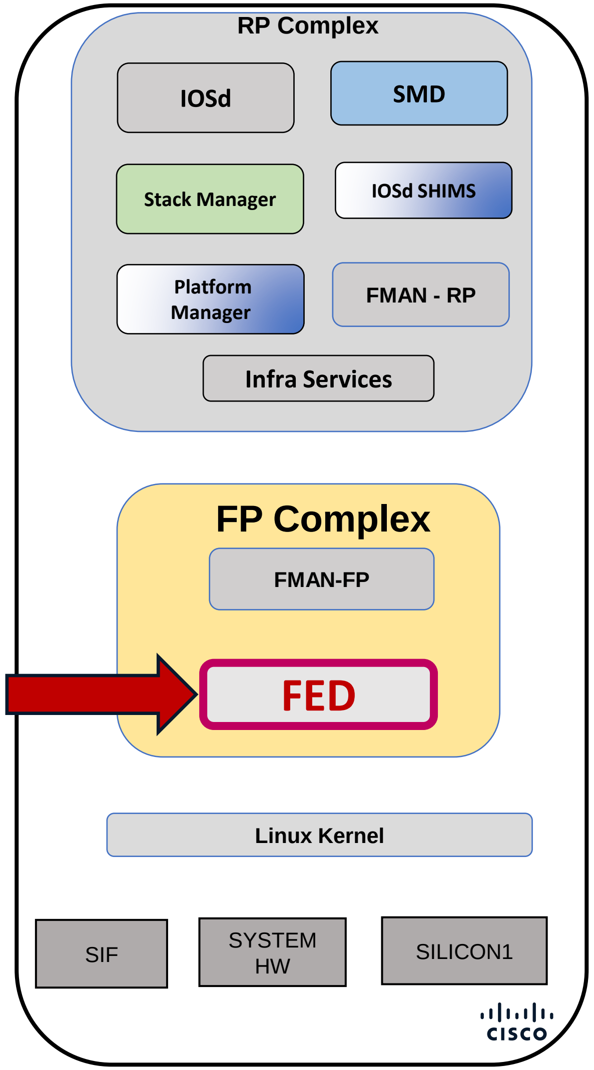
FED – Forwarding Engine Driver

FiftyGigE 1/1/0/3 : Access Vlan 10

interface vlan 10
ip address : 10.10.10.1/24
mac address : f433.9295.bc05
interface id : dec (1177) = hex (0x499)




```
C9600-Silicon1-SVL#show  platform software fed switch active ip route 10.10.10.30/32 detail
IPV4ROUTE_ID:id:0x63a010cfcc68 nobj:(IPNEXTHOP_ID,0x12) 10.10.10.30/32 tblid:0 DA:1
    State:(0) success
   NPD: device:0 lspa_rec:0 api_type:host(1)
   State:(0) success
   Mac address of Host:00b8.b3e2.8f66
   Host L3Port OID:0x817 vrf(gid/oid):0/0x2c1

 SDK Object Details:
   NPD: SDK oid 0x817, devid:0, asic:0
        SVI Object:
        sdk SVI port obj active: 1
        MAC address of interface: f433.9295.bc05

  Events Log
   [2026/01/07 19:14:09.485] NPI IPROUTE CREATE
   [2026/01/07 19:14:09.485] NPD IPROUTE EV ADD IPV4 HOST

ADJ:objid:0x12 (IPv4: 10.10.10.30)  nh_type:NHADJ_NORMAL iif_id:0x499 ether_type:0x8 #child:1
    srcmac:f433.9295.bc05 dstmac:00b8.b3e2.8f66
   State:(0) success
 ~
```



Silicon ONE L3 Forwarding

Directly Connected Routes

SDK – Forward ASIC

FiftyGigE 1/1/0/3 : Access Vlan 10

10.10.10.30
00b8.b3e2.8f66

interface vlan 10
ip address : 10.10.10.1/24
mac address : f433.9295.bc05
interface id : dec (1177) = hex (0x499)

C9600-Silicon1-SVL#show platform hardware fed switch active fwd-asic insight sdk_object(0x817)

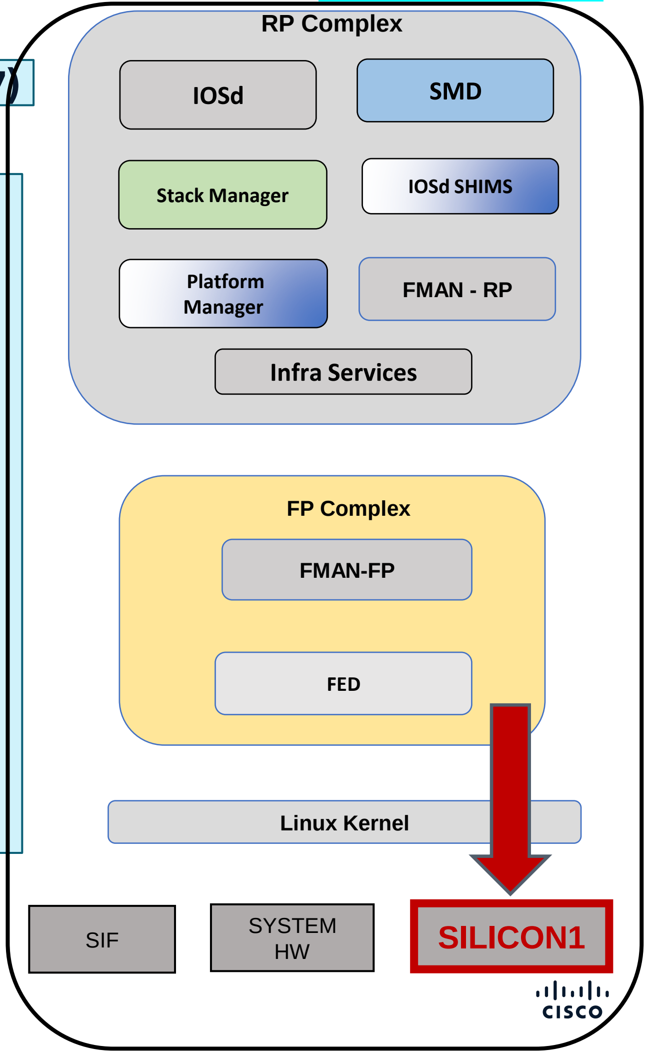
```
# svi_port object
active: true
all_vrf_ipv4_hosts:
~
~
- host_vrf: =oref('vrf', oid=0x2c1)
  ip_address:
    b_addr:
    - =0x1e
    - =0xa
    - =0xa
    - =0xa
    s_addr: =0xa0a0a1e
```

ip address : 10.10.10.30

```
mac_address:
  bytes:
  - =0x66
  - =0x8f
  - =0xe2
  - =0xb3
  - =0xb8
  - =0x0
  flat: =0xb8b3e28f66
  word:
  - =0x8f66
  - =0xb3e2
  - =0xb8
~
~
```

mac address : 00b8.b3e2.8f66

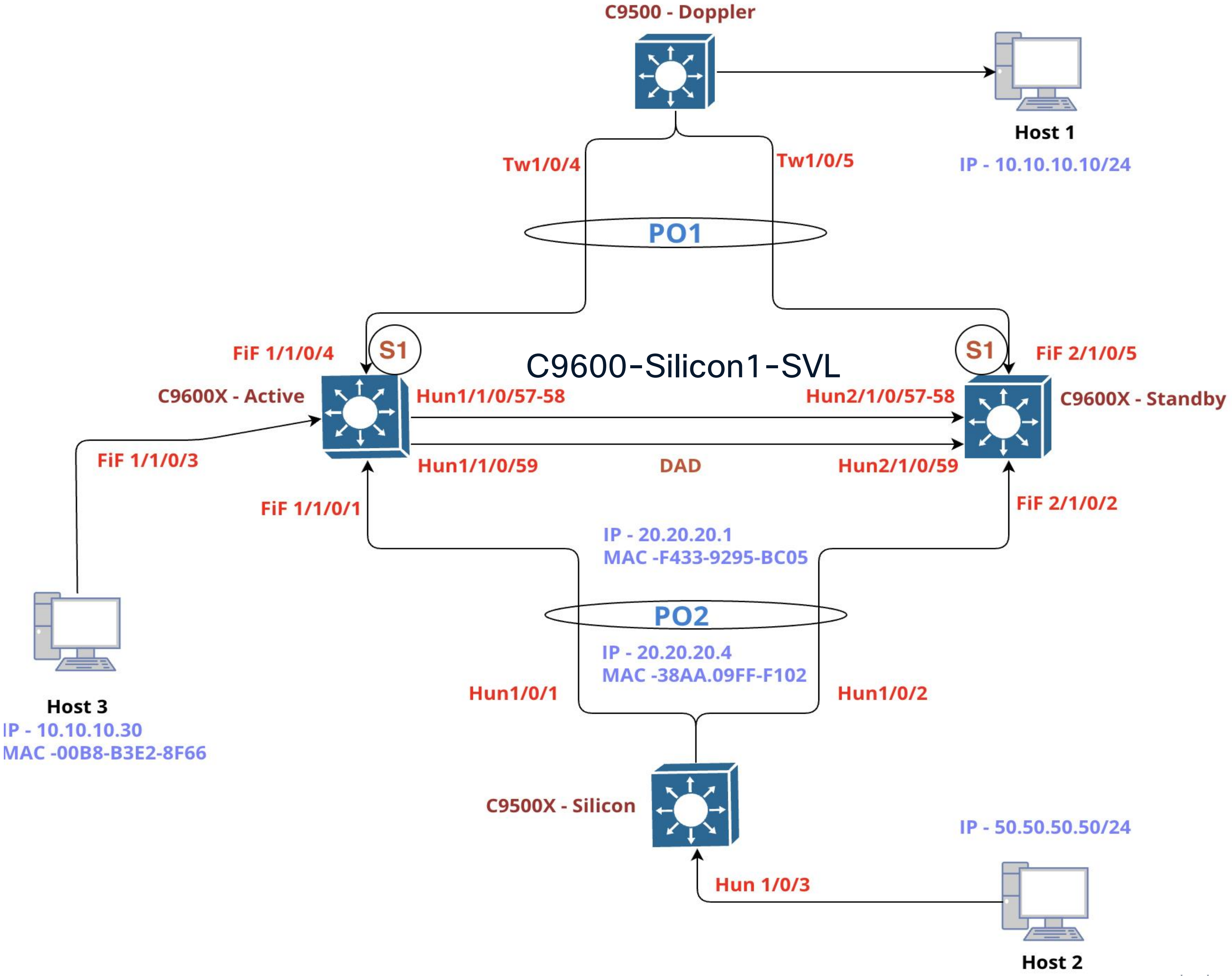
Above information will be present in "global ipv4 hosts table" as well as "all vrf ipv4 hosts table".



Indirect Route via Routing Protocol Programming

Silicon ASIC

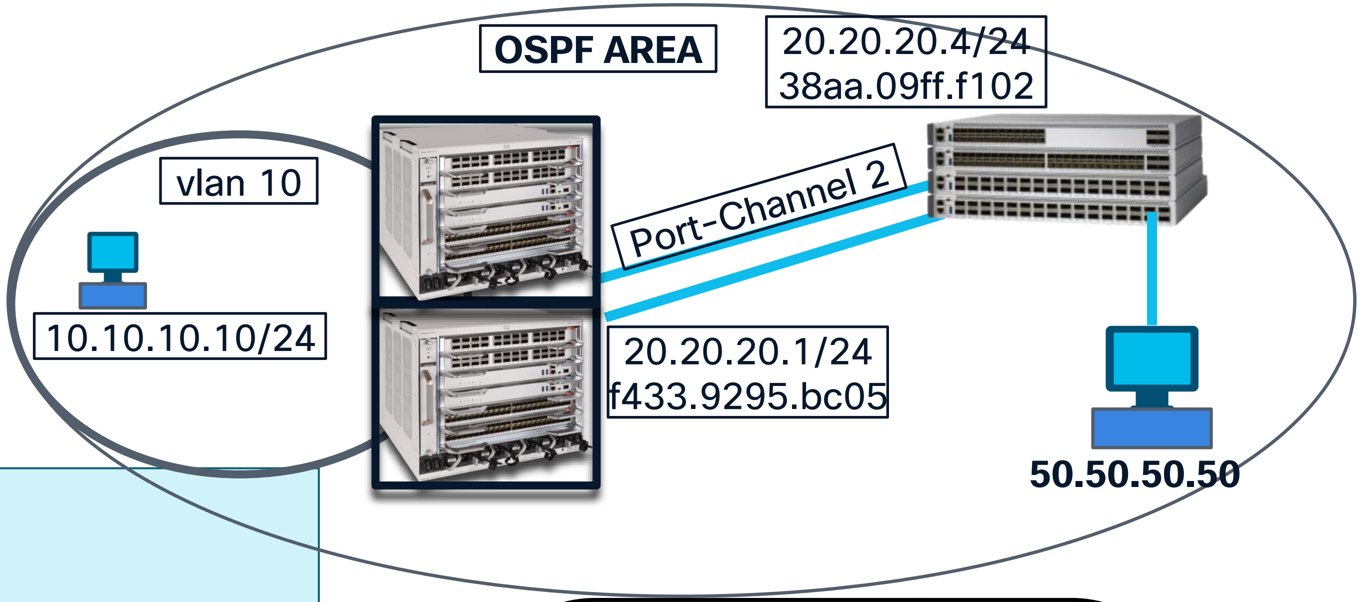
Lab Topology



Silicon ONE L3 Forwarding

Indirect Route : via Routing Protocol

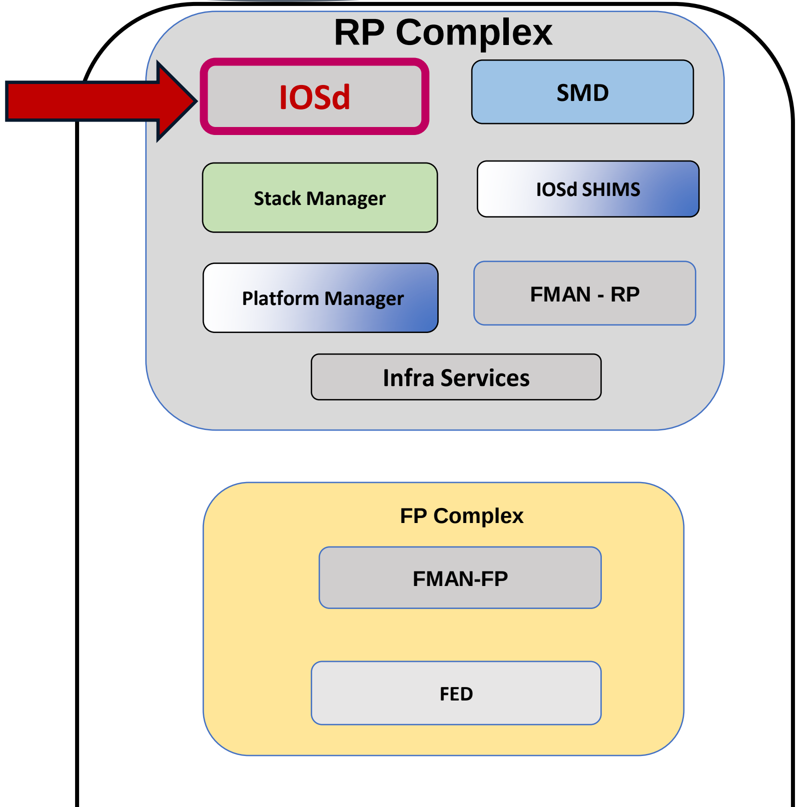
IOSd



C9600-Silicon1-SVL#show ip route 50.50.50.50
Routing entry for **50.50.50.0/24**
Known via "**ospf 1**", distance 110, metric 2, type intra area
Last update from **20.20.20.4 on Port-channel2**, 3d23h ago
Routing Descriptor Blocks:
* **20.20.20.4, from 4.4.4.4, 3d23h ago, via Port-channel2**
Route metric is 2, traffic share count is 1

C9600-Silicon1-SVL#show ip cef 50.50.50.50
50.50.50.0/24
nexthop **20.20.20.4** Port-channel2

C9600-Silicon1-SVL#show ip cef exact-route 10.10.10.10 50.50.50.50
10.10.10.10 -> 50.50.50.50 =>IP adj out of Port-channel2, addr 20.20.20.4

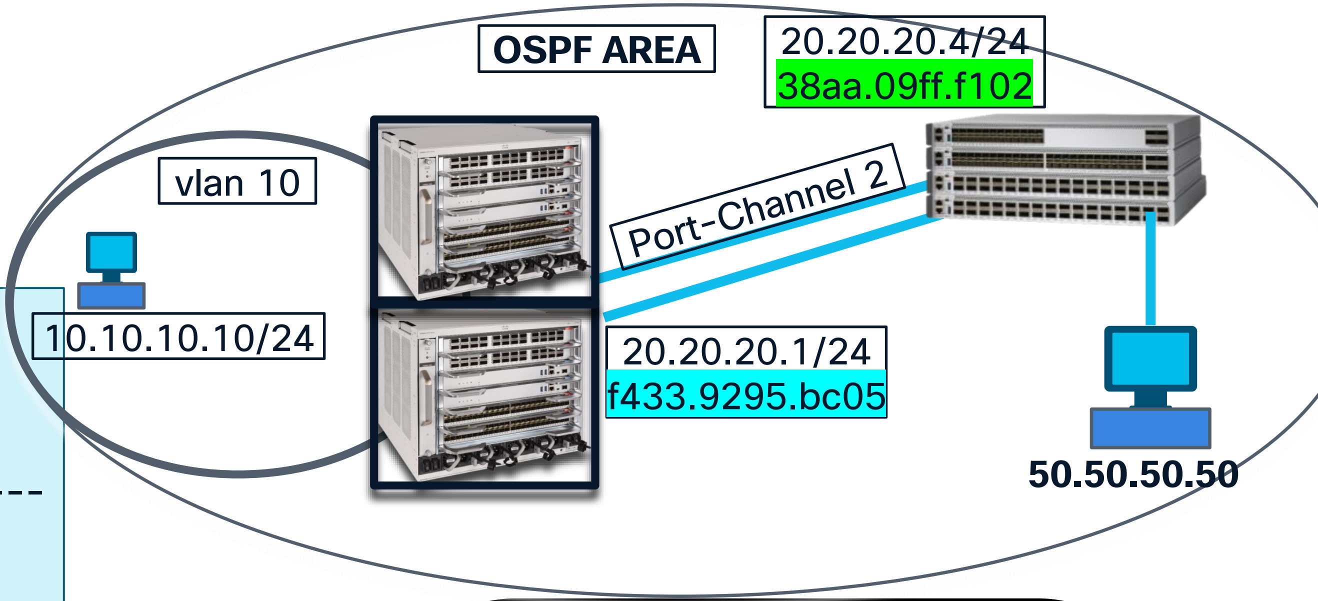


Silicon ONE L3 Forwarding

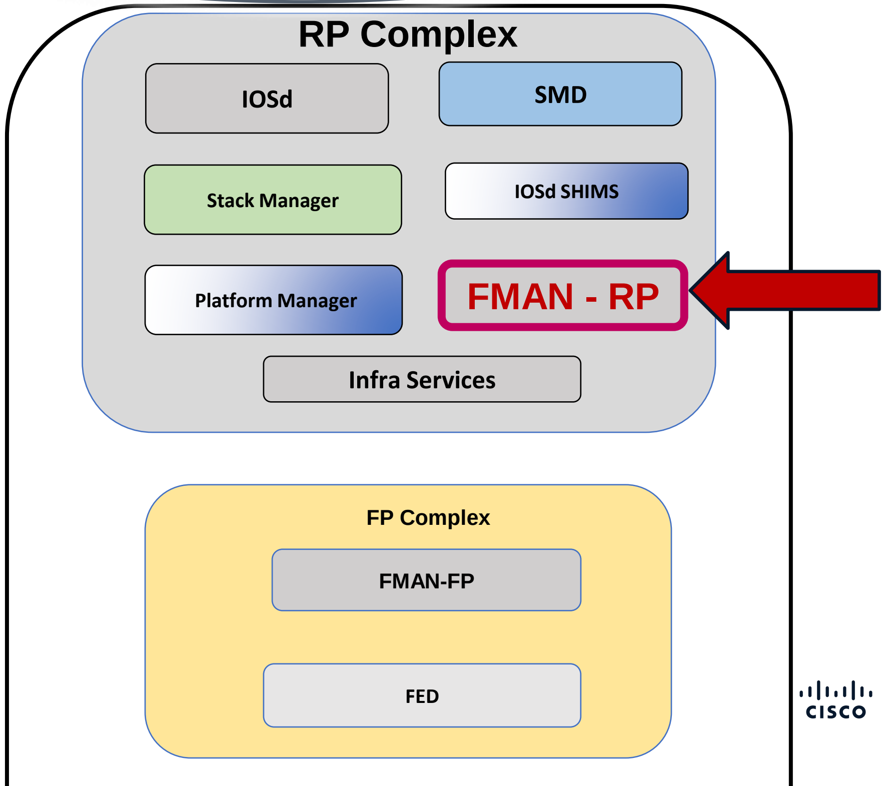
Indirect Route : via Routing Protocol

FMAN-RP

```
C9600-Silicon1-SVL#show  platform software ip switch active R0
cef prefix 50.50.50.0/24
Forwarding Table
Prefix/Len              Next Object          Index
-----------------------------------------------------------
-
50.50.50.0/24      OBJ_ADJACENCY    0x24
```




```
C9600-Silicon1-SVL#show  platform software adjacency switch active R0 index 0x24
Number of adjacency objects: 12

Adjacency id: 0x24 (36)
  Interface: Port-channel2, IF index: 1174, Link Type: MCP_LINK_IP
  Encap: 38:aa:9:ff:f1:2:f4:33:92:95:bc:5:8:0
  Encap Length: 14, Encap Type: MCP_ET_ARPA, MTU: 1500
  Flags: no-l3-inject
  Incomplete behavior type: None
  Fixup: unknown
  Fixup_Flags_2: unknown
 Nexthop addr: 20.20.20.4
  IP FRR MCP_ADJ_IPFRR_NONE 0
  OM handle: 0x34804c61e0
```



Silicon ONE L3 Forwarding

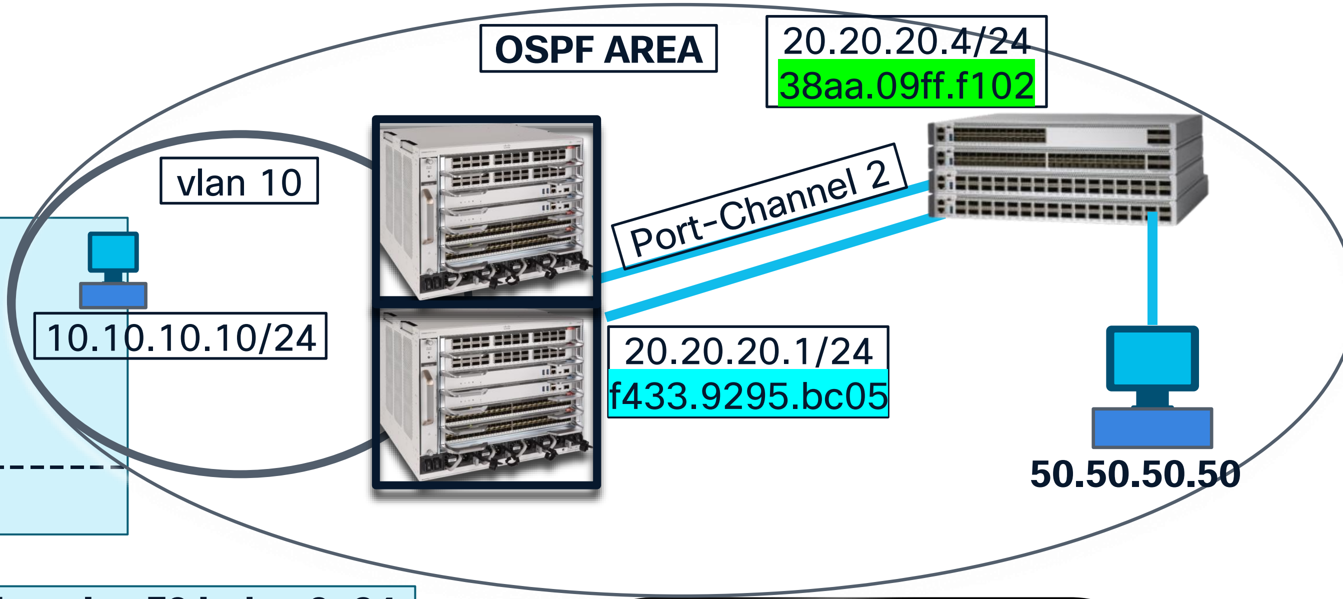
Indirect Route : via Routing Protocol

FMAN-FP

C9600-Silicon1-SVL#show platform software ip switch active F0 cef prefix 50.50.50.0/24

Forwarding Table

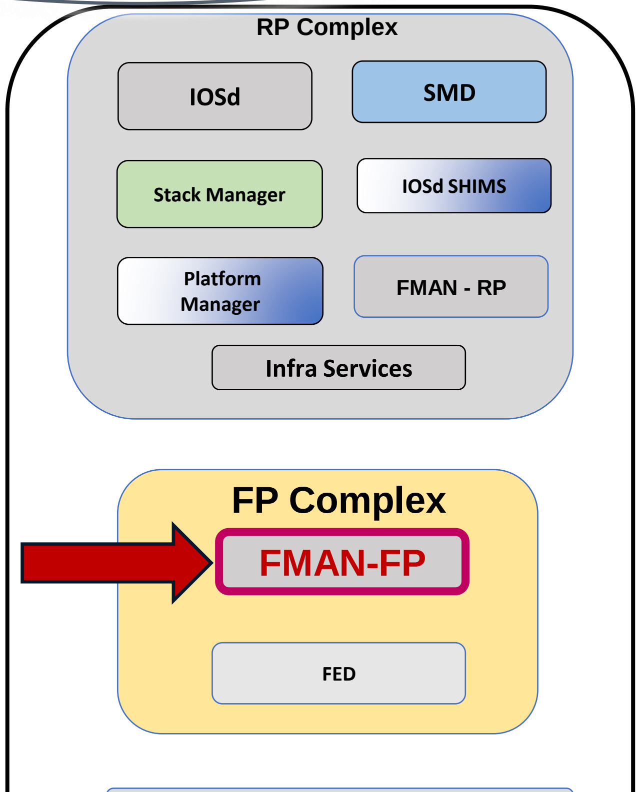
Prefix/Len	Next Object	Index
50.50.50.0/24	**OBJ_ADJACENCY**	**0x24**



C9600-Silicon1-SVL#show platform software adjacency switch active F0 index 0x24

Number of adjacency objects: 13

Adjacency id: 0x24 (36)
Interface: **Port-channel2**, **IF index: 1174**, Link Type: MCP_LINK_IP
Encap: **38:aa:9:ff:f1:2**:**f4:33:92:95:bc:5**:8:0
Encap Length: 14, Encap Type: MCP_ET_ARPA, MTU: 1500
Flags: no-l3-inject
Incomplete behavior type: None
Fixup: unknown
Fixup_Flags_2: unknown
Nexthop addr: 20.20.20.4
IP FRR MCP_ADJ_IPFRR_NONE 0
aom id: 5817, HW handle: (nil) (**created**)



Silicon ONE L3 Forwarding

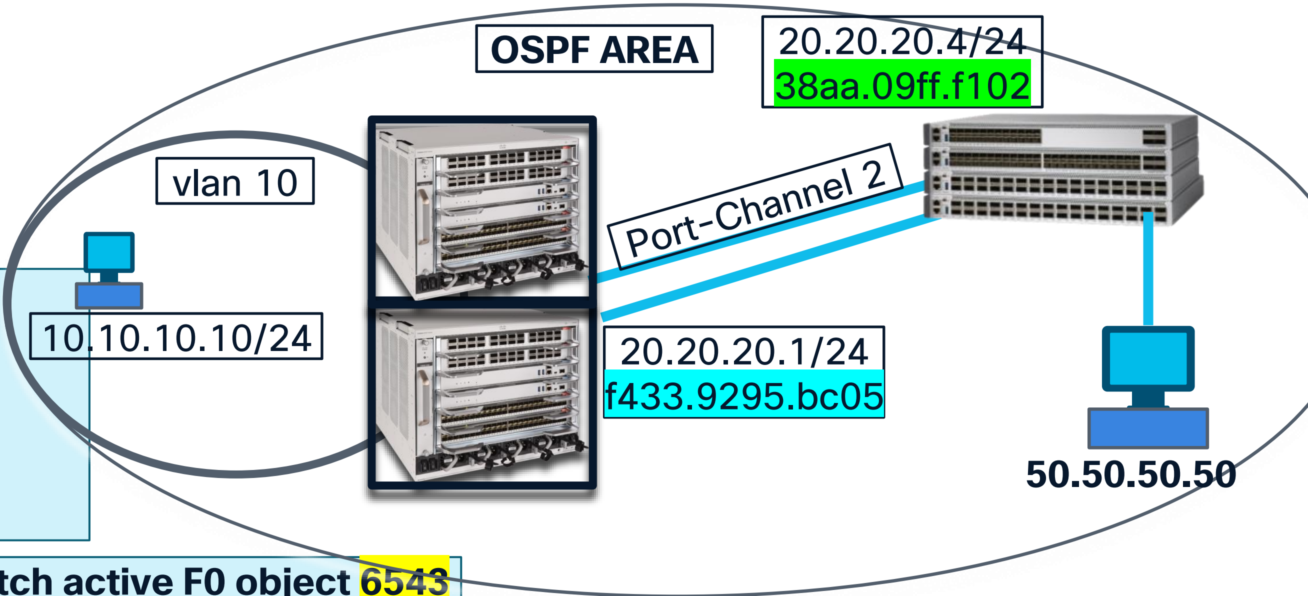
Indirect Route : via Routing Protocol

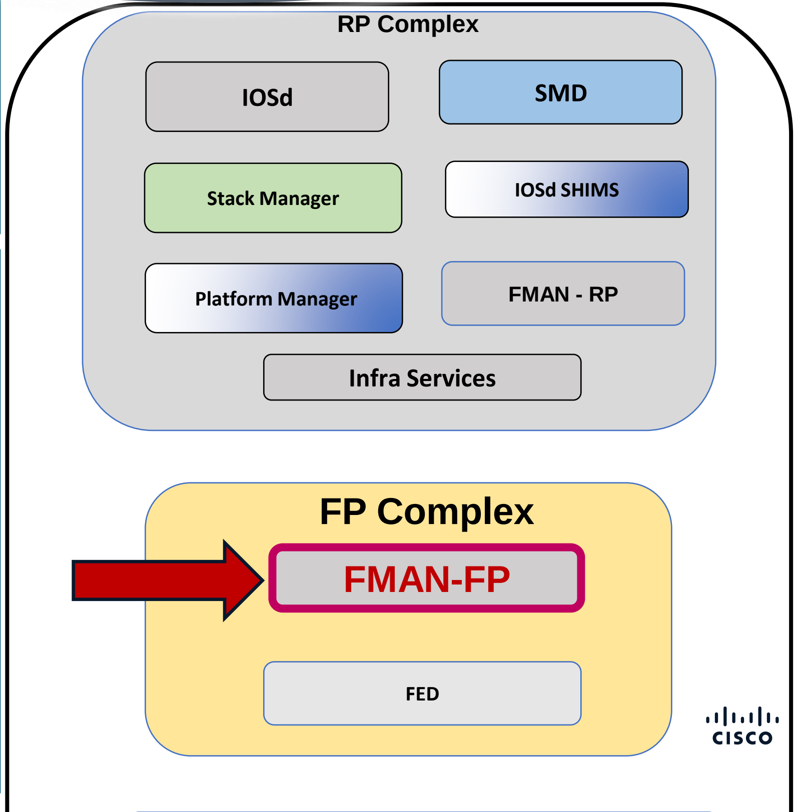
FMAN-FP

```
C9600-Silicon1-SVL#show  platform software ip switch active
F0 cef prefix 50.50.50.0/24 detail
Forwarding Table
50.50.50.0/24 -> OBJ_ADJACENCY (0x24), urpf: 38
Prefix Flags: unknown
aom id: 6543, HW handle: (nil) (created)
```

```
C9600-Silicon1-SVL#show  platform software object-manager switch active F0 object 6543
Object identifier: 6543
  Description: PREFIX 50.50.50.0/24 (Table id 0)
  Obj type id: 72
  Obj type: route-pfx
  Status: Done, Epoch: 0, Client data: 0x10d70598
```

```
~con1-SVL#show  platform software object-manager switch active F0 object 6543 parents
Object identifier: 151
  Description: ipv4 table 0 (Default), vrf id 0
  Status: Done

Object identifier: 5817
  Description: adj 0x24, Flags None
  Status: Done

Object identifier: 5821
  Description: uRPF-list(hdl=0x00000026)
  Status: Done
```





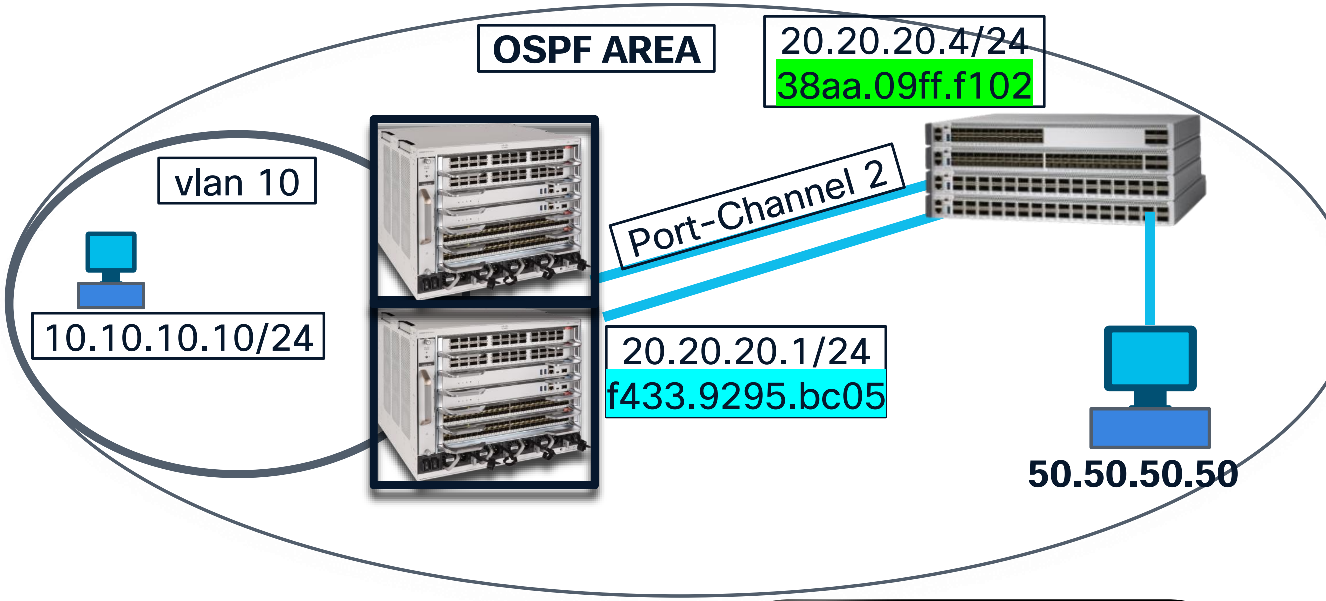
Silicon ONE L3 Forwarding

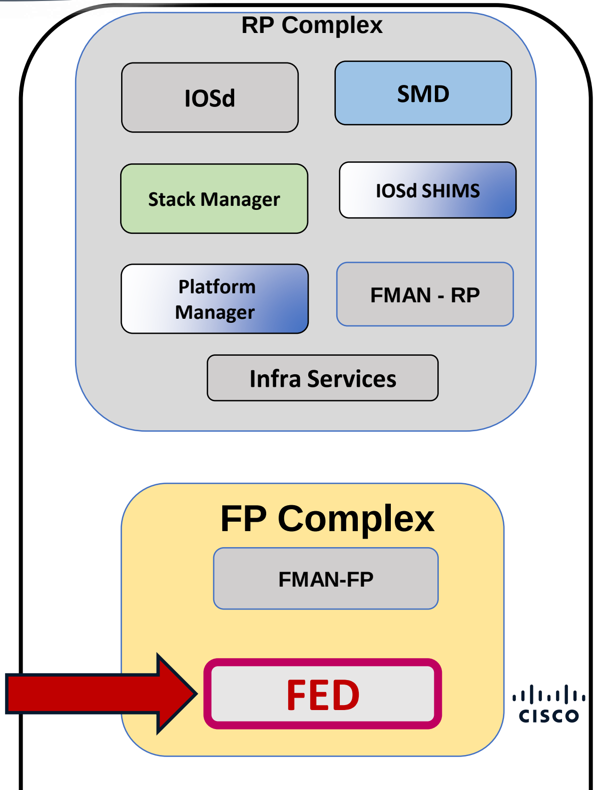
Directly Connected Routes

FED – Forwarding Engine Driver

```
C9600-Silicon1-SVL#show  platform software fed switch active
ifm mappings etherchannel
Mappings Table
Chan        Interface                        IF_ID
-------------------------------------------------------------
1           Port-channel1                    0x00000494
2           Port-channel2                    0x00000496
241         Port-channel241                  0x0000004b
```

```
C9600-Silicon1-SVL#show  platform software fed switch active ip route 50.50.50.0/24
IPV4ROUTE_ID:id:0x63a010d70598 nobj:(IPNEXTHOP_ID,0x24) 50.50.50.0/24 tblid:0 DA:0
  State:(0) success
  NPD: device:0 lspa_rec:0 api_type:route(3)
  State:(0) success
  SDK: is_host:0 l3_dest:0x5a4f8b751000 l3_dest:la_l3_fec_impl_base(oid=1745) vrf(gid/oid):0/0x2c1
 ADJ:objid:0x24 (IPv4: 20.20.20.4)  nh_type:NHADJ_NORMAL iif_id:0x496 ether_type:0x8 #child:6
   srcmac:f433.9295.bc05 dstmac:38aa.09ff.f102
   State:(0) success
  NPD: fec_oid:1745 was_nor_nh:1 cr_def:0 stale:0 l3port_valid:1
  NPD: device:0 nh_gid/oid:9/0x6cf old_gid/oid:0/0x0 parent_oid:2080
  State:(0) success
  SDK: cla_nhtype:0
```

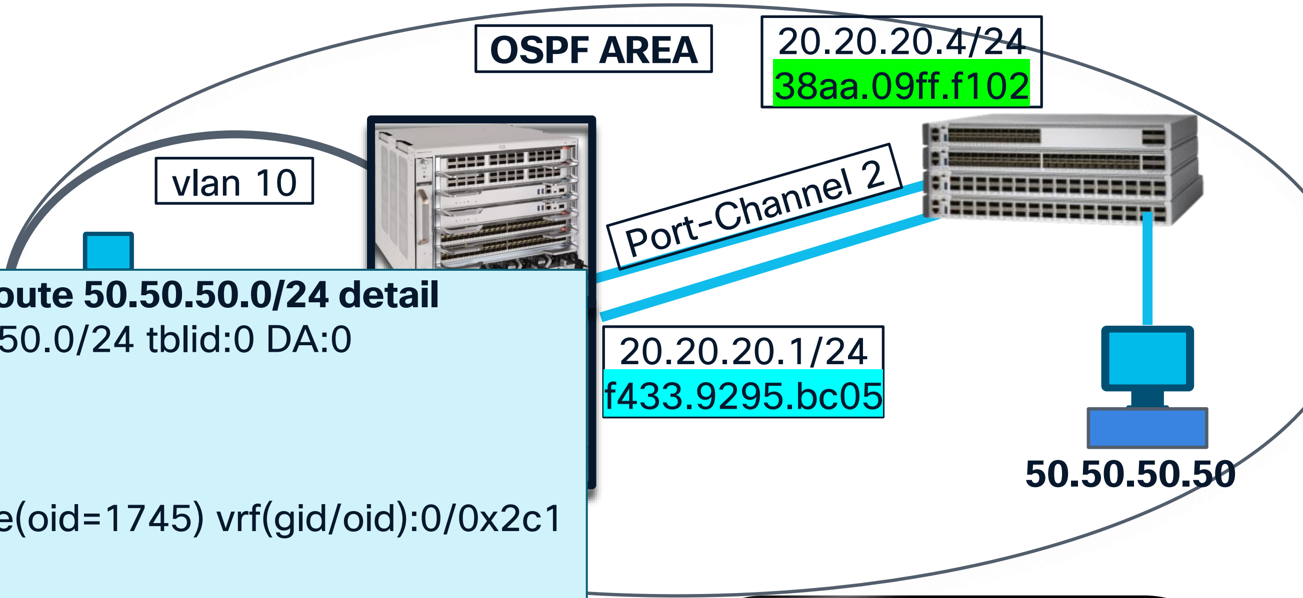




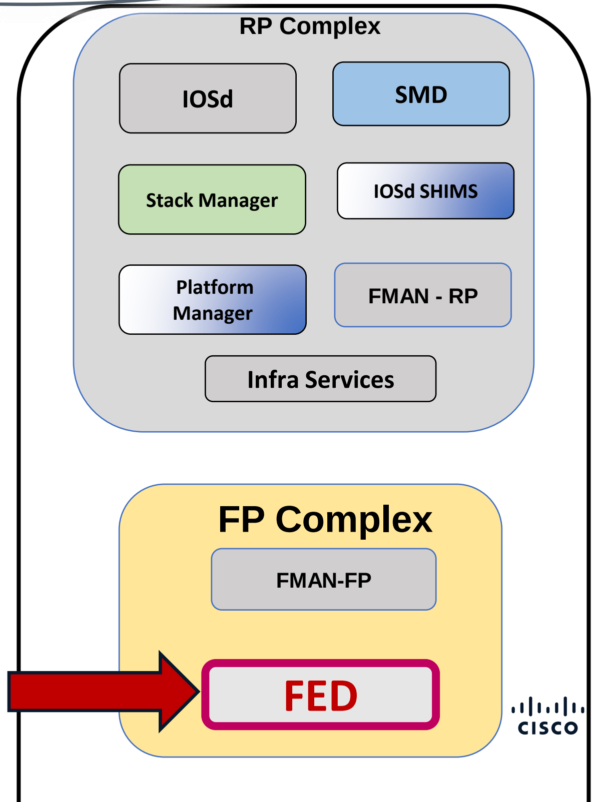
Silicon ONE L3 Forwarding

Indirect Route : via Routing Protocol

FED – Forwarding Engine Driver



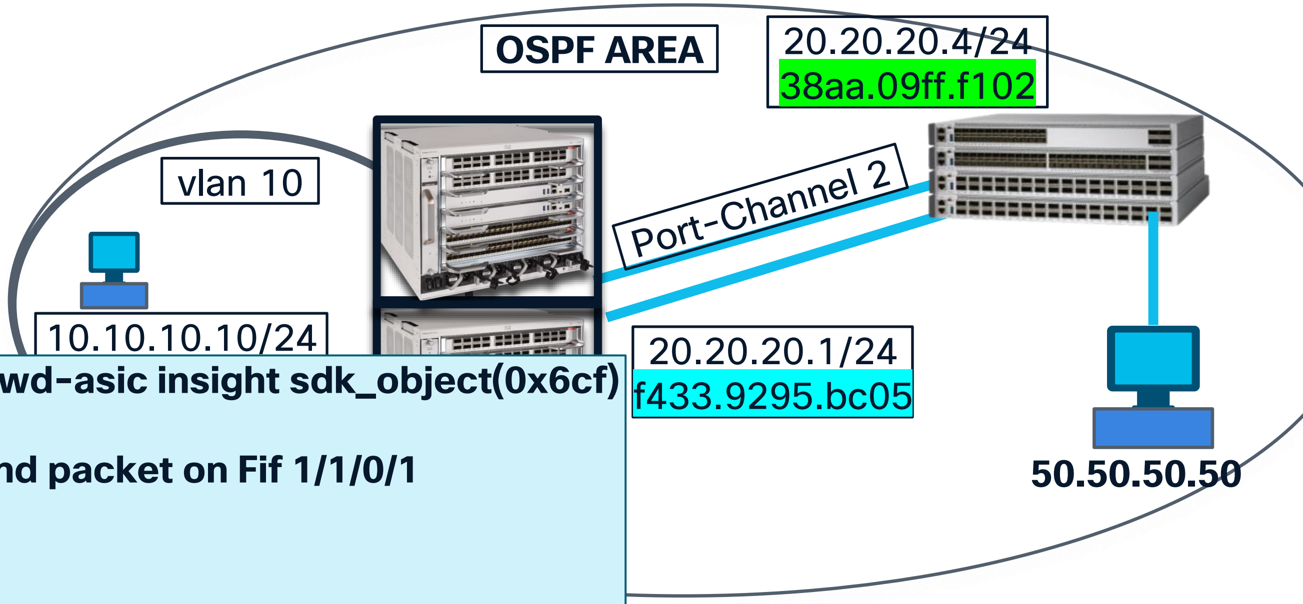

```
C9600-Silicon1-SVL#show  platform software fed switch active ip route 50.50.50.0/24 detail
IPV4ROUTE_ID:id:0x63a010d70598 nobj:(IPNEXTHOP_ID,0x24) 50.50.50.0/24 tblid:0 DA:0
    State:(0) success
   NPD: device:0 lspa_rec:0 api_type:route(3)
   State:(0) success
   SDK: is_host:0 l3_dest:0x5a4f8b751000 l3_dest:la_l3_fec_impl_base(oid=1745) vrf(gid/oid):0/0x2c1

 SDK Object Details:
   NPD: SDK oid 0x6d1, devid:0, asic:0
 ~
  Events Log
   [2026/01/07 17:27:44.926] NPI IPROUTE CREATE
   [2026/01/07 17:27:44.927] NPD IPROUTE EV ADD IPV4 ROUTE

ADJ:objid:0x24 (IPv4: 20.20.20.4)  nh_type:NHADJ_NORMAL iif_id:0x496 ether_type:0x8 #child:6
    srcmac:f433.9295.bc05 dstmac:38aa.09ff.f102
    State:(0) success
~
        Next Hop Object:
         sdk nexthop oid:0x6cf,dev:0, gid:0x9
         macaddr:38aa.09ff.f102, nh_type:normal(0)
         sdk outgoing port oid 0x820, porttype:l3ac(68)
```



Silicon ONE L3 Forwarding

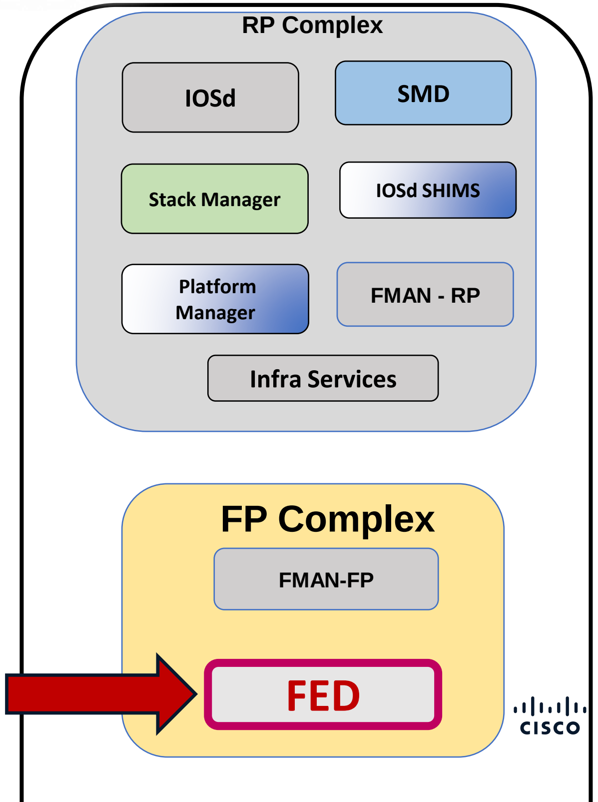
Indirect Route : via Routing Protocol

SDK : Silicon1




```
C9600-Silicon1-SVL#show  platform hardware fed switch active fwd-asic insight sdk_object(0x6cf)
# next_hop object
cookie: Fif1/1/0/1                    <<<<<< Active Switch will send packet on Fif 1/1/0/1
device: =oref('device', oid=0x0)
gid: =0x9
mac:
  bytes:
  - =0x2
  - =0xf1
  - =0xff
  - =0x9
  - =0xaa
  - =0x38
  flat: =0x38aa09fff102               <<<<<< Next Hop Mac
  word:
  - =0xf102
  - =0x9ff
  - =0x38aa
nh_type: =0x0
oid: =0x6cf
router_port: =oref('l3_ac_port', oid=0x820)
```

Port-Channel 2 = Fif 1/1/0/1 + Fif 2/1/0/2



Silicon ONE L3 Forwarding

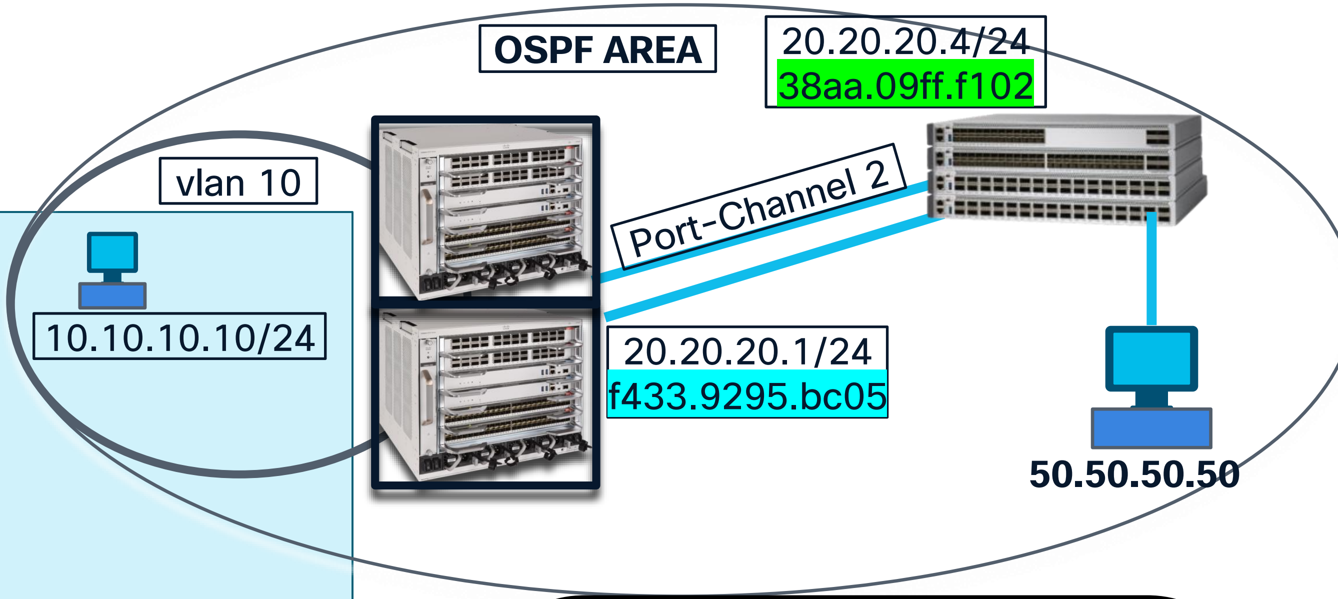
Indirect Route : via Routing Protocol

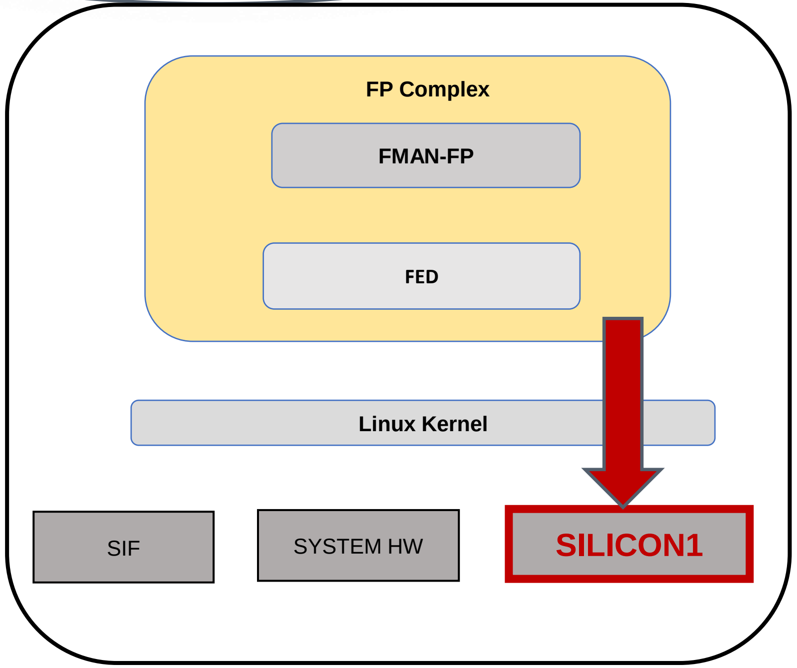
SDK – Forward ASIC

```
C9600-Silicon1-SVL#show  platform hardware fed switch active
fwd-asic insight sdk_object(0x820)

# l3_ac_port object
active: true
all_vrf_ipv4_hosts:
- host_vrf: =oref('vrf', oid=0x2c1)
  ip_address:
    b_addr:
    - =0x4
    - =0x14
    - =0x14
    - =0x14
  s_addr: =0x14141404
  mac_address:
    bytes:
    - =0x2
    - =0xf1
    - =0xff
    - =0x9
    - =0xaa
    - =0x38
  flat: =0x38aa09fff102
```

Next hop ip address : 20.20.20.4

Next hop mac address : 38aa.09ff.f102





Debug / Traces

1. Elevate Trace Level
2. Perform Action (clear prefix, add subnet, redistribute, etc..)
3. Collect Traces

set plat soft trace fed active l3_adj debug
set plat soft trace fed active l3_fib debug
set plat soft trace fed active asic_l3u debug
set plat soft trace fed active l3_intf debug
set plat soft trace fed active fed3_l3u_npd debug

show plat soft trace level fed active | in fed3_td|l3_adj|l3_fib|asic_l3u|l3_intf

asic_l3u Debug
fed3_l3u_npd Debug
l3_adj Debug
l3_fib Debug
l3_intf Debug

Case Study

Destination prefix points to incorrect next-hop VLAN, resulting in traffic being blackholed on c9600.

- Destination prefix 8.8.8.2
- Next-hop to reach this destination prefix is in VLAN 815
- However, in the impacted state, it pointed to VLAN 318

```
SiliconOne#show platform software fed active ip route vrf TEST 8.8.8.2/32
IPV4ROUTE_ID:id:0x5642551037a8 nobj:(IPNEXTHOP_ID,129) 8.8.8.2/32 tblid:3 DA:0
   NPD: lspa_rec:0 api_type:3
   SDK: is_host:0 l3_dest:0x7fec65345ac0 l3_dest:la_l3_fec_impl(oid=2804) vrf(gid/oid):3/590
  ADJ:objid:129 nh_type:NHADJ_NORMAL iif_id:0x482 ether_type:0x8 child:17
      srcmac:6cb2.ae4a.cf85 dstmac:a478.0638.e8d7
     NPD: nh_gid/oid:16/2803 old_gid/oid:0/0 parent_oid:2721 ← parent_oid is 2721 and 2803 is SDK object id
         fec_oid:2804 was_nor_nh:1 cr_def:0 stale:0 l3port_valid:1
     SDK: cla_nhtype:0
```

2721 = 0xaa1 in Hex

Case Study

```
SiliconOne#show platform hardware fed active fwd-asic insight sdk_object(0xaa1)
svi_port object
egress_vlan_tag:
- tci:
    fields:
      dei: =0x0
      pcp: =0x0
      vid: =0x13e   ← VID = VLAN ID in hex, so 0x13e to decimal is 318
    raw: =0x13e0
  tpid: =0x8100
- tci:
```

In the working state, the next-hop VLAN is 815

```
SiliconOne#show platform hardware fed active fwd-asic insight sdk_object(0xaa4) ← 0xaa4 in decimal is 2724 in decimal
# svi_port object
- tci:
    fields:
      dei: =0x0
      pcp: =0x0
      vid: =0x32f ← VID == VLAN in hex 0x32f, so convert 0x32f to decimal is 815
    raw: =0x32f0
```

Case Study

```
SiliconOne#show platform software fed active ifm interfaces svi
Interface                  IF_ID              State
------------------------------------------------------------------------
Vlan1                      0x0000046a         Ready
Vlan317                      0x0000047e         Ready
Vlan318                      0x0000047f         Ready  ← Wrong next hop VLAN
Vlan319                      0x00000480         Ready
Vlan320                      0x00000481         Ready
Vlan815                      0x00000482         Ready  ← Correct next hop VLAN
Vlan816                      0x00000483         Ready
Vlan818                      0x00000484         Ready
Vlan916                      0x00000485         Ready
Vlan918                      0x00000486         Ready
Vlan1022                      0x00000487         Ready
```

Case Study

```
SiliconOne#show platform software fed active ifm if-id 0x0000047f
Interface IF_ID          : 0x000000000000047f
Interface Name           : Vlan318
Interface Block Pointer : 0x7fec64ad5528
Interface Block State   : Ready
Interface State          : Enabled
Interface Status         : UPD,
Interface Ref-Cnt        : 1
Interface Type           : SVI
        Vlan id          : 318 ← VLAN 318
        SNMP IF Index  : 123
        IPv4 MTU         : 9100
 --- <Snip> ----
No Feature Reference count Present
IFM Feature Sub block information
        Port L3 Subblock
                VRF ID .................. [2]
                IPv4 Routing Enabled .... [Yes]
                IPv6 Routing Enabled .... [No]
                MPLS Enabled ............ [No]
                Pimv4 Enabled ........... [Yes]
                Pimv6 Enabled ........... [No]
                IPv4 MTU ................ [9118]
                IPv6 MTU ................ [18]
                L3 srv port gid ......... [4190]
                L3 srv port oid ......... [2721] ← Wrong parent_oid
```

Case Study

```
SiliconOne#show platform software fed active ifm if-id 0x00000482
Interface IF_ID         : 0x0000000000000482
Interface Name          : Vlan815
Interface Block Pointer : 0x7fec64ae7fe8
Interface Block State   : Ready
Interface State         : Enabled
Interface Status        : UPD,
Interface Ref-Cnt       : 1
Interface Type          : SVI
        Vlan id         : 815  ← VLAN 815
        SNMP IF Index   : 126
        IPv4 MTU        : 9100
        IPv6 MTU        : 0
        IPv4 VRF ID     : 0x3
        IPv6 VRF ID     : 0x65535
        Protocol flags : 0x0005 [ ipv4 pim_ipv4 ]
        Misc flags      : 0x0045 [ ipv4 mcast_v4 --- ]
        ICMPv4 flags    : 0x01 [ unreachable ]

            IPv4 MTU ................ [9118]
            IPv6 MTU ................ [18]
            L3 srv port gid ......... [4191]
            L3 srv port oid ......... [2724] ← correct parent_oid
Events Log
```

Quiz

What is the role of the Software Development Kit (SDK) in the context of Cisco Catalyst 9000 Series switches?

1. Refer the exhibit, "All the HW handles empty. It means the MAC address is not programmed in the HW layer." With respect to Silicon One switch, is the statement correct?

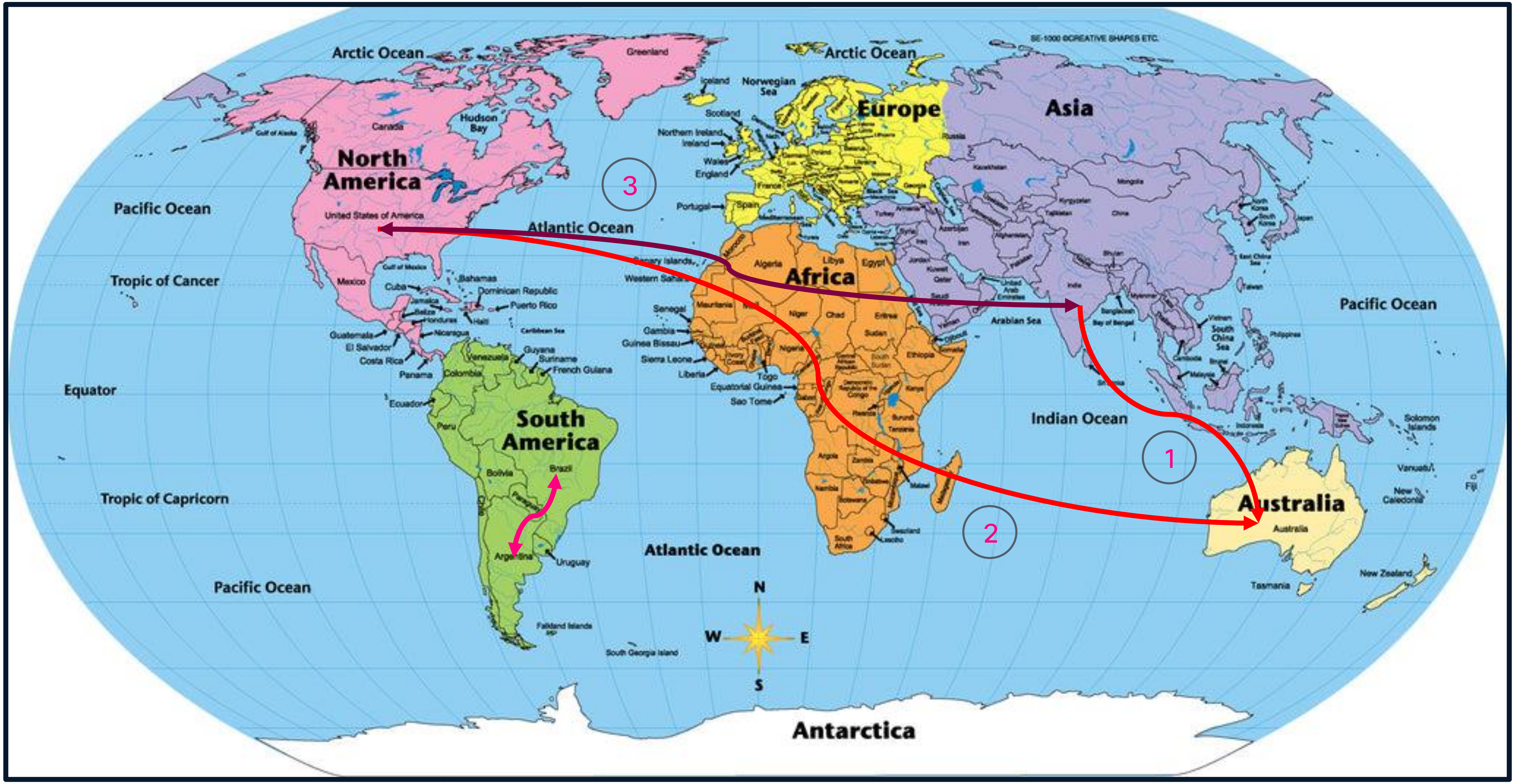
Under SDK output while checking port programming if we observe mac_learning_mode: =0x2 then what this mac learning state indicates?

Silicon 1 is The FUTURE

Enterprise
Campus

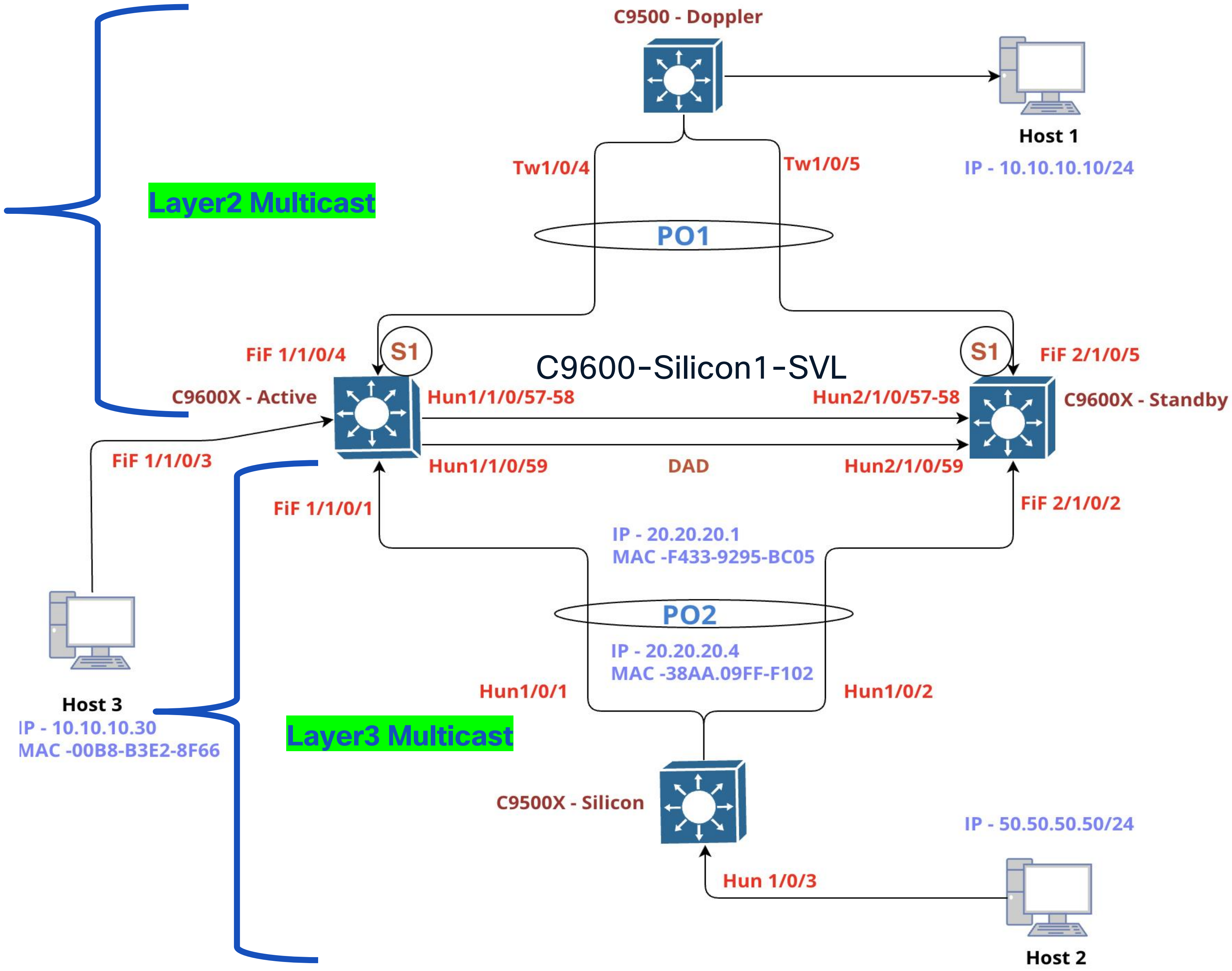
This session will make you FUTURE aware.

Silicon ONE
Multicast Forwarding


Arctic Ocean
Greenland
Arctic Ocean
Europe
Asia
North America
Canada
Hudson Bay
United States of America
Pacific Ocean
Atlantic Ocean
Africa
Tropic of Cancer
Mexico
Pacific Ocean
Equator
South America
Indian Ocean
Australia
Tropic of Capricorn
Atlantic Ocean
Pacific Ocean
Antarctica
N
W
E
S
1
2
3

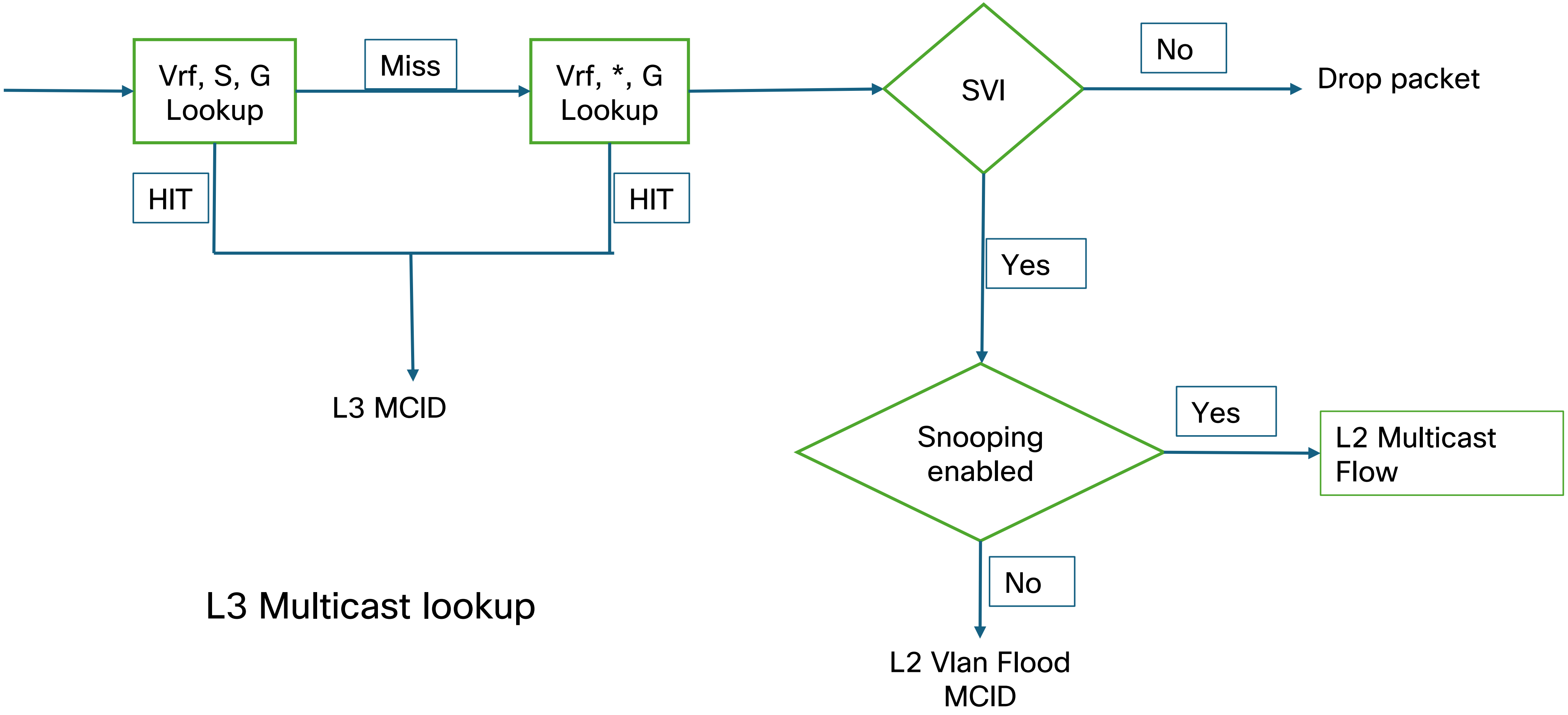
Silicon ASIC

Lab Topology



Multicast Packet Forwarding Paradigm

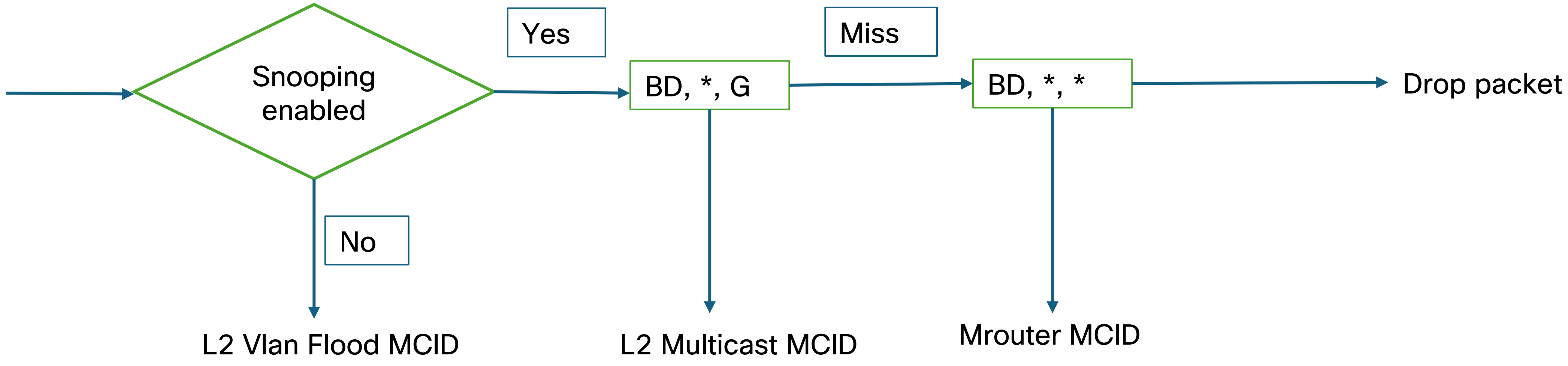
Ingress Pipeline



L3 Multicast lookup

MCID = Multicast ID

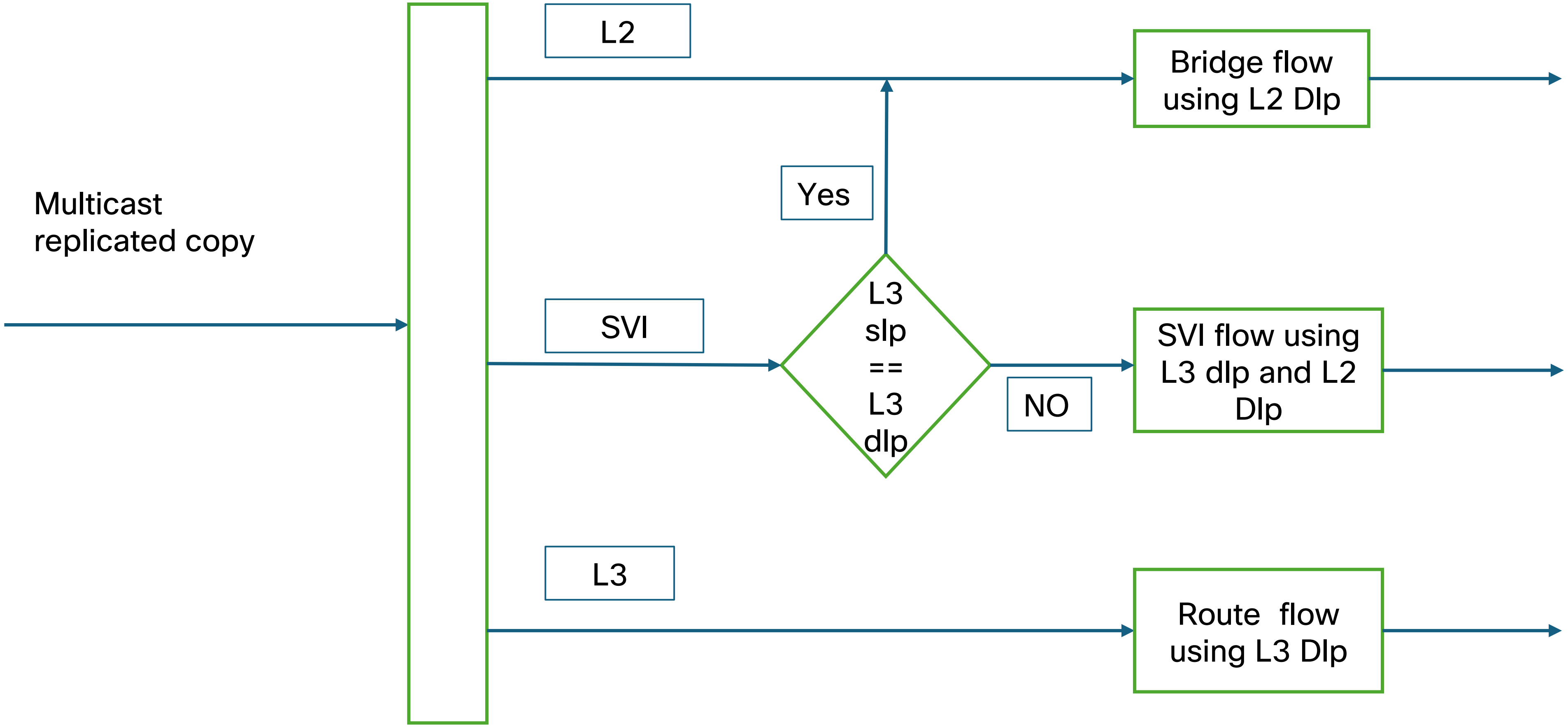
Ingress Pipeline



L2 Multicast lookup

MCID = Multicast ID

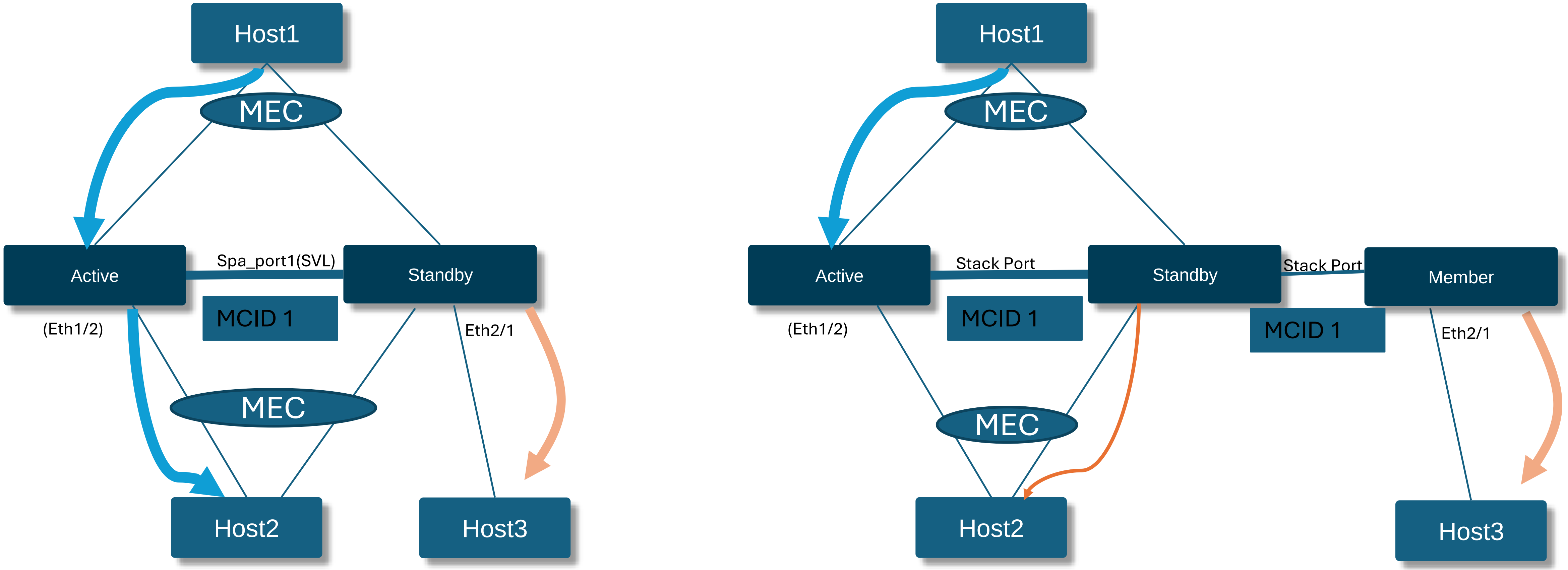
Egress Pipeline



SIp = Source Lookup

DIp = Destination Lookup

Multicast Stacking and SVL packet flow



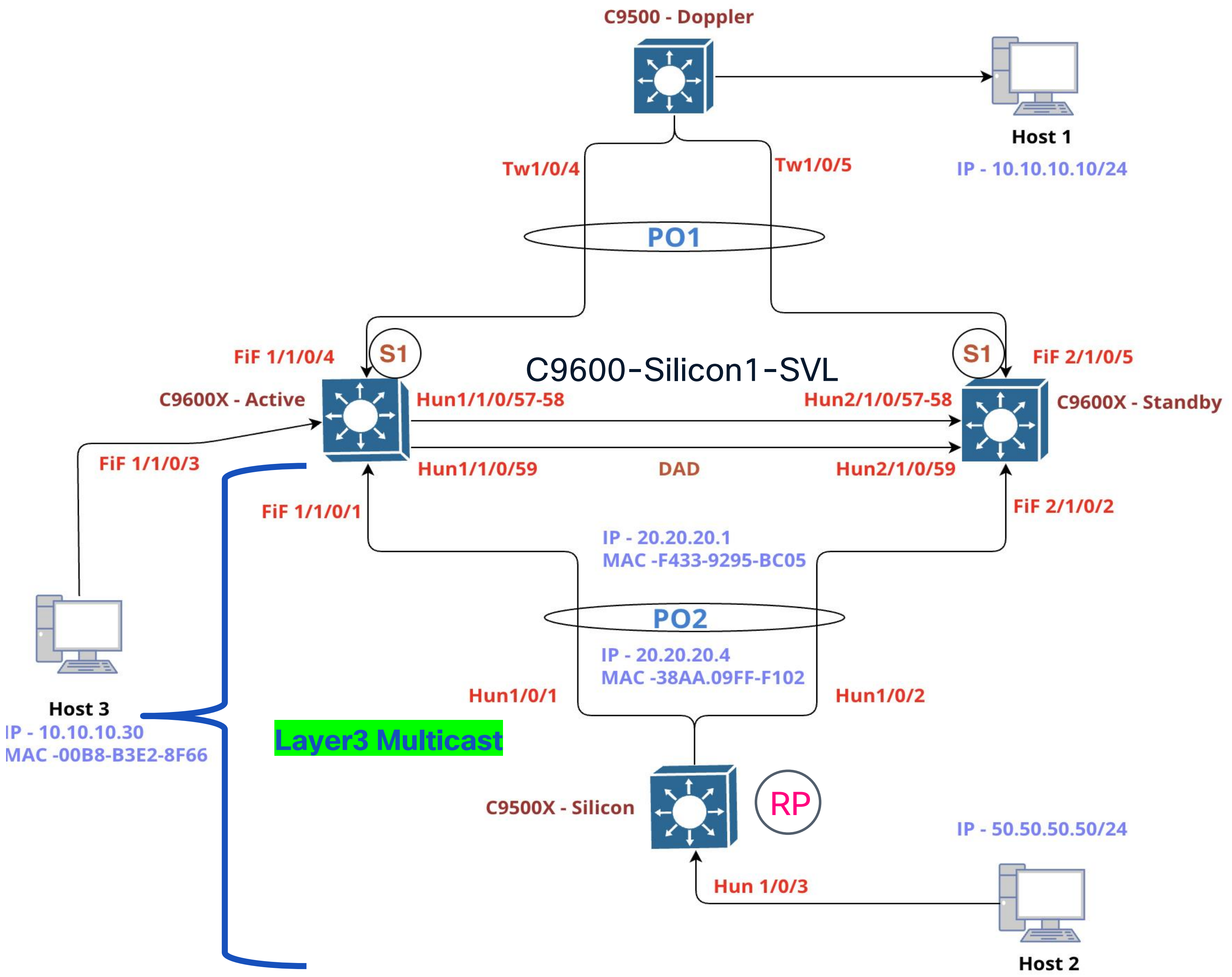
9610 SVL Packet Flow for Multicast

9350 Stacking Packet Flow for Multicast

Silicon ONE L3 Multicast

Silicon ASIC

Lab Topology



Troubleshooting L3 Multicast on Silicon ONE

To debug group specific issue, it's highly recommended to use show tech-support option

- It collects all the details specific to all layers in the switch.
- IOSD, FMAN-RP, FMAN-FP, FED, SDK outputs for specific groups are collected using a single cli.

show tech-support platform layer3 multicast group_ipaddr <grp ip> srcIp <src ip>
show tech-support platform layer3 multicast group_ipv6addr <grp ip> srcIp <src ip>

To collect trace archive

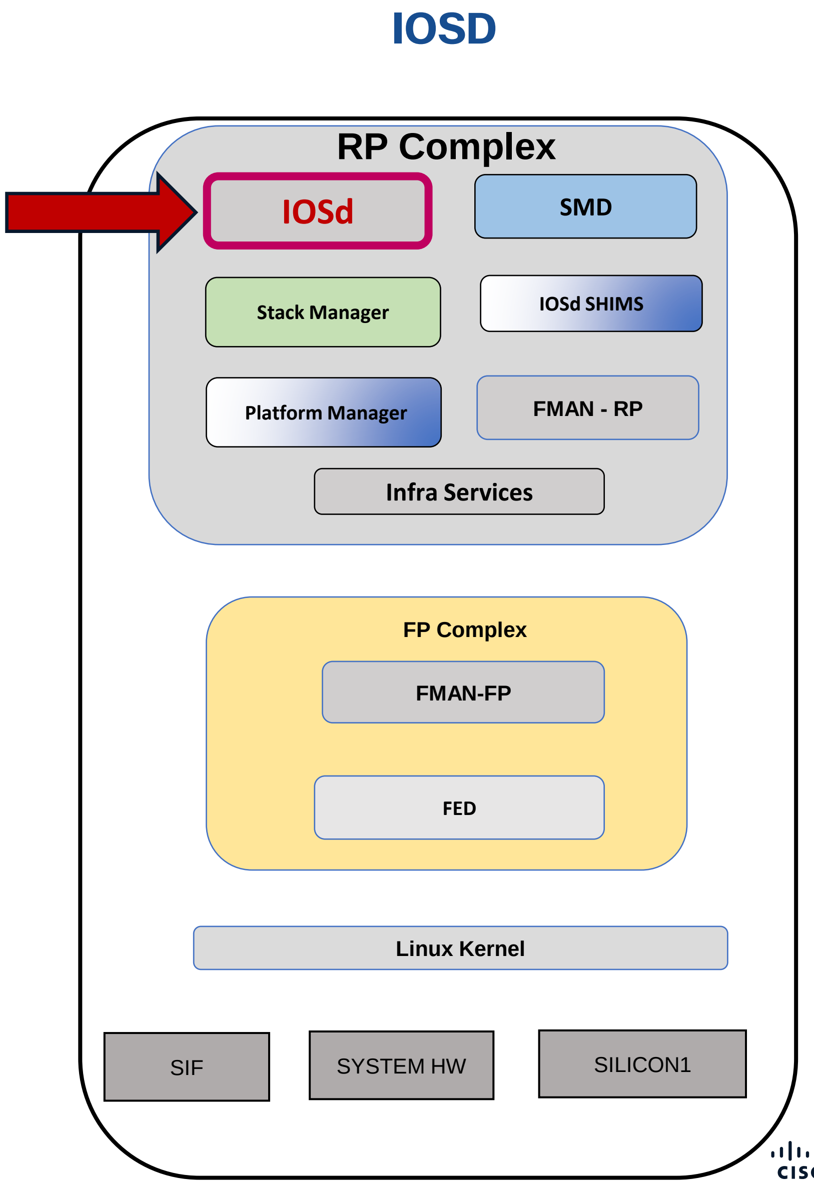
```
request platform software trace archive
```

Silicon ONE L3 Multicast

```
C9500-Silicon1#show ip mroute  239.255.0.1 50.50.50.50
~~
(50.50.50.50, 239.255.0.1), 00:02:34/00:03:02, flags: FT
  Incoming interface: Vlan50, RPF nbr 0.0.0.0
  Outgoing interface list:
    Port-channel2, Forward/Sparse, 00:02:34/00:03:24, flags:
```

```
C9500-Silicon1#show ip mfib  239.255.0.1 50.50.50.50
Forwarding Counts: Pkt Count/Pkts per second/Avg Pkt Size/Kbits per second
Other counts:      Total/RPF failed/Other drops
I/O Item Counts:   HW Pkt Count/FS Pkt Count/PS Pkt Count   Egress Rate in pps
Default
 (50.50.50.50,239.255.0.1) Flags: HW
   SW Forwarding: 1/0/100/0, Other: 0/0/0
   HW Forwarding:   74/0/118/0, Other: 0/0/0
   Vlan50 Flags: A
   Port-channel2 Flags: F NS
     Pkts: 0/0/1    Rate: 0 pps
```

```
C9500-Silicon1#show  ip cef exact-route 10.10.10.10 50.50.50.50
RPF information for ? (50.50.50.50)
  RPF interface: Vlan50
  RPF neighbor: ? (50.50.50.50) - directly connected
  RPF route/mask: 50.50.50.0/24
  RPF type: multicast (connected)
  Doing distance-preferred lookups across tables
  RPF topology: ipv4 multicast base
```

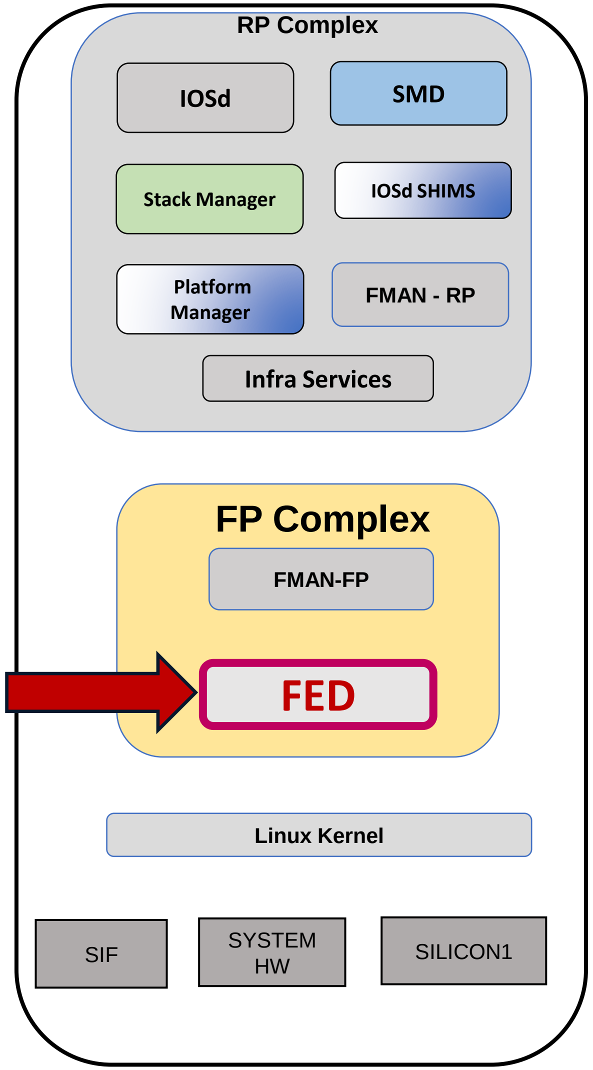


Silicon ONE L3 Multicast

FED – Forwarding Engine Driver

```
C9500-Silicon1#show platform software fed active ip mfib summary
Mfib v4 All Tables summary:
  Multicast Vrf Current Count                 : 1
  Mroute Current Count/Max Reached            : 3/4
  (*,G) Current Count                         : 2
  (S,G) Current Count                         : 1
  (*,G/m) RetryQ Count                        : 0
  (*,G) RetryQ Count                          : 0
  (S,G) RetryQ Count                          : 0
  Oifs Current Count/Max Reached              : 6/9
  Last used Mlist Urid                        : 0x1000000000000012
```

```
C9500-Silicon1#show platform software fed active oifset
L2M Fset Current Count / Max Reached            : 4 / 6
 L2M current vp_oif count / Max Reached         : 0 / 1
 L2M Fset Last Used Urid                        : 0x2000000000000006 (changes/sec:0 approx.)
 L3M Fset Current Count / Max Reached           : 3 / 4
 L3M current oif count /Max Reached             : 4 / 4
 L3M Fset Last Used Urid                        : 0x3000000000000016 (changes/sec:0 approx.)
 L3M Fset Oif Last Used Urid                    : 0x8000000000000019 (changes/sec:0 approx.)
 Oif_Obj Fset Current Count / Max Reached       : 2 / 2
```

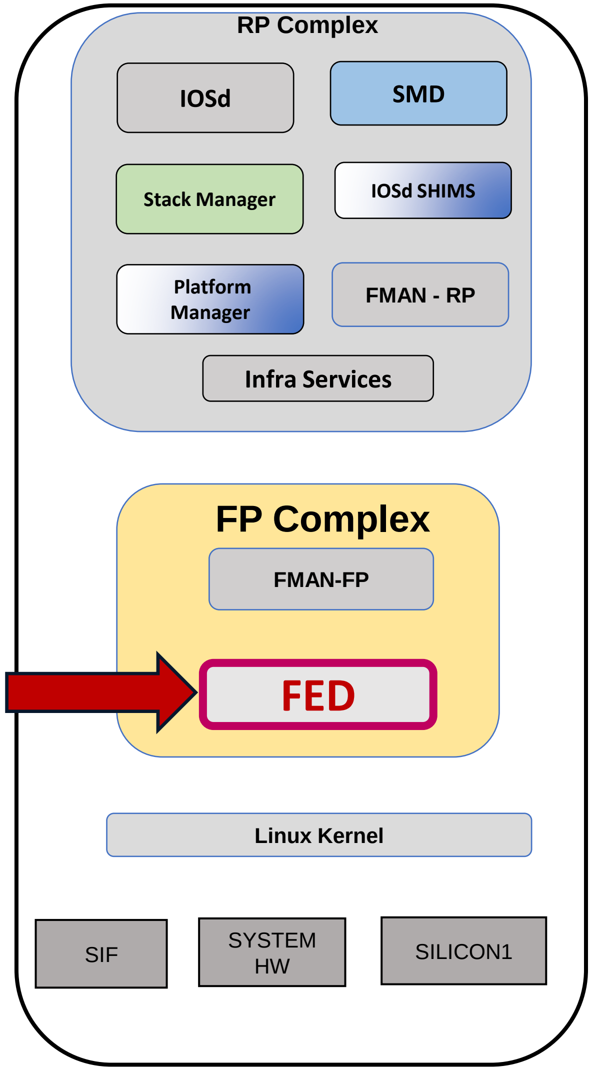


Silicon ONE L3 Multicast

FED – Forwarding Engine Driver

```
C9500-Silicon1# show platform software fed active ip mfib   239.255.0.1/32  50.50.50.50 detail
Mvrf: 0  ( 50.50.50.50, 239.255.0.1 ) Attrs:
   Hw Flag               : InHw
   Mlist Flags           : None
   Mlist_hndl (Id)       : 0x11880700bb0 ( 0x2e )
   Mlist Urid            : 0x1000000000000012
   Fset Urid (Hash)      : 0x3000000000000016 ( 8da59129 )
   Fset Aux Urid         : 0x0
   RPF Adjacency ID      : 0xf8004af1
   CPU Credit            : 0
   Total Packets[StatsMode]: 74 ( 0 pps approx.)[ Egress ]
   Ec_seed               : 1
   npi_mroute_ent        : 0x118807012b0
   svi_fwd_ifs           : 1
   ios_f_ifs/mlist_f_ifs : 1/1
   OIF Count             : 2
   OIF Details:
    AdjID        Interface     ParentIf     HwFlag    Flags    IntfType    MsgType
    0xf8004a81   Po2           Hu1/0/2      ---       F NS     EC_IF       NORMAL
    0xf8004af1   Vl50          ---------    ---       A        SVI_IF      NORMAL
   GID                : 8199
   MCID OID Asic[0]   : 1339
   Hardware Info ASIC[0] :
    IP MCID OID       :1339 (cookie: urid:0x30::16)
    RPF PORT OID      :1317
```

This hash to be used in oifset cli to get the oifset details and also same URID will be shown in all Hardware dump to help get the software entry



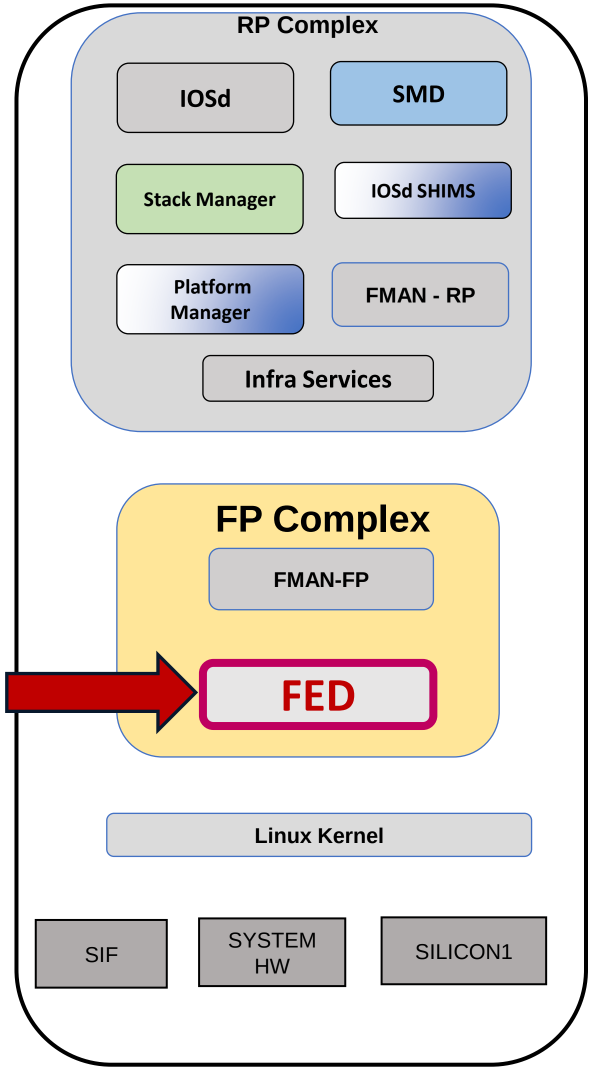
Silicon ONE L3 Multicast

FED – Forwarding Engine Driver

C9500-Silicon1# show platform software fed active oifset urid 0x3000000000000016 detail | ex Users|urid

```
Type: s_g_vrf            State: allocated       MD5:(8da59129ae84a6eb:a49157036092329b)
  Fset Urid                   : 0x3000000000000016
  Remote Port Count           : 0
  Svi Port Count              : 1
  Mioitem Count               : 2
    No. AdjID          Interface         PhysicalIf       IfType        Flags
     1. 0xf8004a81     Po2               Hu1/0/2          ec_if         InHw
        Urids   => Mio:0x80::18 Parent:0x0 child_repl:0x0(------)  adj_obj:0x90::5
        (Asic[0]=> l3_port_oid/port_oid/nh_oid : 1336 / 1306 / 0)
     2. 0xf8004af1     Vl50              -----------      svi_if        InHw IngressRep
        Urids   => Mio:0x80::19 Parent:0x60::8 child_repl:0x20::5(80040362)  adj_obj:0x90::c
        (Asic[0]=> l3_port_oid/port_oid/nh_oid : 1318 / 0 / 0)
  Fset MCID Gid               : 8199
  Asic[0] mcid_oid            : 1339
  Hw IP Mcg Info Asic[0] :
    Idx.    Member Info
     1.    l3_port: 1336
```

Gid and Oid's for all asic's of mcid is displayed here.



Silicon ONE L3 Multicast

FED – Forwarding Engine Driver

C9500-Silicon1# show platform software fed active ip adj | i 0xf8004a81

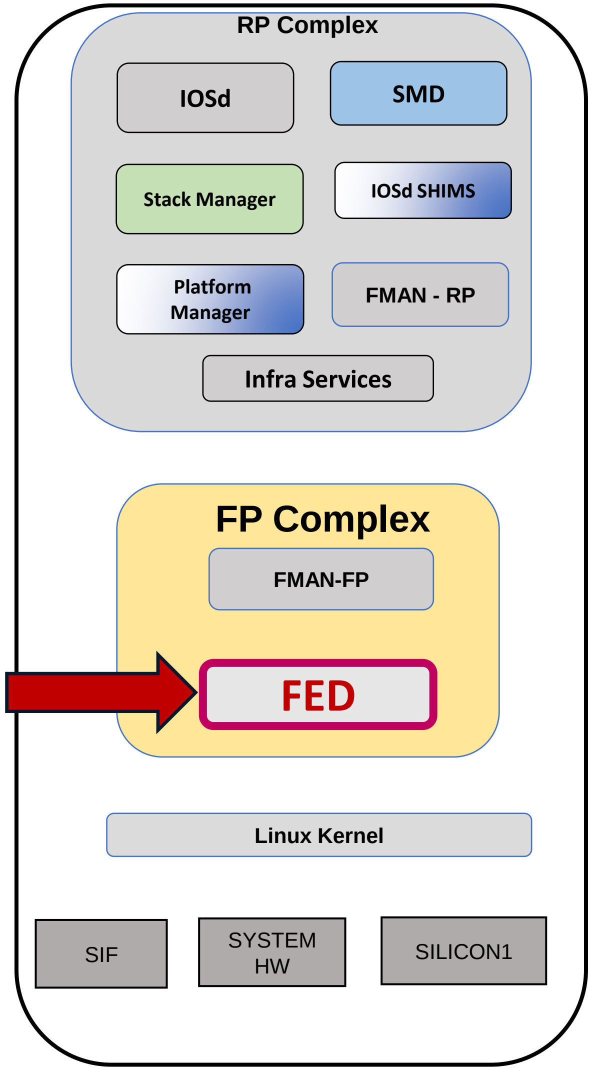
0xf8004a81	225.0.0.0	0100.5e00.0000	NORMAL	**Port-channel2**

C9500-Silicon1# show platform software fed active ip adj | i 0xf8004af1

0xf8004af1	225.0.0.0	0100.5e00.0000	NORMAL	**Vlan50**

C9500-Silicon1# show platform hardware fed active fwd-asic resource utilization | i MC EMDB|MCID|IPV4 VRF S G TABLE|IPV4 VRF DIP EM TABLE

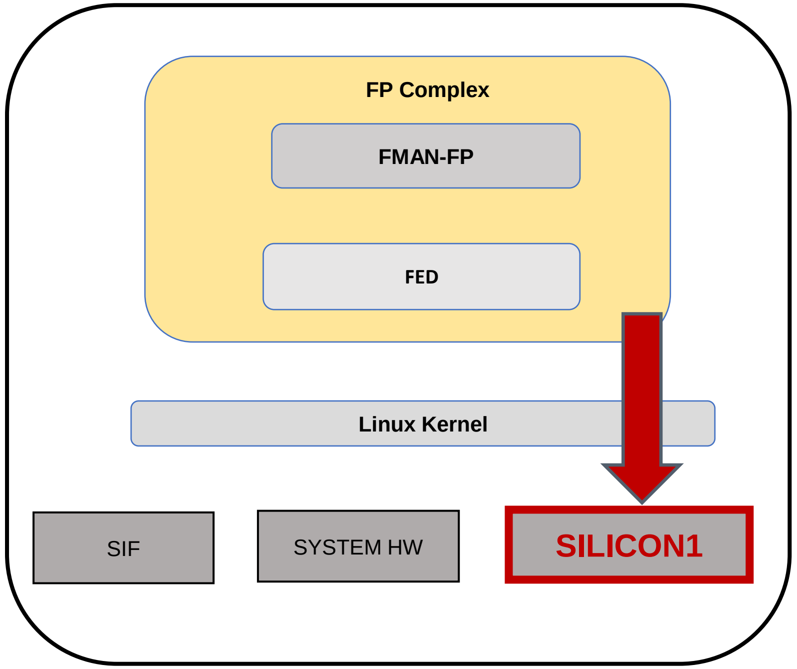
IPV4 VRF S G TABLE	d-0	1	311289
MC LOCAL MCID	d-0	11	262137
MC EMDB	d-0	7	65568
IPV4 VRF DIP EM TABLE	d-0	6	622583



Silicon ONE L3 Multicast

SDK – Forward ASIC

All Insight Commands are linked into the show tech for multicast entry. Recommended to use show tech commands for multicast feature to get all commands output together and for all Asics and Switches.



MCID gid used by multicast route in sdk.

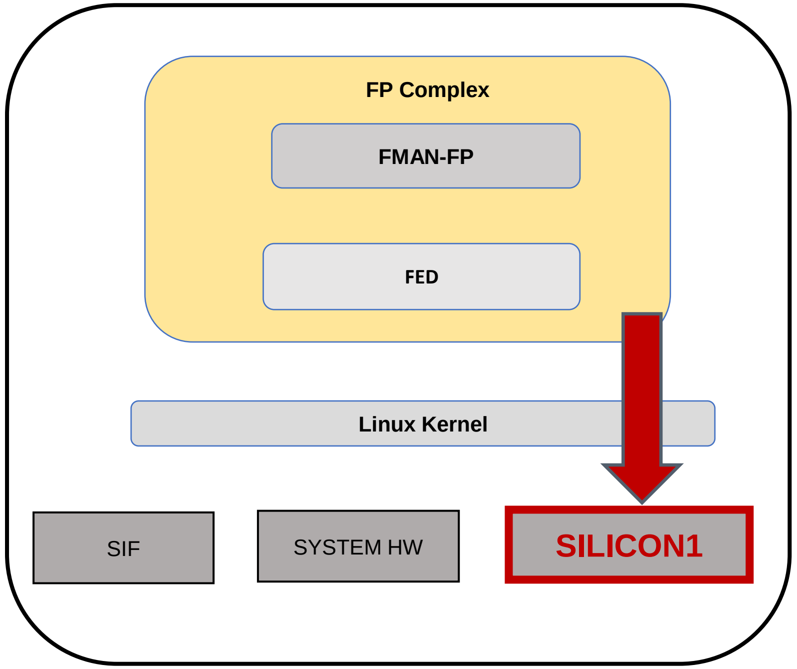
C9500-Silicon1# show platform hardware fed active fwd-asic insight l3m_routes(saddr='50.50.50.50',gaddr='239.255.0.1’)

```
+---------+------------+------------+-------------+-------------+-------------+---------------+-------+-------------+-----
-----------+-------------+--------------+------------+-------------+
| vrf-gid | vrf-cookie | ip-version | saddr       | gaddr       | l3-mcg-gid | l3-mcg-cookie | snoop | rpf-enabled | rpf-path-type | rpf-path-id
| rpf-fail-punt | counter-id | counter-data |
+---------+------------+------------+-------------+-------------+-------------+---------------+-------+-------------+-----
-----------+-------------+--------------+------------+-------------+
| 0       |    N/A     | 4          | 50.50.50.50 | 239.255.0.1 | 8199       | urid:0x30::16 | False |    True     |  L3_SVI_GID  | 14          |      True
| 1340        | [5,590]     |
|         |            |            |             |             |             |               |       |             |              |             |
+---------+------------+------------+-------------+-------------+-------------+---------------+-------+-------------+-----
-----------+-------------+--------------+------------+-------------+
```

Silicon ONE L3 Multicast

SDK – Forward ASIC

All Insight Commands are linked into the show tech for multicast entry. Recommended to use show tech commands for multicast feature to get all commands output together and for all Asics and Switches.



MCID gid used by multicast route in sdk.

C9500-Silicon1# show platform hardware fed active fwd-asic insight l3m_groups(devid=0,l3_mcg_gid='8199')

```
+------------+---------------+-------------+----------------------+-------------------+---------------------+
| l3-mcg-gid | l3-mcg-cookie | num-members | replication-paradigm | egress-counter-id | egress-counter-data |
+------------+---------------+-------------+----------------------+-------------------+---------------------+
| 8199       | urid:0x30::16 | 2           |      INGRESS         | 1340              | [5,590]             |
|            |               |             |                      |                   |                     |
+------------+---------------+-------------+----------------------+-------------------+---------------------+
```

Silicon ONE L3 Multicast

SDK – Forward ASIC

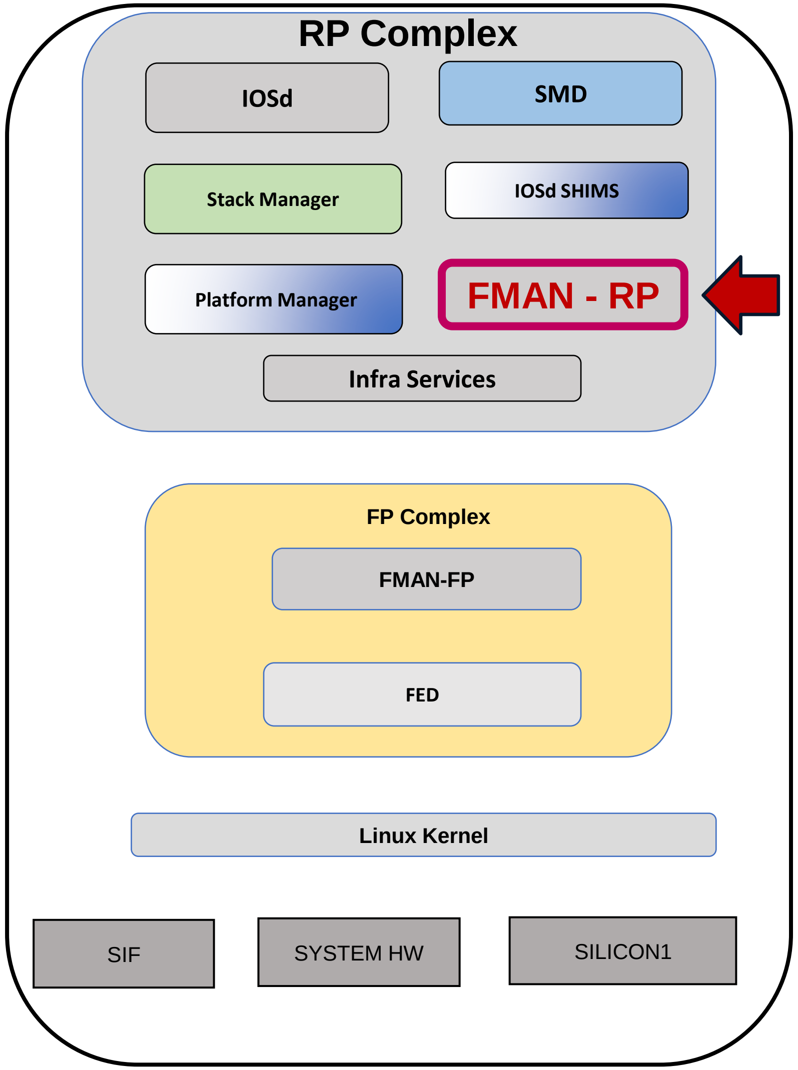
```
C9500-Silicon1# show platform hardware fed active fwd-asic insight l3m_groups(devid=0,l3_mcg_gid='8199')
+------------+-----+--------+-------------+------------+-------------+------------+-----------------
+-------------+--------------------+-----------------+---------------+------------+
| l3-mcg-gid | idx | type   | l3-port-type | l3-port-gid | l2-port-type | l2-port-gid | l2-mcg-gid | l3-mcg-member-gid | next-hop-gid |
mc-fanout-group-oid | mc-gre-encap-oid | stack-port-oid | sysport-gid |
+------------+-----+--------+-------------+------------+-------------+------------+-----------------
+-------------+--------------------+-----------------+---------------+------------+
| 8199       | 1   | NONE   |    NONE     | 0          |    NONE     | 0          | 0          | 0                | 0          | 0                | 0               | 0
| 0          |
|            |     |        |             |            |             |            |            |                  |            |                  |                 |
| 8199       | 2   | L3_AC  | L3_AC_PORT  | 8          |    NONE     | 0          | 0          | 0                | 0          | 0                | 0               | 0
| 62         |
|            |     |        |             |            |             |            |            |                  |            |                  |                 |
| 8199       | 3   | L2_MCG | L3_SVI_PORT | 14         |    NONE     | 0          | 8197       | 0                | 0          | 0                | 0
| 0          | 0          |
|            |     |        |             |            |             |            |            |                  |            |                  |                 |
+------------+-----+--------+-------------+------------+-------------+------------+-----------------
+-------------+--------------------+-----------------+---------------+------------+
```

Silicon ONE L3 Multicast

FMAN-RP

```
C9500-Silicon1#show platform software ip  RP active mfib   group address 239.255.0.1
50.50.50.50
Route flags:
S - Signal;             C - Directly connected;
IA - Inherit A Flag;     L - Local;
BR - Bidir route
239.255.0.1, 50.50.50.50/64 --> OBJ_INTF_LIST (0x2e)
                Obj id: 0x2e, Flags: unknown
                OM handle: 0x34804e4590
```

```
C9500-Silicon1# show platform software mlist  RP active index  0x2e
Number of adjacency objects: 12

Multicast List entries

OCE Flags:
NS - Negate Signalling; IC - Internal copy;
A - Accept;             F - Forward;

OCE              Type              OCE Flags         Interface
-----------------------------------------------------------------------------
0xf8004a81       OBJ_ADJACENCY     NS, F             Port-channel2
0xf8004af1       OBJ_ADJACENCY     A                 Vlan50
```

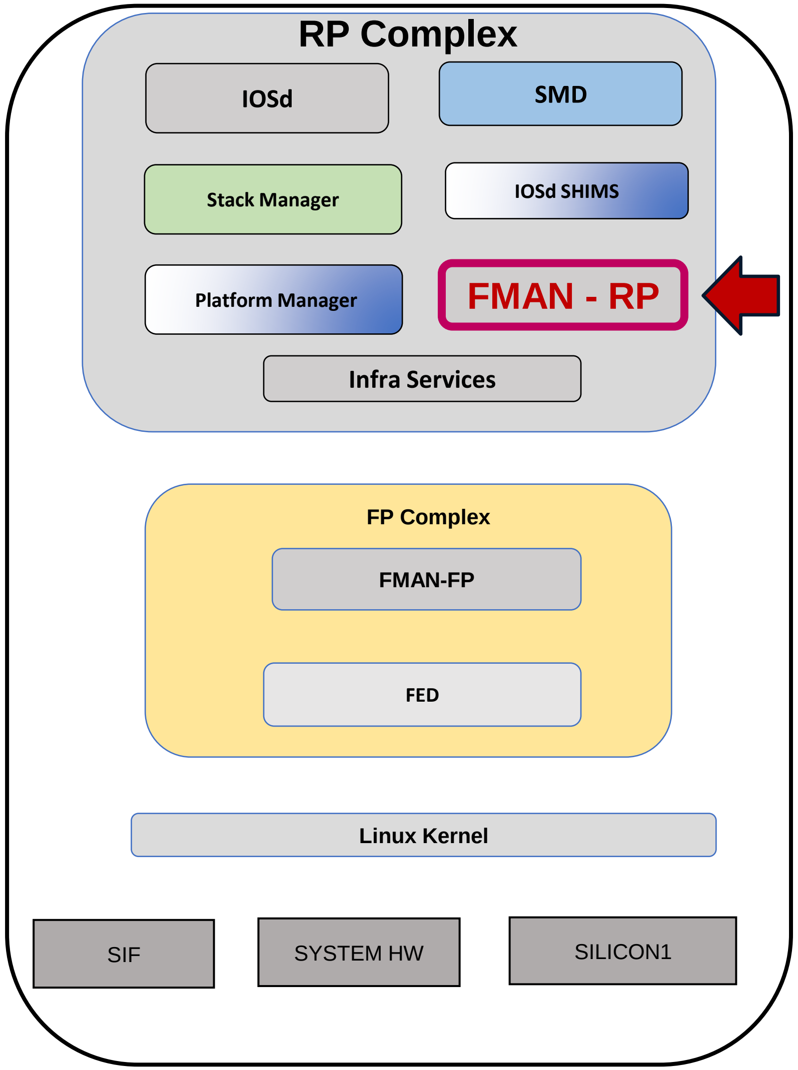


Silicon ONE L3 Multicast

FMAN-RP

```
C9500-Silicon1#show platform software adjacency  RP active index 0xf8004a81

Number of adjacency objects: 13

Adjacency id: 0xf8004a81 (4160768641)
  Interface: Port-channel2, IF index: 1192, Link Type: MCP_LINK_IP
  Encap: 1:0:5e:0:0:0:38:aa:9:ff:f1:2:8:0
  Encap Length: 14, Encap Type: MCP_ET_ARPA, MTU: 1500
!!!! ==== snip ==== !!!!
Nexthop addr: 225.0.0.0
  IP FRR MCP_ADJ_IPFRR_NONE 0
  OM handle: 0x34804d8880
```

```
C9500-Silicon1#show platform software adjacency  RP active index 0xf8004af1
Number of adjacency objects: 13

Adjacency id: 0xf8004af1 (4160768753)
  Interface: Vlan50, IF index: 1199, Link Type: MCP_LINK_IP
  Encap: 1:0:5e:0:0:0:38:aa:9:ff:f1:2:8:0
  Encap Length: 14, Encap Type: MCP_ET_ARPA, MTU: 1500
!!!! ==== snip ==== !!!!
Nexthop addr: 225.0.0.0
  IP FRR MCP_ADJ_IPFRR_NONE 0
  OM handle: 0x34804dfcc0
```

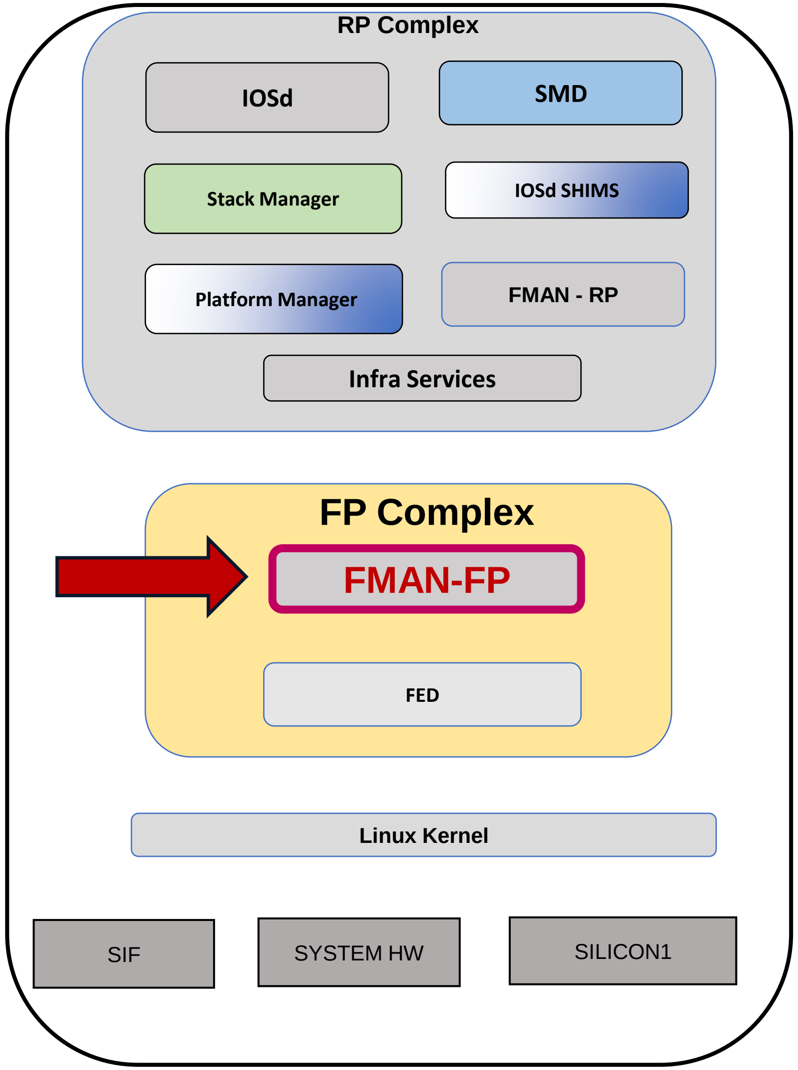


Silicon ONE L3 Multicast

FMAN-FP

```
C9500-Silicon1#show platform software ip  FP active mfib   group address 239.255.0.1 50.50.50.50
Route flags:
S - Signal;             C - Directly connected;
IA - Inherit A Flag;    L - Local;
BR - Bidir route
239.255.0.1, 50.50.50.50/64 --> OBJ_INTF_LIST (0x2e)
               Obj id: 0x2e, Flags: unknown
               aom id: 5486, HW handle: (nil) (created)
```

```
C9500-Silicon1#show platform software object-manager  FP active object 5486
Object identifier: 5486
  Description: PREFIX 50.50.50.50 , 239.255.0.1/64 (Table id 0)
  Obj type id: 73
  Obj type: mroute-pfx
  Status: Done, Epoch: 0, Client data: 0xd4c25928

C9500-Silicon1#show platform software object-manager  FP active object 5486 parents
Object identifier: 5391
  Description: ipv4_mcast table 0 (), vrf id 0
  Status: Done

Object identifier: 5484
  Description: mlist 46
  Status: Done
```

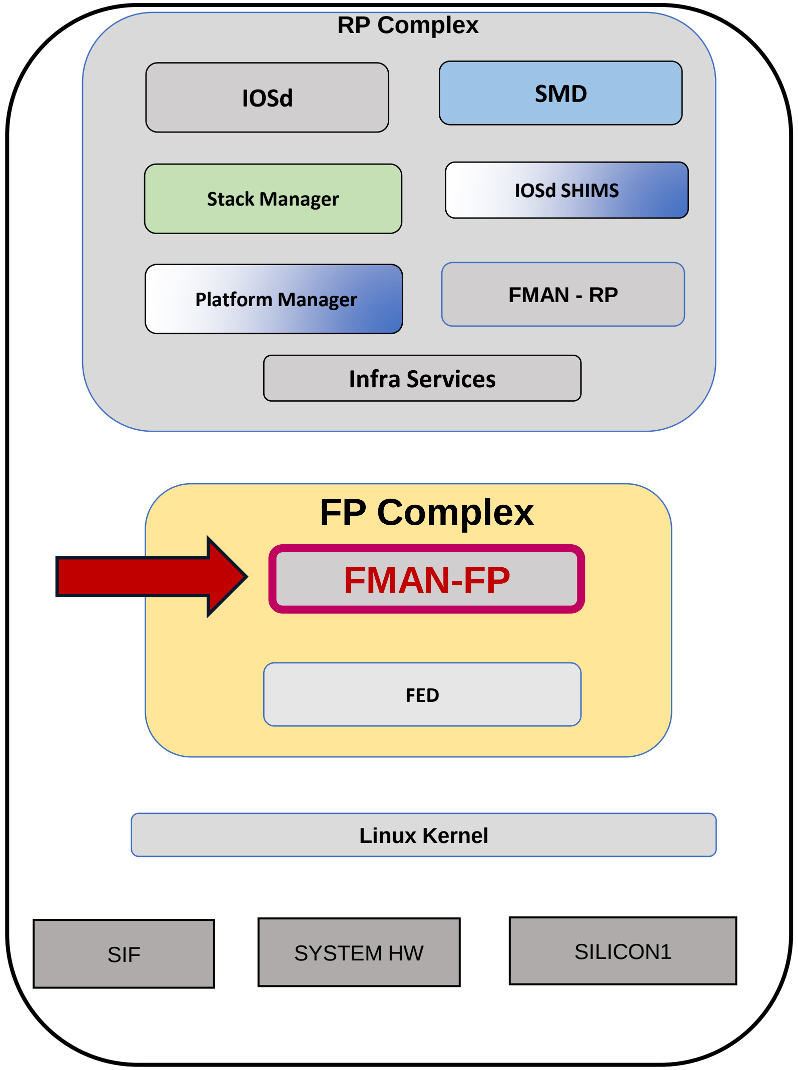


Silicon ONE L3 Multicast

FMAN-FP

```
C9500-Silicon1#show platform software mlist  FP active index 0x2e
Multicast List entries

OCE Flags:
NS - Negate Signalling; IC - Internal copy;
A - Accept;            F - Forward;

OCE              Type               OCE Flags       Interface
------------------------------------------------------------------------------------
0xf8004a81       OBJ_ADJACENCY      NS, F           Port-channel2
0xf8004af1       OBJ_ADJACENCY      A               Vlan50
```

```
C9500-Silicon1#show platform software object-manager FP active statistics
Forwarding Manager Asynchronous Object Manager Statistics

Object update: Pending-issue: 0, Pending-acknowledgement: 0
Batch begin:   Pending-issue: 0, Pending-acknowledgement: 0
Batch end:     Pending-issue: 0, Pending-acknowledgement: 0
Command:       Pending-acknowledgement: 0
Total-objects: 1231
Stale-objects: 0
Resolve-objects: 0
Childless-delete-objects: 0
Backplane-objects: 0
Error-objects: 0
```

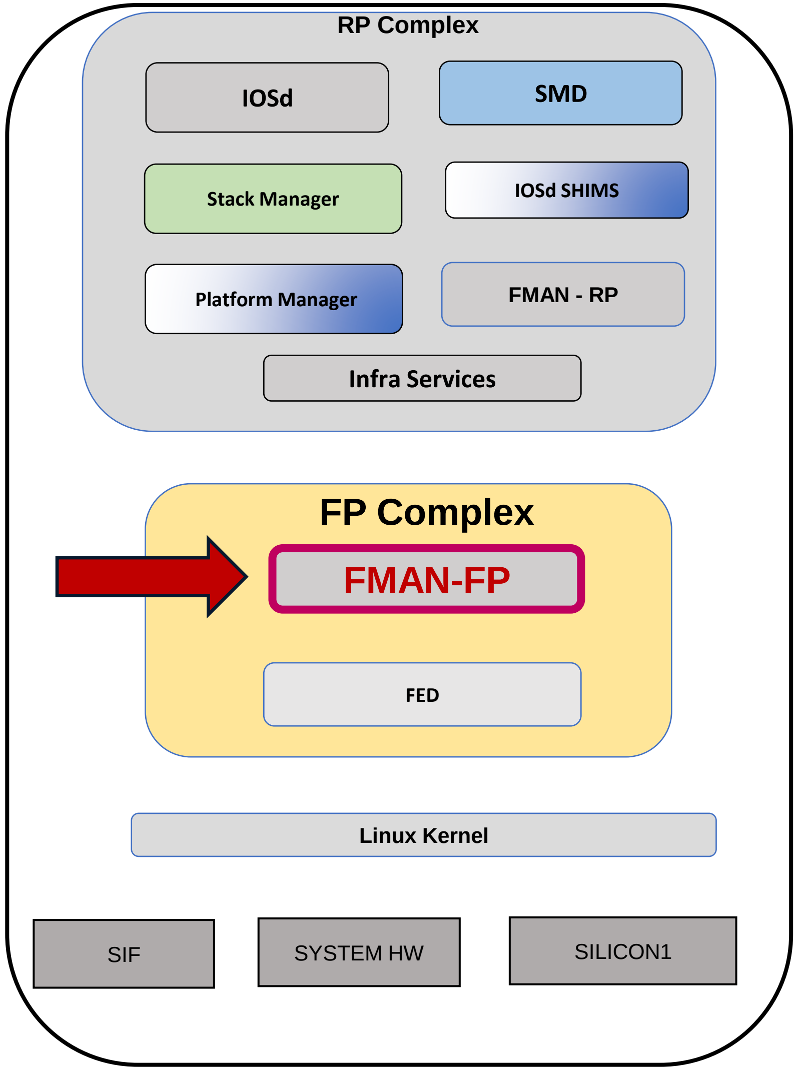


Silicon ONE L3 Multicast

FMAN-FP

```
C9500-Silicon1#show platform software adjacency  FP active index 0xf8004a81
Number of adjacency objects: 13

Adjacency id: 0xf8004a81 (4160768641)
  Interface: Port-channel2, IF index: 1192, Link Type: MCP_LINK_IP
  Encap: 1:0:5e:0:0:0:38:aa:9:ff:f1:2:8:0
  Encap Length: 14, Encap Type: MCP_ET_ARPA, MTU: 1500
!!!! ==== snip ==== !!!!
Nexthop addr: 225.0.0.0
  IP FRR MCP_ADJ_IPFRR_NONE 0
  aom id: 5393, HW handle: (nil) (created)
```

```
C9500-Silicon1#show platform software object-manager  FP active object 5393
Object identifier: 5393
  Description: adj 0xf8004a81, Flags None
  Obj type id: 41
  Obj type: adj
  Status: Done, Epoch: 0, Client data: 0xd4919f18

C9500-Silicon1#show platform software object-manager  FP active object 5393
parents
Object identifier: 366
  Description: intf Port-channel2, handle 1192, hw handle 1192, HW dirty: NONE AOM dirty
NONE
  Status: Done
```

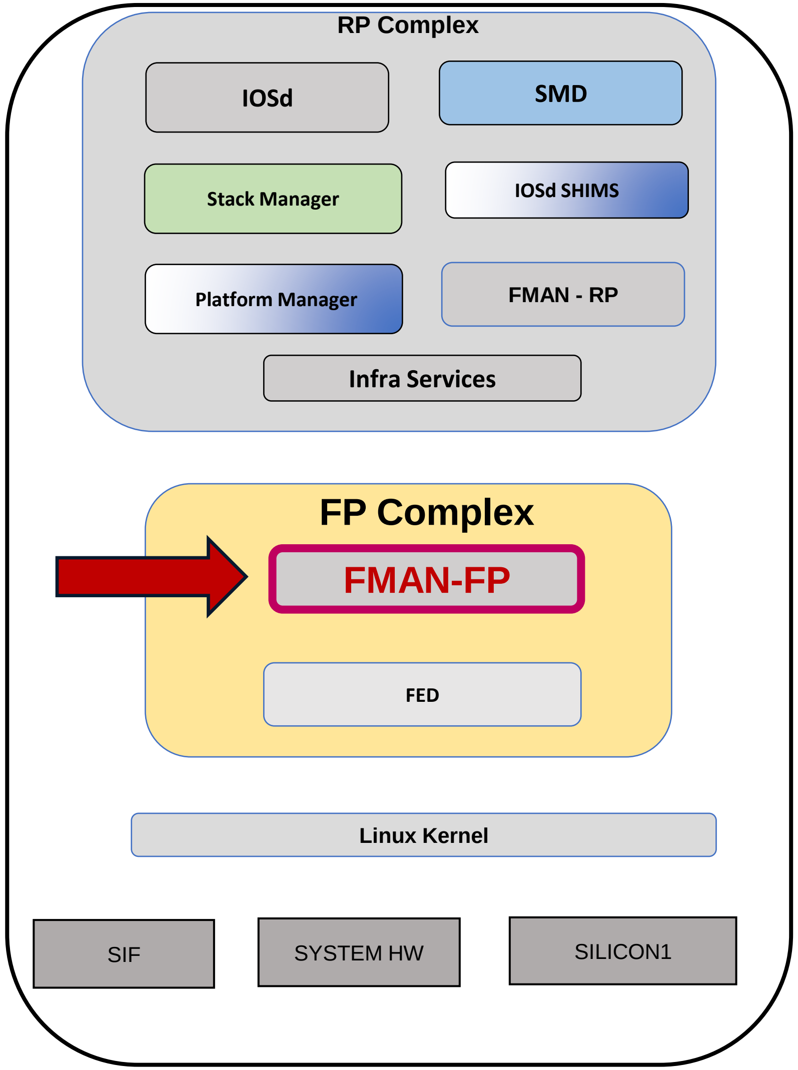


Silicon ONE L3 Multicast

FMAN-FP

```
C9500-Silicon1#show platform software adjacency  FP active index 0xf8004af1
Number of adjacency objects: 13

Adjacency id: 0xf8004af1 (4160768753)
  Interface: Vlan50, IF index: 1199, Link Type: MCP_LINK_IP
  Encap: 1:0:5e:0:0:0:38:aa:9:ff:f1:2:8:0
  Encap Length: 14, Encap Type: MCP_ET_ARPA, MTU: 1500
!!!! ==== snip ==== !!!!
 Nexthop addr: 225.0.0.0
  IP FRR MCP_ADJ_IPFRR_NONE 0
  aom id: 5399, HW handle: (nil) (created)
```

```
C9500-Silicon1#show platform software object-manager  FP active object 5399
Object identifier: 5399
  Description: adj 0xf8004af1, Flags None
  Obj type id: 41
  Obj type: adj
  Status: Done, Epoch: 0, Client data: 0xd4c1bed8

C9500-Silicon1#show platform software object-manager  FP active object 5399
parents
Object identifier: 5361
  Description: intf Vlan50, handle 1199, hw handle 1199, HW dirty: NONE AOM dirty NONE
  Status: Done
```



L3 Multicast : Btrace options for l3mcast and oifset

L3M feature btraces:

set platform software trace fed switch active la_l3m_entry <btrace level>
set platform software trace fed switch active l3m_stats <btrace level>
set platform software trace fed switch active l3m_entry <btrace level>

L3m fset(oifset) btraces:

set platform software trace fed switch active fset_infra <btrace level>
set platform software trace fed switch active fset_l3m <btrace level>
set platform software trace fed switch active la_fset_l3m <btrace level>
set platform software trace fed switch active la_fset_infra <btrace level>

Traces starts with la_<> are npd module and others are NPI module traces. Appropriate traces to be enabled to avoid overflow.

To collect trace archive

request platform software trace archive

L3 Multicast : L3 multicast state's summary

- Summary provides snapshot of l3 multicast across VRFs. It gives multicast route count, historical maximum l3 multicast reached, oif count, historical oif max etc.
- To know the scale supported for the system, look for ‘L3 Multicast entries’ under ‘show sdp prefer’.

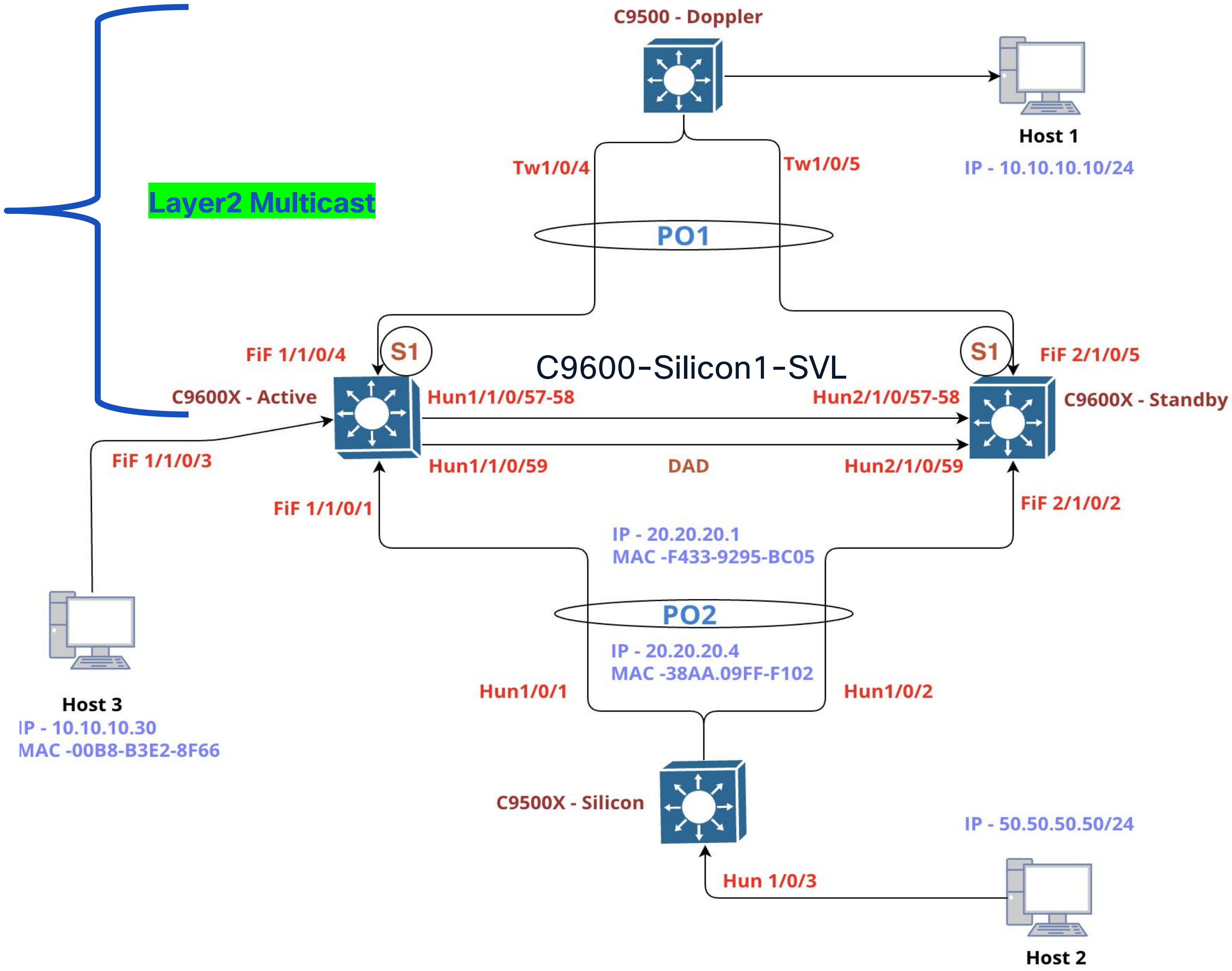
show platform software fed switch active ip mfib summary

```
Mfib v4 All Tables summary:
  Multicast Vrf Current Count              : 1
  Mroute Current Count/Max Reached         : 32001/34001 ← user has over scaled at some point
  (*,G) Current Count                      : 16001
  (S,G) Current Count                      : 16000
  (*,G/m) RetryQ Count                     : 0
  (*,G) RetryQ Count                       : 0
  (S,G) RetryQ Count                       : 0
  Oifs Current Count/Max Reached           : 64002/68002
  Last used Mlist Urid                     : 0x10000000000084f1
```

Silicon ONE L2 Multicast

Silicon ASIC

Lab Topology



Troubleshooting L2 Multicast on Silicon ONE

To debug group specific issue, it's highly recommended to use show tech-support option

- It collects all the details specific to all layers in the switch.
- IOSD, FMAN-RP, FMAN-FP, FED, SDK outputs for specific groups are collected using a single cli.

show tech-support platform igmp-snooping group_ipAddr <group addr> vlan <vlan id>
show tech-support platform mld_snooping group_ipv6Addr <group addr> vlan <vlan id>

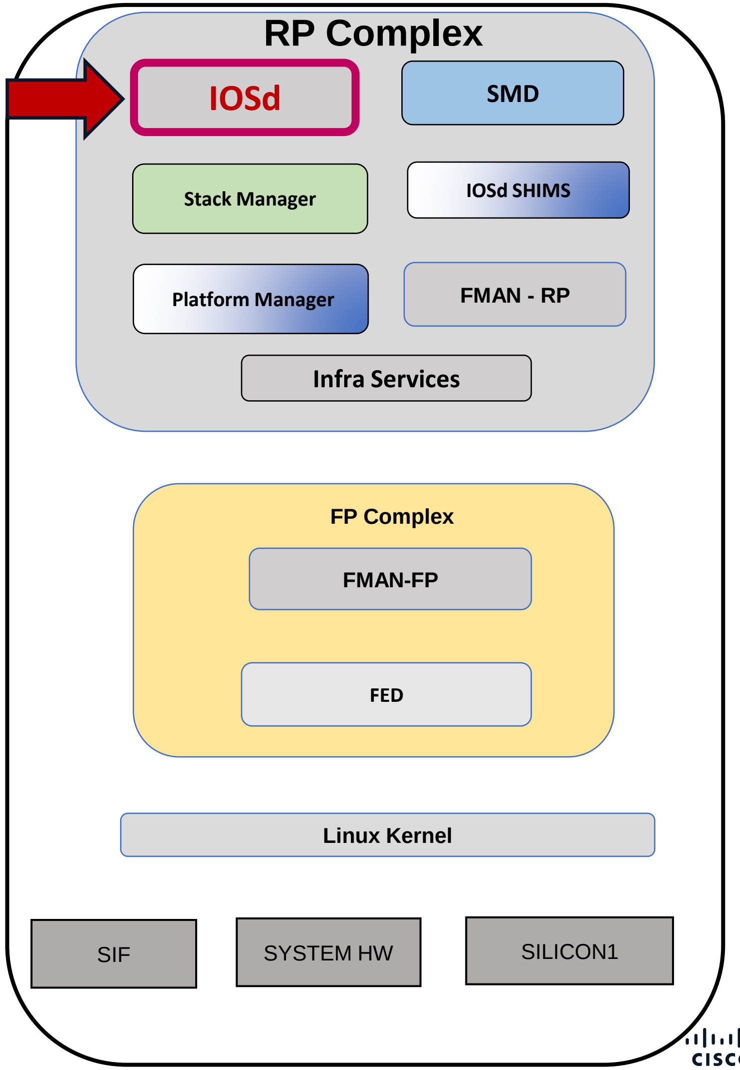
To collect trace archive

request platform software trace archive

Silicon ONE L2 Multicast

```
C9600-Silicon1-SVL#show ip igmp snooping groups vlan 10 239.255.0.1
~~
Flags: I -- IGMP snooping, S -- Static, E -- EVPN sync

Vlan     Group                 Type      Version    Port List
-----------------------------------------------------------------
10   239.255.0.1              I          v2         Po1
C9600-Silicon1-SVL#show ip igmp snooping mrouter vlan 10
Vlan    ports
----    -----
  10   Router

C9600-Silicon1-SVL#show ip igmp snooping querier vlan 10
IP address          : 10.10.10.1
IGMP version          : v2
Port                 : Router
Max response time        : 10s
```

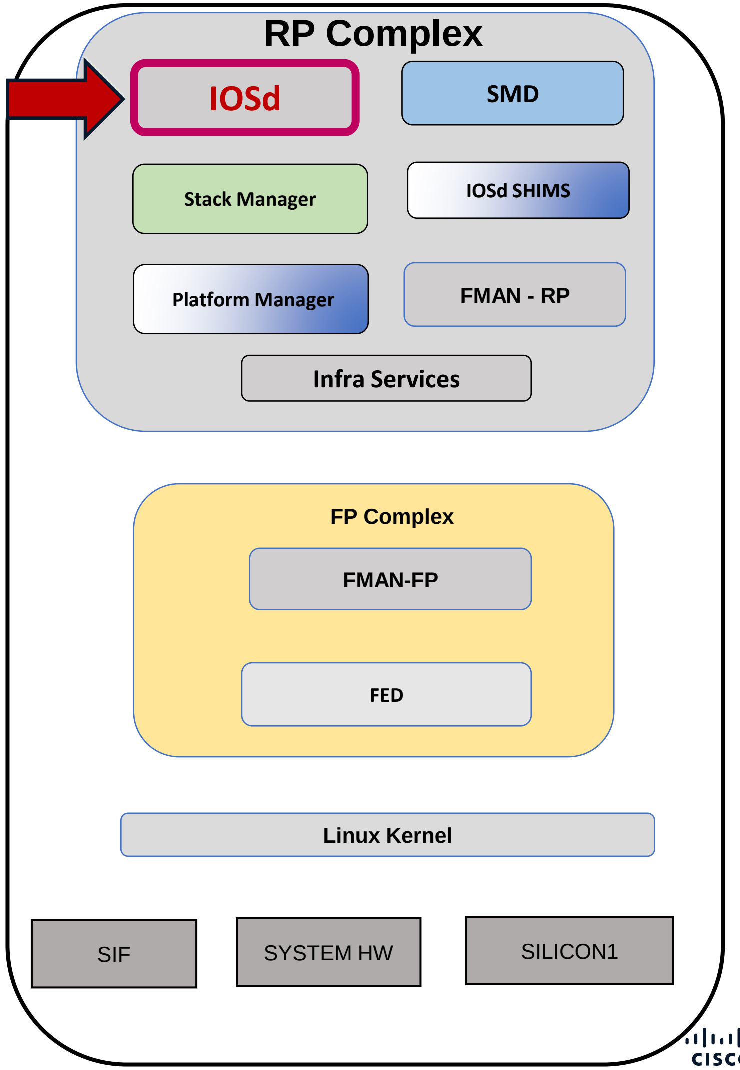
IOSD



Silicon ONE L2 Multicast

```
C9600-Silicon1-SVL#show ip igmp snooping vlan 10
Global IGMP Snooping configuration:
-------------------------------------------
IGMP snooping              : Enabled
Global PIM Snooping        : Disabled
IGMPv3 snooping            : Enabled
Report suppression         : Enabled
TCN solicit query          : Disabled
TCN flood query count      : 2
Robustness variable        : 2
Last member query count    : 2
Last member query interval : 1000

Vlan 10:
--------
IGMP snooping                     : Enabled
Pim Snooping                      : Disabled
IGMPv2 immediate leave            : Disabled
Explicit host tracking            : Enabled
Multicast router learning mode    : pim-dvmrp
CGMP interoperability mode        : IGMP_ONLY
Robustness variable               : 2
Last member query count           : 2
Last member query interval        : 1000
```

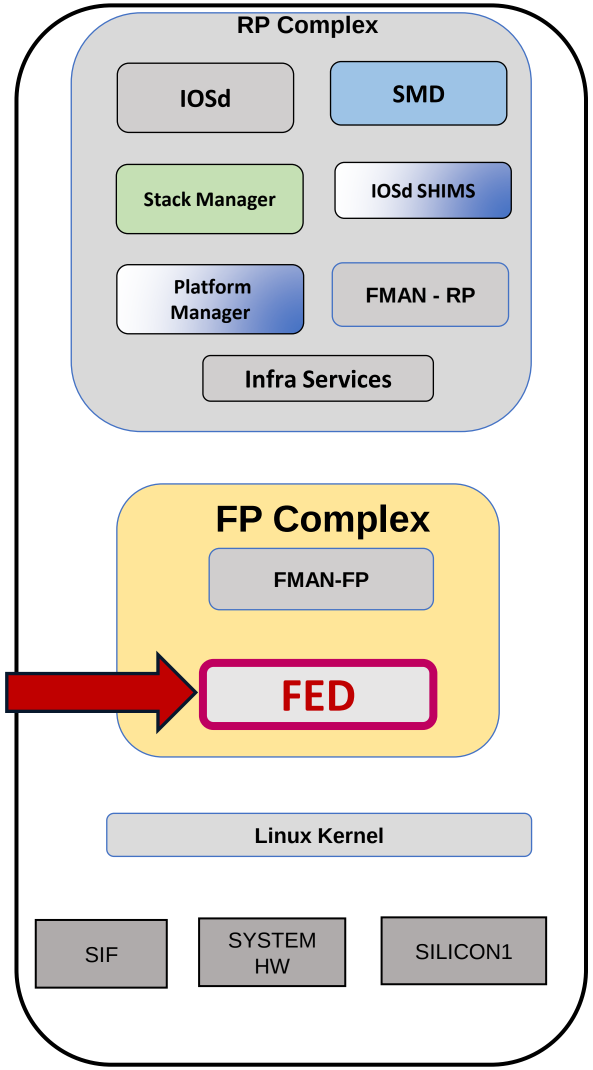
IOSD



Silicon ONE L2 Multicast

FED – Forwarding Engine Driver

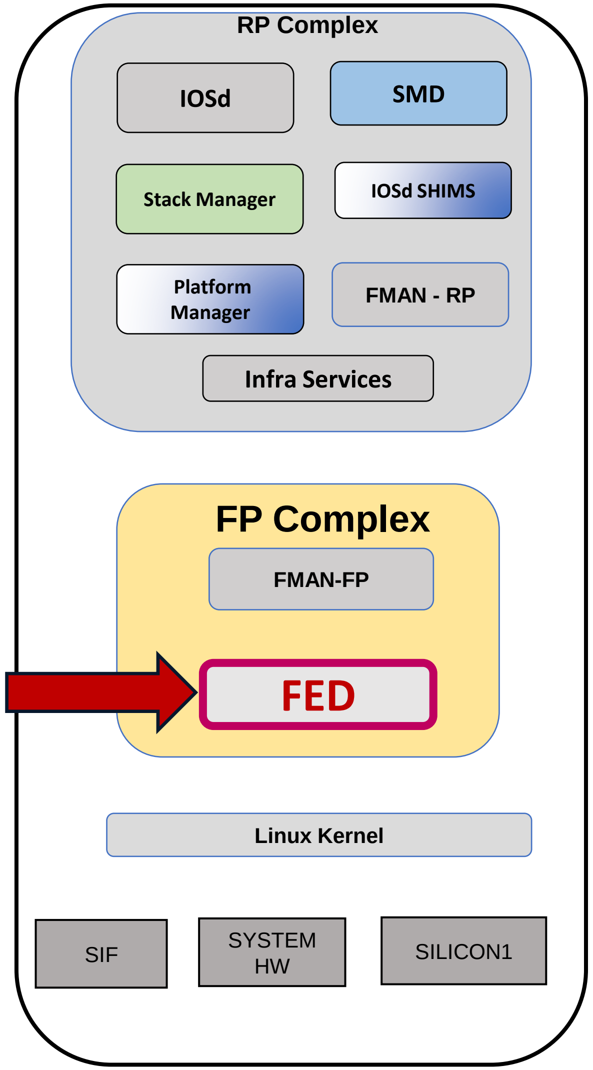
```
C9600-Silicon1-SVL# show platform software fed switch 1 ip igmp snooping vlan 10

(ipv4, vlan: 10)
    Snoop State               : ON
    STP TCN State             : OFF
    Flood Mode                : OFF
    IOS Flood Mode            : OFF
    Pim state                 : ON
    Evpn Proxy                : OFF
    Secondary Vlan            : NO
    Vlan Urid                 : 0x4000000000000003
    Fset Urid ( hash )        : 0x2000000000000003 ( 75ad50f0 )
    Dependant users count     : 0
    Mrouter ports             : 0
    Flood ports               :
        FiftyGigE1/1/0/3
        Port-channel1
    Gid                       : 8195
    Mcid Asic[0]              : 2083
```



Silicon ONE L2 Multicast

FED – Forwarding Engine Driver

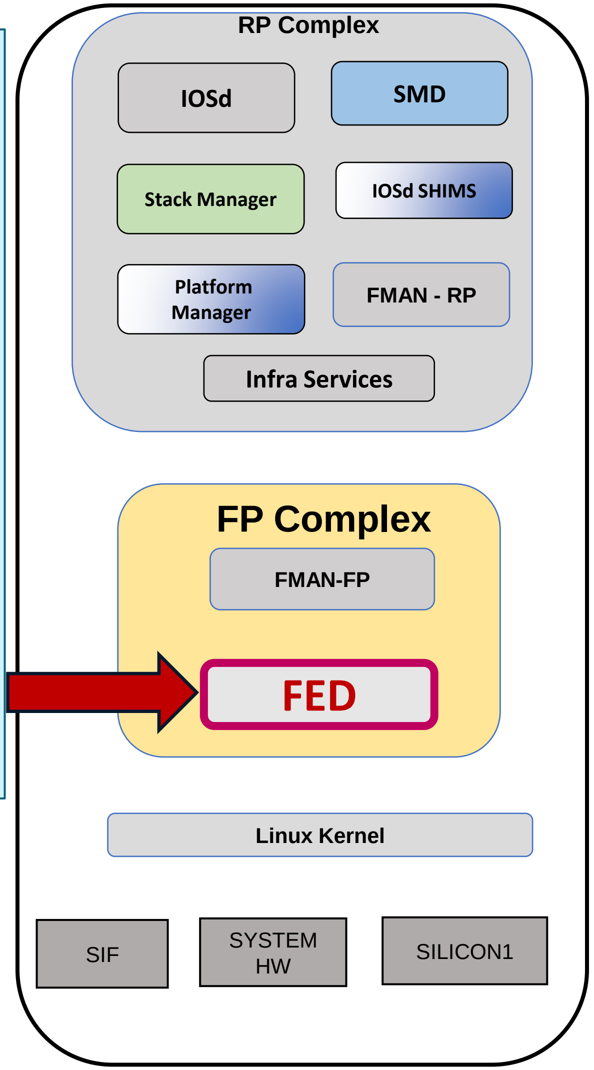
```
C9600-Silicon1-SVL# show platform software fed switch 1 ip igmp snooping groups vlan 10
239.255.0.1 detail

(Vlan: 10, 239.255.0.1)
    Member ports              : 1
       Port-channel1 (Port:FiftyGigE1/1/0/4)
    Dependent Users Count   : 1
    Group Urid             : 0x6000000000000005
    Fset Urid ( Hash )      : 0x2000000000000005 ( e47017ed )
    Fset Aux Urid           : 0x0
    Layer 3 Stub Entry      : No
    Group Flags             : None
    Ec seed                 : 1
    IOS Vp Count            : 1
    Gid                     : 8199
    Mcid Asic[0]            : 2085
    Hw Group Entry Asic[0]:
       Hw Mcid Oid          : 2085 (cookie: urid:0x20::5)
```



Silicon ONE L2 Multicast

FED – Forwarding Engine Driver

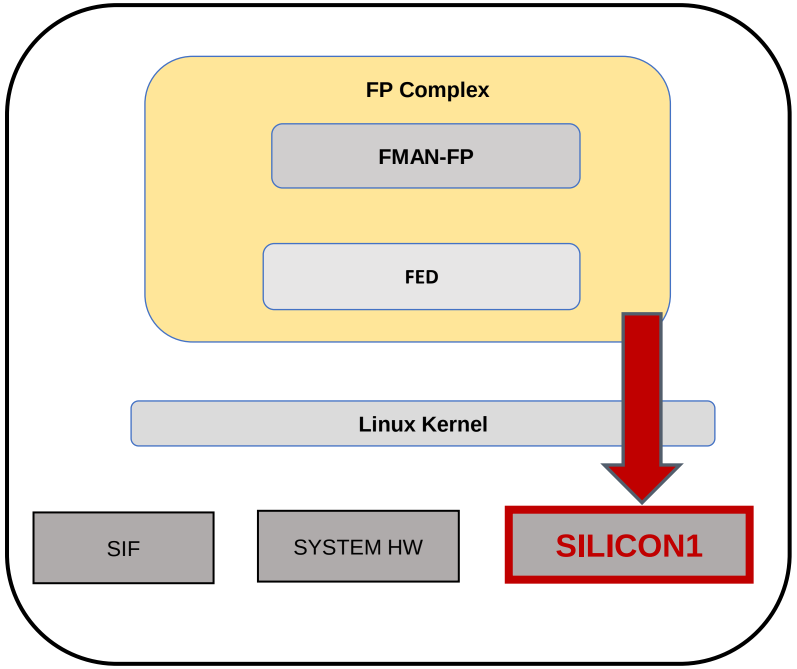
```
C9600-Silicon1-SVL# how platform software fed switch 1 oifset urid 0x20000000000000005 detail |
ex Users|urid

Type: oif_vp_handles      State: allocated      MD5:(e47017edc89b710a:ea2a48a5ee59691e)
  Fset Urid               : 0x20000000000000005
  Parent URID             : 0x20000000000000003
  Remote Port Count       : 0
  VP OIF Count            : 1
   No. VP Interface            Interface              Physical interface
    1. Po1_V10                 Port-channel1          FiftyGigE1/1/0/4
         flags:[ oif_port ]    hw_flags: [ InHw ]
        (Asic[0]: l2_ac_oid: 1936)
  FSET Gid                : 8199
  Asic[0] mcid_oid        : 2085
  Hw L2 Mcg Info Asic[0] :
   Idx.    Member Info
    1.    l2_dest: 1936
```



Silicon ONE L2 Multicast

SDK – Forward ASIC

All Insight Commands are linked into the show tech for multicast entry. Recommended to use show tech commands for multicast feature to get all commands output together and for all Asics and Switches.



MCID gid used by multicast route in sdk.

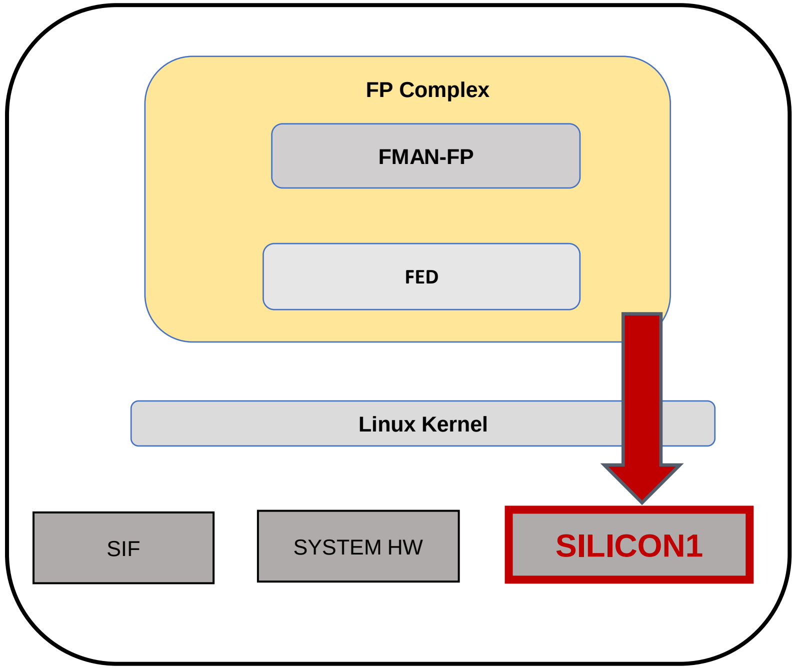
C9600-Silicon1-SVL# show platform hardware fed switch 1 fwd-asic insight l2m_routes(switch_gid='10',gaddr='239.255.0.1')

```
+------------+---------------+------------+-----------------+------------+------------+--------------+
| switch-gid | switch-cookie | ip-version | saddr           | gaddr       | l2-mcg-gid | l2-mcg-cookie |
+------------+---------------+------------+-----------------+------------+------------+--------------+
| 10         |       10      | 4          | 255.255.255.255 | 239.255.0.1 | 8199       | urid:0x20::5 |
|            |               |            |                 |             |            |              |
+------------+---------------+------------+-----------------+------------+------------+--------------+
```

Silicon ONE L2 Multicast

SDK – Forward ASIC

All Insight Commands are linked into the show tech for multicast entry. Recommended to use show tech commands for multicast feature to get all commands output together and for all Asics and Switches.



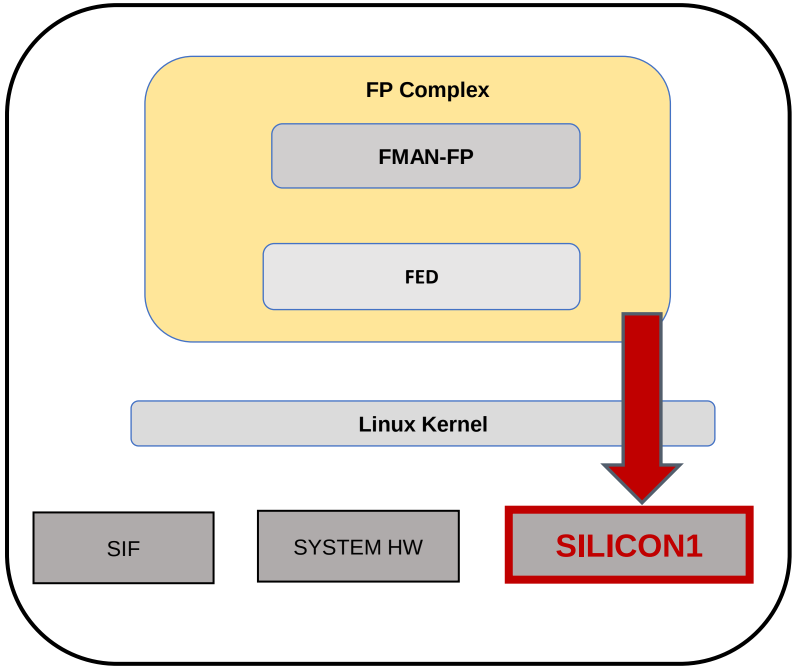
MCID gid used by multicast route in sdk.

C9600-Silicon1-SVL# show platform hardware fed switch 1 fwd-asic insight l2m_groups(devid=0,l2_mcg_gid='8199')

```
+------------+---------------+-------------+---------------------+------------------+---------------------+
| l2-mcg-gid | l2-mcg-cookie | num-members | replication-paradigm | egress-counter-id | egress-counter-data |
+------------+---------------+-------------+---------------------+------------------+---------------------+
| 8199       | urid:0x20::5  | 1           |       EGRESS        | 0                | [0,0]               |
|            |               |             |                     |                  |                     |
+------------+---------------+-------------+---------------------+------------------+---------------------+
```

Silicon ONE L2 Multicast

SDK – Forward ASIC



All Insight Commands are linked into the show tech for multicast entry. Recommended to use show tech commands for multicast feature to get all commands output together and for all Asics and Switches.

MCID gid used by multicast route in sdk.

C9600-Silicon1-SVL# show platform hardware fed switch 1 fwd-asic insight l2m_group_members(devid=0,l2_mcg_gid='8199')

l2-mcg-gid	idx	type	l3-port-type	l2-dest-type	l2-dest-gid	l2-mcg-member-gid	next-hop-gid	stack-port-oid	sysport-gid
8199	1	L2_AC	NONE	AC	132	0	0	0	45

L2 Multicast : Btrace options for L2 multicast and oifset

L2M feature btraces:

set platform software trace fed switch active l2m_entry <btrace level>
set platform software trace fed switch active la_l2m_entry <btrace level>
set platform software trace fed switch standby l2m_entry <btrace level>
set platform software trace fed switch standby la_l2m_entry <btrace level>

Fset l2m btraces:

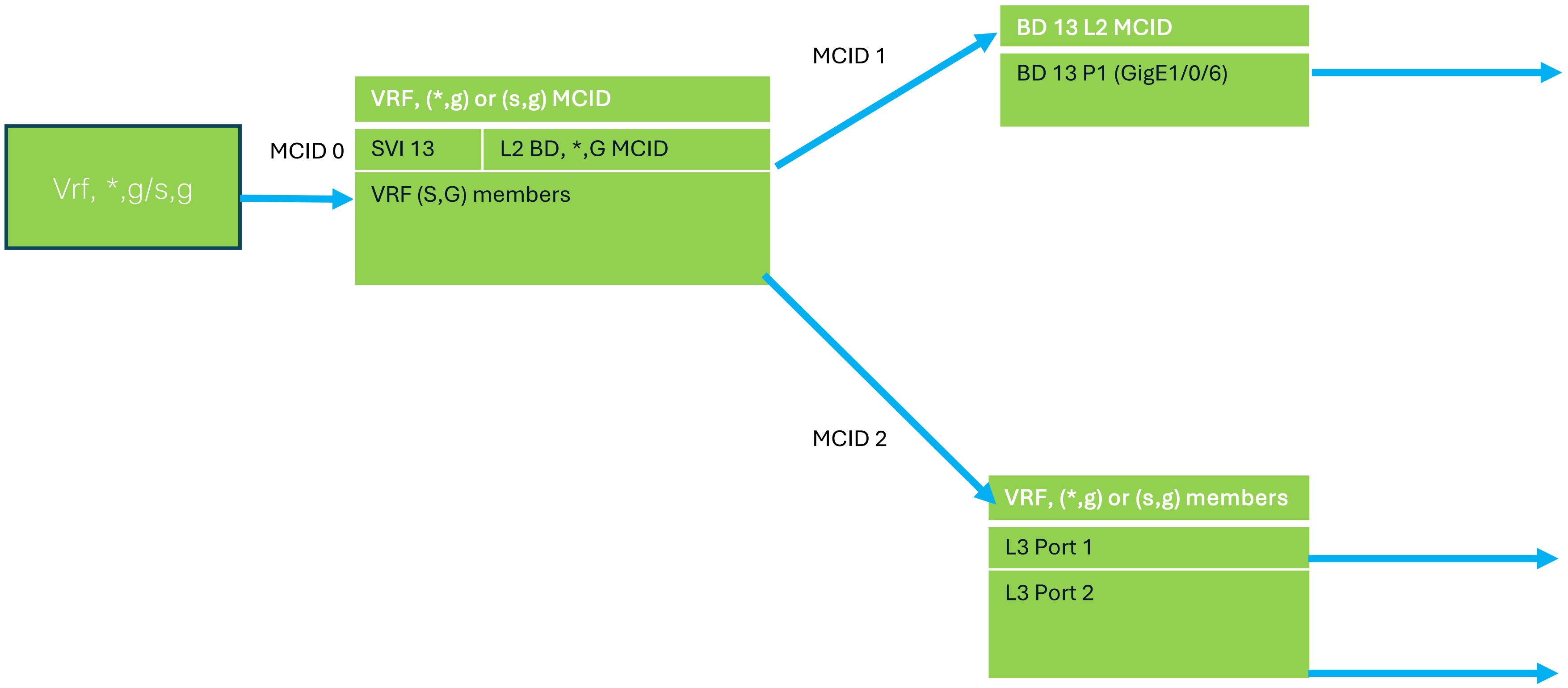
set platform software trace fed switch active fset_infra <btrace level>
set platform software trace fed switch active fset_l2m <btrace level>
set platform software trace fed switch active la_fset_l2m <btrace level>
set platform software trace fed switch active la_fset_infra <btrace level>

Traces starts with la_<> are npd module and others are NPI module traces.
Appropriate traces to be enabled to avoid overflow.

To collect trace archive

request platform software trace archive

SVI debugging in Multicast



- SVI uses ingress replication model in L3M. L2 MCID's acts as the sublist MCID.
- L2 port's to be replicated under a svi oif has to be derived from L3 mcid.
- Silicon One achieve this by linking L2 mcid under L3 mcid.
- L2 mcid to have L2 replication ports

SVI debugging in Multicast

show platform software fed switch active ip mfib 225.0.0.1 10.10.10.2

```
 Mvrf: 0  ( 10.10.10.2, 225.0.0.1 ) Attrs:
   Hw Flag              : InHw  EntryActive
   Mlist Flags          : None
   Mlist_hndl (Id)       : 0x11891253a60 ( 0xd4 )
   Mlist Urid           : 0x100000000001c662
   Fset Urid (Hash)      : 0x300000000000000d ( d140940b )
   RPF Adjacency ID       : 0xf8004e41
   CPU Credit            : 0
   Total Packets[StatsMode]: 39974417 ( 9999 pps approx.)[ Ingress ]
   OIF Count             : 2
   OIF Details:
     AdjID        Interface        ParentIf       HwFlag     Flags
     0xf8004e41     GigE1/0/5        ---------        ---        A
     0xf8004fe1     Vl13           ---------        ---        F NS
```

show platform software fed switch active oifset l3m hash d140940b

```
Type: group_oif_adj_ids    State:
allocated       MD5:(d140940b21efe182:5b1b540db1ab56f3)
   Fset Urid                : 0x300000000000000d
   Remote Port Count            : 1
   Svi Port Count             : 1
   Users Count               : 2
   Mioitem Count               : 2
     No. AdjID         Interface         PhysicalIf       IfType        Flags
      1. 0xf8004fe1      Vl13           -----------        svi_if        InHw Remote IngressRep
          Urids   => Mio:0x80::11 Parent:0x60::7
child_repl:0x20::1ff7(55e10688)  adj_obj:0x90::12
```

L3 mcid of **d140940b** is linked with L2 mcid of **55e10688**. So, L2m's Fset_urid of 0x20::1ff7 is under child_repl of L3m's fset_urid.

show platform software fed switch active ip igmp snooping groups vlan 13 225.0.0.1

```
 (Vlan: 13, 225.0.0.1)
    Member ports             : 1
       GigE1/0/6
    Dependent Users Count   : 1
    Group Urid             : 0x6000000000000007
    Fset Urid ( Hash )      : 0x2000000000001ff7  ( 55e10688
)
    Layer 3 Stub Entry      : No
    Group Flags            : None
    Gid                  : 16371
```

Epcot_E11_E12_SVL#show platform software fed switch active oifset l2m hash **55e10688**

```
Type: oif_vp_handles       State:
allocated       MD5:(55e1068867b09b3f:f7be09893aa6c061)
  Fset Urid              : 0x2000000000001ff7
  Parent URID            : 0x2000000000001821
  Remote Port Count       : 1
  Users Count            : 2
  VP OIF Count            : 1
   No. VP Interface          Interface              Physical
interface
    1. GigE1/0/6_V13          GigE1/0/6      GigE1/0/6
        flags:[ oif_port  ]    hw_flags: [ InHw  remote ]
```

9350: L2Mcast and L3Mcast scale numbers

Feature	Scale supported
IGMP snooping mroutes	8K
MLD snooping mroutes	4K
IPv4 Multicast routes	8K
IPv6 Multicast routes	4K

9610: L2 Multicast and L3 Multicast scale numbers

Feature	Scale supported
IGMP snooping mroutes	16K
MLD snooping mroutes	16K
IPv4 Multicast routes	32K
IPv6 Multicast routes	32K

Multicast Limitations:

Etherchannel loadbalancing Limitations:
- Currently 9350 and 9610 don't support etherchannel loadbalancing in hardware for BUM traffic.
- Hardware loadbalancing is planned to come in 26.2 release.

V6's group addresses key in lookup is last 32 bits only:
- This limitation is present only in 17.18.1. It's fixed in 17.18.2 for 9350 and 9610 switches. 9600x and 9500x(Gibraltar based) switches still have this issue.

Replication limit in Ingress replication mode:
- While MCID is in ingress replication mode, maximum replication possible for a sub-mcid is 64 for 9610 and 63 for 9350 under one slice.

Non-Routable Multicast packets are punted and reinjected from cpu:
- NRM range(224.0.0.0/24 and ff02::/16) packets are punted and reinjected when SVI is present for Vlan in L2. For PIM Hello, it's punted and reinjected in absence of SVI as well in L2.
- This limits rate to CoPP policer packet is mapped to.

Unsupported features:
- PIM BiDir
- PIM Dense mode

Silicon 1 is The FUTURE

Enterprise
Campus

This session will make you FUTURE aware.

Silicon ONE
Tips & Tricks

(1) Embedded Packet Capture

1. Embedded Packet Capture

17.15.1 Onwards

```
switch# monitor capture <name> ?
  control-plane  Control Plane
  interface      Interface
  ...
```

Packet capture on **Layer3** interface

Packet capture on **Layer2** interface

Packet capture on **Control-Plane**

Main characteristic:

- Direction can be ingress, egress or both
- Protocol can be IPv4, IPv6 or MAC
- Filters are defined by ACL rules.
- Limiter – it gives network administrator additional control by defining different capture parameter like, capture buffer size, buffer type (circular/linear), packet capture rate, sampling interval etc.

EPC L2 Usage Workflow

01 Selecting the L2 interface :

```
switch#show run interface hundredGigE 1/0/1
!
interface HundredGigE1/0/1
 switchport mode trunk

end
```

02 Starting the capture :

```
switch#monitor capture TAC interface HundredGigE 1/0/1 both  match any start
```

03 Stopping the capture :

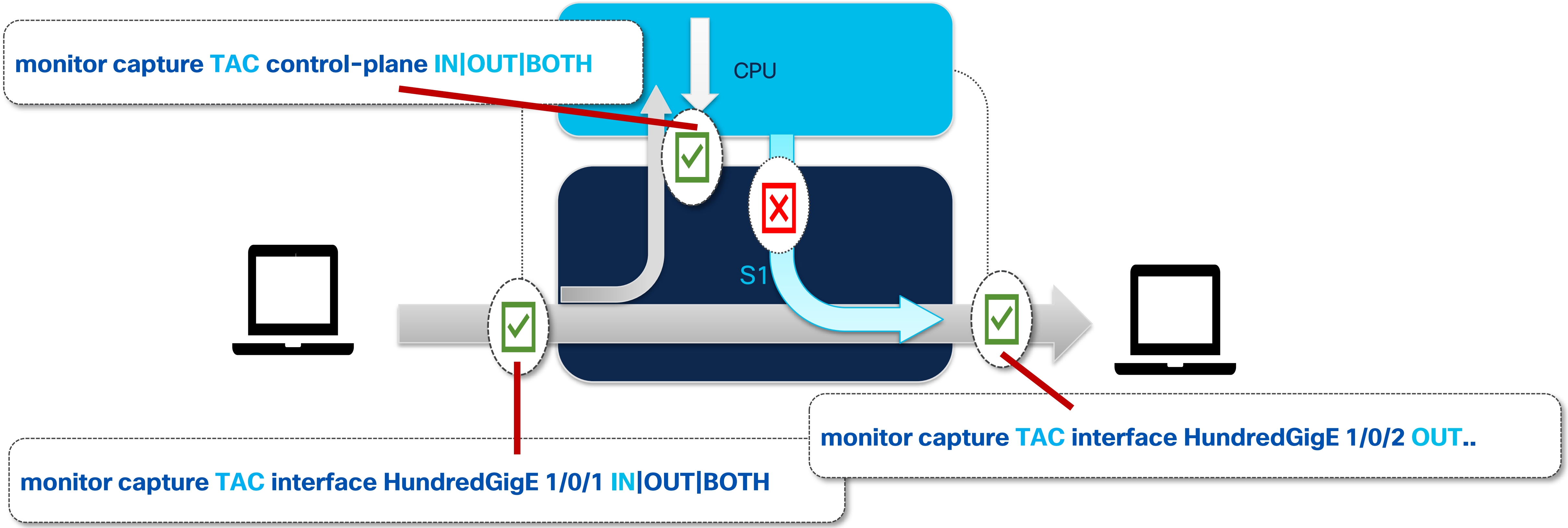
```
switch#monitor capture TAC stop
Capture statistics collected at software:
        Capture duration - 3 seconds
        Packets received - 2
        Packets dropped - 0
        Packets oversized - 0

Number of Bytes dropped at asic not collected

Capture buffer will exists till exported or cleared

Stopped capture point : TAC
```

04 Buffer verification :

```
switch#show monitor capture TAC buffer
Starting the packet display ........ Press Ctrl + Shift + 6 to exit

  1   0.000000 6c:29:d2:9d:36:da -> 09:00:2b:00:00:05 ISIS HELLO 1420 P2P HELLO, System-ID: 0001.0000.0411
  2   0.541957 34:1b:2d:76:fd:02 -> 09:00:2b:00:00:05 ISIS HELLO 1512 P2P HELLO, System-ID: 0100.9800.4003
```

Silicon1 EPC limitation

- EPC doesn't capture **non-IP traffic** injected by the CPU and sent out via an interface in the egress direction (e.g., STP or CDP packets originating from the device cannot be captured).
- However, **IP traffic** can be captured.



EPC Control-Plane Workflow

01 Starting the CP capture:

```
switch#monitor capture TAC control-plane both match  any start
```

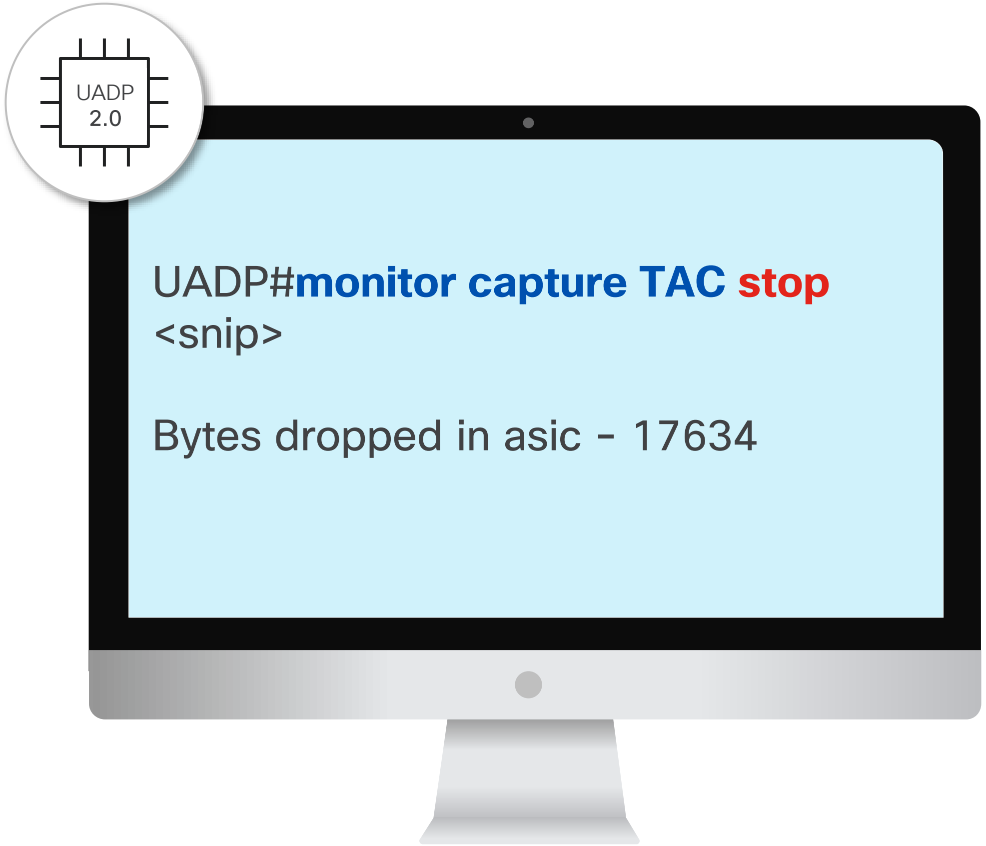
02 Stopping the CP capture:

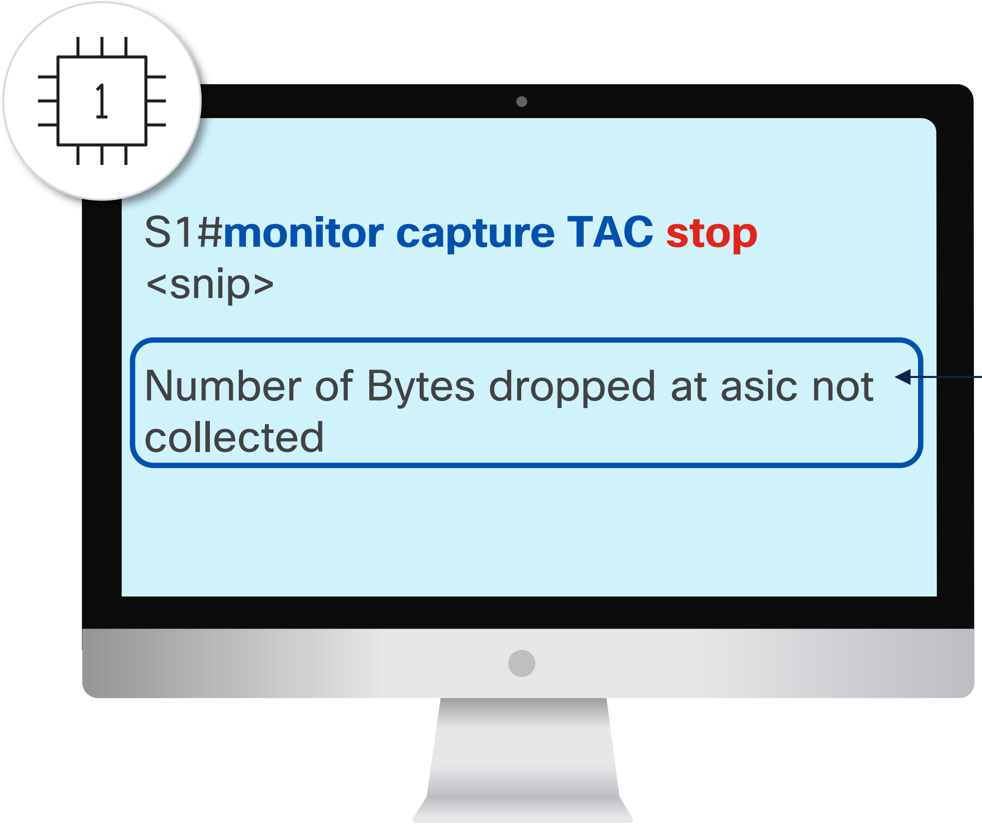
```
switch#monitor capture TAC stop
Capture statistics collected at software:
        Capture duration - 6 seconds
        Packets received - 79
        Packets dropped - 0
        Packets oversized - 0

Number of Bytes dropped at asic not collected

Capture buffer will exists till exported or cleared

Stopped capture point : TAC
```

03 Buffer verification :

```
switch#show monitor capture TAC buffer
Starting the packet display ........ Press Ctrl + Shift + 6 to exit
                                                                    IP Packet
    1   0.000000 192.168.40.166 -> 224.0.0.13    PIMv2 72 Hello
    2   0.152621 6c:29:d2:9d:36:da -> 09:00:2b:00:00:05 ISIS HELLO 1420 P2P
HELLO, System-ID: 0001.0000.0411                                    Non-IP Packet
    3   0.945654 34:1b:2d:76:fc:01 -> 01:00:0c:cc:cc:cd STP 64 RST. Root =
32768/1022/34:1b:2d:76:fc:00
… <snip>
```

Have you captured all packets?

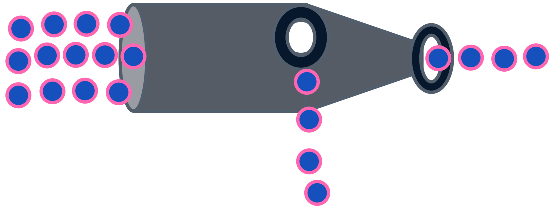


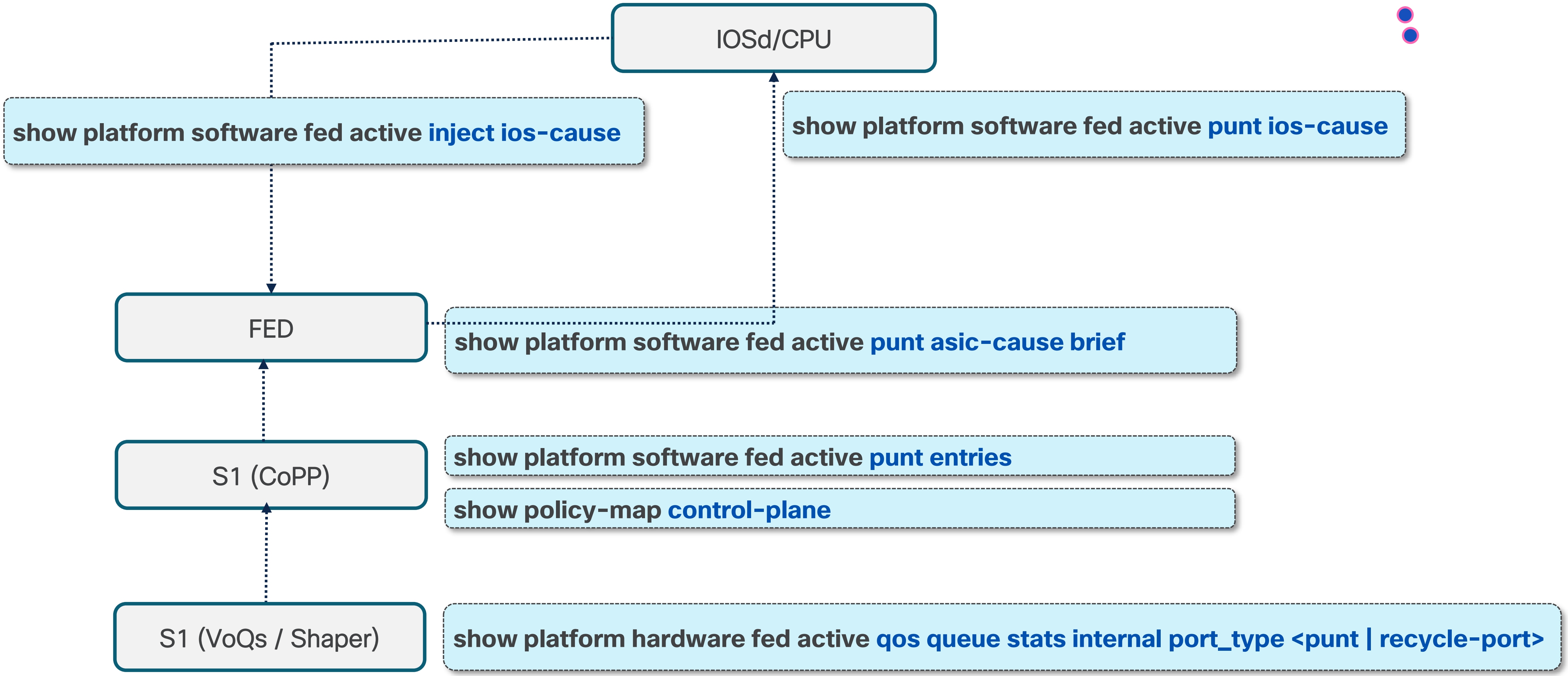


EPC on S1 does not collect hardware drop statistics

(2) Bottlenecks & Drops

CPU/IOS bottlenecks





VoQ & Shaper

 # show platform hardware fed active qos queue stats internal port_type punt

```
=============================================================================================
 VOQ Stats For : Puntport/0 [ 0x1 ]
=============================================================================================
  VOQ ID |          |                Packets                |              Bytes               |
         |----------|--------------------------------------------------------------------------|
         |          |                       VOQ Size In Each Slice                             |
=============================================================================================
         | Asic     |                                  0
         |----------|--------------------------------------------------------------------------|
  0      | Enqueued |                                  0 |                                    0 |
         | Dropped  |                                  0 |                                    0 |
         | Total    |                                  0 |                                    0 |
         |----------|--------------------------------------------------------------------------|
         |   Slice  |          0 |          1 |          2 |          3 |          4 |          5 |
         |          |--------------------------------------------------------------------------|
         |SMS Bytes |          0 |          0 |          0 |          0 |          0 |          0 |
         |HBM Blocks|          0 |          0 |          0 |          0 |          0 |          0 |
         |HBM Bytes |          0 |          0 |          0 |          0 |          0 |          0 |
============================================================================================
         | Asic     |                                  0
         |----------|--------------------------------------------------------------------------|
  1      | Enqueued |                                  0 |                                    0 |
         | Dropped  |                                  0 |                                    0 |
         | Total    |                                  0 |                                    0 |
         |----------|--------------------------------------------------------------------------|
         |   Slice  |          0 |          1 |          2 |          3 |          4 |          5 |
         |          |--------------------------------------------------------------------------|
         |SMS Bytes |          0 |          0 |          0 |          0 |          0 |          0 |
         |HBM Blocks|          0 |          0 |          0 |          0 |          0 |          0 |
         |HBM Bytes |          0 |          0 |          0 |          0 |          0 |          0 |
============================================================================================
<snip>
```

Drops

CoPP

#show policy-map control-plane input class system-cpp-default

```
 Control Plane

  Service-policy input: system-cpp-policy

    Class-map: system-cpp-default (match-any)
      0 packets, 0 bytes
      5 minute offered rate 0000 bps, drop rate 0000 bps
      Match: none
      police:
          rate 2000 pps, burst 11264 packets
          conformed 0 bytes; actions:
          transmit
          exceeded 0 bytes; actions:
          drop
```

Confirmed

Dropped

#show platform software fed active punt entries | i system-cpp-default|Policy

Source	Name	Pri	TC	Policy	CIR-SW	CIR-HW	Pkts(A)	Bytes(A)	Pkts(D)	Bytes(D)
TRAP	ACL Drop(ETH)	1	0	system-cpp-default	2000	1931	0	0	0	0
TRAP	GOLD	0	3	system-cpp-default	2000	1931	0	0	0	0
TRAP	MC snoop lookup miss	2	0	system-cpp-default	0	0	0	0	0	0
TRAP	IPv4 Checksum	3	2	system-cpp-default	2000	1931	0	0	0	0
TRAP	IPV4 UNKNOWN	4	3	system-cpp-default	2000	1931	0	0	0	0
TRAP	ACL Drop(punt to cpu)	1	2	system-cpp-default	2000	1931	0	0	0	0
TRAP	ACL LOG	0	2	system-cpp-default	2000	1931	0	0	0	0
TRAP	RPF(unicast)	2	0	system-cpp-default	0	0	0	0	0	0
TRAP	GLEAN-SUBNET	5	2	system-cpp-default	2000	1931	0	0	0	0
LPTSv4	WCCP IPv4	16	4	system-cpp-default-v4	2000	1931	0	0	0	0
LPTSv4	RSVP IPV4	34	5	system-cpp-default-v4	2000	1931	0	0	0	0

^ This command is a replacement for "internal cpu policer" command available in Doppler platforms ^

FED drops – FED to IOSd

show platform software fed active punt asic-cause brief

ASIC Cause Statistics Brief

```
+------------------------------------------------------------------------------------+
| Source |            Cause        |        Rx            |          Drop            |
|        |                         |    cur    |   delta  |     cur    |    delta    |
+------------------------------------------------------------------------------------+
   UKNWN   UNKNOWN                       646          646          646          646
   INMIR   ARP MIRROR                    386          386            0            0
   INMIR   MCAST *,G or S,G             1171         1171            0            0
    LPTS   ICMP IPv4                      20           20            0            0
    LPTS   OSPF v2 MC                 734034       734034            0            0
    LPTS   OSPF v2 UC                    105          105            0            0
```

switch#**show platform software fed active punt asic-cause brief | i EPC|Source|cur|--**

```
+------------------------------------------------------------------------------------+
| Source |            Cause        |        Rx            |          Drop            |
|        |                         |    cur    |   delta  |     cur    |    delta    |
+------------------------------------------------------------------------------------+
   INMIR   EPC                          24192            0            0            0
   EGMIR   EPC                          14755            0            0            0
```

During EPC
How many packets are getting punted from ASIC/FED to IOS?

IOSd

switch#**show platform software fed active punt ios-cause brief**

```
Statistics for all causes

Cause  Cause Info                       Rcvd                 Dropped
------------------------------------------------------------------------------
0      Reserved                         0                    2265
5      CLNS IS-IS Control               6192285              67197
7      ARP request or response          7828                 91
11     For-us data                      41446629             11083
55     For-us control                   1133900              12952
58     Layer2 bridge domain data pack   706661               8438
75     EPC                              38313                38316
96     Layer2 control protocols         13931094             137840
------------------------------------------------------------------------------
```

- In case of performing EPC on control-plane interface – the "For-us data" Cause will increase and not the "EPC"

EPC packets received to IOS from FED

- It is expected to see a non zero values for this output as in case of capturing Data plane traffic EPC packets punted to IOSd are simply ignored.

(3)Object Statistics

Object Statistics

If the object creation fails, then check the below output.

```
C9600-Silicon1-SVL#show  platform software object-manager switch active f0 statistics
Forwarding Manager Asynchronous Object Manager Statistics

Object update: Pending-issue: 0, Pending-acknowledgement: 0
Batch begin:   Pending-issue: 0, Pending-acknowledgement: 0
Batch end:     Pending-issue: 0, Pending-acknowledgement: 0
Command:       Pending-acknowledgement: 0
Total-objects: 1182
Stale-objects: 0
Resolve-objects: 0
Childless-delete-objects: 1
Backplane-objects: 0
Error-objects: 0
Number of bundles: 129
Paused-types: 8
```

(4)Enabling Wireshark App

Wireshark app

State of the Wireshark App:

#show monitor capture

```
Status Information for Capture TAC
  Target Type:
 Interface: HundredGigE1/0/1, Direction: BOTH
   Status : Inactive
  Filter Details:
    Capture all packets
  Buffer Details:
   Buffer Type: LINEAR (default)
   Buffer Size (in MB): 10
  File Details:
   File not associated
  Limit Details:
   Number of Packets to capture: 10
   Packet Capture duration: 0 (no limit)
   Packet Size to capture: 0 (no limit)
   Maximum number of packets to capture per second: 1
   Packet sampling rate: 0 (no sampling)
```

Configured parameters

#show monitor capture TAC parameter

```
   monitor capture TAC interface HundredGigE1/0/1 BOTH
   monitor capture TAC match any
   monitor capture TAC buffer size 10
   monitor capture TAC limit packets 10 pps 1
```

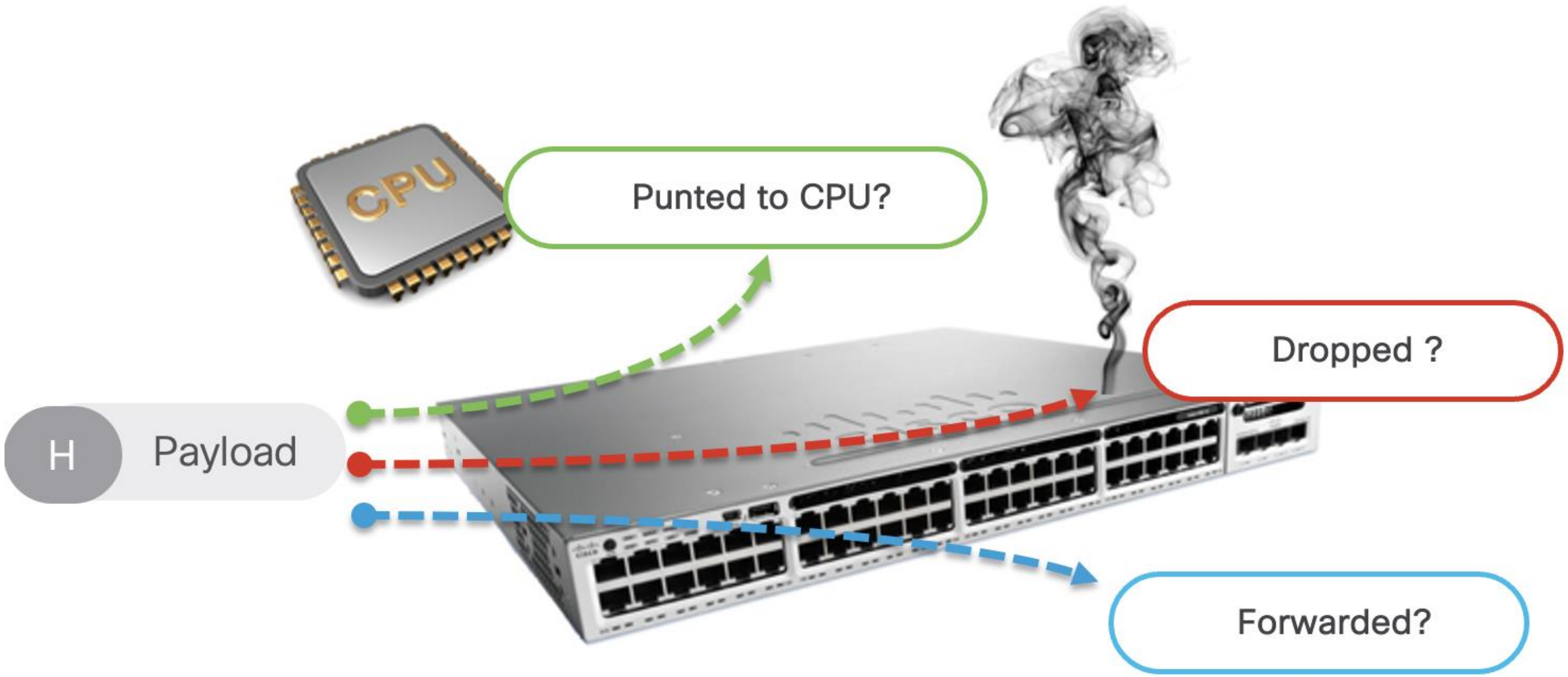
Restarting the Wireshark process:

#test platform software wireshark reset

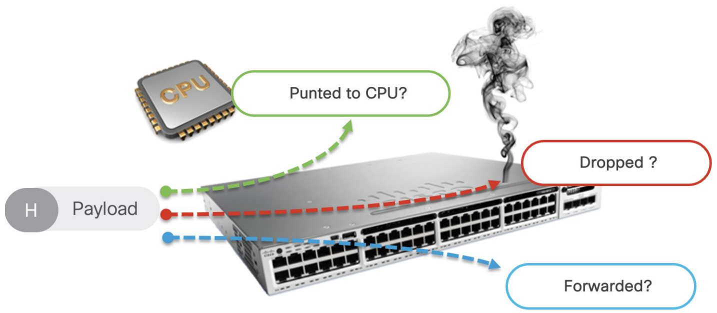
```
PI reset in liaison success
PD reset in Fed success
```

(4)Packet Simulation (SPF)

Why Packet Simulation?



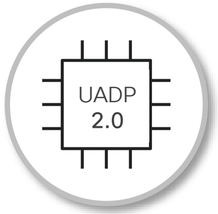
Why Packet Simulation?



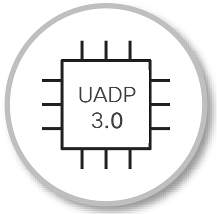
CPU Simulations

SPF: `#show platform hardware fed active forward ...`

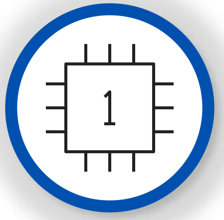
New feature



UADP 2.0
Catalyst 9200
Catalyst 9300
Catalyst 9400
Catalyst 9500

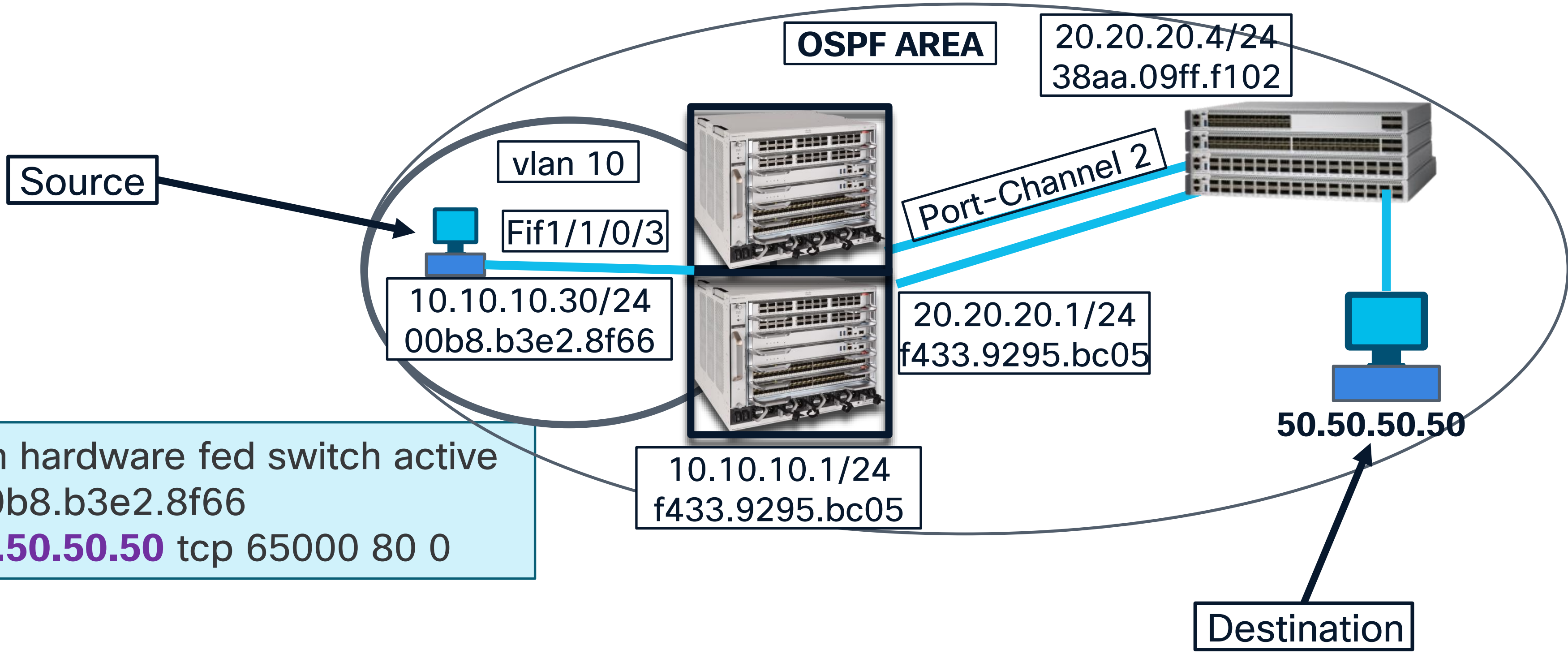


UADP 3.0
Catalyst 9600
Catalyst 9500H
Catalyst 9400X



Silicon ONE
Catalyst 9500X
Catalyst 9600X

SPF Workflow



01 Manual header

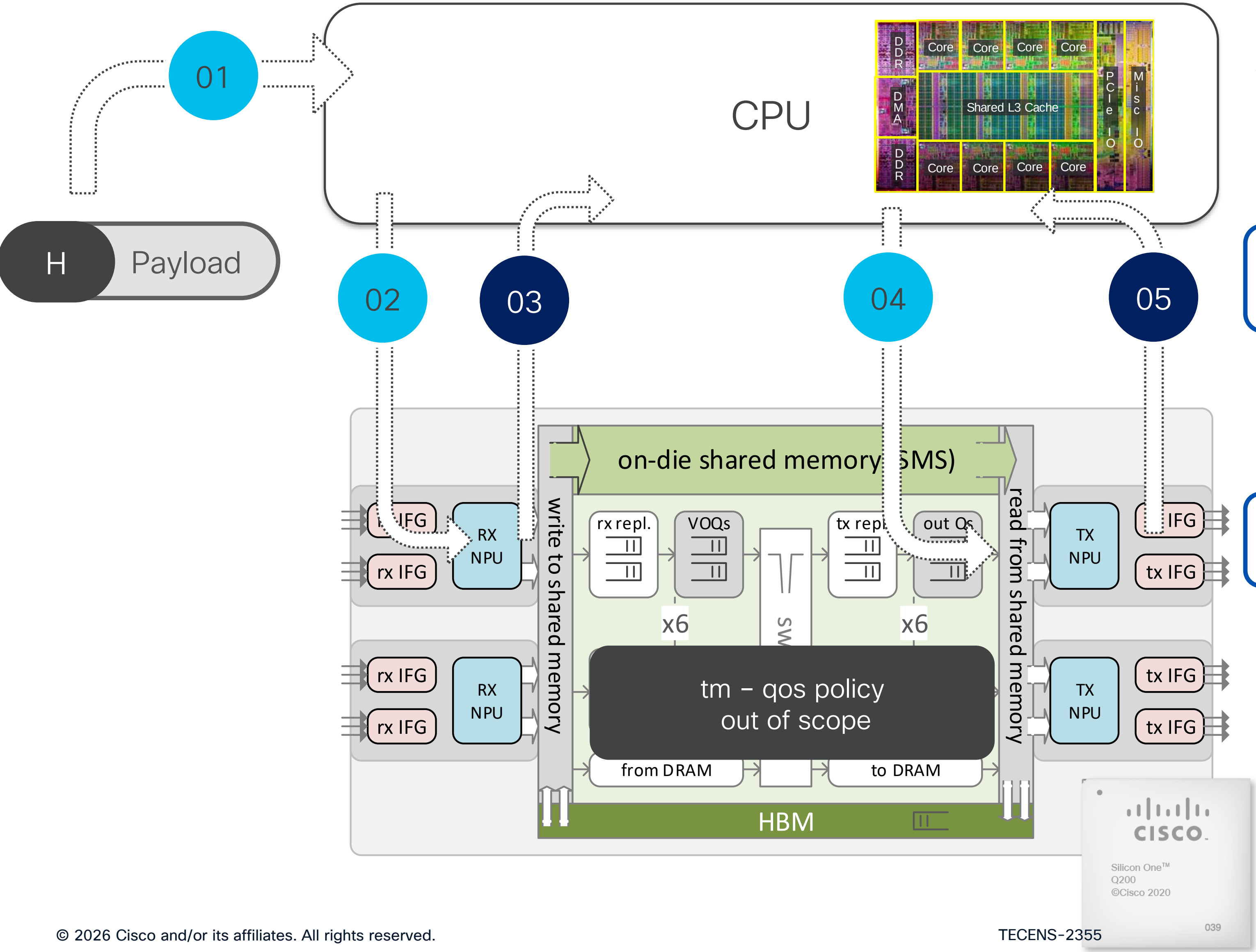
```
C9600-Silicon1-SVL# show  platform hardware fed switch active
forward interface fiftyGigE 1/1/0/3 00b8.b3e2.8f66
f433.9295.bc05 ipv4 10.10.10.30 50.50.50.50 tcp 65000 80 0
```

02 PCAP input

```
Switch#show platform hardware  fed active forward interface fiftyGigE
1/1/0/3 pcap flash:capture.pcap number 2 data
```

The SPF simulation on the S1 platform is almost **instantaneous** and **does not require significant waiting time**, unlike on the UADP-based platforms.

SPF output options



Show Platform Forward on S1:

In the default outputs, only stages 1, 3, and 5 are presented.

show platform hardware fed active forward last summary

In the detailed view, we can see full insights into all five stages, which is mainly relevant for the Cisco team (TAC/Engineering Team)

show platform hardware fed active forward last detail

SPF Command Output

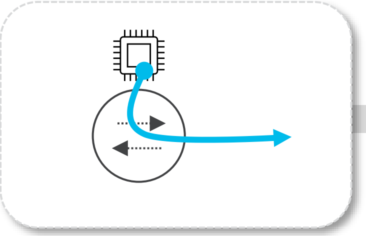
show platform hardware fed switch active forward interface fiftyGigE 1/1/0/3 00b8.b3e2.8f66 f433.9295.bc05 ipv4 10.10.10.30 50.50.50.50 tcp 65000 80 0

```
Packet-trace Id: spf1203511907
  Input packet:
    ###[ Ethernet ]###
     dst      = f4:33:92:95:bc:05
     src      = 00:b8:b3:e2:8f:66
     type     = IPv4
    ###[ IP ]###
       version   = 4
        ~
        ~
       ttl       = 64
       proto     = tcp
       chksum    = 0x244
       src       = 10.10.10.30
       dst       = 50.50.50.50
       \options   \
    ###[ TCP ]###
         sport     = 65000
         dport     = http
~
RX (Ingress):
  Ingress Interface : FiftyGigE1/1/0/3
```

```
Decision:
     #
     # dsp: FiftyGigE1/1/0/1
     # fwd_hdr_type: ipv4
     ~
TX (Egress):
~
hierarchical view:
~
        # destination_sp: FiftyGigE1/1/0/1
~
        ###[ Ethernet ]###
        dst      = 38:aa:09:ff:f1:02
        src      = f4:33:92:95:bc:05
        type     = IPv4
        ###[ IP ]###
~
          proto    = tcp
            chksum    = 0x344
          src      = 10.10.10.30
          dst      = 50.50.50.50
            \options   \
        ###[ TCP ]###
~
```

(5)CPU Captures

CPU punt packet capture



01

Start the packet capture

```
C9600-Silicon1-SVL#debug platform software fed switch active punt packet-capture start
Punt packet capturing started.
```

02

Stop the packet capture

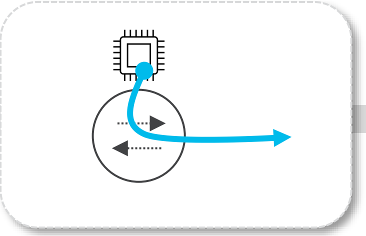
```
C9600-Silicon1-SVL#debug platform software fed switch active punt packet-capture stop
Punt packet capturing stopped. Captured 18 packet(s)
```

CPU punt packet capture

03 Display the packet capture

```
C9600-Silicon1-SVL#show  platform software fed switch active punt packet-capture brief
Punt packet capturing: disabled. Buffer wrapping: disabled
Total captured so far :    18 packet(s)
Capture capacity       :  4096 packet(s)
Max. Meta header size :    88 byte(s)
Max. Packet data size :   128 byte(s)

------ Punt Packet Number: 1, Timestamp: 2026/01/25 20:31:11.580 ------
 interface : phy: Port-channel1 [if-id: 0x00000494], pal: Port-channel1 [if-id: 0x00000494]
 misc info : cause: 96 [Layer2 control protocols], sub-cause: 0
[CPP_PUNT_SUBCAUSE_TRANSPORT_NONE], linktype: LAYER2 [10]
 CE    hdr : dest mac: f4bd.9eff.fff0, src mac: f4bd.9eff.fff1, ethertype: 0x7106
 meta  hdr : Nxt. Hdr: 0x1, Fwd. Hdr: 0, SSP: 0x46e
 meta  hdr : DSP: 0xffff, SLP: 0x80083, DLP: 0x5
 ether hdr : dest mac: 0100.0ccc.cccd, src mac: 4c5d.3ccd.dda4
 ether hdr : length: 50
~
~
~
```

CPU inject packet capture



01

Start the packet capture

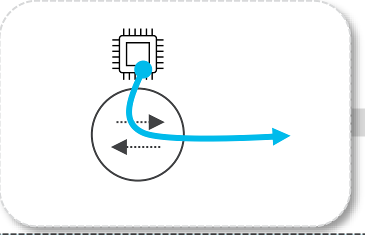
#debug platform software fed active inject packet-capture start
Inject packet capturing started.

02

Stop the packet capture

#debug platform software fed active inject packet-capture stop
Inject packet capturing stopped. Captured 30 packet(s)

CPU inject packet capture



03 Display the packet capture

```
C9600-Silicon1-SVL#show  platform software fed switch active inject packet-capture brief
Inject packet capturing: disabled. Buffer wrapping: disabled
Total captured so far :     10 packet(s)
Capture capacity       :  4096 packet(s)
Max. Meta header size :     88 byte(s)
Max. Packet data size :   128 byte(s)

------ Inject Packet Number: 1, Timestamp: 2026/01/21 15:31:38.434 ------
 interface : phy: FiftyGigE1/1/0/3 [if-id: 0x0000040b], pal: FiftyGigE1/1/0/3 [if-id: 0x0000040b]
 misc info : cause: 1 [L2 control/legacy], sub-cause: 0 [CPP_PI_SUB_CAUSE_NONE], linktype: IP TUNNEL [5]
 CE    hdr : dest mac: 4e41.5000.0111, src mac: 4e41.5000.0010, ethertype: 0X8100
 CE    hdr : vlan: 0x406, ethertype: 0x7103
 meta  hdr : Type: 0 (Inject Down), Encap: 0, Dest. Type: 0xc0
 meta  hdr : Dest. Value: 0x2c, L3 DLP: 0, Down NH: 0
 ether hdr : dest mac: f003.bce9.3402, src mac: f003.bce9.3402
 ether hdr : ethertype: 0x9000
~
```

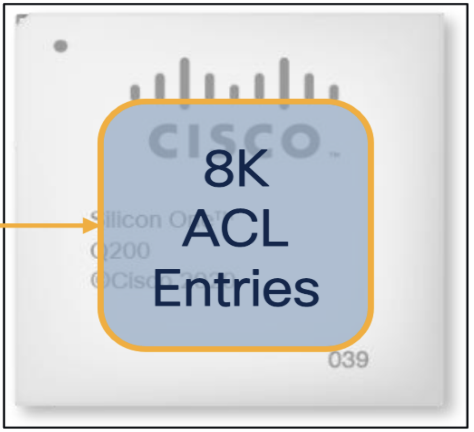
(6)Access List

Access-Lists on Silicon One Q200

Traditional IP ACL

```
ip access-list extended TestACL
 permit ip host 10.5.64.140 any
 permit ip host 10.5.64.86 any
 permit any host 10.5.75.124
 permit ip 10.10.170.32 255.255.255.224 any
 permit ip 10.179.252.0 255.255.252.0 any
 permit 10.5.200.240 255.255.255.240 any
 permit ip any 10.53.199.0 255.255.255.0
 permit ip any 10.56.100.224 255.255.255.240
... Truncated (total 10,000 entries)
 permit host 10.56.223.56
```

```
interface Vlan100
 ip access-group TestACL in
```



TCAM exhausted

permit/deny operations are programmed into the ACL space

10,000 entries programmed in an 8,000 entry TCAM space

ACL - Optimizing the hardware resources

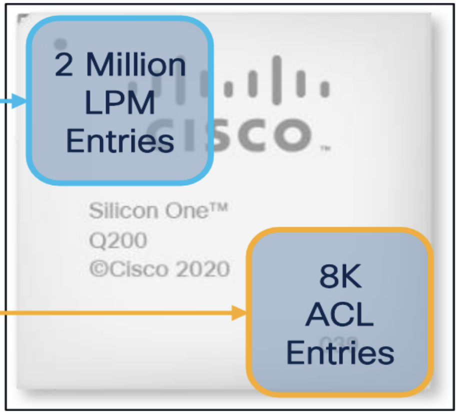
Optimized space utilization using OG-ACL/SG-ACLs*

```
object-group network TestGroup
 host 10.5.64.140
 host 10.5.64.86
 host 10.5.75.124
 10.10.170.32 255.255.255.224
 10.179.252.0 255.255.252.0
 10.5.200.240 255.255.255.240
 10.53.199.0 255.255.255.0
 10.56.100.224 255.255.255.240
... Truncated (10,000 entries)
 host 10.56.223.56
```

```
ip access-list extended SampleACL
 permit ip object-group TestGroup any
 deny ip any any (implicit)
!
```

```
interface Vlan100
 ip access-group SampleACL in
```

Contents of the Object-groups are expanded to the Longest Prefix Match space.



10K entries consumed

2 entries consumed

Comparison with UADP

UADP expands both ACL as well as Object-groups to the ACL storage space.

Entries in object-groups use LPM space instead of TCAM space

(7) TCL Script

Use of TCL Script for addition of file to flash:

Syntax

```
tclsh
puts [open "flash:COMMANDS_TO_EXECUTE.txt" w+] {

Enter Commands to Execute

}

exit
```

Configuration Example

```
C9600-Silicon1-SVL#
C9600-Silicon1-SVL#tclsh
C9600-Silicon1-SVL(tcl)#puts [open "flash:COMMANDS_TO_EXECUTE.txt" w+] {
+>configure terminal
+>interface fiftyGigE 1/1/0/3
+>shutdown
+>end
+>show  run interface fiftyGigE 1/1/0/3
+>}
C9600-Silicon1-SVL(tcl)#
C9600-Silicon1-SVL(tcl)#exit
C9600-Silicon1-SVL#
C9600-Silicon1-SVL#dir
Directory of bootflash:/

357117  -rw-            101  Jan 25 2026 21:37:09 +05:30  COMMANDS_TO_EXECUTE.txt
~
```

(8)Power of EEM Script

Power of EEM (Embedded Event Manager) Script

COMMANDS_TO_EXECUTE.txt file

```
configure terminal
interface fiftyGigE 1/1/0/3
shutdown
end
show  run interface fiftyGigE 1/1/0/3
```

EEM Script

```
event manager applet Cisco_Live authorization bypass
event none maxrun 1000
action 010 cli command "enable"
action 011 cli command "term exec prompt timestamp"
action 012 cli command "term exec prompt expand"
action 020 file open fileout1 flash:OUTPUT.txt a+
action 021 file open filein1 flash:COMMANDS_TO_EXECUTE.txt r
action 022 file read filein1 input
action 030 foreach line $input "\n"
action 031 cli command "$line"
action 032 file write fileout1 "$_cli_result"
action 033 end
action 034 file close fileout1
action 045 file close filein1
```

Power of EEM Script

```
C9600-Silicon1-SVL#event manager run Cisco_Live
```

```
C9600-Silicon1-SVL#more flash:OUTPUT.txt

C9600-Silicon1-SVL#
Enter configuration commands, one per line.  End with CNTL/Z.
C9600-Silicon1-SVL(config)#
C9600-Silicon1-SVL(config-if)#
C9600-Silicon1-SVL(config-if)#
C9600-Silicon1-SVL#Load for five secs: 0%/0%; one minute: 1%; five minutes: 1%
No time source, *22:01:35.406 IST Sun Jan 25 2026
------------------ show running-config interface FiftyGigE1/1/0/3  ------------------

Building configuration...

Current configuration : 127 bytes
!
interface FiftyGigE1/1/0/3
 description To-Host3-Ten 1/1/3
 switchport access vlan 10
 switchport mode access
 shutdown
end

C9600-Silicon1-SVL#
```

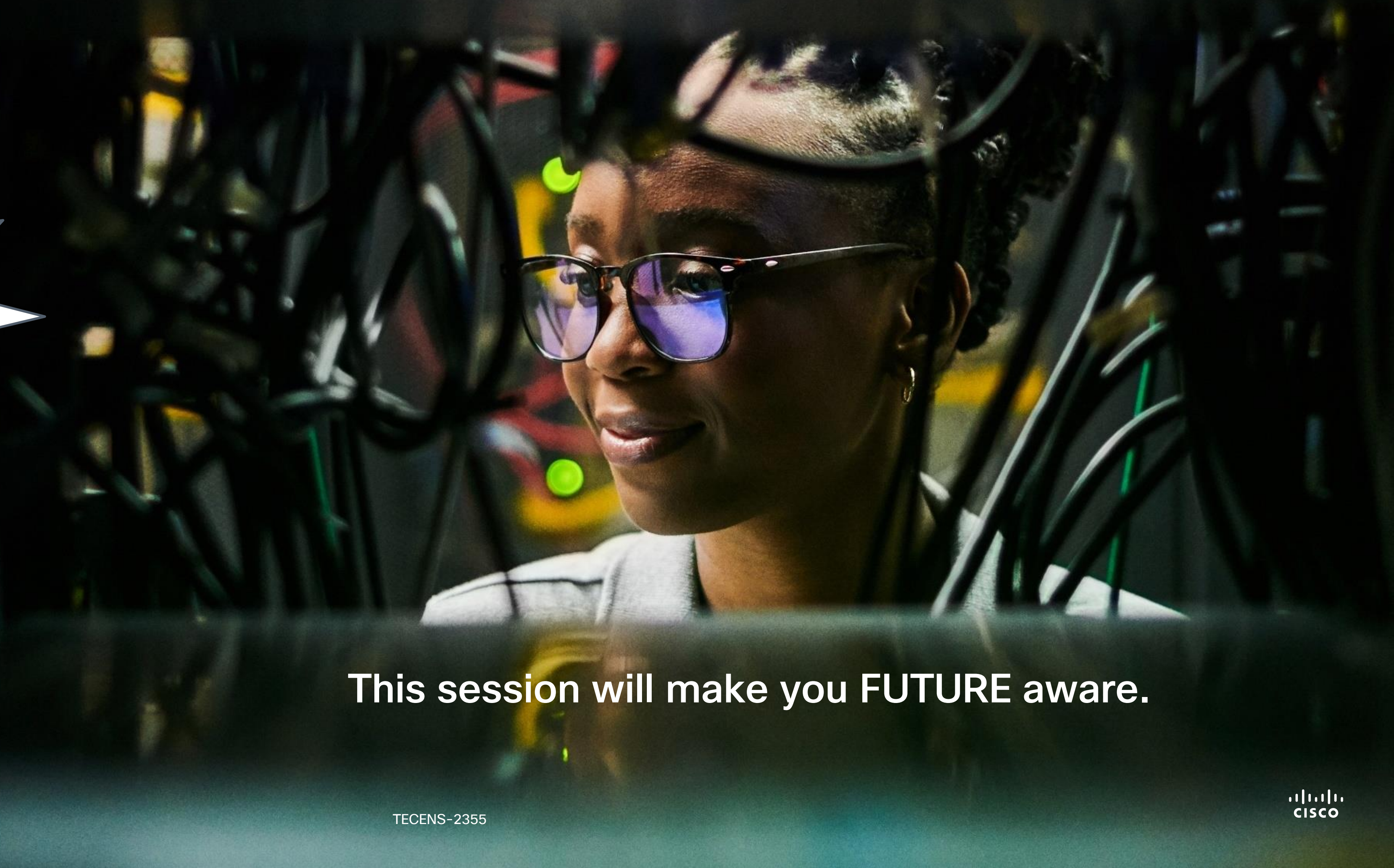
Quiz

What do you mean by punt and inject?

What is the command to collect platform traces on Cisco Catalyst 9000 series switches?

Which command provides all the outputs starting from IOSD to SDK with correlations present w.r.t Layer 3 IPv4 multicast:-

Silicon 1 is The FUTURE



Enterprise Campus

This session will make you FUTURE aware.

Takeaways

Switching Trends & need of New ASIC

High Level Doppler & Polaris Architecture

Doppler L2 and L3 forwarding

Silicon ONE Introduction & Architecture

Silicon ONE L2 and L3 forwarding

Silicon ONE Multicast forwarding

Silicon ONE Tips & Tricks

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- Lieutenant Commander Data

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```
TECENS-2355@survey:~$ ping "score == 5.0"
!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
--- "session score 5.0" ping statistics ---
66 packets transmitted, 66 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 1/1/1 ms

TECENS-2355@survey:~$ ping "score < 5.0"
............................................................
--- "session score < 5.0" ping statistics ---
66 packets transmitted, 0 packets received, 100.0% packet loss
```

Thank you

CISCO Live!